

Laboratory manuals

WHO
laboratory
manual for
typhoid and
paratyphoid
(enteric
fever)



World Health
Organization

WHO laboratory manual for typhoid and paratyphoid (enteric fever)



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Foreword

Enteric fever, a systemic infection caused by *Salmonella* Typhi, Paratyphi A, Paratyphi B or Paratyphi C, is endemic in many low- and middle-income countries. These infections are also not uncommon among returning travellers, particularly those spending time in remote areas or visiting friends or relatives. Clinically, enteric fever is virtually impossible to differentiate from other acute febrile illnesses such as dengue and malaria, among others. As such, accurate laboratory diagnosis is essential to identify the aetiologic agents causing acute febrile illness.

Historically, enteric fever (typhoid in particular) was routinely susceptible to ampicillin, chloramphenicol, and co-trimoxazole. However, resistance to these first-line antimicrobial agents is now common and resistance to second- and third-line drugs (fluoroquinolones, third-generation cephalosporins, and azithromycin) is emerging. Inappropriate therapy may result in poor clinical outcomes and foster antimicrobial resistance. The emergence of drug resistance further stresses the need for timely and accurate surveillance of these pathogens.

Control of typhoid fever relies on accurate and timely diagnosis and treatment; water, sanitation and hygiene (WASH) interventions and vaccination. Laboratory based surveillance is the cornerstone of these control measures. Laboratory data provides data that directly impacts individual patients and contributes to local, national, and global surveillance data, which can be used to guide WASH interventions and vaccination campaigns.

However, for this data to be meaningful, diagnosis of enteric fever based on laboratory results must be accurate. The following manual outlines methods for the isolation, identification, and susceptibility testing of *S. Typhi*, Paratyphi A, Paratyphi B, and Paratyphi C. It is written for a global audience and includes a spectrum of methods ranging from simple yet effective conventional methods that may be implemented in a field setting, to advanced molecular techniques appropriate for a reference centre.

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Acronyms and abbreviations

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ACDP	Advisory Committee on Dangerous Pathogens
AST	Antimicrobial susceptibility testing
BA	Blood agar
BHI	Brain Heart Infusion
CA	Chocolate agar
CDC	Centers for Disease Control and Prevention
CFR	Case fatality ratio
CFU	Colony forming unit
CGE	Center for Genomic Epidemiology
CHRF	Child Health Research Foundation
CLSI	Clinical and Laboratory Standards Institute
COSHH	Control of Substances Hazardous to Health
DE	Dorset Egg
E-Test	Epsilometer Test
EQA	External Quality Assessment
EUCAST	European Committee on Antimicrobial Susceptibility Testing
FTP	File transfer protocol
GN	Gram-negative
HBA	Horse Blood Agar
HE	Hektoen enteric
IQA	Internal Quality Assessment
IQC	Internal Quality Control
KIA	Kligler's Iron Agar
LIA	Lysine Iron agar
LMIC	Low- and middle-income countries
LQA	Laboratory Quality Assurance
LQM	Laboratory quality manual
LSO	Laboratory Safety Officer
MAC	MacConkey agar
MALDI-TOF	Mass Spectroscopy Laser Desorption Ionization Top of Flight

Acronyms and abbreviations

MHA	Mueller-Hinton agar
MHB	Mueller-Hinton broth
MIC	Minimum inhibitory concentration
MIO	Motility-Indol-Ornithine
MLST	Multi-locus sequence type
MS	Mass spectrometry
MSC	Microbiological safety cabinet
MSDS	Material Safety Data Sheets
NA	Nutrient agar
NCTC	National Collection of Type Cultures
NTC	Non-template control
NTS	Non-typhoidal <i>Salmonella</i>
PCR	Polymerase chain reaction
PPE	Personal protective equipment
QA	Quality assurance
QC	Quality controls
QMS	Quality management system
RCA	Root cause analysis
SBA	Sheep blood agar
SEAP	Surveillance of Enteric Fever in Asia Project
SNP	Single nucleotide polymorphism
SOP	Standard Operating Procedures
SPS	Sodium polyanethol sulfonate
ST	Sequence type
TMPD	Tetra-methyl-p-phenylenediamine dihydrochloride
TSA	Trypticase soy agar
TSB	Trypticase soy broth
TSI	Triple sugar iron
TTSS	Two genes encoding type III secretion system
UKHSA	United Kingdom Health Security Agency
WASH	Water, sanitation and hygiene
WGS	Whole-genome sequencing
WHO	World Health Organization
XLD	Xylose-Lysine-Deoxycholate



Chapter 1

Introduction

1.1

Aims and objectives

The aims and objectives of this manual are to describe the microbiological procedures to isolate, identify and confirm the causative agents of enteric fever (*Salmonella enterica* subspecies *enterica* serovar Typhi [*S. Typhi*], *S. Paratyphi A*, *S. Paratyphi B* and *S. Paratyphi C*). This laboratory manual will assist the laboratory worker in the correct procedures to diagnose acute enteric fever cases, monitor for convalescent shedding, and detect asymptomatic carriage. The manual is intended for global use; therefore, methodologies which may not be applicable in low-income countries will be included with the aim of promoting and developing laboratory technologies.

1.2

Microbiology and clinical aspects of enteric fever

Enteric fever is the collective term for typhoid and paratyphoid fevers. Typhoid fever is caused by *S. Typhi*, and paratyphoid fever is caused by *S. Paratyphi A*, *S. Paratyphi B* or rarely *S. Paratyphi C*; these are referred to collectively as typhoidal serovars. Typhoidal serovars only infect humans and higher primates, and enteric fever is a human-restricted disease. Paratyphoid fever is clinically indistinguishable from typhoid fever, and the approach to the diagnosis and clinical management is identical (1,2).

1. Transmission

Transmission occurs by the faecal-oral route, being vehicle-borne through contaminated water or food. There are two predominant patterns of transmission (2,3):

- Short cycle: contamination of food and water in the immediate environment with bacteria shed by carriers, through inadequate hygiene and sanitation measures
- Long cycle: contamination of the broader environment, such as contamination of water supplies by sewage, inadequate treatment of piped water or use of human faeces or untreated sewage as a crop fertilizer.

2. Incubation period

The average incubation period of enteric fever is 9 to 21 days (4), but can range from 3 to 60 days (2).

3. Clinical features and complications

The symptoms and clinical signs of enteric fever are nonspecific and overlap considerably with other causes of acute febrile illness (5). Clinical misdiagnosis is common in low- and middle-income countries (LMIC), particularly in areas endemic for malaria and dengue (6).

Fever is the cardinal symptom and is present in almost all cases (97-100%). Fever is the first indication of illness, rising over a few days (7). Up to 44% of cases experience continuous fever, and a 'stepladder' pattern of fever (characterized by a step-wise increase in fever reaching a peak in the second week of disease) occurs in up to 25% of cases (8,9).

Other clinical features vary in frequency depending on age group. Common generalized systemic features in adults include headache (63%), rigour and chills (47%) and relative bradycardia (47%). An erythematous rash or rose spots occur in $\leq 7\%$ of cases in LMICs, but are more commonly reported in cases in high-income countries (up to 19%) (8). Nausea or vomiting occur in $\sim 50\%$ of all cases, and diarrhoea is more common than constipation, reported in 46% and 24% of cases, respectively. Other gastrointestinal features include abdominal pain or tenderness (54%), splenomegaly (40%) and hepatomegaly (30%). A dry cough is reported in up to 41% of cases. Laboratory features include anaemia in $\sim 30\%$ and thrombocytopenia in 53% of cases, and leukopenia (36%) occurring more commonly than leucocytosis (13%). Up to 28% of cases have evidence of hepatitis.

In children, relative bradycardia occurs less frequently than in adults (11-30%) and is more often observed in African children than children in Asia. Gastrointestinal features are variable and influenced by age: diarrhoea is >2.5 times more common in infants than older children and adults; in children aged 5-10 years, abdominal pain (70%) and nausea or vomiting (45%) are common symptoms and seen more frequently than in children younger than 5 years of age (20% and 32% respectively). Hepatomegaly (44-85%) and splenomegaly (26-90%) are often noted. Respiratory symptoms may dominate the clinical picture with a cough seen in 30-72% of cases. Seizures are more common in children than adults; febrile seizures are seen in susceptible children aged 6 months to 5 years. Anaemia is common among children of all ages ($\sim 70\%$), while thrombocytopenia occurs in $<20\%$ of cases. Children in Africa are at 5- and 15-times greater risk of severe anaemia and thrombocytopenia, respectively, than their counterparts in Asia. Eosinopenia is present in up to 70% of cases. In children of 5-10 years of age, leukocytosis is more common than leukopenia (47% versus 21%), but in children older than 10 years of age leukopenia is more common, as is seen with adults (9). Hepatitis is more common in children younger than 5 years of age (36%) than in older children (26%).

Untreated typhoid fever can lead to severe, life-threatening complications which typically occur after 2-4 weeks of illness. Among hospitalized typhoid fever cases, the prevalence of complications is estimated at 27% for children and 17% for adults. Children with a prolonged duration (≥ 10 days) of symptoms before hospitalization are at three times higher risk of developing complications than those with symptoms for <10 days. Complications reported most frequently include encephalopathy, intestinal perforation, gastrointestinal bleeding and nephritis (10).

Before the advent of antibiotics and in untreated typhoid fever cases, the case fatality ratio (CFR) was historically reported as 10-20% (2). The estimated CFR in endemic countries is $\sim 2.5\%$ but increases to $\sim 4.5\%$ for hospitalized patients. Higher CFR have been reported for antimicrobial resistant enteric fever, at $\sim 6.8\%$ (11). Of all enteric fever complications, intestinal perforation carries the highest CFR of 15% on average, although sub-Saharan African countries tend to report higher CFRs (typically 20-30%) (12-15). The risk of mortality is four times higher in children younger than 5 years of age than in children older than 5 years of age (9).

4. Carriage:

Following acute enteric fever or subclinical infection, several outcomes of the host-pathogen interaction are possible. Clearance occurs when the pathogen is effectively eradicated. Colonization of one or more niches occurs if the pathogen is not cleared; this may be transient or prolonged. Asymptomatic colonization (carriage) results in faecal shedding of pathogens, which can be transmitted to others (16). About 10% of persons who have had an episode of typhoid fever can intermittently shed bacteria in faeces for several weeks after infection and up to 4% of those infected with *S. Typhi* become chronic carriers (16). Carriers serve as biological reservoirs and have important implications for public health. Asymptomatic carriage of *S. Paratyphi* A and *S. Paratyphi* B has also been described, with similar rates of carriage as patients infected with *S. Typhi* (16,17).

Carriers have been defined as follows (18):

1. Presumptive carrier: Evidence of shedding of typhoidal serovars (positive stool culture or polymerase chain reaction [PCR]) of an unknown duration.
2. Definitive carrier:
 - a. Evidence of shedding of typhoidal serovars (positive stool culture or PCR) at least 12 months after finishing an appropriate course of antimicrobial treatment and the resolution of symptoms following a laboratory-confirmed episode of acute disease

OR

- b. Two positive stool samples 12 months apart.
3. Convalescent carrier: Evidence of shedding of typhoidal serovars (positive stool culture or PCR) 1-12 months after finishing an appropriate course of antimicrobial treatment and the resolution of symptoms following a laboratory-confirmed episode of acute disease.

For a more recent explanation on carrier's please refer to Chattaway et al. 2021 (17).

1.3

Case definitions and surveillance of enteric fever

Surveillance requires microbiological culture facilities as enteric fever is largely indistinguishable from other acute febrile illnesses, making syndrome surveillance unhelpful (19).

The objectives of surveillance for enteric fever are to (17,18):

1. Determine the epidemiology and disease burden, to facilitate and support control strategies
2. Facilitate the rapid detection of outbreaks and response to outbreaks
3. Guide vaccination programmes and other control strategies in a country, as there is significant heterogeneity in disease burden across geographical areas and populations
4. Monitor the impact of vaccination on disease and potential changes in epidemiology
5. Evaluate other (non-vaccine) prevention and control measures
6. Monitor antimicrobial resistance patterns, which can inform treatment practices and, in some instances, the need for vaccination campaigns
7. In non-endemic settings, identify imported cases among returned travellers or migrants (which can provide indirect measures of risk in countries visited) and support pre-travel vaccine advice or contact tracing as needed.

Detailed Surveillance Standards for typhoid and other invasive Salmonellosis have been published under the umbrella of World Health Organization Vaccine Preventable Diseases Surveillance Standards (18).

The standardized case definitions are as follows:

1. Suspected case of typhoid or paratyphoid fever for case finding:
 - a. Fever for at least three out of seven consecutive days in an endemic area or following travel from an endemic area

OR

- b. Fever for at least three out of seven consecutive days within 28 days of being in household contact with a confirmed case of typhoid or paratyphoid fever

2. Confirmed case:

- a. Typhoid fever: laboratory confirmation by culture or molecular methods of *S. Typhi* or detection of *S. Typhi* DNA from a normally sterile site
 - b. Paratyphoid fever: Laboratory confirmation by culture or molecular methods of *S. Paratyphi* A, B, or C or detection of *S. Paratyphi* A, B, or C DNA from a normally sterile site
3. Relapse of typhoid or paratyphoid fever: Laboratory confirmation of *S. Typhi* or *S. Paratyphi* from a normally sterile site within one month of completing an appropriate course of antimicrobial treatment and resolution of symptoms.

Due to the difficulties in culturing typhoidal serovars from environmentally sourced samples, there is not enough evidence to currently recommend environmental sampling of water on a routine basis to test for *S. Typhi* and *S. Paratyphi* (over and above routine water testing, which may already be taking place to ensure basic water standards and requirements are met). Environmental sampling can, however, be part of a response to outbreaks (18).

1.4

Epidemiology of enteric fever

S. Typhi emerged about 10,000–43,000 years ago, before agriculture and domestication, and has been a companion within human populations ever since (20,21). Asymptomatic carriers act as reservoirs and facilitate the continued persistence of typhoidal serovars in populations resident in endemic areas, where unsafe water and inadequate sanitation facilities and poor hygiene drive ongoing transmission.

Accurate burden of disease data for enteric fever remains elusive because systematic incidence data are lacking for many LMICs where the disease is endemic. A lack of country- and regional-level surveillance data is one of the main challenges for public health decision-makers (19).

The burden of disease estimates are calculated using data from community- and population-based surveillance programmes and incidence studies, accounting for blood culture insensitivity (3,22). Recently, the global burden of enteric fever was analysed as part of the Global Burden of Disease Study conducted by the Institute for Health Metrics and Evaluation (22). Several recent landmark multinational studies have also provided valuable data on enteric fever incidence and trends across sub-Saharan Africa and Asia (23–25).

Enteric fever has virtually been eliminated from high-income countries as a result of improved water and sanitation infrastructure and public health measures (including improved hygiene and food safety, health education, appropriate antibiotic treatment, and control of asymptomatic carriers) (22,26,27). However, it remains endemic in many LMICs across Asia, Africa and Oceania, where it is a major cause of disability and death (22).

Enteric fever incidence has decreased worldwide over the past few decades (3,28). From 1990 to 2017, fewer cases occurred in all regions and global cases decreased by an estimated 44.6% (22). *S. Typhi* is the most common cause of enteric fever, accounting for an estimated 76.3% of global cases in 2017 (22).

1.4.1 Enteric fever in low- and middle-income countries

While Asia has been known to have high enteric fever incidence and the highest burden of disease, there is growing evidence of a substantial enteric fever burden in Africa. Several islands in Oceania have also reported major typhoid fever problems, and although robust data are lacking, it is likely that the incidence of typhoid fever in this region is now among the highest of any global region (3,28,29). In 2017, an estimated 14.3 million cases of enteric fever occurred worldwide, with 135,900 deaths (a global case fatality of 0.95%). The region with both the highest number of cases and the highest age-standardized incidence rates was south Asia, which accounted for almost 72% of global cases and 70% of global deaths. Southeast Asia, east Asia and Oceania together accounted for 14.1% of global cases, and sub-Saharan Africa accounted for 12.1% of global cases (22).

In areas endemic for enteric fever, children are disproportionately affected. The highest incidence rates typically occur among children of 5 to 15 years of age, but recent studies from Asia and Africa have highlighted the substantial burden of disease among children younger than 5 years of age and infants (2,9,24,25). In 2017, 55.9% of global cases and 59.3% of global deaths occurred among children younger than 15 years. Young children aged 5–9 years had the highest incidence rates as well as the highest mortality rates.

There is increasing evidence of a marked heterogeneity in enteric fever incidence in both time and place, with considerable variation between regions, between countries within a region, and often between sites within the same country (24,28,30). High incidences of enteric fever in urban settlements were thought to be driven in large by a high population density, but a recent multicentre study in sub-Saharan Africa found high incidences of typhoid fever in urban areas with high population density as well as rural areas with low population density (24).

1.4.2 Enteric fever in the context of febrile illness in low- and middle-income countries

Enteric fever is an important cause of non-malarial febrile illness in many LMICs across Asia and Africa. However, definitive diagnosis remains a major challenge in such resource-constrained settings, due mainly because of limited or absent laboratory capacity for blood culture. In the absence of appropriate diagnostic tests, management decisions for patients presenting with non-malarial febrile illness are often based on clinical assessment algorithms (which lead to both under- and over-diagnosis of enteric fever) and rely on empirical antibiotic treatment (31,32). Apart from confirming the diagnosis of enteric fever in the first instance, blood cultures have a critical role in guiding antibiotic therapy; this has become increasingly important with the emergence and spread of drug resistant strains, both to ensure appropriate antibiotic treatment for confirmed enteric fever and to discourage the injudicious use of antibiotics when treatment is not indicated (33).

The relative importance of enteric fever as a cause of non-malarial febrile illness in LMICs varies considerably across regions, across countries within a region, and among age groups.

A review of studies in Africa from 1980 to 2015 reported that the most common cause of non-malarial febrile illness was bacterial infections (53%). Typhoid fever was among the top five most commonly reported bacterial pathogens in northern, eastern, western and central Africa up until 1990, but declined in rank thereafter. Typhoid fever was not ranked among the top five most commonly reported bacterial pathogens in middle or southern Africa at any time point (34). However, regional trends mask the situation at a country level, and other studies highlight that typhoid fever continues to be a common cause of non-malarial febrile illness and bacteraemia in several countries. For example, *S. Typhi* was the most common cause of bacteraemia in children younger than 5 years of age admitted to a district hospital in Tanzania from 2008 to 2016 (35).

A review of studies in Asia from 2012 through to 2017 found that *S. Typhi* was the fourth most common reported etiology of acute febrile illness across the region. However, there is also considerable heterogeneity among countries in the region. A recent multicentre study in Nepal, Bangladesh and Pakistan showed that 11.4% of febrile outpatients had confirmed enteric fever (32). In Bangladesh, enteric fever accounts for a very high proportion (63%) of positive blood cultures in children (25).

1.4.3 Enteric fever in high-income countries

The majority (>85%) of enteric fever cases in high-income countries are imported through international travel (36,37). A global analysis of travellers diagnosed with enteric fever from 2007 to 2018 found that 71% of cases were acquired in south Asia (most commonly India, Nepal, Bangladesh and Pakistan). Almost half (47%) of the cases were caused by *S. Paratyphi* infection, predominantly *S. Paratyphi* A acquired in southern Asia. The high proportion of *S. Paratyphi* A infections in travellers is likely multifactorial; travellers vaccinated against typhoid fever are protected against *S. Typhi* but not *S. Paratyphi* A, and certain Asian countries are known to have increasing prevalence of *S. Paratyphi* A (22,25,36).

1.4.4 *Salmonella Paratyphi A, S. Paratyphi B and S. Paratyphi C*

S. Paratyphi A causes most cases of paratyphoid fever globally, followed by *S. Paratyphi B* and rarely *S. Paratyphi C*. In 2000, paratyphoid fever cases accounted for an estimated 20% of enteric fever cases globally (38). This proportion increased to 23.7% in 2017, driven largely by the increasing burden of *S. Paratyphi A* across some countries in Asia (including Nepal, Cambodia and China) (22,25,39). While there is a paucity of data on the incidence of *S. Paratyphi* in Africa, findings from multicentre population-based surveillance studies and systematic reviews suggest that in Africa, paratyphoid fever accounts for a much lower proportion of enteric fever cases than in Asia; in 2017, an estimated 13.2% of enteric fever cases in Africa were due to paratyphoid fever compared to 25.6% in Asia (22,24,40).

1.4.5 Antimicrobial resistance

Chloramphenicol, ampicillin and co-trimoxazole were historically the first-line antibiotics used to treat typhoid fever. The emergence of multidrug resistant (MDR) *S. Typhi* in the 1980s, defined as resistance to all three first-line antibiotics, was first reported in Asia. The MDR *S. Typhi* strain predominant in Asia (MDR H58 haplotype) was later introduced into Africa and then Oceania, where it subsequently became widespread. With the spread of MDR strains, ciprofloxacin was adopted as the treatment of choice in the late 1980s to early 1990s. However, resistance to ciprofloxacin rapidly appeared (in the early 1990s), with the subsequent emergence of *S. Typhi* strains, which were MDR with fluoroquinolone resistance. Then third-generation cephalosporins, macrolides and carbapenems became increasingly the agents of choice for treating enteric fever (41–43). Extensively drug resistant (XDR) *S. Typhi* (defined as resistance to the first-line antibiotics, a fluoroquinolone and a third-generation cephalosporin) was first reported in Pakistan in 2016, when it caused a large outbreak (44).

There is wide variation in the rates of antimicrobial resistance across regions, and even across countries or areas within regions. A recent multicentre study in sub-Saharan Africa found that 52% of *S. Typhi* strains were MDR and 18% were fluoroquinolone resistant, while a recent multicentre study in Asia reported lower rates of MDR (18%) but much higher rates of fluoroquinolone resistance (>90%) (43).

Longitudinal data from several countries in Asia support this trend of simultaneously decreasing rates of MDR and increasing rates of fluoroquinolone resistance in the region (45,46). XDR *S. Typhi* is geographically confined to Pakistan at present, but travel-related cases have been reported from several countries (43).

Azithromycin is the only remaining oral drug to treat XDR typhoid fever, and the emergence of azithromycin resistant *S. Typhi* in Bangladesh, Pakistan, Nepal and India is a serious concern (47).

MDR is uncommon in *S. Paratyphi A* strains. The multicentre study in Asia that reported a MDR rate of 18% in *S. Typhi* also reported a MDR rate of 1% in *S. Paratyphi A*. Increasing rates of fluoroquinolone resistance are being reported in *S. Paratyphi A* strains in Asia, following the trend of fluoroquinolone resistant *S. Typhi* strains in the region. Azithromycin resistance is also emerging in *S. Paratyphi A*, as reported from Bangladesh, India and Nepal (43).

1.4.6 Outbreaks of enteric fever

A worldwide review of enteric fever outbreaks from 1990–2018 showed that the number and size of reported outbreaks increased over time. Most outbreaks occurred in Asia (51%), followed by Africa (15%) and Oceania (14%), typically in LMICs where enteric fever is endemic. Contaminated water was found to be the most common risk factor, often linked to poor water and sanitation infrastructure and planning (48).

The definition of a suspected enteric fever outbreak should be based on the general definition for a disease outbreak, as follows: an increase in the absolute number of cases occurring in a population over a defined time period, above what would normally be expected for the particular community, geographical area or season. A minimum of two cases confirmed by blood culture should be documented before an outbreak of enteric fever is confirmed. A change in epidemiological, clinical or microbiological patterns, despite having no increase in overall expected occurrence (absolute numbers or incidence rates), should also warrant an outbreak response. In an institutional setting, a lower

threshold of change may be warranted to trigger a response (18). Within a particular community or geographical area, an increase in the number of cases of non-traumatic intestinal perforations or an increase in the proportion of these cases proven to be caused by enteric fever could indicate an outbreak or reflect circulation of a highly drug resistant strain of *S. Typhi* (12,14). Genomic studies in high-income countries have also shown the emergence of successful XDR and MDR clones in association to imported cases and therefore these data can be used as a surrogate sentinel antimicrobial resistance surveillance to assess emerging trends of resistance in other countries and detection of outbreaks (17).

The recommended surveillance case definitions during an outbreak are similar to those for routine surveillance, but with additional features when applicable (18).

1. Suspected case of typhoid or paratyphoid fever:
 - a. At least three out of seven consecutive days of fever (with or without additional clinical features observed in the specific outbreak)

OR

- b. A physician's suspicion of enteric fever (typhoid or paratyphoid)
2. Probable case of typhoid or paratyphoid fever: Meets the suspected case definition, plus an epidemiological link to the outbreak
3. Confirmed case of typhoid or paratyphoid fever: Definition is the same as for routine surveillance

1.5

Treatment of enteric fever

Treatment of a confirmed case of enteric fever should be guided by antimicrobial susceptibility testing (AST). Surveillance and review of trends is important to inform clinical guidelines, particularly with emerging resistance (49) but the choice of empirical treatment pending case confirmation and AST results should be informed by local patterns of AST (18).

1.6

Diagnosis of enteric fever

Enteric fever presents as a nonspecific acute febrile illness. In LMICs, there are often other endemic diseases which also present with fever (including malaria, dengue, yellow fever, scrub typhus, rickettsioses and leptospirosis), in addition to common causes of bacteraemia and other viral infections (34,50). Because the clinical features of enteric fever are not pathognomonic but overlap considerably with such a wide range of diseases, clinical predictive tools have proven unreliable and clinical diagnosis of enteric fever has consistently been shown to have a low positive predictive value (51). A definitive diagnosis of enteric fever therefore requires laboratory tests.

1.7

Laboratory tests and their limitations

Bacterial isolation from bone marrow is historically the gold standard for the definitive diagnosis of enteric fever, with a suggested sensitivity of ~90% after 4 days of culture (52). However, bone marrow biopsies are invasive and specialized procedures; therefore, blood culture is the recommended diagnostic test for enteric fever (18,53) (Chapter 3).

The diagnostic sensitivity of blood culture is influenced by several factors, including the blood sample volume, the timing of the test in relation to the duration of illness, laboratory culture incubation conditions and prior treatment with antibiotics (52,54).

Two recent systematic reviews, which analysed the performance of blood and bone marrow culture in the same subjects, reported estimated blood culture sensitivities of 59% and 66% respectively when compared to bone marrow results (52,54). Lower blood culture sensitivity was reported for lower blood sample volumes (51% for 2 ml samples compared to 65% for 10 ml samples), among patients with prior antibiotic use (34% lower) and among patients with symptoms for >1 week (31% lower) (52) (**Chapter 3**).

Stool culture is not recommended for the diagnosis of acute enteric fever because infections due to *S. Typhi* and *S. Paratyphi A* rarely result in diarrhoea (55) but is useful for monitoring convalescent faecal shedding in patients following acute enteric fever and for detecting chronic carriers (18).

Although widely used in many settings, serologic tests' accuracy continue to be plagued by poor sensitivity and inadequate specificity and are not recommended for the diagnosis of enteric fever (18).

1.8

Role of the laboratory in the diagnosis of enteric fever

1.8.1 Clinical diagnostic microbiology laboratories

The minimum standard recommended for blood culture confirmation of enteric fever is to use a semi-automated system that will support the isolation of other common pathogens (including those associated with vaccine preventable diseases) (**Chapter 5**). If this standard cannot be achieved due to lack of instrumentation, it is recommended that commercially prepared blood culture bottles are used. While a continuously monitored, automated system is preferable, commercial bottles may be incubated manually (i.e., in a 35°C [$\pm$ 2°C] aerobic incubator). While these bottles will need to be manually examined visually (for turbidity) and microscopically (by Gram stain), one will still have the advantage of using bottles that have been prepared according to commercial quality control standards.

Biochemical testing algorithms used for bacterial identification will easily discriminate to a genus level of *Salmonella* spp. Basic slide agglutination on the isolate will need to be performed to indicate *Salmonella* serovars Typhi (O9, Hd, vi), Paratyphi A (O2, Ha), Paratyphi B (O4, Hb) or Paratyphi C (O6,7, Hc), though further testing will be needed to confirm (17,56,57).

AST should be routinely performed. Susceptibility testing should be conducted using quality control guidelines, such as the European Committee on Antimicrobial Susceptibility Testing (www.eucast.org) or the Clinical and Laboratory Standards Institute (www.clsi.org) (**Chapter 6**).

1.8.2 Reference laboratories

Reference laboratories should ideally have the capacity to serotype *Salmonella* using conventional or molecular methods, subtype *Salmonella* using standardized, internationally comparable molecular epidemiology techniques, and conduct confirmatory susceptibility testing, including molecular characterization of resistance mechanisms (18).

1.9

Innovations in enteric fever diagnosis and analysis

Developing improved diagnostic tests for enteric fever remains a critical need, and research into a range of diagnostic approaches is ongoing. Areas of research focus include PCR techniques (58), serologic assays and point-of-care diagnostic tests (33).

Whole-genome sequencing is the most powerful molecular subtyping tool available, and genomic data are used for molecular epidemiology and surveillance, particularly relating to antimicrobial resistance mechanisms. Sharing of genomic data within and between countries provides the opportunity for enhanced analysis to inform enteric fever control policies, and to monitor for the emergence and spread of high-risk clones (17,59,60).

1.10

Prevention and control of enteric fever

As recently as the early 1900s, enteric fever was still a scourge in some high-income countries, but following the implementation of improved water, sanitation and hygiene (WASH) and other public health measures, the disease virtually disappeared in these areas. These same interventions are still the core components of a bundle of effective measures for the prevention and control of enteric fever (43).

Prevention and control measures include (26):

1. Water and sanitation infrastructure: provision of safe potable water and improved sanitation
2. Public health measures: accurate and timely diagnosis, appropriate antimicrobial treatment, identification and treatment of chronic carriers and contacts, vaccination
3. Food safety: regulations and enforcement
4. Health education: handwashing, food safety practice

Three types of typhoid vaccines are currently licensed for use: typhoid conjugate vaccine, unconjugated Vi polysaccharide vaccine, and live attenuated Ty21a vaccine. The WHO recommends their programmatic use for the control of typhoid fever when implemented together with interventions that target WASH, health education and other public health measures (2). These typhoid vaccines do not confer protection against *S. Paratyphi A*, but several bivalent *S. Typhi* and *S. Paratyphi A* vaccines are currently in development (61).

Chapter 2

Role and management of the diagnostic laboratory

Clinical laboratory services play a crucial role in both individual and population-based health care. The prime objective of laboratory medicine is the reporting of accurate and timely test results to the requesting clinician. Approximately 70% of clinical medicine decision-making is predicated upon, or confirmed by or documented by medical laboratory test results (62–64).

Laboratory tests are used to:

1. Establish the clinical status of a patient
2. Diagnose infections and disease
3. Monitor its progress
4. Establish the response to treatment.

Microbiologists play a critical role in collating data for both clinical and public health decision-making. Precision and accuracy in the microbiological diagnosis of typhoid infections guides the choice of treatment and management of the patient. Additional information on serogroup of the causative agents helps guide the response effects in an affected population. Similarly, epidemiological surveillance is critical to guide appropriate antimicrobial therapy through the identification of resistant profiles (65). Fundamentally, the laboratory service aims to reduce morbidity and mortality in patients with Salmonellosis. According to the Surveillance Standards published by the WHO on “Typhoid and other invasive Salmonellosis”, the need for a clinical laboratory is highlighted confirm cases (18):

<https://www.who.int/publications/m/item/vaccine-preventable-diseases-surveillance-standards-typhoid>

The organization of the laboratory is embodied within several controlled documents:

1. Organizational chart
2. Policy manual
3. Safety manual
4. Quality manual
5. Procedure manuals (Standard Operating Procedures [SOPs] etc.).

These documents need to be created to describe the integration of personnel, equipment, safe operation of processes, supplies and facilities for efficient and effective delivery within the laboratory operation.

2.1

Pre-requisites

The following pre-requisites contribute to maintaining good mental health and well-being of the laboratory staff, which is of primary concern. A contented workforce leads to a productive service. Pre-requisites are also in place for achieving and maintaining recognized standard of accreditation.

1. Clear designation of authority and responsibility
2. Precise identification of processes to be performed and objectives to be achieved
3. Accurate assessment of available personnel, skills, space, equipment and supplies/provisions needed to perform procedures

2.2

Organizational chart (Hierarchical, Horizontal and Matrix)

1. Establishes core authority and responsibilities of personnel
2. Clarifies different authorities and responsibilities for the Head/Chief of department, the Laboratory Manager, Safety Officer, Quality Manager (if structure includes one), Technical Supervisors, scientific staff, and support staff
3. Identifies hierarchies and relationships between positions
4. Clarifies reporting lines (which can differ from line management)
5. Clarifies expectations of staff
6. Ideally, should be prominently displayed and available to all staff

2.2.1 Roles and responsibilities

The laboratory worker (scientist) should be expected to, depending upon role, meet the standards of the recognized professional body:

1. Demonstrate Good Laboratory Practice
2. Follow the code of conduct working in Biosafety Level 2 facilities
3. Complete the required occupational health surveillance
4. Be aware of the adverse health effects of the substances being handled, by reading the appropriate Control of Substances Hazardous to Health Regulations (COSHH)
5. Be trained to a standard deemed competent to perform work activities
6. Abide by all safety rules, guidelines and work instructions as directed by the laboratory manager or supervisor
7. Follow the safe work practices outlined in this manual
8. Be forthcoming to report any unsafe conditions and any malfunction of equipment, errors in methods or the management system
9. Report all accidents/incidents/concerns to their laboratory manager/supervisor.

The laboratory manager should be responsible for:

1. The recruitment of appropriate personnel
2. The overall safety of the laboratory. It is required that formal safety and housekeeping inspections be conducted regularly and that the outcome of these inspections and any resulting corrective actions are filed and kept for record
3. The local environment to be suitable for the purpose and well maintained
4. Ensuring all laboratory staff (both scientific and support) complete all mandatory and role-based training
5. An appropriate health and safety orientation is given, to individuals when they are first assigned to a laboratory
6. All laboratory staff to be aware of health and safety rules specific to the laboratory and that such rules are followed
7. Training has been provided for special or unusual hazards in non-routine work to laboratory scientific staff
8. Keeping a record of health and safety related training
9. Suitable and appropriate risk assessments are completed for laboratory activities
10. Completion of a job hazard profile for laboratory personnel in their area before any work commences
11. Identifying the need for occupational health surveillance and ensuring that relevant staff are identified for such
12. Appropriate emergency equipment e.g., emergency eyewash, emergency shower, fire extinguisher etc. is readily available, is operational and fit for purpose

Laboratory Safety Officer (LSO) is responsible for:

1. The implementation of laboratory safety procedures
2. The development of safe laboratory procedures (including vaccination of laboratory workers as appropriate and in consultation with medical professionals)
3. Helping to identify and implement appropriate arrangements for the continual safe operation of the laboratory. This may include, for example, investigating incidents, identifying incident trends, and recommending engineering or work practice control changes to the laboratory manager
4. Conducting regular health and safety inspections and audits, which should include subject areas, such as safe working practices, fire and electrical safety, ergonomics, chemical hygiene, general housekeeping, waste management, personal protective equipment, and infection prevention
5. Providing staff with information, instructions and training on safety matters, including spillage training, fire drills, and other safety related topics

2.3

Policy manual(s)

1. Policies are the laws and rules followed for the organization of the laboratory
2. Employee handbook is published by human resources department of the institution
3. Health & Safety Policy is created and signed by the Chief Executive Officer of the organization. Local safety manual would be written by the department with the laboratory safety manager/officer
4. Quality manual (includes quality assurance) compiled by the laboratory and quality manager guided by the ISO standard laboratory is accredited to
5. Information technology and purchasing manuals originate from the laboratory personnel

6. The laboratory (user) manual/directory is the responsibility of the laboratory lead senior management and contains information for the laboratory's users (i.e., clinicians/nursing staff):
 - a. Names and contact numbers of all laboratory staff
 - b. Laboratory staff – role, responsibilities and level of expertise
3. Accreditation information – e.g., UKAS ISO 15189
4. Location, hours of service
5. Services provided
6. Test ordering information
7. Specimen requirements (storage conditions, volume, quality, type, appropriate sample for diagnostic test request)
8. Result reporting systems (manual [report books], hard copy clinical forms, PC system)
9. Listing of individual tests including test name, methodology, result interpretation, reference ranges, quality controls (QC), sensitivity, specificity, predictive values (positive/negative), diagnostic efficiency and Turn Around Times

2.4

Procedure manuals

These articles of controlled documentation contain the instructions for performing specific diagnostic tests, e.g., AST (**Chapter 6**) or other laboratory specific tasks (SOPS), risk assessments, biological risk assessment, COSHHs, equipment monitoring required specifications and maintenance documentation.

2.5

Safe working practice

Many of the information provided here can be found in detail in UKHSA manuals (66).

Good Laboratory Practice for laboratory personnel

Laboratory personnel should be made aware of controlled documentation in place, these being Codes of Practice, SOPs, Risk Assessments and COSHH that cover specific laboratory requirements for associated procedures. If the laboratory worker does not understand any of these requirements, they must seek guidance from the laboratory manager or LSO.

Good safety practice and preventive actions to avoid or reduce the risk of a laboratory acquired infection must be followed by all laboratory workers. The following are some of the instructions for good safety practice:

1. Preparation

1. Fully fastened laboratory coats/gowns **MUST** be worn as soon as one enters the laboratory. This prevents clothing from contamination and jewellery, if worn, from possible entanglement. Additional personal protective equipment (PPE) (i.e., gloves, safety spectacles, face shield) is worn as instructed by the risk assessment associated for the procedure.
2. Wear sensible, low heeled, closed shoes. Wear protective footwear (e.g., steel capped) as instructed for specific activities (for example transport of gas cylinders and other heavy apparatus)
3. Always tie back long hair
4. Plan and organize workload beforehand, to ensure all materials and equipment required are available (and operational)

5. Chemicals and substances used for diagnostic tests can be hazardous to health, via inhalation, hand to mouth contact, through skin punctures, cuts or abrasions or by liquids splashing onto mucous membranes. Working with microorganism possess levels of pathogenicity (see **Chapter 2.8 Microbiological hazards**), therefore:
 - a. **BEFORE** starting any kind of laboratory associated work, cover any cuts or wounds with a waterproof dressing, and wear gloves for all laboratory processes (change as necessary)
 - b. **BEFORE** leaving the laboratory, remove gloves, wash hands as per the routine shown in **Figure 1**, and replace the dressing with a new one.
6. In the event of an incident, the location of first aid boxes, eye wash stations, firefighting equipment (fire extinguishers), fire alarms, evacuation routes, fire exits, call points and assembly locations must be **KNOWN**.

2. Laboratory requirements and maintenance

The laboratory must be maintained in a clean, clutter-free state and aisles and exits kept clear (to prevent slip, fall and trip hazards).

1. **ENSURE** that all emergency exits are kept clear, and any fire doors are kept closed (unless on an automatic closure device)
2. **ALWAYS** exercise care when manoeuvring around laboratory to avoid collision with other staff or stationary equipment while working
3. Ensure there is adequate bench space to carry out the process safely, for example, to avoid potential spill hazards
4. With all processes, an ergonomic assessment should be part of the laboratory risk assessment to ascertain whether space may be an issue
5. All reagents/chemicals/substances **MUST** be clearly labelled (with appropriate hazard warning labels) according to local quality management systems
6. All chemicals and reagents are stored appropriately with an inventory list on the door of the cupboard/store room:
 - a. Flammables in fire-proof cupboards (i.e., metal cupboards)
 - b. Acids and alkalis stored separately
 - c. Poisons in a lockable cupboard
7. Retain and store viable bacterial cultures and chemicals in line with local procedures
8. Keep all unnecessary excess paperwork, books and writing implements away from work surfaces and store in clean designated areas
9. **ENSURE** all materials and apparatus are safe to use. Items must be made safe (decontaminated) by approved methods before they leave the laboratory area. Dispose of all unwanted items in accordance with local safety procedures
10. Safety equipment e.g., certified microbiological safety cabinets and fume hoods **MUST** be used – whenever they are indicated by local procedures, e.g., handling infectious agents, chemical or substances accordingly
11. Prior to leaving the laboratory, remove your protective clothing and **ALWAYS** wash hands thoroughly following routine as illustrated in **Figure 1** for a period of one minute.

12. For effective infection control measures **DO NOT**:

- a. Bring personal effects (handbags/wallets) into the laboratory
- b. Bring or store food and drink into the laboratory, laboratory refrigerators or cold stores
- c. Eat, chew gum, put pens and pencils to the mouth, drink, apply cosmetics or use personal mobile phones

All these restrictions are to avoid the likelihood of a laboratory acquired infection or contaminating body surfaces or personal clothing.

3. **Incidents and corrective actions/resolution**

1. In cases when your protective clothing has become contaminated with infectious or hazardous material, **STOP WORK** and carefully remove contaminated clothing into a container* for autoclaving, wash hands thoroughly (**Figure 1**). Report to supervisor. Replace with a clean lab coat. *Label container to indicate items need to be returned
2. In the event hands or worn gloves become contaminated with hazardous substances including chemicals, blood, body fluids or biological material, **STOP WORK** immediately, discard the gloves and wash hands thoroughly (**Figure 1**)
3. If worn gloves are punctured during use, **STOP WORK**, discard gloves, wash hands thoroughly (**Figure 1**), report to supervisor and wear new clean gloves
4. **ALWAYS** report any sickness acquired and injuries inflicted during working hours, first and foremost, to your manager. Any cuts or skin punctures received during working hours must be encouraged to bleed, by gentle squeezing and washing it under running water. **DO NOT** scrub the wound. The wound must be treated and dressed. Seek attention and advice from the onsite first aider
5. **REPORT** any accidental equipment breakage and/or spillage of potentially infectious material immediately. Any decontamination processes must be supervised by a trained senior member of laboratory staff
6. **ALWAYS** report spills, accidents, or other incidents to your laboratory manager, supervisor or LSO
7. **REPORT** any evidence or suspicion of rats, mice, insects and other pests in or around the laboratory to the laboratory manager
8. **KNOW** the location of first aid boxes, eye wash stations, firefighting equipment and fire alarm call points

FIGURE
1

Hand washing technique with soap and water (conducted over one minute)



Source: Image taken from (66).

2.6

Quality assurance

Quality assurance (QA) is the total process whereby the quality of laboratory results can be guaranteed (67–69). The reliability of results is improved by undertaking QA, therefore, minimizing the variability of results. The variability may arise from biological or analytical sources that are fundamental in all qualitative and quantitative investigations.

QA encompasses several components including QA, QC, quality management, proficiency testing, performance measurement, process improvement, and outcomes management. QA is an essential component of the analytical testing phase of the total testing process (**Chapter 9**).

2.7

Risk assessment and control

2.7.1 Risk assessments

A risk assessment is the written findings of the process whereby hazards in the workplace are identified and control measures put in place to reduce the risk from those hazards as far as reasonably practicable.

Risk assessments must only be undertaken by suitably trained and competent nominated assessors.

Note: The specific control measures identified by the risk assessment should be highlighted in your local SOPs.

It is important that staff carrying out these procedures have a full understanding of the hazard control measures in place for that procedure. This should be explained during initial training and competency assessment.

Any new method introduced into routine diagnostic use must be risk assessed first before any work commences. If any method is changed in any way, then the risk assessment must be reviewed and any changes to control measures must be highlighted in the amended SOP.

In summary, risk assessments must be carried out before implementing or introducing into the workplace for all of the following:

1. New procedures
2. New equipment
3. New chemicals or reagent
4. Any change to an existing procedure, including relocation

2.7.2 Risk assessment policy, procedure and guidance

A policy, procedure and guidance that explains the approach for the various types of risk assessment is required. The policy covers organizational and individual responsibilities, as well as how the risk assessment system process is implemented. The policy highlights the risk control hierarchy that is fundamental to the process:

5. Identify the hazard – e.g., infectious (whether route of infection is by inhalation, contact or ingestion), flammable, toxic, mutagenic, and eliminate the hazards wherever feasible
6. Assess the risk, likelihood of injury or illness and possible severity of injury or illness. Substitute materials, agents, conditions or activities with the potential to cause harm, wherever feasible
7. Control the risk where the potential for harm cannot be eliminated or substituted, use all available engineering measures to control the risks (e.g., microbiological safety cabinets and fume hoods). Use hierarchy of controls (elimination/prevention, substitution, isolation, engineering, administration, PPE) and Typhi vaccination
8. Review the controls, establish adequate managerial controls – reduce the numbers of people at risk and/or duration of exposure through implementation of SOPs, and training and competence through safe systems of work
9. Minimize any remaining risk for laboratory workers (for example with the appropriate use PPE and administration of vaccination/immunization)

2.8

Microbiological hazards

Exposure to viable pathogenic organisms can result in infection and disease. This will depend upon factors, such as the nature of the organism, its concentration, dose and the route of exposure. Infection may be localized or more general and the symptoms may manifest rapidly or as a delayed reaction, lasting for months to even years. The symptoms may be trivial, serious or permanent, and can be in some cases difficult to treat. Therefore, it is vital to prevent virulent microorganisms from entering the body.

Many organisms are considered to be low hazard but may cause allergy or disease in persons of lowered immunity or may revert to a pathogenic state.

Cultures may also become contaminated by harmful organisms or incorrectly labelled during diagnostic testing. Consequently, all microorganisms should be handled with care, not just those known to be pathogenic.

Other sources of hazard are the products of microorganisms, such as biotoxins, enzymes, spores or cysts, some of which persist in the environment or are heat labile.

Biological material including blood or serum may carry infectious agents, such as hepatitis B or HIV; therefore, these hazards can be encountered in all laboratories, where biological materials are processed, and not just within microbiological laboratories.

2.9

Hazard groups

Because investigations into LAI showed that certain microorganisms were more likely to be involved than the others, several attempts have been made to classify human and animal pathogens according to the risks they present to the health of laboratory workers and to the human and animal community should they escape from the laboratory.

Classification frameworks for biological agents developed by various national committees show a worldwide agreement on a four-group classification system, which aligns with the level of risk posed to human health. However, deviations may exist between, and within states in the allocation of agents to a particular hazard or risk group. As an example, the 'Advisory Committee on Dangerous Pathogens (ACDP)' in the United Kingdom have classified biological agents into four hazard groups where Hazard Group 1 agents are not considered to pose a risk to human health and Hazard Group 4 agents present the greatest risk. Definitions of the relevant hazard groups are given below:

Hazard Group 1: An organism that is unlikely to cause human disease.

Hazard Group 2: An organism that can cause human disease and might be a hazard to the laboratory worker, but it is unlikely to spread to the community and there is usually effective prophylaxis or treatment available.

Hazard Group 3: An organism that can cause severe human disease and may be a serious hazard to the laboratory worker; it may spread to the community, but there is usually effective prophylaxis or treatment available.

Hazard Group 4: An organism that can cause severe human disease and is a serious hazard to the laboratory worker; it is likely to spread to the community and there is usually no effective prophylaxis or treatment available.

You can use the following site to find out more about the classification systems in different countries: <https://www.biosafety.be/content/contained-use-international-classifications-schemes-micro-organisms-based-their-biological>

2.10 Containment levels

Work with biological material must only be completed in a laboratory with a suitable containment level. The level of containment to be used is determined as part of the risk assessment for the activity.

Generally, as a guideline the Hazard Group of the organism indicates the minimum containment level it must be handled, such that:

1. Hazard Group 2 at Containment Level 2
2. Hazard Group 3 at Containment Level 3
3. Hazard Group 4 at Containment Level 4

There may be, however, derogations or enhancements from this depending upon the nature of the biological material and work. Therefore, the risk assessment and associated SOP must always be consulted. If a laboratory worker is in doubt about the containment level requirements, they must contact their manager, LSO, supervisor, or health and safety advisor immediately for further advice.

2.11 Routes of exposure

If a person is exposed or believes to have been exposed to the organism in any of the following ways, follow the guidelines in **Chapter 2.5 point 3** and report to your laboratory manager, supervisor or LSO.

1. Contact

Splashing of liquids can cause contamination of the skin, clothing and working areas. The majority of microorganisms can enter the body through cuts or scratches, or via the mucous membranes of the eye, nose or mouth. Work surfaces (fomites), equipment, clothing and other surfaces may remain contaminated for several days or longer – even years, particularly when the contaminants are bacterial spores.

2. Inhalation

Inhalation and infection via the eye are common routes of laboratory infection as many organisms easily become airborne during handling. The infectious aerosol hazard is very important. Many common procedures produce minute droplets (invisible to the naked eye) of infective fluid, which become airborne during handling and may be inhaled. Aerosols may remain suspended in the air for several hours.

Aerosols can be produced by any process that involves splashing or turbulence, such as pipetting (particularly when the volume of fluid is forcibly expelled), shaking, vortexing, centrifuging, siphoning, opening ampoules, or stoppered tubes, releasing gas pressure from culture tubes and even opening of screw-capped containers such as universal containers. The risk of leakage from loose or damaged capped containers, particularly when they are shaken, can inevitably lead to the formation of aerosols.

Manipulation should always be conducted with minimum risk of aerosol generation and if this cannot be avoided then the process must be ‘contained’ i.e., process in a microbiological safety cabinet.

Dry or dusty forms of organisms (such as sporulating moulds and freeze-dried preparations) can easily become airborne. Therefore, handling of such organisms should be contained in a microbiological safety cabinet.

3. Ingestion

Ingestion may also result from contaminated fingers, pipettes, and leaking closures on bottles. The placing of anything in the mouth within the laboratory is therefore forbidden. Avoid touching the face at all times. Care should be taken to eliminate contamination of telephones, books, and anything else handled with gloved hands. Mouth pipetting is prohibited.

4. Injections

Needles, sharp metal, glass, teeth and claws can effectively deliver infected material into the blood. Needles, blades and scalpels should only be used where blunt alternatives are unsuitable. Blades and needles must never be re-sheathed after use. Wash any cuts or needle-stick injuries inflicted during working hours with soap and under running water (<https://www.cdc.gov/niosh/docs/2000-108/>). Any injuries should be reported to a first aider.

2.12

Biosafety of enteric pathogens

Salmonella spp. are classified according to regulations for dangerous pathogens, which may vary between countries. For example, in the United Kingdom, *Salmonella* Typhi is classified as Hazard Group 3, whereas in the United States and Canada it is categorized as a Biosafety Level 2 pathogen.

Each laboratory should have its own local safety manual which describes the essential biosafety, chemical, fire and electrical safety requirements to protect staff, the community and the environment.

All staff should be familiar with the contents of the manual and should adhere to the policies accordingly. All new staff are required to read the manual and be made aware of the risks involved in working in a clinical laboratory before commencing the tasks. They must comply with PPE regulations for that laboratory and wear suitable PPE when handling these pathogens. They must be competent in the relevant SOPs, safety protocols and risk assessments.

The biological agents that cause typhoid fever are transmissible via ingestion, contact and fomites and **not** via inhalation, therefore, guidelines need to be reviewed in order to handle these pathogens in a manner protecting the workforce first and foremost.

All staff that routinely handle cultures of *S. Typhi* should be offered vaccination (including booster vaccinations), according to the respective national immunization guidelines (64), as the risk of occupational exposure to microbiology laboratory staff is recognized (65).

Staff handling primary cultures should be aware that blood specimens requested for culture is not specific and other pathogens, including those infectious agents transmissible via aerosolization (e.g., *Brucella* or *N. meningitidis*) may be encountered. Therefore, a risk assessment must consider the full spectrum of organisms, which may be isolated and identified from clinical specimens.

Chapter 3

Collection, storage and transporting specimens from suspected cases of acute enteric fever and suspected carriers

Enteric fever is often clinically indistinguishable from other acute febrile illnesses, including malaria, dengue, and invasive non-typhoidal Salmonellosis (iNTS). As such, isolating or detection of *Salmonella* Typhi/ Paratyphi A or in fact any iNTS from a normally sterile site is needed to confirm the infection (**Chapter 4**). In addition, it is essential to identify carriers (post-acute or chronic) by isolating *S. Typhi/ Paratyphi A* (or detection of Typhi/ Paratyphi A DNA) from stool samples (or occasionally urine) (65,71). While this manual focuses on the laboratory diagnosis of typhoid and other invasive *Salmonella* infections, clinicians must consider all potential causes of acute febrile illness in their region and base testing algorithms on local epidemiology and best practice guidelines.

3.1

Sample collection for laboratory examination (acute enteric fever)

Historically, bone marrow culture was believed to offer the highest diagnostic sensitivity but these earlier studies often compared bone marrow culture to blood cultures performed using small sample volumes and in-house prepared media (65). Recent studies using commercially prepared blood culture media have shown similar sensitivity to bone marrow culture. Blood culture is the preferred specimen for diagnosing acute enteric fever as it is more practical due to it being less invasive, less painful, no special equipment required and the fact that bone marrow extraction requires specialized personnel to perform it. For this reason, bone marrow culture is not routinely recommended for the diagnosis of enteric fever but if a sample is being collected for other studies as part of a broader diagnostic workup, it can be submitted for culture as well (72).

While a positive blood culture is considered 100% specific for acute typhoid/paratyphoid fever, several factors will impact the sensitivity of blood culture and could produce a false negative result. Proper patient preparation, collecting an adequate volume of blood, and using commercially prepared blood culture bottles can help maximize sensitivity (see **Annex A3.1**) (72,73). Sensitivity can be further improved by collecting at least two sets of blood cultures (1 aerobic and 1 anaerobic bottle per set), however, this may deem impractical in many endemic areas and often only a single, aerobic bottle is collected. It must be kept in mind that a single, negative blood culture does not exclude enteric fever, and additional diagnostic tests may be required. Ideally, blood (or any type) cultures should be collected before administering antimicrobials. However, treatment for critically ill patients should not be delayed pending blood culture collection.

Correct preparation of the venepuncture site can have a significant impact on blood culture sensitivity (72,73). Without proper preparation, blood cultures may become contaminated with skin flora, and thus, obfuscate the detection of true pathogens and confound the interpretation of the results (potentially leading to inappropriate therapy). The intended venepuncture site should be cleaned before disinfection (see **Annex A3.1.4** and **Figure A3.1**).

In acute enteric fever, bacteraemia may be as low as 1–2 cfu/1 ml. As a result the clinical sensitivity of blood culture has been reported to be around 60%. To ensure maximum sensitivity, a sufficient sample must be collected. Ideally, 8–12 ml of blood is collected for culture from patients weighing >14 kg and inoculated into an adult size aerobic blood culture bottle. If <4 ml blood is obtained (e.g., from severely dehydrated patients) the adult blood sample should be inoculated into a paediatric bottle. For paediatric patients weighing less than <13.9 kg, ideally 1–3 ml of blood should be collected for culture and inoculated into an aerobic, paediatric sized, blood culture bottle. The optimal volume may vary slightly by manufacturer (typically a ratio of between 1:5 – 1:10 parts blood to media is used). When commercial products are used, it is critical to adhere to the manufacturer's recommended fill volumes (even if they may differ from the volumes mentioned above) (see **Annex A3.1.3b**) (74).

Blood culture bottles should be inoculated as soon as possible after venepuncture, taking essential steps to minimize contamination. Commercially prepared bottles are recommended as they are subjected to quality analysis by the manufacturer. Additionally, some commercial bottles are compatible with butterfly adaptors, which allow blood to flow directly into the bottle, ensuring appropriate fill volume and minimizing the need to manipulate contaminated sharps. Before inoculation, the septum of the blood culture bottle should be disinfected with 70% alcohol (other disinfectants are not used as they may degrade the rubber septum). If a needle and syringe are used for blood collection, the same needle used for venepuncture should be used to inoculate the blood culture bottle and the bottle should be inoculated on a flat surface. To prevent needle-stick injuries, bottles must not be held by hand during inoculation. While historical protocols recommended changing the needle before inoculating the blood culture bottle, more recent studies have shown that this practice had minimal impact on contamination but significantly increases the chances for needle-stick injuries. As such, the needle should not be changed before inoculating the blood culture bottle (75–77).

Following inoculation, blood cultures are gently mixed by hand and the inoculation time (and any additionally required metadata) is noted on the bottle. If commercial blood culture bottles are used, do not place any stickers or markings on the barcode which is pre-printed on the bottle. The bottle is then transported to the laboratory at room temperature (~25°C) (see **Annex A3.1.5**). Extremes of both heat and cold should be avoided. If a bottle cannot be transported to the laboratory immediately, it should be maintained at 25°C (and not placed in a fridge or incubator) pending additional instruction from the laboratory.

3.2

Collecting stool samples for laboratory examination (carriers)

Stool samples are the preferred specimens for identifying post-acute/chronic carriers. Stool samples may also be positive in cases of acute enteric fever; however, results must be interpreted with care, as a positive stool culture in an acutely ill patient may reflect acute enteric fever or indicate chronic carriage in an acutely ill individual with a different disease (65). As the faecal shedding of *Salmonella* by carriers is typically sporadic, multiple cultures are required to identify or exclude carriers. Typically, at least three stool samples collected at least 24 hours apart (and after the completion of antimicrobial therapy/resolution of symptoms) must test negative to exclude an individual as a potential carrier.

The sensitivity of stool culture can be maximized by using a combination of direct culture and selective enrichment (65,78). Stool samples are collected in a clean container, which should be free of urine, soap, or disinfectant residue (see **Annex A3.2.2**). If the specimen can be processed immediately, transport medium is not required, but if processing the sample is going to be delayed by more than 1 hour, the sample should be collected on a sterile swab and inoculated into transport medium (Cary-Blair or Amies). Alternatively, a variety of liquid transport systems are commercially available. If the patient is unable to produce a stool specimen, a rectal swab may be used (see **Annex A3.2.2 point 5**).

Individuals working in sensitive occupations (e.g., food workers, health care workers) who have recovered from *Salmonella* infections (including typhoid and paratyphoid) often must be cleared by local health authorities prior to resuming work. Local and national guidelines may vary significantly regarding the number and timing of samples for testing. As a general guideline, initial screening of carriers should be performed by testing three stool samples collected >24 hours apart with the first sample taken >7 days after the completion of antimicrobial therapy (78–80). However, if national or local guidelines specify different time frames, those guidelines will take precedent.

3.3

Transportation

Shipping infectious or potentially infectious substances are subject to an array of national and international regulations. A full discussion of packing and shipping requirements is beyond the scope of this manual and the reader is advised to consult their relevant, local, regulatory agency. While the ultimate consideration is given to safe transport, it is also critical to take any necessary steps to maximize specimen viability. Stool samples travelling <3 days should be typically shipped with cold packs. However, if the extended transportation times are anticipated or excessive ambient temperature is expected, stool samples should be frozen and shipped on dry ice (if possible) or cold packs chilled to -80°C (see **Annex A3.2.2 point 7 & 8**) (80).

Chapter 4

Isolation from clinical specimens

Salmonella are non-fastidious bacteria which readily grow on a variety of media. If laboratory testing is performed solely to rule-in or rule-out *Salmonella*, simplified algorithms can be used. However, as enteric fever frequently presents as a nonspecific, acute febrile illness, isolation methods, which support a wider range of potential pathogens, may be preferred (81).

4.1

Isolation from blood culture

Following inoculation, blood culture bottles are incubated at 35°C ($\pm$ 2°C) aerobically. If a continuous monitoring (automated) blood culture system is used, bottle manipulation is not required until the instrument detects growth (or reports the bottle as negative). If an automated system is not used, the bottle is incubated upright at 35°C $\pm$ 2° / aerobically without CO₂ enrichment. To minimize contamination, a continuously monitored blood culture bottle should only be accessed if flagged positive by the instrument. In most cases, bottles reported as negative or no growth by the instrument, may be discarded without additional testing. However, if transport to the laboratory was delayed or the bottle was incubated prior to being placed in the instrument, growth may not be reliably detected by the instrument. In these situations, it is critical to remove a sample of fluid from the bottle. The sample is removed by first disinfecting the bottle top with 70% ethanol or isopropanol. A sterile needle and syringe are then used to remove approximately 0.5-1 ml of fluid. Although *Salmonella* are not infectious via aerosol, other pathogens such as *N. meningitidis* or *Brucella* that also cause febrile illness, will grow in blood culture media, and are highly infectious via aerosol. For this reason, bottles should always be manipulated within a certified biosafety cabinet. If a biosafety cabinet is not available, manipulations should be performed behind a plexiglass shield if possible, using maximum precautions to reduce aerosols. A drop of fluid (approximately 50 μ l) from the syringe is placed on a slide, air dried, Gram-stained and examined under a brightfield microscope (see **Annex 4.3**), and an additional drop is plated onto 5% horse (HBA) or sheep blood agar (SBA) or chocolate agar (CA) (MacConkey [MAC] agar may also be included), which is then incubated at 35°C $\pm$ 2°C aerobically, (blood and chocolate plates are ideally incubated with 5% CO₂ enrichment) for 18-24 hours (see **Annex 4 for agar preparations**). CO₂ enrichment is not required for *Salmonella* but is required for many other bacteria associated with acute febrile illness (e.g., *S. pneumoniae* and *Neisseria*).

Macroscopic (turbidity) and microscopic examinations are used to detect growth in manually incubated bottles. Following 24 hours of incubation, a sample of broth is removed as described above, and used to prepare a Gram-stained slide (see **Annex 4.3**). If organisms are not observed, the bottle is re-incubated and another slide is prepared after 48 hours of incubation. If organisms are not present after 48 hours, the bottle is returned to the incubator. If the broth becomes turbid, additional slides can be examined, if turbidity is not evident, a final slide can be prepared after 7 days of incubation. If organisms are observed at any point, a drop (~50 μ l per plate) of fluid from the bottle is plated onto 5% HBA or SBA or CA (MAC agar may also be included), which is then incubated at 35°C ($\pm$ 2°C)/aerobic atmosphere (with blood and chocolate plates, ideally incubated with 5% CO₂ enrichment)/18-24 hours. (81–83).

Salmonella will produce smooth to dewdrop cream-coloured colonies on CA and BA with no haemolysis on the latter. On MAC agar *Salmonella* will produce smooth lactose negative (i.e., white-cream) colonies (see **Table 1**, and **Annex Figure A3.1**). Gram stain and oxidase testing are performed on *Salmonella*-like colonies. Gram-negative, oxidase-negative bacilli are subjected to identification algorithms as outlined in **Chapter 5** and **Annex A5**. Gram-positive and/or oxidase-positive organisms are not *Salmonella*; however, they may or may not be clinically significant. Additional testing of non-*Salmonella* isolates may or may not be indicated. Testing should be performed as appropriate, based on laboratory and clinical practice guidelines (84).

4.2 Isolation from stool samples

Stool is a non-sterile specimen which contains large numbers of commensal flora. Successful isolation requires algorithms which suppress faecal microbiota but enriches the *Salmonella* population. To maximize recovery, specimens are typically inoculated onto two selective media and into a selective enrichment broth. The choice of media is based on local availability, but generally consists of one moderately selective medium (e.g., MAC or *Salmonella*-Shigella), one highly selective medium (Hektoen enteric [HE] agar or Xylose-Lysine-Deoxycholate [XLD]), and one selective enrichment broth (Selenite F or Selenite Cystine) (see **Annex A4.2** media for recipes).

Clinical samples are inoculated onto selective media by applying a loopful of primary inoculum streaked out onto the agar, then incubated at $35^{\circ}\text{C} \pm 2^{\circ}\text{C}$ (without CO_2 enrichment) and examined after 18-24 hours for the presence of *Salmonella*-like colonies (see **Table 1** and **Annex Figure A4.1, A4.2** for colony images). If *Salmonella*-like colonies are not observed, the plates are incubated for a further 24 hours (78).

TABLE
1

Appearance of *Salmonella*-like colonies on different culture plates.

	5% BA	MAC	CA	XLD & HE
Colour	Grey / non-haemolytic	Off-white	Grey	Clear with or without a black centre
Size	2-3 mm	2-3 mm	2-3 mm	2-3 mm
Appearance	Smooth creamy Buff or dewdrop	Smooth creamy	Smooth creamy	Smooth

* BA: blood agar; MAC: MacConkey agar; CA: chocolate agar; XLD: Xylose-lysine deoxycholate; HE: Hektoen enteric.



The faecal sample is also simultaneously inoculated into selective enrichment broth such as selenite broth (F or cystine) (**Annex A4.2**). The purpose of this broth is to suppress commensal flora such as *E. coli* and encourage growth of *Salmonella*, thereby increasing the likelihood of recovering *Salmonella* on solid media. It is critical to inoculate selenite at a ratio of 1 part sample to 10 parts broth, as excessive sample volume will overwhelm the selective properties of the broth. The broth may be easily inoculated by placing a swab into the faecal sample and transferring the swab into 9-10 ml of selenite. Following inoculation, the broth is incubated at $35^{\circ}\text{C} \pm 2^{\circ}\text{C}$ / aerobic atmosphere (without CO_2 enrichment) / 18-24 hours and then sub-cultured to a highly selective agar medium (HE or XLD). The selective medium is incubated at $35^{\circ} \pm 2^{\circ}\text{C}$ / aerobic atmosphere (without CO_2 enrichment) / 18-24 hours and examined for *Salmonella*-like colonies (**Table 1, Annex Figure A4.1, A4.2**).

Salmonella-like colonies are then subjected to identification algorithms as outlined in **Chapter 5**. Gram staining and/or oxidase testing may be performed if desired (**Annex A4.3**), however, this is not always necessary as the above media are highly selective for Enterobacterales.

Note: This algorithm will also support the growth of *Shigella* spp. The isolation of *Shigella* from stool samples is considered a significant finding and *Shigella*-like colonies should be subjected to confirmatory testing (78).

Chapter 5

Basic microbiology for confirming *S. Typhi*, *Paratyphi A*, *B* and *C*

Typhoid and paratyphoid fever (enteric fever) are caused by *Salmonella enterica* subspecies *enterica* serovar *Typhi* (*S. Typhi*) and *S. Paratyphi A*, *B* or rarely *C*, respectively (18,85). Typhoid and paratyphoid fever are difficult to distinguish from each other and from other common febrile illnesses including, but not limited to invasive non-typhoidal Salmonellosis (iNTS) (65). All isolates of *Salmonella* from blood, and faecal samples, require further testing to differentiate *S. Typhi*, *S. Paratyphi A*, *B* and *C*, and non-typhoidal *Salmonella* (NTS) serovars. Laboratory testing may be comprised of biochemical and/or serological tests, as well as molecular techniques (see **Chapter 7** and **8**). Accurate identification of *S. Paratyphi B* and *C* requires specialized media, antisera and/or molecular techniques, which may not be available outside a Reference Laboratory. As such, this chapter will focus on the phenotypic, conventional (non-molecular) laboratory testing and identification of *S. Typhi* and *S. Paratyphi A*, the differentiation of these serovars from each other and from *Salmonella* serovars other than *S. Typhi* and *S. Paratyphi A*.

5.1

Suitable samples for confirmation of *Salmonella* spp.

1. Serological agglutination tests can be performed on the purity plate subculture or from growth on Triple Sugar Iron (TSI) Agar or Kligler's Iron Agar (KIA) (see **Annex A5.2.5** and **A5.2.6**).
2. Confirmatory biochemical tests must be performed from a single, well-isolated colony. Pure cultures are also required for AST, whole-genome sequencing (WGS) (see **Chapter 8**) and long-term storage (see **Annex 5.4**), if being performed. If there is a question regarding the purity of the colony, a purity plate subculture derived from a single colony should be prepared and used for testing, following the steps below:
 - a. Prepare purity plate sub-cultures by taking a single colony, morphologically resembling *Salmonella* from the selective Agar plate (see **Chapter 4**) with a sterile, straight wire.
 - b. Then subculture onto a non-selective solid medium, such as HBA/SBA or NA (Nutrient Agar) (Appendices **A4.1.1.1** and **A5.5.1**, respectively).
 - c. Streak for single colonies and incubate at 35 ($\pm$ 2° C) / aerobic atmosphere (without CO₂ enrichment) / 18-24 hours.

Note: A selective plate may contain mixed populations of bacteria, therefore, take care to touch a single colony only and avoid touching multiple colonies.

5.2

Screening tests for *Salmonella* spp

5.2.1 Gram stain and motility

Gram stain and motility are of limited to identify *Salmonella*. Gram staining is generally not performed on stool culture isolates as the selective media (MAC, XLD, HE) inhibit Gram-positive organisms, and growth on these media is indicative of Gram-negative organisms. A Gram stain indicating Gram-negative bacilli may suggest *Salmonella* sp., however, this would be consistent with other *Enterobacterales*, such as *E. coli*, *Klebsiella* spp., and *Proteus* spp. Gram stain is more critical with specimens from sterile sites (e.g., blood) and this may impact empirical therapy. A Gram stain result other than Gram-negative bacilli, however, excludes *Salmonella* spp., including *S. Typhi* and *S. Paratyphi* A. *Salmonella* is generally motile; however, weakly motile or non-motile variants may be encountered (see **Annex 5.3, Table A5.1**).

5.2.2 Biochemical screening tests

A number of basic laboratory tests can be easily used to screen *Salmonella*-like colonies and determine if further testing is warranted. These tests include the appearance of colonies on MAC, XLD agar, oxidase, urease and reaction/s in TSI or KIA agar (see **Annex A4.1.1.1C** and **A5.2.1**). Expected/exclusion reactions for *Salmonella* using these tests are defined in **Annex 5.1 Table A5.1**. Alternatively, if resources allow, a more comprehensive confirmatory biochemical panel may be performed (84).

5.3

Confirmatory tests for *Salmonella* spp

5.3.1 Biochemical tests to identify *Salmonella*

Additional biochemical tests, to those listed in **Annex 5.1, Table A5.1**, can be incorporated into a biochemical testing algorithm (see **Annex 5.1, Figure A5.1**) to identify and support differentiation of *S. Typhi*, and *S. Paratyphi* A from NTS and differentiate these serovars from each other. Media, inoculation procedures and the expected reactions are shown in **Annex A5.1.1**, and **Tables A5.1** and **A5.2**. Growth and appearance of *S. Typhi*, *S. Paratyphi* A and *S. Newport* (representing NTS) in the biochemical panel are demonstrated in **Annex 5.1, Figures A5.2-A5.4**, respectively.

Inoculation, QC and interpretation of the biochemical panel including oxidase, Simmon's citrate, Lysine Iron agar (LIA), Motility-Indol-Ornithine (MIO) agar, TSI agar and Christensen's urea test medium are shown in **Annex A5.2.1** to **A5.2.5**.

5.3.2 Commercial identification systems

Commercial systems, e.g., API (analytical profile index) strips or semi-automated instruments (Phoenix, MicroScan and Vitek® [Biomérieux, France]), generally perform well when manufacturer's guidelines are strictly followed (86–88). These instruments also work best with phenotypically "typical" strains. Atypical strains, or deviations from manufacturer guidelines may lead to ambiguous or incorrect results.

5.3.3 Mass Spectroscopy Laser Desorption Ionization Top of Flight (MALDI-TOF)

MALDI-TOF mass spectrometry (MS) has emerged as a rapid and accurate means of identifying many bacteria, offering substantial cost and time saving over phenotypic methods.

The two main platforms are the Bruker Biotyper® (Bruker Daltonics, Germany) and Vitek® MS (bioMérieux). Studies evaluating in vitro diagnostic libraries for MALDI-TOF MS demonstrate accurate identification of *Salmonella* spp. to the genus level (89,90). However, neither system can reliably differentiate *S. Typhi*, *S. Paratyphi A* and NTS; additional testing must be performed to identify the specific serovar.

5.3.4 Molecular assays

To confirm and identify *Salmonella* isolates specific PCR assays can be used, e.g., *ttr* gene or WGS and bioinformatic techniques, such as kmer identification. Furthermore, the sequence read data is available for additional analyses, such as *in silico* serotyping, detection of resistance genes/mutations and phylogeny studies (see **Chapters 6 and 7**). Note that currently, phenotypic identification of *Salmonella* sp. in one or two days offers a quicker turnaround time than WGS, which may take 4 days or more.

5.3.5 Other faecal pathogens

While *S. Typhi*, *S. Paratyphi A* and NTS may be isolated from faecal samples, diarrhoeal illness may be caused by a number of other viral, bacterial, and parasitic enteric pathogens. Some selective media used to isolate *Salmonella* from stool samples, including typhoidal or paratyphoidal serovars, will also favour growth of *Shigella* e.g., XLD and MAC agar. Therefore, isolating non-*Salmonella*, non-*Shigella* enteric pathogens requires the incorporation of the appropriate pathogen specific selective media, growth conditions and biochemical test panels into the test algorithm (84). Growth and appearance of *S. sonnei* and *S. flexneri* in the biochemical test panel are shown in **Annex A5.1, Figures A5.5 and A5.6**, respectively.

5.3.6 Other faecal organisms

Although selective agars can suppress the growth of commensal enteric organisms, some strains may ‘breakthrough’ and grow on these media. To exclude faecal bacterial organisms, including Enterobacterales that may grow on selective agar, an appropriate panel of biochemical tests can aid in identification. Growth and appearance of *Proteus mirabilis*, *Citrobacter freundii*, *E. coli* and *Pseudomonas aeruginosa* (i.e., commonly isolated faecal non-pathogens) in the biochemical test panel are shown in **Annex A5.1 Figures A5.7 to A5.10**, respectively.

5.4 Conventional *Salmonella* serotyping

5.4.1 Identifying phenotypic profile

Blood isolates displaying a phenotypic profile consistent with *S. Typhi* or *S. Paratyphi A*, according to the algorithm presented in **Annex A5.1, Figure A5.1**, may be reported as such without additional confirmation, particularly where serotyping antisera is unavailable. However, determining the serotype and identifying O (lipopolysaccharide [LPS]), H (flagellar) and Vi (capsular) antigens is necessary if a limited biochemical panel was performed (see **Chapter 5.2.2**). Full antigenic characterization is also required to identify the serovar of NTS (56).

Due to the diversity of organisms in stool samples, it is recommended that faecal isolates are confirmed using both biochemical and serotyping tests or molecular assays. In addition, due to high antigenic homology among the Enterobacterales (in particular, *Salmonella*, *Citrobacter*, *E. coli* and *Proteus*), reactivity with polyvalent *Salmonella* antisera is not considered confirmatory without additional biochemical or molecular confirmation or serotyping with monospecific sera.

While a full description of *Salmonella* serotyping is beyond the scope of this manual, an abbreviated algorithm to assist with the identification of *S. Typhi*, *S. Paratyphi A*, B or C is shown in **Annex 5.3, Table A5.15**. More detailed information can be found online in standard operation procedures, such as those published by UKHSA (91).

5.4.2 Suitable culture for serological agglutination testing

The agglutination tests can be performed on a pure culture taken from non-selective agar e.g., HBA, SBA, NA or TSI/KIA. Selective media (e.g., HE, XLD, etc.) may produce false positive and/or negative agglutination with typing sera, as such, *Salmonella*-like colonies from selective media must be sub-cultured to a non-selective media prior to serotyping.

5.4.3 Antigenic formulae of enteric fever organisms and NTS

The *Salmonella* that cause typhoid fever and paratyphoid fever, their antigenic compositions and serogroups are indicated in **Annex 5.3, Table A5.15**.

5.4.4 Determination of O and Vi antigens

Serologic identification of *S. Typhi* and *S. Paratyphi A* begins by screening suspect isolates using O:2 (Group A), O:9 (Group D), and Vi antisera. *S. Paratyphi A* is the only Group A serovar encountered with any frequency that agglutinates with Group A antiserum in the context of an appropriate TSI or LIA reaction consistent with *S. Paratyphi A* (see **Annex A5.1, Figure A5.3**).

S. Typhi may react with O:9 and/or Vi antisera; as the Vi antigen is a capsular antigen, isolates which produce copious amounts of Vi (typically freshly isolated cultures) may fail to agglutinate with O:9. Vi antigen can be removed by harvesting a sufficient culture from the purity plate to prepare a turbid (milky) suspension, in a test-tube containing approximately 1 ml sterile saline then placing the tube in a boiling water bath for 10 minutes and repeating agglutination tests on the boiled suspension to detect the heat stable O:9 antigen. Unlike Group A, numerous Group D serovars (including but not limited to *S. Typhi*) are commonly encountered human pathogens, as such, agglutination with O:9 alone is not a diagnosis of *S. Typhi*.

Salmonella isolates may also fail to produce LPS (or produce incomplete LPS) and are termed “rough”. Rough isolates may fail to agglutinate with antisera or agglutinate spontaneously (auto-agglutination) with all tested antisera. Reporting of isolates displaying phenotypic profiles consistent with *S. Typhi*, *S. Paratyphi A*, or NTS should not be delayed pending a lack of antisera or ambiguous reactivity with antisera. Agglutination reactions are graded according to the level of antigen-antibody reaction or ‘clumping’ observed; grading and visual representation of the O slide agglutination reactions are shown in **Annex A5.3, Table A5.15** and **Figure A5.18**, respectively. Materials and methods are outlined in **Annex A5.3.1**.

Notes:

1. *S. Typhi* may typically be O:9 and/or Vi positive. Strains that strongly react with Vi antiserum may mask the O antigens and fail to react with O:9 antiserum. Vi can be removed after boiling; see details above and in **Annex A5.3.1.5**.
2. *S. Dublin* and *Citrobacter freundii* possess Vi antigens. However, the phase 1 H antigens of *S. Dublin* are g, p, whereas the phase I H antigen of *S. Typhi* is d.
3. Vi antigen of *S. Typhi* is more frequently detected in freshly isolated cultures than in sub-cultures.

5.4.5 Determination of H antigens

Following O antigen determination, the H antigens are tested. Testing for H antigens may be carried out by slide agglutination using the same culture used for O antigen testing. However, optimum results are obtained from highly motile cultures. The small amount of liquid which typically accumulates at the bottom of TSI/KIA/LIA slants is often a good source of highly motile cultures. Screening for H antigens associated with *S. Paratyphi* A and *S. Typhi* is carried out using H:a and H:d antisera, respectively. *S. Paratyphi* A will react with H:a antisera (rare strains may exhibit a H:1.5 antigen and not react with H:a sera). Virtually all *S. Typhi* isolates will react with H:d antiserum; however, occasionally H:j or H:z66 strains may be encountered (primarily among isolates from the Asia-Pacific region)(56).

5.5

Identification of *S. Typhi*, *S. Paratyphi* A, *S. Paratyphi* B, *S. Paratyphi* C and iNTS

5.5.1 *Salmonella Typhi*

Suspect *S. Typhi* isolates can be confirmed by the characteristic phenotypic profile using either:

1. Biochemical tests: TSI/KIA, citrate, indole, urease, MIO, LIA (see **Annex A5.2**)
2. An automated identification system giving high confidence identification
3. Screening biochemical panel (see **Chapter 5.2.2**) plus serotyping: O (Group D), Vi and H-type d (see **Annex 5.3.1**)

Note: *S. Typhi* does not produce gas but not all O:9 positive gas negative serovars are *S. Typhi*.

5.5.2 *Salmonella Paratyphi* A

Suspect *S. Paratyphi* A isolates can be confirmed by the characteristic phenotypic profile using either:

1. Biochemical tests: TSI/KIA, citrate, indole, urea, ornithine, lysine (see **Annex A5.2**)
2. An automated identification system giving a high confidence identification
3. Screening biochemical panel (see **Chapter 5.2.2**) plus serotyping: O (Group A), and H-type a

Note: Laboratories overlook *S. Paratyphi* A because its atypical phenotypic profile may be confused with (an inactive) *E. coli* or perhaps *Shigella* spp. (92).

5.5.3 *Salmonella Paratyphi* B

S. Paratyphi B and *S. Paratyphi* B variant L(+)-tartrate (d-tartrate) + (also known as biovariant Java) have the biochemical profile of a typical NTS and same antigenic formula i.e. serotyping with O (Group B) and H-type b.

These biovariants are distinguished phenotypically by their tartrate reaction (see **Annex A5.1, Table A5.3**).

S. Java is tartrate-positive and causes gastrointestinal illness, whereas *S. Paratyphi* B is tartrate-negative and causes typhoidal Salmonellosis. The antigenic formula and biochemical characteristics of *S. Paratyphi* B, *S. Java* and an antigenically similar subspecies II serovar are shown in **Annex A5.1, Table A5.3**.

5.5.4 *Salmonella Paratyphi C*

The antigenic formula 6,7:c:1,5 is shared by different *Salmonella* serovars: Paratyphi C, Typhisuis, Choleraesuis (56). Furthermore, three biotype subdivisions based on the basis H₂S production, and on the utilization of mucate and dulcitol have been described within the *S. Choleraesuis* serovar: Choleraesuis sensu stricto, Choleraesuis var. Kunzendorf, and Choleraesuis var. Decatur (93) (see **Annex 5.1, Table A5.4**).

S. Paratyphi C is associated with enteric fever in humans, *S. Typhisuis* with chronic paratyphoid/caseous lymphadenitis in swine (94) and *S. Choleraesuis* may cause serious outbreaks of Salmonellosis and paratyphoid in pigs (95), with clinical outcomes, such as enterocolitis and septicemia (96), often resulting in fatal systemic disease (97). Serovar Choleraesuis is currently highly prevalent in north America and Asia, but it is rare in Australia and the European Union (EU)(98).

5.5.5 Non-typhoidal *Salmonella*

Suspect *Salmonella* isolates can be confirmed to the genus level by the characteristic phenotypic profile using either:

1. Biochemical tests: TSI/KIA, citrate, indole, urea, ornithine, lysine
2. An automated identification system giving a high confidence identification

An isolate which is biochemically identified as a *Salmonella* should be referred to a Reference Laboratory for serotyping (traditional or molecular) and other subtyping methodologies as appropriate. However, the reporting of patient results should not be delayed pending serotyping results.

5.6

Laboratory reporting

5.6.1 Preliminary report

A preliminary report can be issued as soon as *Salmonella* identification is obtained. If the biochemical algorithm provided is followed, the isolate may be reported as:

‘*Salmonella* Typhi’

‘*Salmonella* Paratyphi A or B or C’ or

‘*Salmonella* sp. not serovar Typhi or Paratyphi A or B or C’

In most situations, the identification of *S. Typhi* or *S. Paratyphi A or B or C* will be based on phenotypic traits determined either by traditional or commercial systems and complimented by reactivity with *Salmonella* O grouping antisera.

Report heading must state ‘Preliminary Report’, as most hospital laboratories lack the capacity to fully type NTS. If a culture is sent to a Reference Laboratory for confirmation, enter a comment indicating: “Isolate has been sent for confirmatory identification to [name and location]”.

5.6.2 Confirmed identification

Report heading must state ‘Final Report’. If the final report follows a preliminary report, add a statement to clearly identify the previous report e.g., “Update of preliminary results issued for <insert Sample number or Report number> – dated DD-MM-YYYY”.

5.7

Notification of *S. Typhi*, *S. Paratyphi A*, *S. Paratyphi B*, *S. Paratyphi C* and *Salmonella* spp

Notify all reportable enteric pathogens to the appropriate public health agency as required by the local regulatory and government jurisdictions. Immediately notify the local health department of suspected *S. Typhi* and *S. Paratyphi A* or *B* or *C* because of their epidemiological consequences.

5.8

Widal test

This serological test relies on the agglutination reaction between known antigens for *S. Typhi* and *S. Paratyphi A* and antibody present in the patient's serum, e.g., agglutination with antigens O:9, Vi and H-d, may suggest a *S. Typhi* infection. However, although serologic tests are commonly used in many settings, current evidence suggests that these tests are limited by poor sensitivity and inadequate specificity, and thus, inappropriate for use in routine surveillance or for diagnosis (18).

5.9

Storage of *Salmonella* isolates

Storage of isolates is recommended to allow further testing (serotyping, genotyping and AST) at a later date, if required. Due to the complexity of tests, some tests may need to be performed at a Reference Laboratory. The isolate earmarked for storage should consist of fresh growth, derived from a single colony, and harvested from a non-selective purity plate.

Samples should be stored (see **Annex 5.4**) either:

1. Short-term (2 weeks):

At room temperature on suitable media. Motility stabs or slants should be sealed with parafilm. Media with minimal additional carbohydrates e.g., trypticase soy agar or NA is recommended to be stored at room temperature

or

2. Long-term (2-3 years):

a. As freeze-dried cultures

or

b. If available, frozen in Trypticase Soy Broth (TSB) with 15-20% glycerol stored at less than -70°C or in cryobeads stored at -80°C

Stored isolates should be labelled with their unique identifier that links with the patient information held in the Laboratory Information Management System.

5.10

Waste disposal and decontamination

Laboratory guidelines should be used to minimize the risk associated with the disposal of laboratory waste, especially low infectious dose *Salmonella* serovars, such as *S. Typhi* and *S. Paratyphi A*. All the waste generated from isolating and identifying typhoidal or NTS should be categorized as biohazard (infectious) and waste must be disposed of in a container labelled with the biohazard sign, including all the culture media (both solid and liquid) inoculated with bacterial specimens.

Any sharp or glass objects, such as microscope slides and pipette tips should be discarded separately in an appropriate, approved, rigid-walled (sharps labelled) container. Do not overfill sharps containers i.e., no more than two thirds full.

Discarded containers should be positioned adjacent to the work area and easily accessible for discarding infectious materials. Infectious materials from the container should be autoclaved or incinerated before disposal. After sterilization, liquid waste can be drained off through the existing drainage facility or discarded in the specific waste disposal area.

Chapter 6

Antimicrobial susceptibility testing of *Salmonella enterica* serovars Typhi and Paratyphi A, B and C

6.1 Background

Treatment with an appropriate antimicrobial agent is crucial for patients with typhoid. With the global spread of multidrug resistance (MDR) (99–101), resistance to ampicillin, chloramphenicol and co-trimoxazole, recent emergence of extensively drug resistant (XDR) strains in Pakistan (44), MDR and resistance to ciprofloxacin and ceftriaxone, and a growing number of azithromycin resistant mutants (R717Q/L mutation in AcrB gene) in Bangladesh (102), we risk losing our key defences for treating typhoid fever (49). The ease of travel is no longer confining these strains to endemic regions and recently there have been at least eight additional travel-related XDR and several azithromycin resistant AcrB R717Q/L mutants reported worldwide (43,103). Therefore, *S. Typhi* and *S. Paratyphi* isolates should undergo AST as soon as possible. The AST methods presented in this chapter are based on the Clinical and Laboratory Standards Institute (CLSI) (formerly known as National Committee on Clinical Laboratory Standards – NCCLS) (<https://clsi.org>) and European Committee on Antimicrobial Susceptibility Testing (EUCAST) (<http://www.eucast.org>) guidelines. If performed according to the description below, these methods will provide data that can reliably predict in vivo effectiveness of the drug in question. However, any deviation from the method can invalidate the results. For this reason, if laboratories lack the resources to perform the test as described in the protocol, isolates should be forwarded to laboratories with the capacity and competence to perform AST.

6.2 Different methods for AST

Several methods are employed in different clinical laboratories, with the decision often based on the local availability of resources. These data are used both for patient management and to generate data for evidence-based health policy decisions. These approaches include disk diffusion (Kirby-Bauer), broth microdilution and gradient strip diffusion (e.g. E-test). Several other commercial systems are also available, which use culture or molecular based methods to characterize antimicrobial resistance among any given bacterial species. However, the commercial systems mentioned later have been excluded in this manual and only the first three techniques recommended either by CLSI and/or EUCAST will be discussed in the later sections.

6.2.1 Selection of antimicrobial agents and special considerations

Testing some bacteria against certain antimicrobial agents may determine misleading results because the in vitro results do not necessarily correlate with in vivo activity. *Salmonella* (including ser. Typhi and Paratyphi A, B or C) isolates, for instance, are usually susceptible to aminoglycosides (e.g., gentamicin, kanamycin) and first- and second-generation cephalosporins using the disk diffusion test, but treatment with these drugs is often not effective [EUCAST 2020]. It is also worth noting that sometimes the result of one AST will indicate the need for additional tests to confirm an expected result. For example, when an isolate of *S. Typhi* or *S. Paratyphi* is resistant to nalidixic acid, it will usually exhibit reduced susceptibility to ciprofloxacin; this scenario may translate in the clinical setting, the need for a higher dose or extended course of treatment. Isolates exhibiting resistance to nalidixic acid should be tested for susceptibility to ciprofloxacin. More recently, pefloxacin has been recommended as surrogate for ciprofloxacin. Pefloxacin may be tested in lieu of ciprofloxacin with the ciprofloxacin result extrapolated from the pefloxacin result. The following antimicrobial agents can be used for susceptibility testing of *S.*

Typhi and *S. Paratyphi* A, B and C (**Table 2**). Interpretive criteria and clinical breakpoints suggested for AST can be found in the:

1. CLSI (Enterobacterales) guidelines

<https://clsi.org/standards/products/microbiology/documents/m100/>

or

2. EUCAST (Enterobacterales) guidelines

http://www.eucast.org/clinical_breakpoints/

TABLE
2

Antimicrobial agents to be used for the susceptibility profiling of *S. Typhi* and *S. Paratyphi* A, B and C.

The given potency applies only for the disk diffusion method as concentration/potency of any antibiotic varies, which can be adjusted in case of E-strips and broth microdilution.

ANTIMICROBIAL AGENTS	POTENCY (µg/disk)	NOTES	CLINICAL BREAKPOINTS
Ampicillin (AMP)	10 µg	Priority Antibiotics * Can also consider Ceftazidime or Cefotaxime on availability. ** The preferred test for assessing fluoroquinolone susceptibility or resistance is a ciprofloxacin MIC test.	Please see the most up-to-date CLSI or EUCAST guidelines for breakpoints and updates
Co-trimoxazole (SXT)	1.25/23.75 µg		
Chloramphenicol (C)	30 µg		
Ceftriaxone (CRO)*	30 µg		
Ciprofloxacin (CIP)**	5 µg		
Pefloxacin (PEF)	5 µg	If not susceptible to “Priority Antibiotics” ** Can also consider Meropenem or Ertapenem	
Azithromycin (AZM)	15 µg		
Imipenem (IMP)**	10 µg		
Tetracycline (TE)	30 µg	Only suited for “epidemiological studies”	
Amoxicillin/Clavulanic acid (AMC)	20/10 µg		

6.2.2 Agar disk diffusion (Kirby-Bauer method)

After being extensively scrutinized by W. M. Kirby and his colleague, A. W. Bauer, the standardized single disk method was initially introduced in 1966 (104). Currently, this protocol (with some modifications), with the help of the WHO, CLSI and EUCAST has become a standardized method for AST all over the world. Mueller-Hinton agar (MHA) medium is the only medium that has been validated thoroughly and should always be used for susceptibility testing for *S. Typhi* and *S. Paratyphi* (105) (as suggested by the EUCAST guidelines (106), see **Annex 6.1**). As seemingly minor deviations in pH or cation ratios may significantly impact results, only prepared plates or dehydrated media from a reputable commercial source should be used. Laboratories should not attempt to prepare MHA from the individual constituent ingredients. All media (including commercially prepared plates) must undergo required QC testing before and during use.

6.2.3 Determination of minimum inhibitory concentration (MIC)

The MIC is the lowest concentration of a molecule (typically an antimicrobial) needed to prevent the growth of a given microorganism. It should not be confused with minimum bactericidal concentration, which generally is the concentration required to exterminate a bacterium. MIC can be determined in vitro by preparing gradient (different concentration) of a specific antimicrobial, followed by inoculation and incubation of bacterial cells. After being challenged with varying concentrations of a particular antimicrobial agent, results are usually interpreted by observing the point where bacterial growth has been completely inhibited. The objective of this test is to ensure that the antimicrobial agents are selected in an evidence-based manner, which in turn maximizes the chances of successful treatment (**Annex 6**).

6.2.4 Broth microdilution

The principal behind this semi-quantitative test is like any among its kind, but the growth medium recommended is slightly different. Broth microdilutions use non-selective cation-adjusted Mueller-Hinton broth for in vitro determination of MIC. A microtitre plate consisting of 96 wells is generally used to test several microorganisms at a time, making the procedure convenient, economical and time saving. Inoculum preparation for this test is prepared as previously described but, antimicrobial agent preparation and inoculation procedure is distinct. In order to prepare antimicrobial agents for this test, it is recommended to follow the guidelines provided by CLSI or EUCAST (see links in **Chapter 6.2.1** and also **Annex A6.3**).

6.2.5 E-test

Previously known as the Epsilometer test, E-test (and similar products produced by other vendors) is a tool used in many laboratories to determine the MIC of any specific strain of bacteria. First described in the 1980s, it uses a slender nonporous plastic strip that is impregnated with dried antimicrobial agents of varying concentration, which creates a gradient when diffused through the solid bacteriological agar surface. Due to the ease, accuracy, and reproducibility of the method (**Annex A6.2.1**), it is highly regarded in some parts of the world despite being more expensive than the conventional broth microdilution (**Annex A6.2**).

6.3

Reporting the results

After obtaining zone diameter or MIC value, information must be stored in a secure server using a commercial or in-house software programme. If such a system is not available, a paper-based logbook can be maintained to record all the antimicrobial susceptibility data against any given isolate. Variables such as isolate/case identification ID, name, age, gender of the patient, organism name, antimicrobial tested zone size (in mm) / MIC (in µg/ml) and interpretive category must be kept at minimum. Reporting of the result can vary by organism and the site of infection. Therefore, the selection of antimicrobial agents test panel and final report must only be given after consulting with the assigned medical microbiologist.

Chapter 7

Molecular testing via conventional and Real-Time Polymerase Chain Reaction

7.1

Background

Rapid and accurate differentiation of *Salmonella* spp. causing enteric fever from NTS is essential for clinical management of cases, laboratory risk management and implementation of public health measures. The PCR is a quick assay which detects and amplifies a gene target of interest if present in a sample where results can be achieved in a few hours. The assay described below identifies the genus *Salmonella* by detecting the *ttr* gene, encoding tetrathionate reductase present in all *Salmonella* (107) and defines typhoidal *Salmonella* by the detection of *Salmonella* enterica serovar Typhi- and Paratyphi-specific gene combinations. For example, the *tviB* gene, encoding a Vi polysaccharide capsule, which is present in *S. Typhi* and *S. Paratyphi* C (108) but not in *S. Paratyphi* A or *S. Paratyphi* B, can identify a subset of typhoidal *Salmonella* but does not distinguish *S. Typhi* or *S. Paratyphi* C (109). In order to differentiate *Salmonella* serovars causing enteric fever, additional genes are required. The additional genes include: *SPA2308* gene (encoding a hypothetical protein) for the detection of *S. Paratyphi* A; *STY0201* target (also known as the *staG* gene, encoding a putative fimbrial protein) for *S. Typhi* (110); *SPC0869* gene (encoding a hypothetical protein, shown to be present only in *S. Paratyphi* C (109) for *S. Paratyphi* C. *S. Paratyphi* B can be distinguished from *S. Paratyphi* B biovar. Java (NTS) using two genes encoding type III secretion system effector proteins, *sseJ* and *srfJ* (111), with *S. Paratyphi* B possessing only *srfJ* but *S. Paratyphi* B biovar. Java possessing both *sseJ* and *srfJ*. These genes have been validated as a combination approach to differentially identify TS as a single gene approach is not appropriate (58). It is known that 'typhoidal' genes can be acquired in NTS strains but will not match the specific typhoidal *Salmonella* gene profile. It is therefore important that inference of typhoidal and NTS identification is carried out using the whole suite of genes and not one single target. This chapter and corresponding **Annex 7** describes the procedure for differentiation between typhoidal and NTS and the identification of *S. Typhi*, *S. Paratyphi* A, *S. Paratyphi* B and *S. Paratyphi* C by a combination of seven genes: *ttr*, *sseJ*, *srfJ*, *tviB*, *SPC0869*, *SPA2308* and *staG*.

7.2

Sample types

This PCR SOP is based on testing of cultures, though cultures are pure, this PCR will also work on cultures contaminated with other pathogens. The type of sample you are testing will determine what DNA extraction method you will use; isolated culture will provide the highest amount of DNA to test for the PCR and a crude lysate (1 µl of culture in 490 µl of water and boil for 20 minutes) will provide sufficient DNA for this test. However, for other sample types, specialized kits and procedures will be required and there are multiple kits on the market that specialize in DNA extraction such as clinical samples including a blood, faeces and bacterial isolates such as cultures. It is recommended for users to research the most appropriate kit for the type of sample they want to process. Ensure you follow manufacturer's instructions and include a positive control with your DNA extraction batch to ensure the extraction has worked correctly and a negative control to check for contamination. You will need to validate any method you implement in the laboratory before you routinely use it to diagnose *S. Typhi* even when using commercial kits or published methods. If you are undertaking a PCR from a primary specimen, you should also include an internal positive control for each sample.

Chapter 8

Advanced molecular testing – Analysis of whole-genome sequencing data

The use of WGS as a method for the identification and typing of microorganisms has become more accessible over recent years and is now used across the globe in many clinical and reference laboratories. The advantage of using WGS methods over conventional phenotypic methods (which can be interpretative) and PCR (which can vary in sensitivity and specificity) is the accuracy and robustness of the method. The read data can also be used for fine typing to detect outbreaks and antimicrobial resistance markers and replaces multiple laboratory methods. The use of sequencing technologies is currently transforming almost every aspect of biological science. In relation to infectious diseases, the advances are rapidly changing our scientific discoveries, as well as diagnostic and outbreak investigations. The ability to generate and analyse sequencing data and take advantage of the rapid progress, is however, not equally distributed between institutions and countries. For those laboratories who do not have the capacity to sequence samples, DNA can be shipped to a regional or commercial laboratory. As well as the availability of multiple commercial DNA extraction, sequencing kits and sequencing services, publicly available platforms have been developed to analyse the data. This has given many laboratories the capacity to identify and type *Salmonella* without the need for bespoke bioinformatics pipelines or specialized bioinformaticians to process the data. This chapter and more so the accompanying **Annex 8** is a guide for identification and typing of *Salmonella* using raw sequence data files on the freely available genomic platforms Enterobase (<https://enterobase.warwick.ac.uk/>) and Centre of Genomic Epidemiology (<http://www.genomicepidemiology.org/>). Both platforms have multiple applications, so only a couple of key functions will be described, for further information, guides or training opportunities, it is recommended you go directly to the website or contact the administrator of each platform.

There are multiple other platforms including Pathogenwatch (<https://pathogen.watch/>) and TyphiNET (<https://typhinet.herokuapp.com/>) that are available to use and can be accessed using search engines on the Internet (Google). Many have guides available on their websites and it is worth researching the different platforms available to suit your needs.

8.1 Sample types

Typing platforms have been designed to process data from a pure culture of *Salmonella*, this is required to enable fine typing of the strain. Refer to manufacturing instructions of your sequencing equipment for the data output file type to assess if suitable for the platforms; though most platforms are set-up to accept different sequencing output files. Although methods and platforms for metagenomics (sequencing of mixed bacterial populations, such as DNA from faeces) are being developed, many of the processes require bioinformatic expertise and are not discussed in this chapter or annex.

Chapter 9

Quality management

International standards such as ISO 9001/ISO 15189 underline the requirements for a quality management system (QMS). It ensures that an organization (in this case a diagnostic medical microbiology laboratory) continues to offer quality services and operates correctly under its regulatory measures.

It is therefore strongly recommended that tests undertaken within clinical diagnostic laboratories should be assured by a comprehensive QA scheme, which should be subject to third party accreditation to the internationally recognized standard (i.e., ISO/IEC 15189, ISO 17025, ISO9001, CAP) for laboratory competence.

QMS / quality assurance system (QAS)

The QMS covers the provision and control of:

1. SOPs: planned maintenance and calibration of equipment, reagents and material certification
2. Documentation and record keeping including monitoring of response times and audits
3. Education, training and competency of staff
4. Internal Quality Control (IQC), Internal Quality Assessment (IQA) and External Quality Assessment (EQA)

General requirements for the competence of testing and calibration laboratories

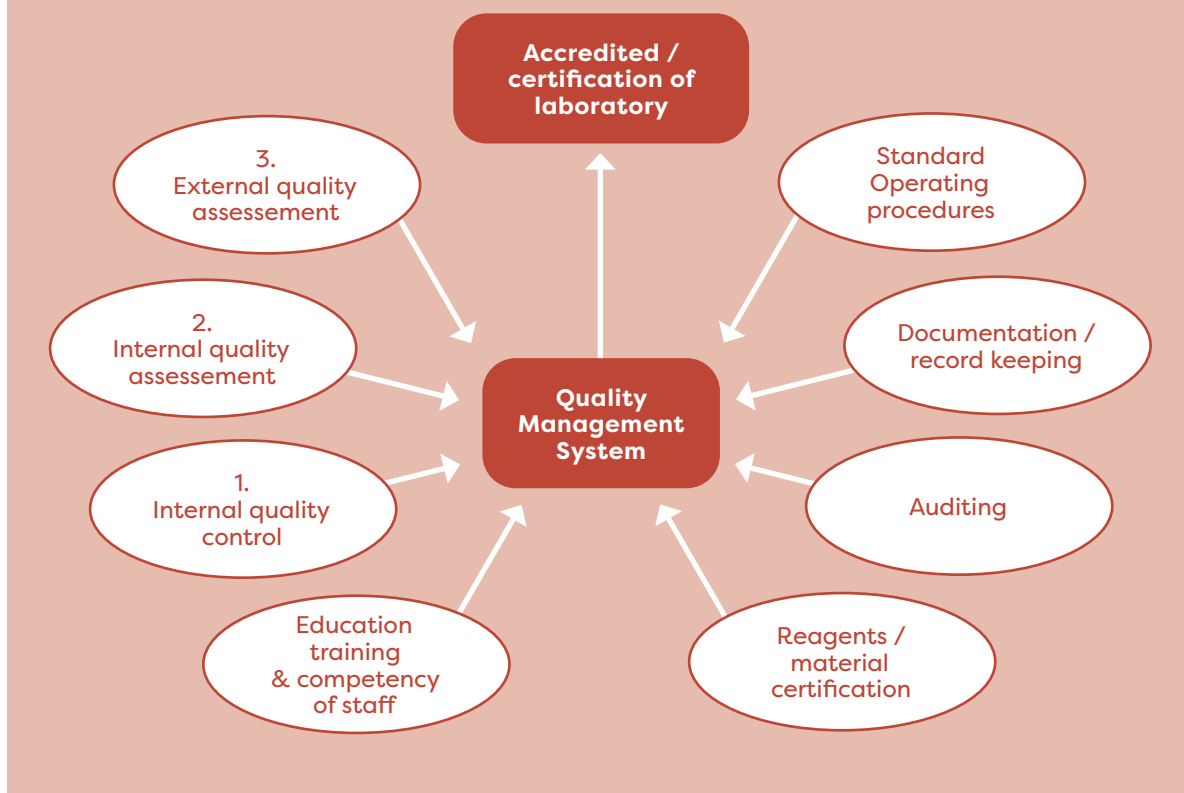
Clinical diagnostic laboratories first and foremost must adhere to the standard of a containment level acceptable for processing the organisms (see **Chapter 2.8** and **2.9**) For clinical samples, potentially containing the causative agent of typhoid fever, national guidelines need to be consulted but it is generally advised that the pathogen(s) be processed within containment level 2 facilities.

Many laboratories are formally accredited (certification) to acknowledge conformance with defined quality standards, such as ISO 17025 (112). Accreditation to ISO 15189 (113) for medical laboratories is a requirement for quality and competence and includes participation in an EQA for a diagnostic test or process if one is available. If not, IQA can partially supplement the requirement. The interplay between the different components of QMS is summarized in **Figure 2**.

The structure and management of the laboratory must be organized so that quality management policies can be established and implemented. The laboratory should prepare an organizational chart that reflects the hierarchy and lines of authority with functions and responsibilities for each post. The current duties and responsibilities of staff should be specified within written job descriptions including the training required and necessary experience. The commitment of management (directors, laboratory managers etc.) is crucial, which should also include the appointment of a dedicated quality manager to ensure the implementation, monitoring and maintenance of quality policies (68).

FIGURE
2

Schematic of components of a quality management system.



9.1

Basis of laboratory QA

Laboratory quality assurance (LQA) is the total process whereby the quality of laboratory results, from specimen collection to analysis of tests and reporting of results, can be guaranteed and enable clinical diagnosis and appropriate treatment (114). LQA helps define the procedures, documents and controls used to enhance the quality and integrity of the activities and the final results generated. Components of QA include good laboratory practice, IQC, audit, validation, IQA, accreditation, evaluation, education and EQA.

Objectives of a well-organized LQA system include:

1. Preventing risks
2. Detecting deviations from the norm
3. Correcting and preventing further errors
4. Improving efficiency
5. Ensuring data quality and integrity

Most European laboratories must comply with accreditation or certifications standards and regulations through third party assessments (regulatory authorities and accreditation bodies) for national recognition and status. The responsibility ultimately lies with the head or chief of the laboratory to establish, implement and ensure compliance with LQA, although all laboratory personnel should understand and adhere to most aspects of the LQA. Elements that constitute a successful QA system are described below.

9.2 Staff and staffing levels

All diagnostic and reference laboratories for *Salmonella* should have staff who are appropriately qualified, competent and experienced to perform the tasks safely and accurately and ensure precise results are reported. A clear organigram should be prepared to illustrate the hierarchy and lines of responsibility and ideally include the director or chief of the laboratory, the head of each unit if appropriate (e.g., identification, toxin testing, serology, molecular typing), the scientific and technical staff, auxiliary and administrative support. Each role should have a job description describing their roles and responsibilities, with academic qualifications, training, skills and experience required for each post.

In addition, staff who routinely handle cultures of *Salmonella* Typhi, should be offered vaccination (including booster vaccinations) according to the respective national immunization guidelines. A risk assessment must be performed to determine if additional precautions and expanded vaccinations are necessary for staff who routinely manipulate primary cultures of clinical material (18,115).

Staffing levels should be adequate to perform all levels of service that a clinical laboratory undertakes without compromising safety or the integrity of the results generated. There should be at least one person with a minimum of 12 months experience in the specialized techniques used and the services provided. At least one additional person should work alongside the experienced person to gain experience training, understanding and competency to build capacity and cover staff absences. In addition, new staff should receive thorough and comprehensive training, and all staff should be encouraged to attend both internal and external training courses, according to the needs of the staff and the laboratory. Such training and test witnessing should be documented to record the skills gained as part of the staffs' continuing personal development and education.

9.3 Space allocation

A dedicated enteric laboratory should have adequate space to safely perform all activities and is usually within a state or government centre or institute, which is recognized by the country's Ministry of Health, or equivalent. The infrastructure and management must be in place to allow enough rooms to separate tasks from infectious and non-infectious work. Space permitting, specific areas and preferably specific rooms should be designated for:

1. Reagents and consumables storage
2. Washing, preparation and sterilization
3. Specimen receipt, registration and recording
4. Bacteriology activities (e.g., conventional identification, AST, molecular methods e.g., DNA extraction)
5. Specialized activities (e.g., unidirectional pathway for PCR preparation, PCR amplification, gel electrophoresis)
6. Documentation, archiving and control
7. The administrative area

Some of these can be shared within other departments of the centre or institute for cost-effectiveness, considering it is safe to do so and does not jeopardize the health of other staff employed, or cross-contaminate reagents.

General characteristics that the clinical laboratory should comply with are:

1. Adequate lighting and ventilation appropriate for the activities performed. The workbench surfaces should be smooth, nonporous, easy to clean and resistant to chemicals
2. Safety systems to combat fire, electrical emergencies, biological and/or chemical spillages and other potential local events
3. Appropriate PPE
4. Sustainable hot and cold running water, and electricity to cover use, for the duration of the activities. These amenities should be sufficient for essential operation of equipment, such as incubators, microbiological safety cabinets, freezers, etc. An added consideration as a backup supply, a standby generator is readily available, especially in areas where power supply is unreliable
5. Adequate space to store both supplies for immediate use and long-term storage some distance away from active working areas
6. Hand washbasins, with running water, should be provided in each laboratory room, ideally located near to the exit door
7. An autoclave within the same building as the laboratory
8. Facilities for storage of outer garments and personal items, and for eating and drinking, outside the working areas
9. Adherence to biosafety and other safety standards during installation and operation of equipment

9.4 Standard Operating Procedures

SOPs describe in detail the activities performed in the laboratory to:

1. Provide uniformity, consistency and reliability in each of the activities performed in the laboratory
2. Reduce systematic errors
3. Provide guidance for new staff who are learning the procedures

The SOPs should be written by competent and experienced staff, approved by the laboratory manager or director, reviewed by those competent in the procedure, and read/acknowledged by the relevant staff. To ensure clarity, SOPs can be divided into sequentially numbered sections and subdivided as appropriate; a common format consists of (**Annex A2**):

1. Title of procedure
2. Unique code or SOP number
3. The author and authorizer of the SOP
4. An 'effective from' and 'review' date
5. Introduction/scope and purpose of the process being described
6. A summary or brief description of the procedure
7. A chronological list of related safety documents, including Material Safety Data Sheets (MSDS), COSHH (biological & chemical) and risk assessments and other cross-referenced SOPs
8. Materials, reagents and equipment used for the procedure

9. A chronological description of method, written in a format, understandable to those with little or no experience
10. Expected results, acceptance and rejection criteria and the recommended interpretations
11. Any quality control or assurance set-up alongside the test specimens for the procedure
12. An acknowledgement form indicating the SOP has been signed and understood

SOPs should be clear and concise, as this will encourage staff to read, understand and comply with processes. Any changes to SOPs should be implemented and finalized in the same manner as a new SOP.

9.5 Documentation and equipment

This applies to the set of quality manuals, SOPs, forms, reports and record of data that serve as evidence of LQA and permit the traceability of data. The laboratory should have the necessary equipment for the accurate performance of all tests performed and monitored where required (e.g., incubator temperatures, airflow of safety cabinets).

Precise and accurate management of purchasing and inventory of reagents, media, and supplies can produce cost savings in addition to contributing towards accurate and timely reporting of laboratory results (68,69). The procedures should be written and implemented to assure that all reagents and supplies are correctly selected, purchased, used and stored in a manner that preserves integrity and reliability. The inventory should be kept up to date including information on receipt, storage and issuance. Package inserts and MSDS should be kept/held on record.

Documents provide written information about policies, processes, and testing procedures and should be stored in the laboratory quality manual (LQM) for each laboratory (116). The LQM should serve as a basis for writing the laboratory SOPs, which need to reflect the laboratory's role and capacity. The SOPs, QC/QA procedures, specimen testing request forms, report forms and other laboratory forms are all important components of the quality manual, which documents the QMS.

An SOP should be written for each of the procedures conducted in the laboratory, for example, specimen collection, transport, storage, waste disposal, Gram stain, microscopy, biochemical tests, culture for isolation, identification, AST, subtyping methods (serological or molecular methods), reagents and media preparation, equipment use and maintenance, and SOP writing (68).

Documentation should be designed and implemented to capture and record evidence of actions and results, for example, request forms, report forms, logbooks, QC results, patient reports, critical communications and notices from hospitals or public health authorities. For the purpose of traceability, entries should clearly state the date (and time) and operator who performed the task.

9.6 Reference materials and reagents

The control strains recommended for phenotypic identification and antimicrobial susceptibility tests are available from a variety of commercial sources. Recommended strains for the QC of phenotypic identification media may be found in the relevant chapters. Required QC strains for AST (see **Annex A6.1.1**) may be obtained from the most current CLSI (www.CLSI.org) or EUCAST guidelines (www.EUCAST.org).

When a reference strain is received, the organism should be cultured for purity and to ensure appropriate reactivity (phenotypic or molecular). The isolate can be stored at $\leq 80^{\circ}\text{C}$ in TSB with 20% glycerol (TSB with glycerol) in multiple, individual cryovials. Other long-term storage examples include inoculating into cryobeads (following manufactures instructions), stored at -80°C . Where -80°C freezers are not available, cultures for *Salmonella* (but not fastidious organisms) have been known to retain viability on cryobeads as low as -40°C , but this isn't best practice. Other medium term storage methods include inoculating the culture onto Dorset egg slants on screwcap bijoux containers, these can be kept at room temperature ($16-22^{\circ}\text{C}$), though these can dry out if not maintained correctly. On the first workday of the month, the QC strains are plated and inoculated onto NA slants (slopes) or Tryptic Soy Agar slants. These slants are used for the week's QC needs, stored at room temperature and sub-cultured on Friday. The slants can be sub-cultured weekly for one month. To avoid a laboratory adapted organism of the QC isolates, a new cryovial should be retrieved from the freezer to prepare the QC strains, ready for the next set of diagnostic testing for the next month.

9.6.1 Reagents are chemical or biological materials used to perform specific assays

Laboratories should hold a reserve stock to efficiently perform the assays and guarantee provision of a reputable *Salmonella* (Typhoid) diagnostic/typing service. In general, it is recommended to check the quality of a reagent, using the appropriate controls with each new lot number or delivery. Reagents used for AST have very specific QC requirements which is determined by a number of factors, including the frequency of testing and previous QC results. Specific information can be found in CLSI (www.CLSI.org) and/or EUCAST guidelines (www.EUCAST.org).

Organism identification and characterization requires a variety of instruments/equipment, such as brightfield microscopes, incubators, autoclaves, biosafety cabinets, refrigerators, freezers, water baths, automated identification and antibiotic susceptibility testing systems. Choosing the right equipment, correct installation and assuring that the staff using the equipment are trained and deemed competent. To assure that the new equipment functions to required specifications, appropriate and regular maintenance is crucial. Equipment manuals should be easily accessible within the laboratory area for reference. An inventory of equipment including records of maintenance and repair must be maintained.

A logbook or folder (regularly updated) should be kept with the following details for all reference materials and reagents:

1. Name and catalogue number of the reference material/reagent
2. Manufacturer and supplier
3. Lot/batch number
4. Date of analysis or QC certificate
5. Expiry date
6. Concentration (where applicable)
7. Initials of the person responsible
8. Location of storage e.g., flammable/toxic
9. Temperature monitoring of storage conditions

9.7

Audits and accreditation

Audits are a way of independently examining the documentation and processes to assess whether they are of a certain standard and are appropriate for the service provided. Audits may be internal, performed by staff that do not have direct involvement in the processes being assessed, or external by regulating authorities or accreditation bodies for QA purposes. Audits should be viewed as a way of maintaining and/or improving the quality of the service by identifying any weaknesses and undertaking corrective actions and should be regularly conducted and recorded accordingly.

There are three types of audits that can be conducted:

1. **Vertical:** one department audited against all elements of a standard or procedure
2. **Horizontal:** all departments involved are audited against one standard or procedure e.g. document control
3. **Examination:** an observation of an activity e.g., Gram stain

Traceability is an important concept allows an audit to be traced back to the laboratory report.

Accreditation provides documentation that the laboratory has the capability to detect, identify and perform susceptibility testing of *Salmonella*. The process also provides a learning opportunity, a mechanism for identifying resources, training needs and a measure of progress.

There is no defined WHO mechanism for the accreditation of national *Salmonella* or typhoid laboratories. Accreditation is usually achieved according to the criteria established within each country or region.

9.8

Training workshops and internal quality assurance/external quality assurance (IQA/EQA)

One of the most important laboratory resources is competent, trained and motivated staff. Continuous education opportunities should be offered and available to the staff and recorded, especially if and when new tests or methods are introduced or changed. Regular competency assessment and proficiency testing should be conducted and documented. Competency assessment can be carried out by direct observation of the operator, evaluating records, and/or analysing IQC or EQA results.

Numerous workshops have been held in the held in the African, South-East Asia, European, Eastern Mediterranean and Western Pacific Regions by WHO during the last 15 years, under the auspices of WHO headquarters and the Regional Offices of each region, as well as the US-CDC. These workshops are key to maintain awareness and microbiological expertise. It is essential to maintain this level of training globally on a regular basis.

9.9

Benefits of EQA

The EQA allows the participating laboratory to assess its own performance with specimens distributed for investigation in comparison with the expected results and compare performance with other sites. Participation in an EQA can:

1. Highlight issues at an early stage with inadequacies with kits and procedures
2. Provide objective evidence and efficacy of testing pathways
3. Monitor internal QC procedures
4. Provide an educational stimulus for improvement
5. Identify any staff training needs

It is important to maintain this high standard and to continue offering training and EQA programmes within this specialized area of microbiology globally.

9.10

Quality Assurance

Quality assurance (QA) is usually composed of three complimentary processes, as described below.

9.10.1 INTERNAL QUALITY CONTROL

Internal quality control (IQC) is carried out at regular intervals e.g., weekly, within the laboratory to ensure the systems are performing to pre-defined specifications. IQC assesses and monitors test performance in real-time to ensure consistent performance and reproducible, valid results. IQC monitors processes or measures reproducibility, but not necessarily the accuracy, in order to effectively monitor accuracy labs should participate in an external quality assurance (EQA). The majority of these procedures involves comparing the sample test results in conjunction with the predetermined expected result of control material or international standards (where available) to validate the test results and the process used to attain the results.

For example, test NCTC/ATCC reference organisms at the same time as the test organism to ensure the conditions e.g., inoculum, temperature and media are assessed and meet the predetermined criteria.

9.10.2 INTERNAL QUALITY ASSURANCE

Clinical specimens are anonymously returned into the diagnostic laboratory processes to check that the media, reagents and equipment are performing within specifications – demonstrating that results are reproducible i.e., the ‘same today as they were yesterday’ approach.

Repeat testing confirms the processes are:

1. Operating according to predetermined acceptance criteria
2. Monitors performance of staff/operators
3. Monitors consistency/reproducibility of test procedures

9.10.3 EXTERNAL QUALITY ASSURANCE

External quality assurance (EQA) is the challenge of the effectiveness of a laboratory's quality system with specimens of known but undisclosed content, allowing you to see if your results are the same as another laboratory performing the same test. Thus, EQA provides inter-laboratory comparability, standardization of diagnostic testing and fundamentally is an educational tool (www.ukneqasmicro.org.uk).

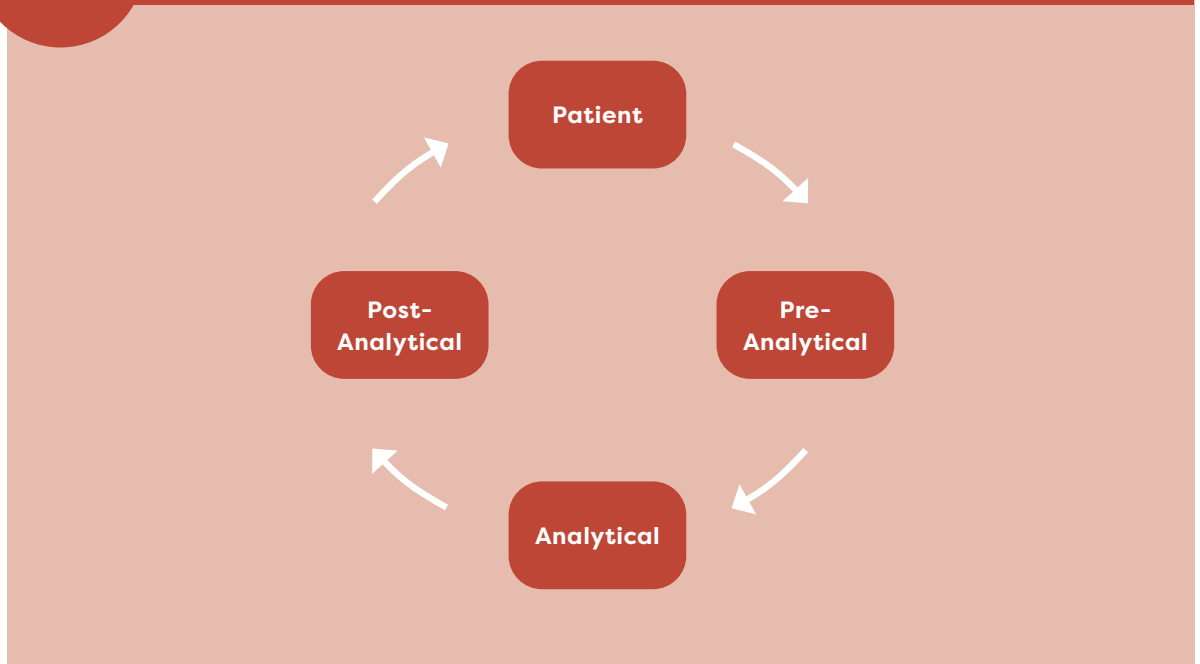
A system for objectively checking the laboratory's performance using an external provider provides reassurance that all components of the quality system are working.

9.10.4 The total testing process

Laboratory processes can be grouped into pre-examination (pre-analytical), examination (analytical), and post-examination (post-analytical) categories (**Figure 3**) (68). Quality management measures should be applied to the entire workflow that begins with the patient and ends in interpreting and reporting results to the clinician. The QMS is not only concerned with monitoring QC/QA programmes, but should also include administrative considerations that may indirectly influence the quality and efficiency of the laboratory operation (67).

FIGURE
3

Stages of the total testing process.



9.10.5 Laboratory errors

The majority of errors occur in the pre-analytical and post-analytical stages and not the analytical, (**Figure 4**) (117,118). However, below are common errors that occur at each of the stages described (119). It should be noted that errors can occur at any point in the total testing process. Identifying critical control points in SOPs and implementing preventive controls to mitigate errors are both important. It is the processes and procedures in the analytical phase that are assessed by participating in an EQA scheme.

Pre-analytical errors

1. Sample mix up, e.g., incorrect labelling of samples
2. Inappropriate or incorrect assignment of tests

Analytical errors

1. Sample mix up
2. Inappropriate tests carried out
3. Diagnostic tests performed incorrectly
4. Automation failure
5. Misidentification of the target organism
6. Contaminant tested instead of target organism
7. Test system not calibrated
8. Reagents prepared incorrectly, stored inappropriately, or used after expiry date
9. Equipment malfunction
10. IQC failed but not considered

Post-analytical errors

1. Result data entry — transcription/transposition error
2. Failure to communicate critical value
3. Provider misinterprets lab result
4. Misinterpretation of results by requester
5. Verbal miscommunication of results
6. Reporting the contaminant(s)

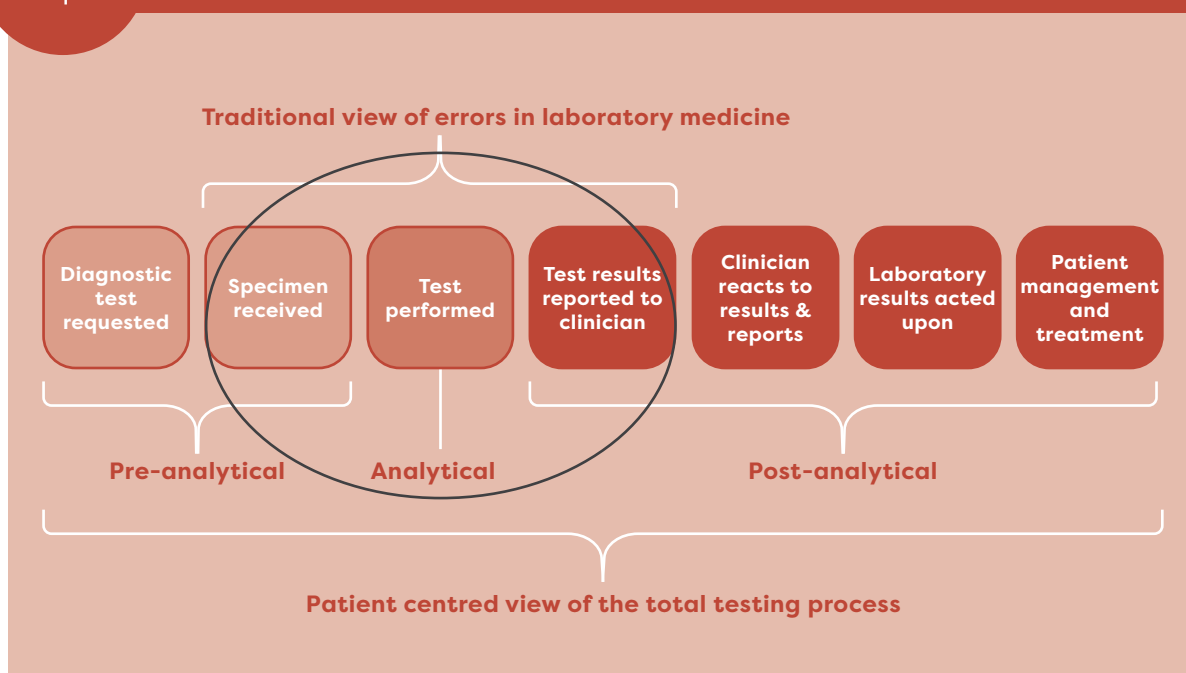
Possible issues with post-analytical stage

1. Turnaround times – delay in date results are reported to the requesting clinician
2. Clinician or another provider fails to retrieve test result
3. Results are misplaced or lost

FIGURE

4

Total testing process with traditional view of errors.



Consequences of laboratory errors – non-conformances

1. Misdiagnosis
2. Mismanagement of patient
3. Incorrect/inappropriate treatment
4. Lack of treatment
5. Delay in treatment
6. Exclusion from or erroneous inclusion in an outbreak investigation

Note: 24-30% of laboratory errors impact patient care, 3-12% may cause potential or actual patient harm (120) .

Investigations for non-conformances

Non-conformances (Nonconformity) is the failure to meet one or more of the existing requirements in ISO/IEC 15189, ISO 17025, ISO9001 and CAP. When a laboratory finds itself outside of regulatory boundaries with laboratory errors, the problem must be resolved before continuing to provide laboratory services. A non-conformance can be found at any level of a laboratories' operations from QC and manufacturing to personnel procedures.

Root cause analysis (RCA) and corrective action

The goal of RCA is to identify the root, or underlying cause of the laboratory error. The next step is to implement long-term corrective actions to eliminate the root cause and prevent recurrence of the error/problem. RCA involves developing recommendations for improvements to reduce the probability of a similar incident occurring in the future.

Questions to pose when performing an RCA (known as the 5 W'S).

1. What happened?
2. When did the error happen?
3. Who was involved?
4. Why did it happen?
5. What preventive measures need to be implemented to prevent re-occurrence of the error?

Internal audits can be conducted by laboratory staff to proactively identify weaknesses and undertake corrective actions. All findings must be documented, SOPs updated and communicated to staff accordingly, and filed for future reference. Fishbone diagram can be used as a tool to visualize the RCA.

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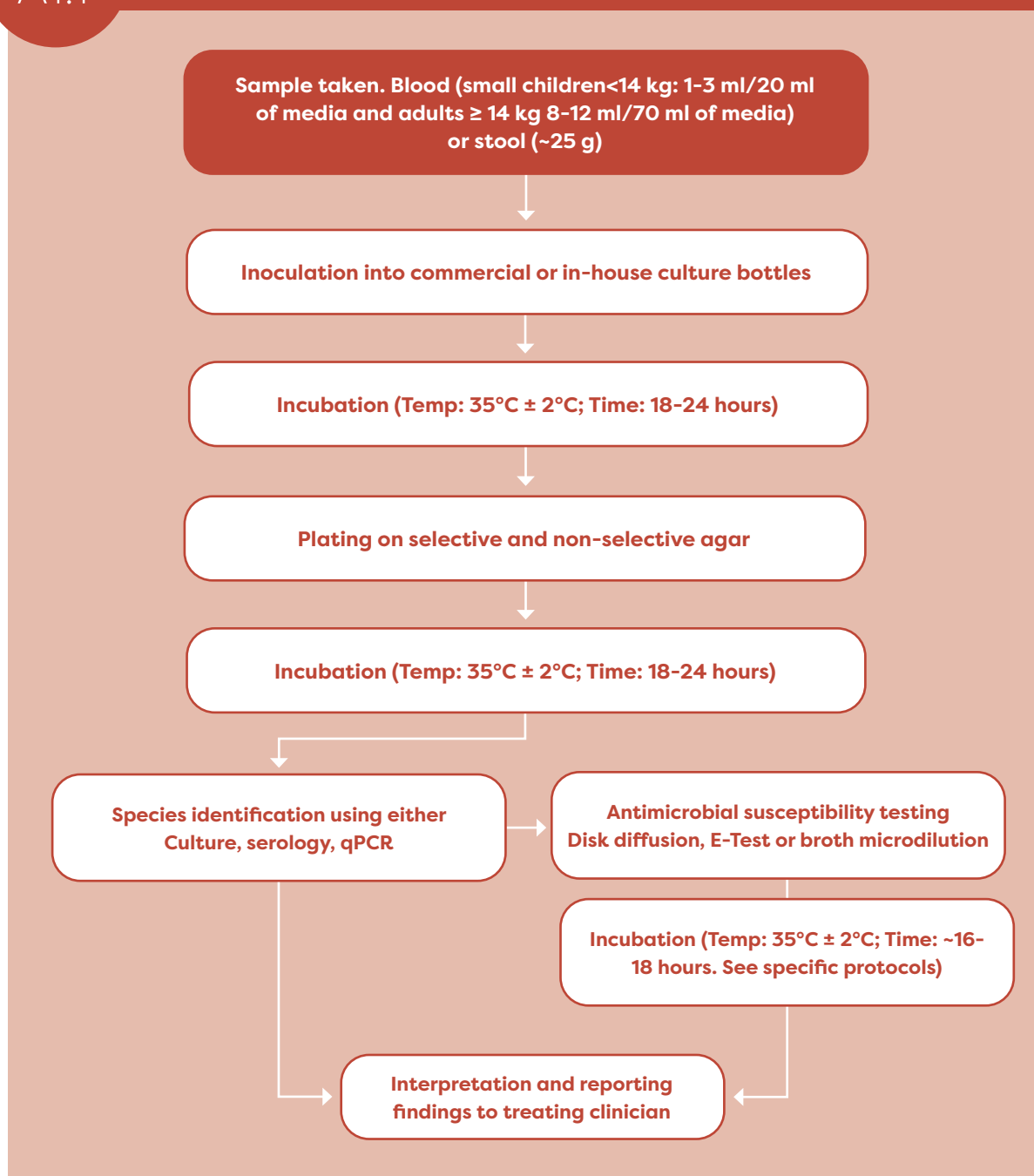
The references that appear in the following chapters will appear at the end of each Annex and do not relate to the references in the main text.

Annex 1

Protocol for sampling, isolating, confirming and diagnosing typhoid and paratyphoid fevers

FIGURE
A1.1

Flow diagram of sampling, isolating, confirming and diagnosing typhoid and paratyphoid fevers that are covered in this manual.



Annex 2

Example of Standard Operating Procedure

A2.1 Standard Operating Procedure (SOP)

A SOP is a set of step by step instructions compiled to help laboratory staff follow and carry out routine procedures. SOPs aim to achieve efficiency, quality output and uniformity of performance, while reducing miscommunication and failure to comply with regulatory standards. With respect to health and safety, an SOP is important as it instructs on a safe system of work for that procedure.

A copy of the relevant SOPs for the procedures carried out in that particular laboratory (together with a copy of the corresponding risk assessment) is kept either in each laboratory or, accessible in other areas as per your local arrangements. This is important for reference and for training in a new procedure.

The following pages include an example of an SOP.

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Page 2 of 4

Overwrite this text with the title of the SOP

1. Purpose

The bullets in this template are to be used at all times. You may NOT delete one of these headings. If the heading does not apply to the particular SOP, type "None applicable".

The template does not always suit the SOP, particularly those SOPs that are more to do with policy rather than procedure. You may wish to add other headings. This is acceptable as long as you maintain a clear and concise style of writing. Please maintain the overall style in the new document.

Overwrite all of this text in bullet 1 with the reason why this SOP is being written. For example: "This SOP is to ensure that all SOPs whether new or under review follow a standardised format". Each section of the SOP is governed by a particular word processing style. This section is "Body Text First Indent". To access a particular style click on the arrow next to the style box (The style box is usually next to the font box). Select the style you want preferably before typing. If you have already typed the text, highlight all the text that you wish to apply the style to.

The English used throughout the SOP should be precise, concise and understandable. Ensure there are no spelling or grammatical errors. It is essential that a logical order of events be maintained throughout the SOP. In addition, it is preferable to give the instruction(s) first and then elaborate regarding those instructions.

1.1 In the case of a laboratory method, enter the principle of the test into this point number (1.1).

2. Area of Application

Describe the section(s) where this procedure will be carried out, e.g. laboratory.

3. Definitions

Cross-refer to the standard document that defines common abbreviations used in the department. Use this area to define uncommon words, both scientific and non-scientific. This section can also be used to expand on any abbreviations not on the standard document used in the SOP.

4. Safety

Describe all instructions necessary for the safe operation of the procedure, e.g. carry out all testing in a fume cabinet. In addition, list all relevant COSHHs, Biological COSHHs and Risks in the table below.

COSHHs and Risks

Number	Title

5. Cross References

List the cross references in the table. Also list any references used to produce the SOP. If there are no references, type 'not applicable'. Cross references are the other departmental SOPs, and any forms, worksheets etc required for the procedure. Procedural references can include

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Page 3 of 4

textbooks, journals, other publications, standards, other non-departmental SOPs, regulations and guidelines.

SOPs and DOCs

Number	Title

Forms/worksheets

Number	Main information recorded	Filed

Procedural references

6. Equipment

List the equipment used in the procedure.

7. Procedure

List the personnel responsible for carrying out the SOP. Give positions of the personnel, NOT names.

7.1 Use this section to list the procedural steps. Ensure that all critical steps are included. Ensure that the steps given are in a logical sequence. Ensure the procedure reflects practice NOT theory. In the case of a laboratory method, this point number (7.1) must outline the internal quality control procedures used during the method.

7.2 If you wish to use a numbered list, select "List Number" from the style box. Adjust the numbering level by clicking on the right indent button. There are four levels available, the main one for the sections, e.g. references; this level, followed by:

7.2.1 This level, and then a bullet point.

- As shown.
- And finally a dash.

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COSHH:
Risk:

Annex 3

Collection, storage and transporting specimens from suspected cases

A3.1 How to collect blood samples

All patients meeting the case definition for suspected enteric fever should have a blood culture done. **Blood culture specimens should be collected before antibiotic administration** otherwise the laboratory may be unable to culture the bacterium, provide confirmation of diagnosis and information on antimicrobial susceptibility of the organism. However, necessary care for critically ill patients should not be delayed pending specimen collection.

A3.1.1 Fundamentals of blood culture

The gold standard for laboratory diagnosis of typhoid fever is the isolation of *Salmonella* Typhi from a normally sterile site (typically blood)

Several factors affect sensitivity:

1. Antimicrobial use prior to sample collection
2. Timing since symptom onset (blood cultures tend to be positive in the first week and stool is typically positive later in the course of disease)
3. Volume of blood collected (varies between adult and paediatric patients)
4. Quality control of blood culture bottles of media
5. Aseptic technique before performing the venepuncture to collect the patient blood.

Quality control: Before taking the blood sample, inspect the blood culture bottle for any signs of damage, contamination, or deterioration. If the date of expiry of the bottle has been exceeded, the broth appears cloudy or darkened, the bottle is damaged. Also if the plastic flip-off cap is missing, or the rubber stopper is bulging or depressed then do NOT use the bottle. Obtain a new bottle.

A3.1.2 Supplies needed to collect blood for culture

In most sentinel sites, blood culturing will probably be conducted by using a semi-automated system (e.g., BACTEC™/ BacT/ALERT®). If a semi-automated system is not available, commercial blood culture bottles should be used (these contain a defined medium that is designed to recover the majority of organisms from blood and the bottles are subjected to rigorous QA). Inoculated bottles can be incubated in a standard incubator at 35 ± 2 °C.

For each patient, one aerobic bottle (minimum) should be inoculated. To collect blood for culture, the following supplies are needed:

1. Liquid soap and clean water
2. 70% isopropyl or ethyl alcohol pad
3. Either: 10% povidone, 2% iodine tincture, or single use alcoholic-chlorhexidine pad
4. Needle and syringe for taking blood sample
5. Nitrile or powder free latex gloves
6. Sterile gauze
7. Adhesive bandage
8. Cotton wool ball
9. Tourniquet
10. Local anaesthetic patch or cream (optional: may be used if available)
11. Aerobic blood culture bottle containing a broth medium capable of supporting growth of common obligate aerobic and facultative organisms
12. Unique Patient ID labels for use on:
 - a. Blood culture bottles
 - b. Case report form
 - c. Specimen collection and feedback form
 - d. Any other blood collection tubes used. The same ID number MUST be used for any other specimen for the same patient (cerebrospinal fluid, pleural effusion)
13. Self-sealing plastic bags with two compartments: self-sealing section for blood samples and a separate section for forms/paperwork
14. Disposable puncture-proof container for needles (safety boxes)

A3.1.3 Essential measures for collecting blood for culture

Only experienced clinical staff should collect blood for culture. The use of a careful, aseptic technique will not only reduce the risk of contamination of the blood culture but also minimize the risk of needle-stick injuries.

The quality of the blood collection technique affects both the sensitivity and the reliability of any system of blood culture. Success in detecting clinically significant pathogens relies upon:

1. **Timing of blood collection:** Blood should be collected **before** administering antimicrobial agents whenever possible; this should not delay necessary care for critically ill patients. The blood cultures should be taken as early as possible in the course of the disease as the number of organisms circulating in the blood stream declines over time (1).
2. **Volume of blood:** The bacterial load in typhoid is low with on average 1 cfu/ml (2). For this reason, it is important to collect an adequate volume of blood for inoculation into blood culture bottles. Inadequate blood volumes decrease the likelihood of detecting clinically significant infections in any blood culture system. Despite the common reluctance to draw blood from younger patients, low-level bacteraemia that might be missed by relying upon small blood volumes can be quite common in typhoid fever cases.
3. **Factors to know for blood volume:**
 1. The bacterial load in blood in acute typhoid is low (≈ 1 cfu/ml of blood) (2) and is maximal during the first week of illness (1)
 2. Small children (<14 kg): 1-3 ml/20 ml of media
 3. Adults (≥ 14 kg): 8-12 ml/70 ml of media
 4. Use the above-mentioned volumes and weight ranges
 5. The optimal volume may vary slightly by manufacturer (typically a ratio of between 1:5 – 1:10 parts blood to media). When commercial products are used, it is critical to adhere to the manufacturer's recommended fill volumes (even if they may differ from the volumes mentioned above).
4. **To ensure quality, blood culture volume should be verified:** This may be accomplished by weighing bottles prior to incubation and comparing the weight against the weight of a bottle (same brand and size) inoculated with a known volume of blood. Alternatively, visual comparison may be performed by comparing the fluid level of the patient sample to reference standards prepared by inoculating control bottles with a known volume of crystal violet or safranin red.
5. **Thorough skin disinfection at the site of venepuncture:** Contamination of blood cultures with bacteria that commonly reside on the skin can complicate patient care and consume unnecessary resources. Thorough disinfection of the venepuncture site involves initially cleaning the site of visible oils and dirt. Visibly soiled skin should be cleaned with soap and clean water. The site is then decontaminated with 70% alcohol swabbing in concentric circles away from the puncture site. The skin is then prepared using either 10% povidone (which is allowed to dry for >1 minute), 2% iodine tincture (which is allowed to dry for > 30 seconds) or alcoholic-chlorexidine (which is allowed to dry for > 30 seconds)(3,4). **See below for a more detailed protocol for drawing blood A3.1.4**
6. **The ratio of blood to culture medium in the blood culture bottle:** Blood culture systems depend upon an optimal broth-to-blood ratio. Commercial blood culture bottles contain concentrated nutrient broth, additional growth factors, antibiotic-binding resin, and an anticoagulant sodium polyanethol sulfonate (SPS). Inoculating the bottles with too little blood might not only exclude the possibility of detecting low-level bacteraemia, but also inhibit bacterial growth due to insufficiently diluted media. Filling bottles beyond their optimal volume can lead to the detection of CO₂ emitted by blood cells rather than growing bacteria. While under-filling culture bottles can cause significant numbers of false negatives.

Note: If bottles from a particular ward or collected by a specific phlebotomist are regularly under-filled or contaminated, refresher training should be offered.

A3.1.4 Recommended procedure for taking blood cultures

The following procedure should be followed when collecting blood for culture (5):

1. Explain the procedure to the patient or guardian.
2. Complete the top (boxed) part of the Specimen Collection and Feedback Form.
3. Place one of the unique Patient ID labels onto the upper right corner of the form.
4. The same unique Patient ID label **MUST** be placed on the blood culture bottle and each tube used to collect blood from the patient. Ensure the collection date, time and initials of the person that collected the blood is written on each blood collection tube and blood culture bottle.

Do not place the Patient ID label over the barcode or lot number.

5. Select an arm for venepuncture. When any dirt is visible on the patient's arm, clean with soap.
6. Apply a tourniquet to the selected upper arm to restrict the flow of venous blood. The most prominent vein of the forearm is usually chosen.
7. Disinfect the skin at the proposed venepuncture (needle insertion) site by (**Figure A3.1**):
 - a. Disinfect first by scrubbing vigorously enough to create friction with a 70% isopropyl alcohol solution (higher concentrations are not effective and should not be used). Repeat until any visible traces of dirt are removed and the pad remains white.
 - b. An iodine solution should then be used and left on the skin for an appropriate length of time before and during the blood draw:
 - a. a tincture of 2% iodine requires 30 seconds before drawing blood
 - b. iodophors, such as 10% povidone-iodine require a one minute contact time.
 - c. Alternatively, the skin may be scrubbed with a single use alcoholic-chlorhexidine pad.

If the vein is palpated again, repeat the skin disinfection.

- d. For patients with iodine sensitivity, scrub with 70% isopropyl alcohol alone for 60 seconds. Blood should always be drawn directly from a vein rather than through a central venous access catheter (5).

FIGURE
A3.1 Skin disinfection



(A and B) Use a 70% isopropyl alcohol solution at the venepuncture site. (C) Once iodine solution has been left for the desired time, the vein can be entered. Source: Images courtesy of CDC photo by Matthew Mikoleit.

8. Remove the protective plastic flip-top cover from the culture bottle.
 9. Carefully disinfect the top (rubber septum) of each bottle with a 70% alcohol wipe. (Use of iodine or other disinfectants for this purpose is NOT recommended.)
 10. Insert the needle, bevel-upwards, into the vein. Once the vein is entered, withdraw the required amount of blood by pulling back the plunger of the syringe at a slow, steady pace. Air must not be pumped into a vein. If additional blood specimens are needed (e.g. for chemistry analysis) these should be drawn after the blood culture is collected. After the desired amount of blood is obtained, release the tourniquet and place a sterile cotton ball over the insertion site while holding the needle in place. Withdraw the needle and ask the patient or guardian to hold the cotton ball firmly in place until the wound has stopped bleeding.
 - a. Alternatively, some blood culture bottles are compatible with butterfly & Vacutainer systems which eliminate both the need to collect blood using a syringe and needle and the need to manually inoculate the blood culture bottle.
 11. Inoculate blood culture bottle with appropriate volume of blood (see Annex A3.1.6) and swirl gently to mix.
- Note: Blood must be inoculated within one minute of collection to avoid coagulation.**
12. Inject remaining blood into other tubes as needed
 13. Dispose of the needle and syringe in a puncture-resistant, disposable container. Do not re-cap the needle
 14. To prevent irritation, remove residual iodine from the venepuncture site after collection by cleaning the skin with 70% isopropyl alcohol. Bandage and ask the patient or guardian to continue to place pressure on the upright arm for a few minutes
 15. Following inoculation, gently rotate the bottle to ensure the sample is completely mixed
 16. Fill tubes from a single venepuncture (if required) in the following order ⁵:
 - a. Blood culture bottles
 - b. Citrate tube
 - c. Plain tube (serum sample)
 - d. Heparin tube
 - e. EDTA tube
 17. Place the inoculated blood culture bottles, other specimen tubes, and the Laboratory Form back in the empty plastic bag and re-seal. If available, use dual compartment plastic self-sealing bags to contain and transport the culture bottles and tubes separately from the paperwork. The bag and specimens should then be placed in a hard walled shipping container for transportation.

A3.1.5 Transport of blood culture to laboratory

1. Blood culture bottles should arrive at laboratory as soon as possible
 - a. Blood culture bottles should arrive no later than 2 days after collection
 - b. SPS / citrated blood is discouraged. Must arrive within 12 hours
2. Samples should be kept at room temperature
3. Do NOT refrigerate blood culture bottles
4. Do not vent bottles prior to transportation
5. Do NOT incubate samples in the 35°C incubator without approval from the receiving laboratory
6. If incubated, note incubation time

A3.1.6 Incubation

1. Some commercially prepared bottles must be vented after inoculation and during incubation. If venting is required, it will be noted in the package insert accompanying the bottles. Follow manufacturer's instructions for venting if required
2. If a commercial system (e.g., BacTec or BacT/ALERT) is used, follow manufacturer instructions for loading the bottle into the instrument for incubation
3. If a commercial system is not used, the bottle should be incubated upright at $35 \pm 2^\circ\text{C}$

A3.2

How to collect stool specimens for culture

Stool samples are the preferred specimens for identifying post-acute/chronic carriers. Typically, at least three stool samples collected at least 24 hours apart (and after the completion of antimicrobial therapy/resolution of symptoms) must test negative to exclude an individual as a potential carrier. However, the timing and number of specimens may vary by jurisdiction and local sanitary guidelines should be consulted.

A3.2.1 Items needed

1. Clean, wide-mouth container. The container must be free of urine, disinfectant or detergent residue
2. Sterile swabs
3. Gloves, nitrile or powder free latex
4. Autoclave bags for disposal of waste
5. Transport medium (Cary-Blair is preferred)
6. Patient ID labels

A3.2.2 Steps in stool collection:

1. Explain procedure to patient or guardian and obtain patient consent.
2. Instruct the patient to provide approximately 25 grams of stool (approximately the size of a walnut).
3. The stool specimen should be collected into a wide-mouth container that is free of urine, detergent or disinfectant residue.
4. Clearly label the container with the "Patient ID sticker", "Patient Name", "Laboratory Number", "Facility Name", and "Date/Time of Collection"

Note: If a stool sample is not possible, follow the below steps to perform a rectal swab.

5. Rectal swab:
- a. If the patient is unable to produce a faecal specimen, a rectal swab may be collected. Rectal swabs are collected by moistening a sterile swab in sterile transport media, inserting the swab 2-3 cm (1-1.5 inches) into the rectal sphincter, and gently rotating.
 - b. Visible faecal matter should be present on the swab, if it is not present then repeat the procedure using the same swab.
 - c. If the specimen can be cultured within 2 hours, place in tube and it should be kept cool (cold packs) and transported directly to the laboratory for culture.
 - d. If the specimen cannot be cultured within 2 hours, coat two sterile swabs in faecal material (sample any areas containing blood or intestinal epithelium) and place them into a tube of Cary-Blair medium (other media, such as Amies' or Stuart's may be used, however, the recovery of *V. cholerae* will be diminished. Buffered glycerol saline should not be used).
 - e. Tightly cap the tube of transport medium
 - f. Label the tube with the swab and transport medium with the "Patient ID Sticker", "Patient Name", "Laboratory Number", "Facility Name", and "Date/Time of Collection"
6. Specimens in transport media should be kept refrigerated and submitted to the laboratory within 48-72 hours. Refrigeration reduces the pH and can kill *Shigella* species if present in the stool.
7. If the specimen will not reach the laboratory within 72 hours, the specimen should be frozen (< -70 °C is preferable) until transport.

References for Annex 3

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Annex 4

Isolating bacteria from clinical specimens

A4.1 Sub-culturing blood cultures

A4.1.1 Sample extraction of specimen cultured in blood bottles

Note: Venting and/or opening blood culture bottles has the potential to generate aerosols and potentially release organisms that are highly infectious via aerosol (e.g., *Brucella* or *Neisseria meningitidis*). Ideally, bottles should be vented/opened in a biological safety cabinet.

Place the blood specimen bottle upright and clean the rubber stopper with an alcohol swab. To release any build-up of pressure in the bottle, insert a sterile venting needle through the alcohol-swabbed stopper. After the pressure has been released remove the venting needle and discard appropriately. The broth can then be sampled using a sterile needle and syringe.

For maximum yield of isolates, all negative cultures can be sub-cultured after 5 days incubation before they are discarded.

Blood specimens can be cultured on either:

1. Blood agar (BA, 5% sheep [SBA] or horse [HBA])
2. Chocolate agar (CA)
3. MacConkey agar (MAC)

A4.1.1.1 Preparation of bacteriological Agar media for blood samples

Important note: Detailed instructions have been provided for preparation of media from first principals using individual components. However, follow the manufacturer's instructions where commercially available powders are employed.

1. **BA plate: trypticase soy agar (TSA) + 5% sheep or horse blood**

Media preparation

1. Prepare the TSA media according to the manufacturer's instructions
 - a. Ideally, prepare 500 ml of molten agar in a 1-2 L flask
 - b. The media should be heated and fully dissolved with no powder on the walls of the vessel before autoclaving
2. Autoclave at 121 °C for 20 minutes.
3. Cool to 60°C in a water bath. Keep in water bath to prevent solidification until ready to do the following steps.
4. Add 5% sterile, defibrinated sheep or horse blood
 - a. For example, 5 ml blood can be added to 100 ml of TSA agar
 - b. For different volumes of agar, adjust the amount of sheep or horse blood added to 5% (e.g., 50 ml of blood per L of medium). Do NOT use human blood.
5. Dispense 20 ml into 15x100 mm Petri dishes. Allow the media to solidify and condensation to dry for 18-24 hours.
6. Place the plates in sterile plastic bags and store inverted (lid side down) at 4°C until use. Label with the type of agar plate and date prepared.
7. As a sterility test, incubate an uninoculated plate at 35 °C ± 2 °C with ~5% CO₂ enrichment (or in a candle-jar) / 48 hours. If colonies grow the plate is contaminated and should be discarded.

2. **CA plate**

Media preparation

1. Prepare the CA medium according to the manufacturer's instructions
2. Autoclave at 121 °C for 15 minutes
3. Heat-lyse a volume of horse or sheep blood that is 5% of the total volume of agar media being prepared very slowly to 56 °C in a water bath.
4. Dispense 20 ml into 15x100 mm Petri dishes. Allow the media to solidify and condensation to dry.
5. Place the plates in sterile plastic bags and store inverted at 4°C until use. Label with type of agar and date prepared.
6. As a sterility test, incubate an uninoculated plate at 35 °C ± 2 °C with ~5% CO₂ enrichment (or in a candle-jar) / 48 hours. If colonies grow the plate is contaminated and should be discarded.

3. **MAC agar plate**

Media preparation

1. Prepare MAC agar medium according to manufacturer's instructions.
2. Sterilize the medium by autoclaving at 121 °C for 15 minutes.
3. Cool to 50 °C in a water bath.
4. Dispense 20 ml into 15x100 mm Petri dishes. Allow the media to solidify and condensation to dry.
5. Place the plates in sterile plastic bags and store inverted at 4°C until use. Label with the type of agar and date prepared.
6. As a sterility test, incubate an uninoculated plate at 35 °C ± 2 °C with ~5% CO₂ enrichment (or in a candle-jar) / for 48 hours. If colonies grow the plate has been contaminated and should be discarded.

D. Nutrient agar

See Annex A5.5.1

A4.1.2 Plating of blood specimen

Transfer approximately two drops (100ul) per plate of the blood culture broth onto the Agar plate and streak to grow for isolated colonies. Label the plates with the specimen, date and any other information needed to correctly identify the plate.

1. Incubate the selected agar plate at $35\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$ / aerobic atmosphere with ~5% CO_2 enrichment (or in a candle-jar) / 18-24 hours.
2. Inspect plate for growth and identify the colonies (see **Figure A4.1** for examples of *Salmonella* growth).

Note: for colony characteristics see Chapter 4, Table 1.

A4.2

Sub-culturing stool cultures

A4.2.1 Preparation of bacteriological media for stool samples

1. XLD agar

1. Prepare XLD agar medium according to manufacturer's instructions.
2. Sterilize the medium by autoclaving at $118\text{ }^{\circ}\text{C}$ for 10 minutes. Note: some products should not be autoclaved. See manufacturer's protocol.
3. Cool to $45\text{--}50\text{ }^{\circ}\text{C}$ in a water bath.
4. Dispense 20 ml into 15x100 mm Petri dishes. Allow the media to solidify and condensation to dry.
5. As a sterility test, incubate an uninoculated plate at $35\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$ with ~5% CO_2 enrichment (or in a candle-jar) / for 48 hours. If colonies grow the plate has been contaminated and should be discarded.
6. Test samples of the finished product for performance using stable, typical control cultures.

2. HE agar

1. Prepare HE agar medium according to manufacturer's instructions.
2. Heat to boiling to dissolve product but do not autoclave (see manufacturer's guidelines for precise protocol).
3. Cool to $45\text{--}50\text{ }^{\circ}\text{C}$.
4. Dispense 20 ml into 15x100 mm Petri dishes. Allow the media to solidify and condensation to dry.
5. As a sterility test, incubate an uninoculated plate at $35\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$ with ~5% CO_2 enrichment (or in a candle-jar) / for 48 hours. If colonies grow the plate has been contaminated and should be discarded.
6. Test samples of the finished product for performance using stable, typical control cultures.

3. Selenite F/Cystine broth

1. Prepare the Selenite F or Selenite Cysteine broth according to manufacturer's instructions. **Care must be taken when preparing Selenite as it is a toxic compound.**
2. Heat to boiling to dissolve product but do not overheat or autoclave (see manufacturer's guidelines for precise protocol).
3. Test samples of the finished product for performance using stable, typical control cultures.

A4.2.2 Plating of stool cultures

Depending on the media used, refer to manufacturer's guidelines if necessary.

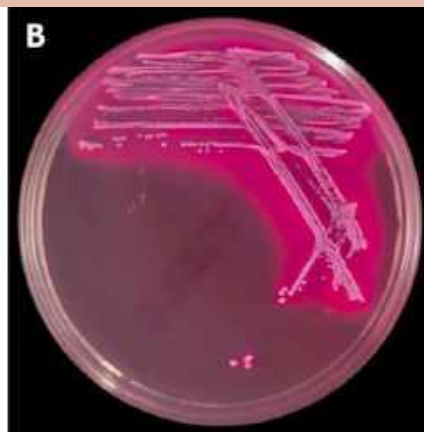
1. For faeces and other solid materials, suspend 1-2 g of the specimen in the broth (approximately 10-15% by volume) and emulsify with an inoculating needle, if necessary.
2. Incubate tubes with loosened caps at $35 \pm 2^\circ\text{C}$ for up to 24 hours.
3. Sub-cultures should be made after 12-18 hours of incubation, if possible. Coliforms will tend to overgrow the pathogens if incubated longer than 24 hours.

FIGURE
A4.1

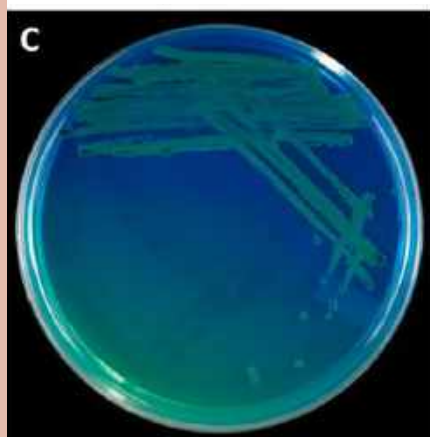
Morphology of *Salmonella* spp, *Salmonella* serovars Typhi, Paratyphi A and *E. coli* on various selective and differential agar media after 18-24 incubation at 35 °C.



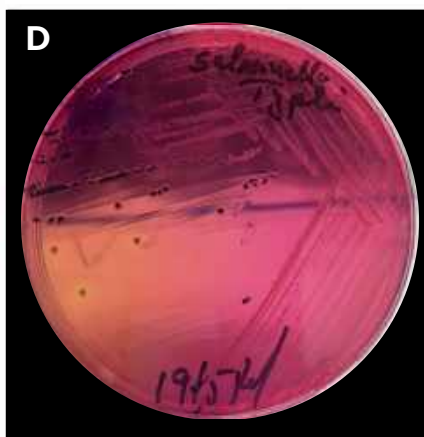
MacConkey agar with *Salmonella* spp.: Lactose negative (clear) colonies (*Shigella* will produce similar colonies)



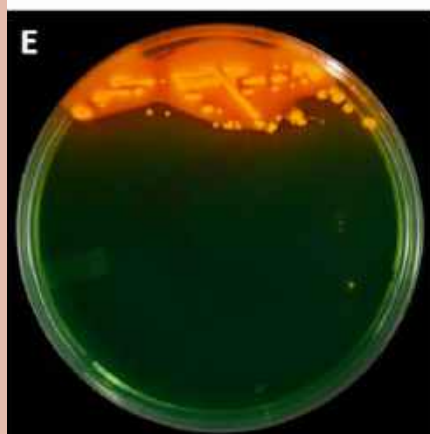
MacConkey agar with *E. coli*: Lactose positive (pink) colonies



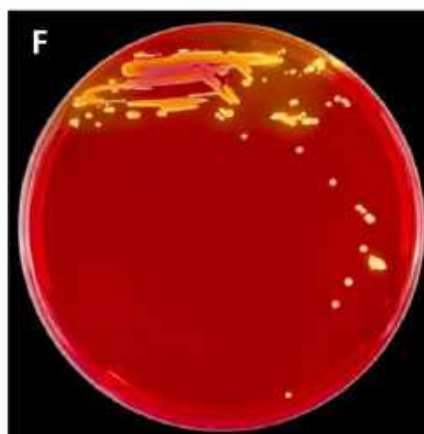
Hektoen enteric Agar with *Salmonella* Typhi: Clear colonies with black centres



XLD agar with *Salmonella* Typhi: Clear colonies with black centres



Hektoen enteric agar with *E. coli*: Yellow colonies



XLD agar with *E. coli*: Yellow colonies

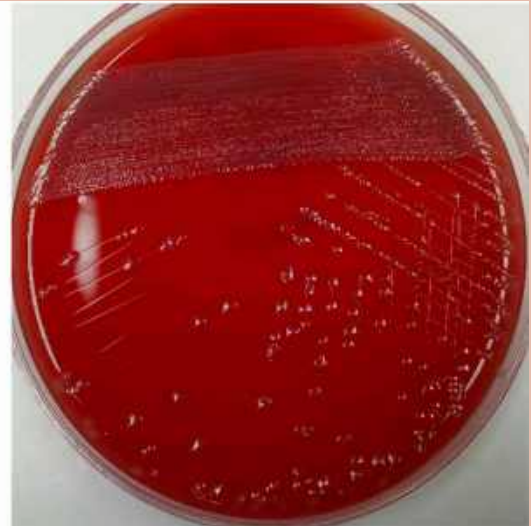
Source: All photos by Jim Gathany, CDC.

FIGURE
A4.2

Colony morphology of *Salmonella* Typhi on various agars.



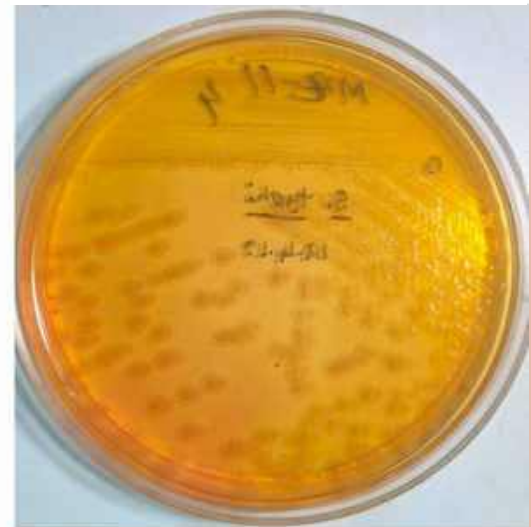
S. Typhi on SS agar



S. Typhi on blood agar



S. Typhi on chocolate agar



S. Typhi on MacConkey agar

Source: Image courtesy of Child Health research Foundation (CHRF).

A4.3 Gram stain

A4.3.1 Purpose

Gram staining is a differential staining procedure most commonly used for microscopic examination of bacteria (1).

A4.3.2 Background

The staining reaction reflects underlying differences in cell wall structure, which are relevant for antimicrobial susceptibility testing (AST) as well as identification.

A4.3.3 Composition of Gram stain reagents

(see A4.3.5 Method)

A4.3.4 Quality control and Interpretation of Gram stain reagents

TABLE
A4.1

The following Gram stain reactions must be observed with the QC organisms.

QC ORGANISM	RESULT	REACTION
<i>Escherichia coli</i>	Negative	pink bacilli (rods)
<i>Staphylococcus aureus</i>	Positive	purple cocci, in grape-like clusters

Note: Equivalent strains may be substituted based on local availability.

A4.3.5 Methods for Gram staining

1. Prepare a smear from the blood culture bottle (if there are a sufficient quantity of cells) or from a pure bacterial culture where a drop of fluid or a minute amount of organism is emulsified in saline/water, respectively.
2. Fix the smear with either heat or methanol fixation.
3. Flood the slide with the primary stain, crystal violet (10 g of 90% dye in 500 ml methanol).
4. Leave for at least 15 seconds.
5. Wash the slide to remove excess crystal violet.
6. Flood the slide with mordant Gram's iodine (6 g of I₂ and 12 g KI in 1.8 L of H₂O), this increases the affinity of the primary stain to the bacterial cell.

7. Leave for at least 15 seconds.
8. Wash the slide to remove excess iodine.
9. Flood with decolourizing agent, alcohol-acetone (400 ml of acetone in 1.2 L of 95% ethanol); the primary stain will be retained by Gram-positive cells and removed from Gram-negative cells by the decolourizing agent.
10. The decolourizer will initially be purple or brown due to the crystal violet and iodine. As soon as the decolourizer flowing off the slide turns clear (indicating that unbound stain and iodine has been removed). Wash the slide immediately with water. Over decolourization is a common cause of error.
11. Flood the slide with safranin counterstain (10 g of dye in 1 L of distilled or deionized water) for 30 seconds.
12. Wash the slide with water. Blot dry and examine under 1,000x magnification with oil on a light microscope.

A4.3.6 Limitations of Gram stain

1. Older cultures and organisms that exhibit the effects of antibiotics may be especially difficult to interpret.
2. Heat fixing smears may produce artefacts, create aerosols, may not adhere the specimen adequately to the slide.

References for Annex 4

1. Atlas R, Snyder J. In: Reagents, Stains and Media, p. 331-361.: Bacteriology. Carroll KC, Pfaller MA, Landry ML, McAdam AJ, Patel R, Richter SS, Warnock DW. (ed.) Manual of Clinical Microbiology, 11th edition. ASM Press, Washington D.C., USA. 2019.

Annex 5

Confirmation and identification of isolates

A5.1 Biochemical tests and procedures

A5.1.1 Setting up and interpreting biochemical tests (1)

Procedure

1. Label all biochemical tubes in the test series with the unique patient sample ID.
2. Carefully touch part of the colony to be identified with a sterile needle (or small loop). Alternately, a colony may be emulsified into 1 ml sterile saline and the suspension used as the inoculum for biochemical tests.
3. If citrate agar is used it should be inoculated first to avoid carbohydrate residue from being carried over from the other media. Open the screw cap and inoculate the media (see **Annex Table A5.2**).
4. Inoculate the remaining tests as described in **Annex Table A5.2** and in the method for inoculation in **A5.2.1 to A5.2.6**.
5. Incubate the biochemical tests with loosened screw caps at $35^{\circ}\text{C} \pm 2^{\circ}\text{C}$ in atmospheric air (without CO_2 enrichment) for 18-24 hours.
6. Read and record phenotypic reactions on an appropriate worksheet.

Result interpretation

Identify *S. Typhi*, *S. Paratyphi A* or NTS by comparing with expected results in **Annex Table A5.2**.

Notes:

1. Do not use an inoculating loop to inoculate a motility indole ornithine tube. While stabbing the butt, mechanical splitting of the medium may occur, causing a false positive result for gas production.
2. Inoculate from the same single colony for all tests.
3. Incubate all tests in normal atmosphere (without CO_2 enrichment).
4. Take readings within 18-24 hours. Some reactions start reversion after 24 hours due to the continuous growth of the organism.
5. Compare with an uninoculated tube as negative control before interpreting the results.

TABLE
A5.1

Useful biochemical tests and expected and exclusion reactions for *Salmonella* spp

TEST MEDIA	EXPECTED RESULTS (EXCLUSION)
MAC agar	colourless / non-lactose fermenter (pink/lactose fermenter: excludes <i>Salmonella</i>)
Oxidase test reagent	no colour change/negative (purple colour/positive: excludes <i>Salmonella</i>)
Christensen's urease agar	no colour change/negative (pink colour/positive: excludes <i>Salmonella</i>)
TSI or KIA	(i) alkaline (red) slope and acid (yellow) butt (ii) black slope and butt (iii) gas production [for <i>S. Typhi</i> and <i>S. Paratyphi A</i> reactions see Annex Table A5.2, Figures A5.2 and A5.3]

MAC, MacConkey agar; *TSI*, Triple sugar iron; *KIA*, Kligler's iron agar

TABLE
A5.2

Inoculation of phenotypic media and expected reactions for *S. Typhi*, *S. Paratyphi A*, and NTS. Source: Data taken from (2).

TEST	INOCULATION	NTS (Majority)	<i>S. Typhi</i>	<i>S. Paratyphi A</i>
TSI (slant)¹	Streak Slant / Single Stab	K	K	K
TSI (butt)¹	Streak Slant / Single Stab	A	A	A
TSI (H₂S)²	Streak Slant / Single Stab	+	trace	—
TSI (gas)³	Streak Slant / Single Stab	+	—	+
LIA (Lysine)⁴	Streak Slant / Single Stab	+	+	—
MIO (Motility)⁵	Single Stab	+	+	+
MIO (Ornithine)⁶	Streak Slant / Single Stab	+	—	+
MIO (Indole)⁷	Streak Slant / Single Stab	—	—	—
Urea⁸	Streak Slant	—	—	—
Citrate (Simmons')⁹	Streak Slant	+	—	—

Phenotypic tests are incubated at 35°C ± 2 °C and reactions recorded after 24–18 hours. TSI, Triple sugar iron agar (interchangeable with KIA); LIA, Lysine iron agar; MIO, Motility-Indol-Ornithine agar; Urea, Urea agar; Citrate (Simmons'), Simmons' Citrate agar.

1. *A*, acid (yellow colour) or *K*, alkaline (red colour).
2. *H₂S* (in TSI, KIA or LIA): Production of hydrogen sulfide (*H₂S*) evidenced by blackening of agar.
3. *Gas*: Gas production evidenced by splitting of agar or presence of bubbles.
4. *LIA*: Lysine decarboxylation (positive) evidenced by purple colour in butt of tube.
5. *MIO (Motility)*: Motility (positive) evidenced by diffuse growth throughout agar.
6. *MIO (Ornithine)*: Ornithine decarboxylation (positive) evidenced by purple colour in bottom 3/4 of tube.
7. *MIO (Indol)*: Indol production (positive) evidenced by red colour development after addition of Kovacs' reagent.
8. *Urea*: Urease activity (positive) indicated by intense pink colour development on slant.
9. *Citrate*: Citrate utilization (positive) evidenced by blue colour development on slant.

TABLE
A5.3

Expected biochemical reactions of *S. Paratyphi* B, *S. Paratyphi* B bv Java and *S. subsp.* II ser. (Sofia).

ORGANISM	ANTIGENIC FORMULA	D-TARTRATE	ONPG	MALONATE
<i>S. Paratyphi</i> B	1,4,[5],12 : b: 1,2	—	—	—
<i>S. Paratyphi</i> B bv. Java	1,4,[5],12 : b: 1,2	+	—	—
<i>S. subsp.</i> II ser. 1,4,[5],12,[27]: b : [enx] (Sofia)	1,4,[5],12,[27]: b : [enx]	d	d	+

d, variable result

TABLE
A5.4

Expected biochemical reactions of variants with antigenic formula 6,7:c:1,5.

<i>Salmonella</i> serovar	Paratyphi C	Choleraesuis	Choleraesuis var. Kunzendorf	Choleraesuis var. Decatur	Typhisuis ^a
Antigenic Formula	6,7,[Vi] : c : 1,5			6,7 : c : 1,5	
H₂S	+	—	+	+	— or +w
Dulcitol	+	—	—	+	—
Mucate	—	—	— (+*)	+	—
d-tartrate	+	+	+	+	—
Lysine decarboxylase	+	+	+	+	—
Mannitol	+	+	+	+	—
Arabinose	+	—	—	+	+
Trehalose	(+) or +	—	—	+	+

(+) = *delayed positive* (44)

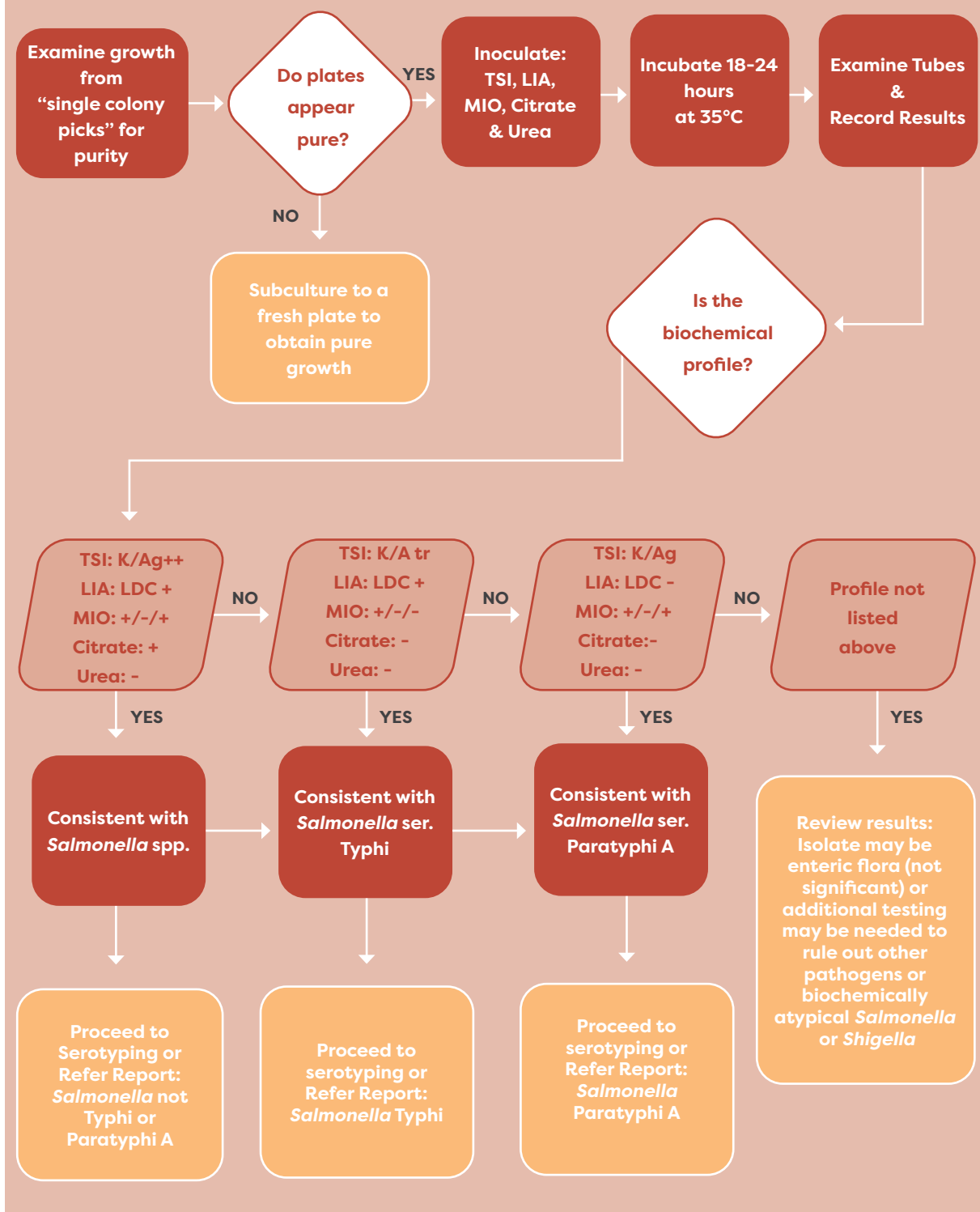
* *Choleraesuis* b.v. *Kunzendorf* *Australia* variant

w, *weakly positive reaction*

a, *Typhisuis* colonies are small (6)

FIGURE
A5.1

Biochemical testing algorithm for *S. Typhi*, *S. Paratyphi A*, and non-typhoidal *Salmonella*. Source: Data taken from (2).





The following figures (**Figures A5.2-A5.10**) show the growth and appearance of various *Enterobacterales*, for examples *S. Typhi*, *S. Paratyphi A*, NTS, and other *Enterobacterales* in the media used in the flow chart (**Figure A5.1**): Triple Sugar Iron agar (TSI), Urea agar (Urea), Lysine Iron agar (LIA), Simmons' Citrate agar (Citrate), and Motility-Indol-Ornithine agar (MIO) (2). While these specific media may not be available in every setting, the phenotypic reaction is more critical than the specific media used to obtain it. For example, if MIO is not available, MIU (motility-indole-urea) may be used and supplemented with ornithine decarboxylase broth (note: MIU details not provided in the Annex). Similarly, an automated system (e.g., API, Microbact, Phoenix, Vitek® etc.) can be used in place of the conventional, tubed media.

FIGURE
A5.2

Expected biochemical reactions for *Salmonella Typhi*.

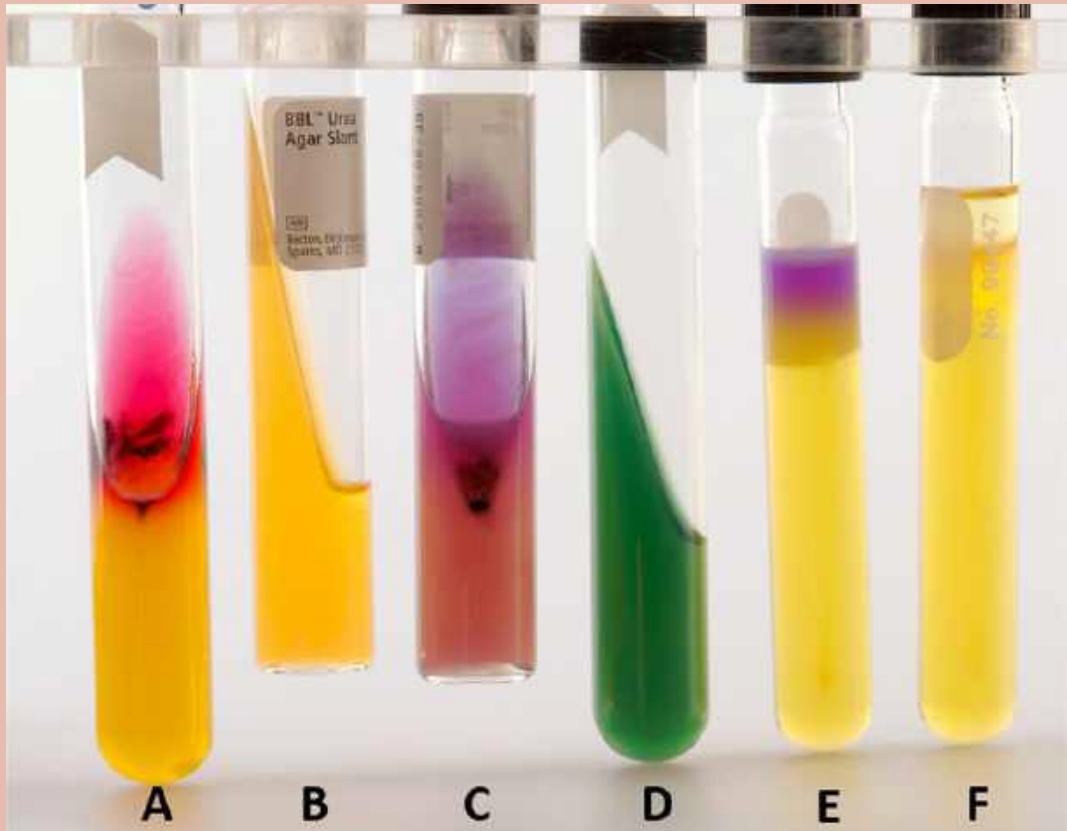


Photo courtesy of the US-CDC.

- A) TSI: alkaline slant / acid butt / no gas / trace H_2S (K / A^{TR})
- B) Urea: urease negative
- C) LIA: lysine decarboxylase positive / trace H_2S
- D) Citrate: negative
- E) MIO: motile / ornithine decarboxylase negative
- F) MIO with Indole Reagent: indole negative

FIGURE
A5.3

Expected biochemical reactions for *Salmonella Paratyphi A*



Photo courtesy of the US-CDC.

- A) TSI: alkaline slant / acid butt / gas production / H₂S negative (K / A, g)
- B) Urea: urease negative
- C) LIA: lysine decarboxylase negative / H₂S negative
- D) Citrate: negative
- E) MIO: motile / ornithine decarboxylase positive
- F) MIO with Indole Reagent: indole negative

FIGURE
A5.4

**Expected biochemical reactions for *Salmonella* Newport
(representative of most non-typhoidal serovars of *S. enterica*)**

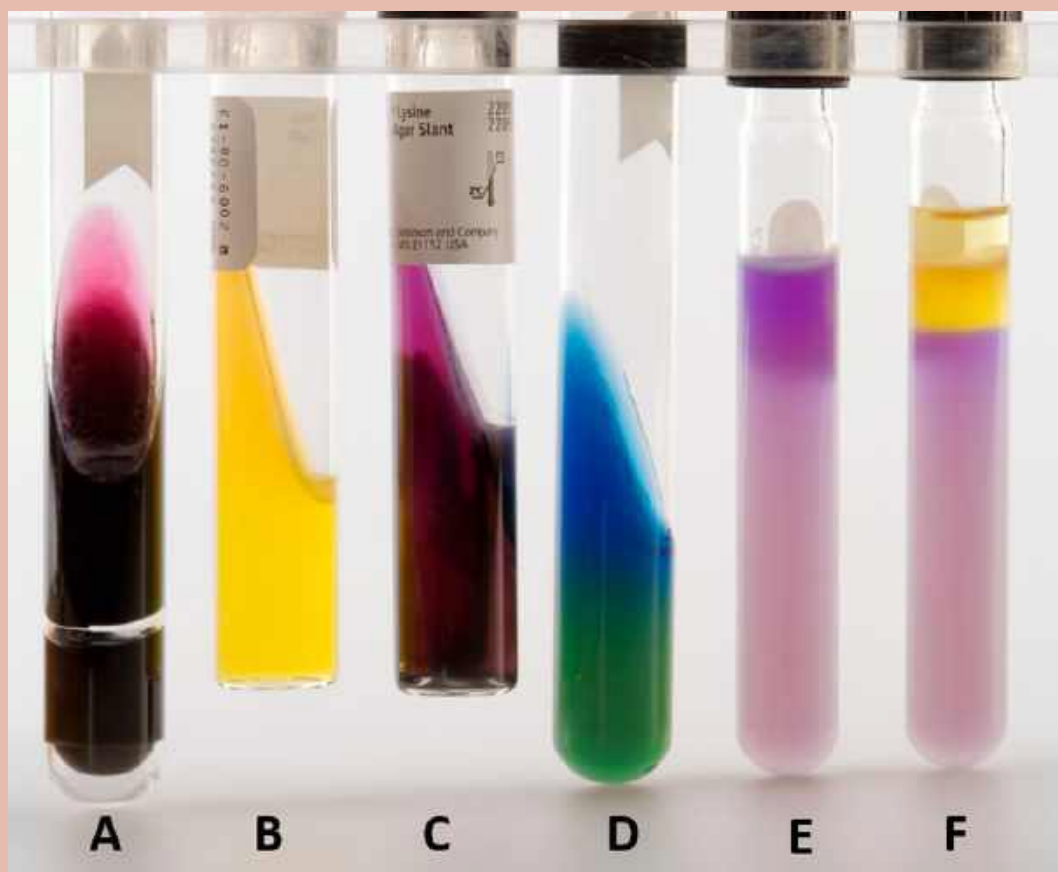


Photo courtesy of the US-CDC.

- A) TSI: alkaline slant / acid butt / gas production / H₂S positive (K / Ag ++)
- B) Urea: urease negative
- C) LIA: lysine decarboxylase positive / H₂S positive
- D) Citrate: positive
- E) MIO: motile / ornithine decarboxylase positive
- F) MIO with Indole Reagent: indole negative

FIGURE
A5.5

Expected biochemical reactions for *Shigella sonnei*



Photo courtesy of the US-CDC.

- A) TSI: alkaline slant / acid butt / no gas / H₂S negative (K / A)
- B) Urea: urease negative
- C) LIA: lysine decarboxylase negative / H₂S negative
- D) Citrate: negative
- E) MIO: non-motile / ornithine decarboxylase positive
- F) MIO with Indole Reagent: indole negative

FIGURE
A5.6

Expected biochemical reactions for *Shigella flexneri* (also representative of most serotypes of *S. dysenteriae* and *S. boydii*).



Photo courtesy of the US-CDC.

- A) TSI: alkaline slant / acid butt / no gas / H₂S negative (K / A)
- B) Urea: urease negative
- C) LIA: lysine decarboxylase negative / H₂S negative
- D) Citrate: negative
- E) MIO: non-motile / ornithine decarboxylase positive
- F) MIO with Indole Reagent: indole negative

FIGURE
A5.7

Expected biochemical reactions for *Proteus mirabilis*.



Photo courtesy of the US-CDC.

- A) TSI: alkaline slant / alkaline butt / no gas / H₂S positive (K / A +++)
- B) Urea: urease positive (often in <6 hours)
- C) LIA: lysine deaminase positive / no H₂S
- D) Citrate: negative (some strains positive)
- E) MIO: motile / ornithine decarboxylase positive
- F) MIO with Indole Reagent: indole negative

FIGURE
A5.8

Expected biochemical reactions for *Citrobacter freundii*.



Photo courtesy of the US-CDC.

- A) TSI: alkaline slant / acid butt / gas / H₂S positive (K / A, g +++)
- B) Urea: urease negative (some strains positive)
- C) LIA: lysine decarboxylase negative / H₂S positive
- D) Citrate: positive
- E) MIO: motile / ornithine decarboxylase negative
- F) MIO with Indole Reagent: indole negative

FIGURE
A5.9

Expected biochemical reactions for *Escherichia coli*.



Photo courtesy of the US-CDC.

- A) TSI: acid slant / acid butt / copious gas / H₂S negative (A / A, g)
- B) Urea: urease negative
- C) LIA: lysine decarboxylase positive / H₂S negative
- D) Citrate: negative
- E) MIO: motile / ornithine decarboxylase positive
- F) MIO with Indole Reagent: indole positive

FIGURE
A5.10

Expected biochemical reactions for *Pseudomonas aeruginosa*.



Photo courtesy of the US-CDC.

- A) TSI: no fermentation / no gas / H₂S negative (K / K)
- B) Urea: urease negative
- C) LIA: lysine decarboxylase negative / H₂S negative
- D) Citrate: positive
- E) MIO: non-motile / ornithine decarboxylase positive
- F) MIO with Indole Reagent: indole negative

A5.2

Composition, preparation, inoculation, quality control, and interpretation of phenotypic media

Important note: Detailed instructions have been provided for preparation of media from first principals using individual components. However, follow the manufacturer's instructions where commercially available powders are employed.

A5.2.1 Oxidase test (3,4)

A5.2.1.1 Purpose

The oxidase test is used in microbiology to determine if a bacterium produces cytochrome oxidase enzymes. The test is used as an aid in the identification and differentiation of a number of bacterial species.

A5.2.1.2 Background

The test uses a disk or filter paper impregnated with a reagent such as *N, N, N', N'*-tetramethyl-*p*-phenylenediamine dihydrochloride (TMPD) which is a redox indicator.

The oxidase catalyses the transport of electrons from donor compounds (NADH) to electron acceptors (usually oxygen). The cytochrome system is usually only present in aerobic organisms, which are capable of using oxygen as the final hydrogen receptor. The end-product of this metabolism is either water or hydrogen peroxide (broken down by catalase).

A5.2.1.3 Composition and preparation of oxidase reagent (or paper)

Reagent: 1% solution of TMPD in distilled water

Reagent impregnated filter paper:

1. Soak sheet/s (or sterile discs) of Whatman's No. 1 filter paper with freshly prepared 1% solution of TMPD (< 1 week old).
2. Drain for about 30 seconds and allow to dry.
3. Cut filter paper into 1 cm x 4 cm strips.
4. Store in a dark bottle tightly sealed with a screw cap. Freeze dry the strips, if possible.

Note: Oxidase reagent is unstable and should be freshly prepared for use.

A5.2.1.4 Materials for oxidase test

Medium: None.

Storage: Keep TMPD solution, impregnated filter paper/disks with reagent in cool dark place (away from light) e.g., inside screw-capped vial wrapped in silver foil. Keep the reagent away from light. Do not use stored filter paper strips that develop a purple colour.

Equipment and consumables: Oxidase reagent or paper strips (see [Annex A5.2.1.3](#)); platinum or plastic loop (not nichrome wire); sterile transfer pipettes.

Specimen: Fresh 24-hour culture of organism to be tested

A5.2.1.5 Quality control of oxidase test

TABLE
A5.5

The following oxidase test reactions must be observed with the QC organisms.

QC ORGANISM	RESULT*	REACTION
<i>Pseudomonas aeruginosa</i>	Positive	Intense purple-blue colour on paper or disk
<i>Escherichia coli</i>	Negative	No colour change

**a positive reaction is indicated by an intense deep purple colour within 10–5 seconds, a “delayed positive” reaction by colouration in 60–10 seconds, a negative reaction by absence of colouration or by colouration later than 60 seconds. Recommendation to perform and document daily QC for the oxidase disks or strips.*

A5.2.1.6 Methods for oxidase test

A. Dry filter paper method

FIGURE
A5.11

Visual example of oxidase test reaction outcomes: dry method.



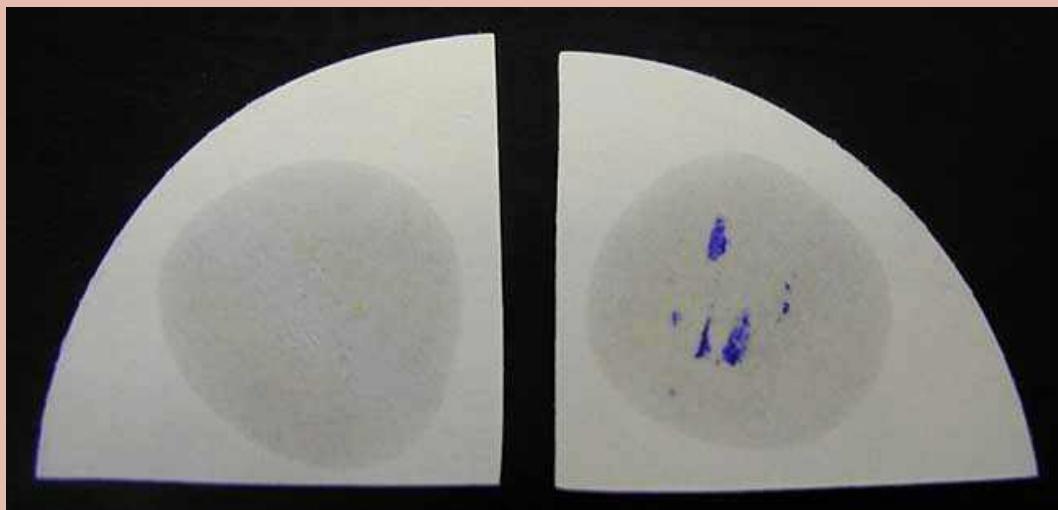
Source: Image taken from (4).

1. Remove one strip of oxidase paper from storage and place in a Petri dish and moisten with sterile distilled water.
2. Pick a colony to be tested with a platinum (or plastic loop) and smear over the moist area.
3. Examine the paper strip for development of a deep purple colour within 10-5 seconds and record results (see Figure A5.11).

B. Wet filter paper method

FIGURE
A5.12

Visual example of oxidase test reaction outcomes: wet method



Source: Image taken from (4).

1. Soak a strip of filter paper with a little freshly made 1% TMDP reagent solution.
2. Pick a colony to be tested with a platinum (or plastic loop) and smear over the moist area.
3. Examine the paper strip for development of a deep purple colour within 10-5 seconds and record results (see Figure A5.12).

C. Direct plate method

1. Add 2-3 drops of reagent directly to suspect colonies on a non-selective agar plate. (Do not use a BA plate. Do not flood the plate with the reagent).
2. Observe for colour change within 10 seconds.

A5.2.1.7 Interpretation of results and reporting of oxidase test

(see **A5.2.1.5 Quality control of oxidase test**)

A5.2.1.8 Limitations of oxidase test

1. The reagents used in the oxidase test have been shown to auto-oxidize, so it is important to use fresh reagents, no older than 1 week.
2. Bacteria taken from media containing high levels of glucose may inhibit oxidase activity and give false negative reactions, therefore, it is recommended to test colonies grown on media without excess sugar, such as nutrient agar (NA) or TSA.
3. Care should be taken when performing the oxidase test on colonies selected from blood-containing media. Red blood cells contain cytochrome oxidase which may give rise to false positive results.
4. Bacteria grown on media containing dyes or tellurite may give aberrant results.
5. The test reagents will effectively kill the microorganisms, so sub-culturing should be done prior to adding any reagent to an active culture.
6. Oxidase reactions of Gram-negative bacilli should be determined on non-selective and non-differential media to ensure valid results.
7. The oxidase test can be used in the presumptive differentiation and identification of Gram-negative bacilli. Oxidase-positive organisms should be examined by Gram stain to determine morphology and Gram reaction. Additional biochemical tests are recommended for complete identification.
8. Use of a nichrome or other iron-containing loop may yield false positive reactions. Platinum (or plastic) loops are recommended.
9. It is recommended to use colonies that are 18-24 hours old. Older colonies will produce weaker reactions.
10. Any colour changes appearing after 20 seconds should be disregarded.

A5.2.2 Simmons' citrate test (1, 2, 5-8)

A5.2.2.1 Purpose

Simmons' citrate agar is a selective and differential medium that tests the ability of an organism to metabolize (sodium) citrate as the sole source of carbon in a medium containing inorganic ammonium salts.

Citrate utilization is one of several phenotypic assays (biochemical tests) used in the identification or characterization of bacteria, particularly to distinguish members of the order *Enterobacterales* and other Gram-negative rods on the basis of citrate utilization.

A5.2.2.2 Background

In most bacteria, citrate is metabolized via the tricarboxylic acid cycle but when oxidative reactions are limited, some bacteria can use the citrate fermentation pathway.

Bacteria that can grow on this medium produce an enzyme, citrate-permease, capable of transporting the citrate across the cell membrane. When the citrate is used sodium ions are left in the medium, these either react with carbon dioxide (a product of metabolism) to form sodium carbonate or with water to form sodium hydroxide. In either case, when the pH of the medium rises above 7.6 (becomes alkaline) the bromothymol blue indicator changes the colour of the Simmons' citrate medium from green to blue.

Simmons citrate agar may be used to differentiate citrate-positive NTS from the citrate-negative *S. Typhi* and *S. Paratyphi A*.

A5.2.2.3 Composition of Simmons' citrate medium

Ingredients	Weight or volume
Magnesium sulfate	0.2 g
Ammonium dihydrogen phosphate	1 g
Dipotassium phosphate	1 g
Sodium citrate	2 g
Sodium chloride	5 g
Bromothymol blue	0.08 g
Agar	15 g
Distilled Water	1,000 ml

Final pH (at 25 °C) 6.8±0.2

A5.2.2.4 Preparation of Simmons' citrate medium

1. Suspend all components (24.28 g) in 1,000 ml distilled water.
2. Heat, to boiling, to dissolve the medium completely.
3. Mix well and dispense 4-5 ml into 16 mm tubes or 2-3 ml into screw-capped bijoux bottles.
4. Sterilize by autoclaving at 121 °C for 15 minutes.
5. Cool in slanted position (long slant, shallow butt). The medium may also be dispensed as an agar plate.
6. The uninoculated medium will be a deep forest green due to the pH of the sample and the bromothymol blue indicator. QC testing is performed with each new lot or shipment of media.

A5.2.2.5 Materials for Simmons' citrate test

Medium: Simmons' citrate agar

Storage conditions: Store at 4 °C. Shelf life ~6 to 8 weeks. Do not use beyond the expiration date.

Equipment and consumables: Sterile inoculating loop or needle.

Specimen: Fresh 24-hour culture of organism to be identified.

A5.2.2.6 Quality control of Simmons' citrate test

TABLE
A5.6

The following Simmons' Citrate reactions must be observed with the QC organisms.

QC Organism	Result	Reaction
<i>Enterobacter aerogenes</i>	Positive	intense blue colour (initially the colour change may only occur in the Agar slant)
<i>Serratia</i> spp. and most <i>Citrobacter</i> , <i>Klebsiella</i> , <i>Proteus</i> , <i>Providencia</i> spp.		
<i>Escherichia coli</i>	Negative	Agar remains green (unchanged)
<i>Shigella</i> spp.		
<i>Yersinia</i> spp.		

Note: Phenotypically equivalent strains may be substituted based on local availability

A5.2.2.7 Method for the inoculation of the Simmons' citrate test

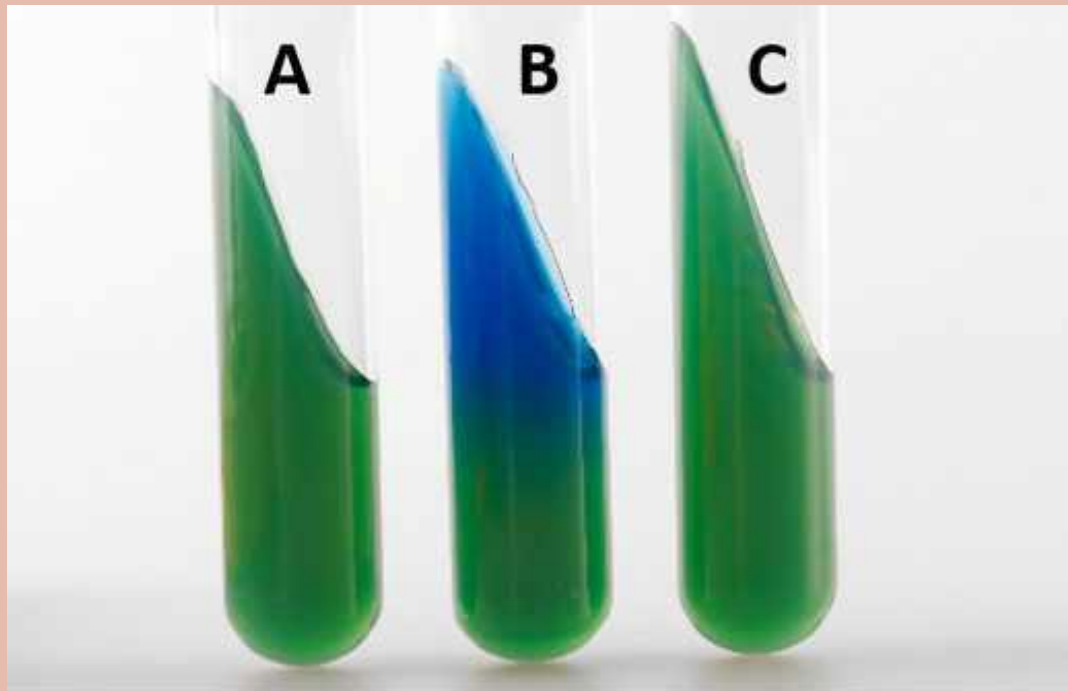
1. Harvest a small amount of growth with a sterile (1 µl) loop.
2. Inoculate slants with growth from a pure culture using the light inoculum (do not use a heavy inoculum).
3. Incubate tubes, with caps loosened, at $35 \pm 2^\circ\text{C}$ / in an aerobic atmosphere (without CO₂ enrichment).
4. Examine tubes and record results at 24 hours, 48 hours, and 5-7 days.

Note: Using cellular debris or residual media may produce false positive results. As such, it is critical to avoid over inoculating the agar and to use a fresh loop or needle.

A5.2.2.8 Interpretation of Simmons' citrate test results and reporting

FIGURE
A5.13

Visual examples of Simmons' citrate test reaction outcomes.



- A) Citrate: uninoculated
- B) Citrate: positive; *Salmonella* Newport
- C) Citrate: negative; *E. coli*

Source: Photo courtesy of the Photo courtesy of the US-CDC(8).

A5.2.2.9 Limitations of Simmons' citrate agar

1. The pH affects the performance of the medium and must be correctly monitored.
2. Before using water, ensure pH of water is 6.5 to 7.0. Initial colour of the medium may deviate from expected colour, if the above precaution is ignored.
3. The medium should be freshly prepared because in dry conditions, changes in colour may appear even before inoculation, especially at the bottom of the slant.
4. Some organisms are capable of growth on citrate and do not produce a colour change. Growth is considered a positive citrate utilization test, even in the absence of a colour change.
5. The inoculum should be from a pure, overnight culture grown on a solid medium and not from a broth suspension.
6. Failure to incubate tubes with loose caps may result in false negative results.
7. Use a light inoculum to streak the slant; a heavy inoculum may result in false positive results.
8. When inoculating multiple biochemicals from the same culture, inoculate this medium first, or flame inoculating needle prior to streaking this medium. Carryover of glucose or other nutrients onto this medium may result in false positive results.

A5.2.3 Lysine Iron agar (2,5,9,10)

A5.2.3.1 Purpose

LIA is a differential medium used to test organisms for the ability to deaminate or decarboxylate lysine and to produce hydrogen sulfide (H_2S). This test medium is used for differentiation of *Enterobacterales* and other Gram-negative bacteria.

A5.2.3.2 Background

LIA is one of several phenotypic assays (biochemical tests) used to identify and characterize bacteria, particularly, members of the Order *Enterobacterales*.

Lysine decarboxylation: For lysine decarboxylation to occur, enzymes require pyridoxal phosphate for their action and are produced under anaerobic conditions at a low pH (6.0) in a rich medium with specific substrate induction. CO_2 is released and the pH of the medium rises due to the formation of amine.

Iron (H_2S) production: To detect H_2S , some bacteria can reduce sodium thiosulphate to sulfide, which reacts with iron salts to form black ferrous sulfide.

LIA contains lysine, peptones, a small amount of glucose, ferric ammonium citrate, and sodium thiosulfate.

The medium has an aerobic slant and an anaerobic butt. The medium is stabbed to the base of the butt and streaked on slant.

When glucose is fermented, the butt of the medium becomes acidic (yellow).

If the organism produces lysine decarboxylase, the butt of the medium reverts to the alkaline state (purple).

If the decarboxylase is not produced, the butt remains acidic (yellow).

If deamination does not occur, the LIA slant remains purple. Bromocresol purple, the pH indicator, is yellow at or below pH 5.2 and purple at or above pH 6.8.

Cultures that produce hydrogen sulfide cause blackening of the medium due to ferrous sulfide production.

A5.2.3.3 Composition of LIA medium

Ingredients	Weight or volume
Peptone	5 g
Yeast extract	3 g
Dextrose (Glucose)	1 g
L-Lysine	10 g
Ferric ammonium citrate	0.5 g
Sodium thiosulphate	0.04 g
Bromocresol purple	0.02 g
Agar	15 g
Distilled Water	1,000 ml

Final pH (at 25 °C): 6.7±0.2

A5.2.3.4 Preparation of LIA medium

1. Suspend 34.56 g in 1,000 ml distilled water.
2. Heat to boiling to completely dissolve the medium.
3. Dispense into tubes and sterilize by autoclaving at 121 °C for 15 minutes.
4. Cool the tubes in slanted position to form slants with deep butts.

A5.2.3.5 Materials for LIA test

Medium: LIA agar slant/slope

Storage conditions: Store at 4 °C. Do not use beyond the expiration date.

Equipment and consumables: sterile inoculating needle or 1 µl loop.

Specimen: Fresh 24-hour culture of organism to be identified.

A5.2.3.6 Quality control of LIA test

QC testing is performed with each new lot or shipment of media.

Prior to use, the media is tested for sterility i.e., no growth on uninoculated medium following 48 hours incubation at 35 °C.

Each new lot or shipment of media must also produce expected reactions with QC organisms.

TABLE
A5.7

The following LIA test reactions must be observed with the QC organisms.

QC ORGANISM	SLANT	BUTT	H ₂ S
<i>Proteus mirabilis</i>	red*	yellow	- / +
<i>S. Typhimurium</i>	purple	purple	+
<i>Shigella flexneri</i>	purple	yellow	—
<i>Citrobacter freundii</i>	purple	yellow	+
<i>Escherichia coli</i>	purple	purple	—

Note: Phenotypically equivalent strains may be substituted based on local availability.

**Proteus* and *Providencia* spp. may be presumptively identified on this medium, as they produce a red slant and an alkaline butt.

A5.2.3.7 Method for inoculation of LIA

1. Harvest a small amount of growth with a sterile, straight inoculating needle (or 1 µl loop).
2. Lightly inoculate the surface of the agar slant.
3. Stab inoculate by making a single vertical stab through the centre of the medium down to the bottom of the tube.
4. Incubate tubes, with caps loosened, at 35 °C ± 2 °C /in an aerobic atmosphere (without CO₂ enrichment).
5. Examine tubes for growth and colour changes in the slant and butt and blackening at the apex of the slant.
6. Record results at 24 hours, 48 hours, and 5-7 days (unless H₂S production occurs sooner).

Note: Do not use a 'wide' loop to stab the agar as it will split the agar and give false positive indications of gas production.

A5.2.3.8 Interpretation of LIA test results and reporting

TABLE
A5.8

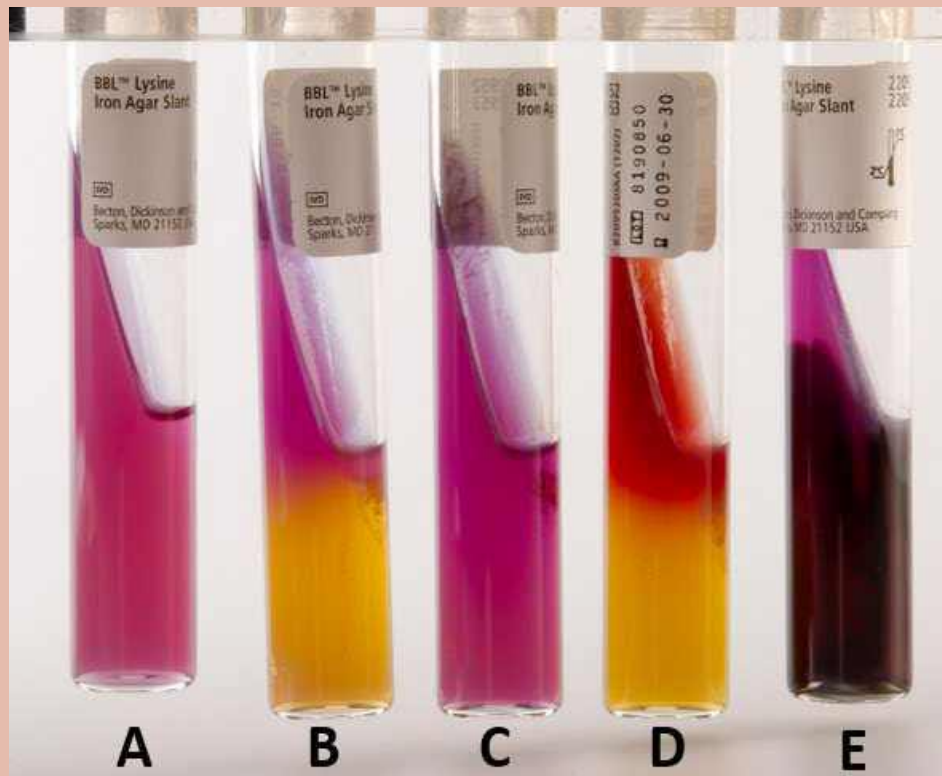
Interpretation of LIA test reactions.

TEST	POSITIVE	NEGATIVE
H ₂ S production:	black colour along the streak / stab line or throughout the medium	no black colour
LDC (butt)	purple	yellow
Lysine deamination (slant)	red	purple

LDC, lysine decarboxylase (detected in the butt of the tube).

FIGURE
A5.14

Visual examples of LIA test reaction outcomes.



- A) Uninoculated LIA
- B) LDC negative / lysine deaminase negative / H_2S negative; *Citrobacter freundii*
- C) LDC positive / lysine deaminase negative / H_2S negative; *Salmonella* Typhi
- D) LDC negative / lysine deaminase positive / H_2S negative; *Proteus mirabilis*
- E) LDC positive / lysine deaminase negative / H_2S positive; *Salmonella* Newport

Source: Photo courtesy of the Photo courtesy of the US-CDC.

A5.2.3.9 Limitations OF LIA TEST

1. It is important to stab the butt of the medium. Failure to stab the butt invalidates this test.
2. LIA can detect H₂S production, however, this media is less sensitive than TSI agar or peptone iron agar.
3. Many lysine decarboxylase negative organisms (e.g., *Citrobacter* or *Proteus*) fail to produce H₂S on LIA i.e., will not blacken the medium.
4. Certain species or strains may give delayed reactions or completely fail to ferment the carbohydrate in the stated manner.
5. Gas production may be irregular or suppressed with organisms other than *Citrobacter* spp.

A5.2.4 Motility-Indol-Ornithine AGAR (2,6,9,11-13)

A5.2.4.1 Purpose

MIO agar is used to demonstrate motility, ornithine decarboxylase activity, and indole production for the differentiation of *Enterobacterales*. This assay is used for phenotypic characterization of bacteria.

A5.2.4.2 Background

The demonstration of motility, ornithine decarboxylase activity, and indole production are performed in a single tube.

Motility: The semi-solid agar medium is inoculated via a single stab (see **A5.2.4.7 Methods**), which allows motile bacteria to swarm and extend through the medium for some distance from the original point of inoculation but a non-motile organism will grow only at the site of inoculation.

Indole: This medium tests for the ability of an organism to split indole from the tryptophan molecule using the enzyme tryptophanase; MIO medium is rich in tryptophan. The indole can be detected by the addition of Kovac's reagent, which involves a chemical combination producing a distinct colour.

Ornithine decarboxylase: see lysine decarboxylation section in **A5.2.3.2 LIA background**

A5.2.4.3 Composition of MIO medium

Ingredients	Weight or volume
Peptone	10 g
Tryptone	10 g
L-Ornithine Monohydrochloride	5 g
Bromocresol Purple	0.02 g
Yeast Extract	3 g
Dextrose	1 g
Agar	2 g
Distilled Water	1,000 ml

Final pH (at 25 °C): 6.7±0.2

A5.2.4.4 Preparation of MIO medium

1. Suspend 31.02 g in 1,000 ml distilled water.
2. Mix well and heat to boiling to completely dissolve the medium (one minute).
3. Dispense into screw-capped tubes and sterilize by autoclaving at 121 °C for 15 minutes.

A5.2.4.5 Materials for MIO test

Medium: MIO agar

Reagent: Kovac's Indole reagent (purchased ready to use)

Storage conditions: Store at 4 °C. Avoid freezing prior to use. Minimize exposure to light. Do not use if media shows signs of microbial contamination, discolouration, drying or other deterioration. Do not use beyond expiration date.

Equipment and consumables: Sterile inoculating needle.

Specimen: Fresh 24-hour culture of organism to be identified.

A5.2.4.6 Quality control for MIO test

QC testing is performed with each new lot or shipment of media.

Prior to use, the media is tested for sterility i.e., no growth in uninoculated medium following 48 hours of incubation at 35 °C.

Each new lot or shipment of media must also produce expected reactions with QC organisms (Table A5.9).

TABLE
A5.9

The following MIO test reactions must be observed with the QC organisms.

QC ORGANISM	MOTILITY	ODC	INDOLE
<i>E. coli</i>	+	+	+
<i>Morganella morganii</i>	+	+	+
<i>Shigella flexneri</i> <i>Klebsiella pneumoniae</i>	—	—	—
<i>Enterobacter aerogenes</i> <i>Proteus mirabilis</i>	+	+	—
<i>Salmonella Typhi</i>	+	—	—

Note: Phenotypically equivalent strains may be substituted based on local availability.

A5.2.4.7 Method for inoculation of MIO test

1. Harvest a small amount of growth with a sterile, straight inoculating needle.
2. Lightly inoculate the surface of the agar slant.
3. Stab inoculate by making a single vertical stab straight into the centre of the medium towards the butt and stop approximately 1 cm from the bottom of the tube.
4. Incubate tubes, with caps loosened, at $35 \pm 2^\circ\text{C}$ / in aerobic atmosphere (without CO_2 enrichment).
5. Examine tubes 18-24 hours for motility and ornithine decarboxylase activity.
6. Record results.
7. After the motility and ornithine decarboxylase results are interpreted, add 3-4 drops of Kovac's Indole Reagent.
8. Interpret and record the indole result.

Notes:

1. Do not use a loop to stab the agar as it will split the agar and give false positive indications of gas production.

2. Do not make multiple stabs into the agar and do not twist the needle into the media.

3. Important : Kovac's Indole reagent will cause the agar to turn yellow.

Record motility and ornithine prior to adding the indole reagent.

A5.2.4.8 Interpretation of MIO test results and reporting

Read motility and decarboxylase activity prior to the addition of the reagent for the detection of indole production.

To interpret the results of the MIO test, use **Table A5.9**, **Table A5.10** and **Figure A5.15** as a reference guide.

TABLE
A5.10

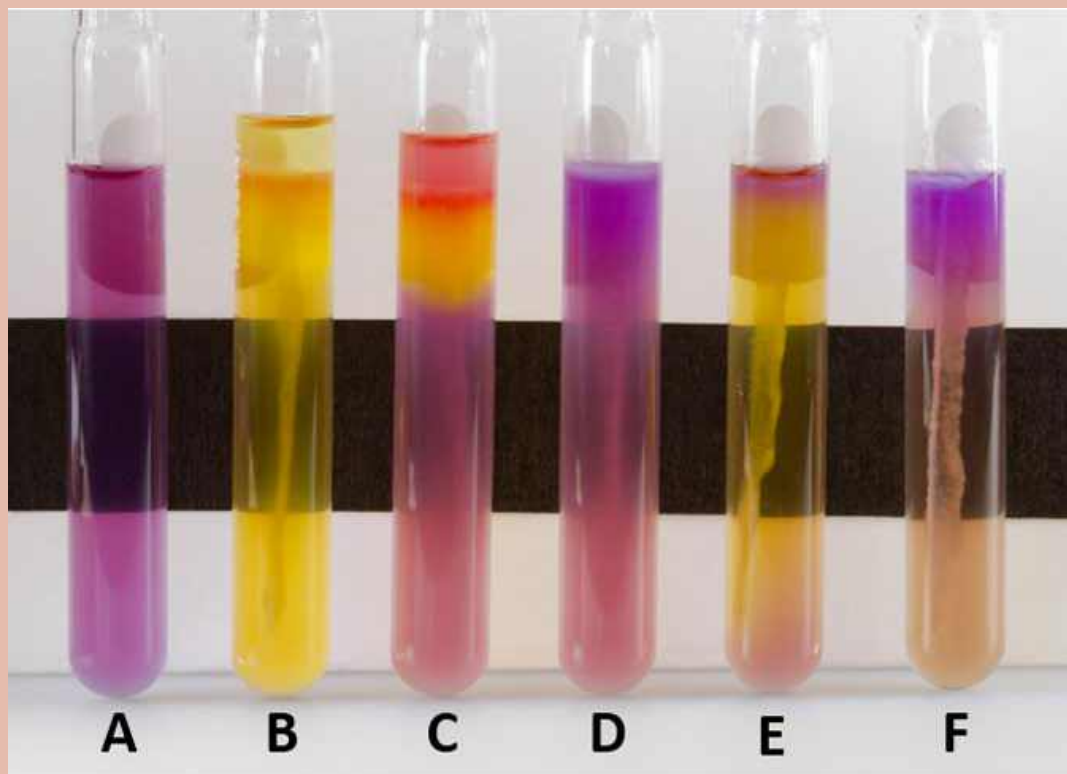
Interpretation of the MIO test reactions.

TEST	POSITIVE	NEGATIVE
Motility	Visible growth extending away from the stab line of inoculation. Typically, the agar will become visibly turbid.	Growth only along the stab line of inoculation. The agar remains clear. Isolates which only produce small tufts of growth along stab line (similar to bristles on a brush) are considered non-motile.
ODC	Decarboxylation of ornithine is indicated by turbid purple (to a faded yellow-purple) colour in the middle of the tube. These tubes are distinctly purple; however, they will be a lighter shade of purple than their uninoculated counterparts.	The agar in the middle of the tube turns yellow. Only the colour of the agar in the middle of the tube should be noted. Oxidation may cause the agar on the surface of the tube to turn purple this is not significant.
Indole	Indole production is shown following the addition of 3-4 drops of Kovac's reagent to the surface of the agar which turns pink-red colour.	No colour change is observed following the addition of Kovac's reagent which remains orange-yellow colour.

ODC, ornithine decarboxylase

FIGURE
A5.15

Visual examples of MIO test reaction outcomes.



- A) Uninoculated MIO tube
- B) Indole negative; *Shigella flexneri*
- C) Indole positive; *E.coli*
- D) Motile / ODC positive; *Salmonella* Newport
- E) Non-motile / ODC negative; *Shigella flexneri*
- F) Non-motile / ODC positive; *Shigella sonnei*

Source: Photo courtesy of the Photo courtesy of the US-CDC.

A5.2.5 Triple Sugar Iron agar (2,9,14-16)

A5.2.5.1 Purpose

TSI agar is a solid culture medium used to demonstrate the ability of a bacteria to produce acid production from carbohydrate (sugar) substrates, with or without gas production, and to produce H₂S from iron. TSI is used for the differentiation of *Enterobacterales* and other Gram-negative rods.

A5.2.5.2 Background

TSI is one of several phenotypic assays (biochemical tests) used to identify or characterize bacteria based on which carbohydrates/sugars (glucose, lactose and sucrose) they can or cannot ferment and their ability to produce hydrogen sulfide.

Identifications are based on the interpretation of multiple phenotypic tests.

TSI contains three sugars – 0.1% glucose, 1% lactose and 1% sucrose – with a pH indicator to visualize acid production (yellow colouration) and ferrous sulfate to demonstrate H₂S production (by blackening of the medium).

In the aerobic regions of the medium (surface and slant), glucose at the relatively low concentration of 0.1% is rapidly oxidized to CO₂ and H₂O. Anaerobically in the butt, glucose is fermented to organic acids and sometimes CO₂ and H₂. Acid and gas are detected. Lactose (and/or sucrose) in the TSI medium may be catabolized but only after glucose has been consumed. Since lactose and sucrose in TSI are present in high concentrations, their catabolism results in high levels of acid throughout the slant and butt. Addition of sucrose in TSI agar permits earlier detection of coliform bacteria (compared with KIA) that ferment sucrose more rapidly than lactose and aids the identification of certain Gram-negative bacteria that could ferment sucrose but not lactose. Gas may also be produced from lactose (and/or sucrose) shown by splitting of the agar. Another basic understanding is the TSI tube contains ‘butt’ (poorly oxygenated area on the bottom) ‘slant’ (or slope) (angled well-oxygenated area on the top).

Note: Kligler’s Iron agar contains 0.1% glucose and 1% lactose only (see Annex A5.2.6)

A5.2.5.3 Composition of TSI medium

Ingredients	Weight or volume
Beef extract	3.0 g
Yeast extract	3.0 g
Peptone	20.0 g
Glucose	1.0 g
Lactose	10.0 g
Sucrose	10.0 g
Ferrous Sulfate or Ferrous Ammonium Sulfate	0.2 g
NaCl	5.0 g
Sodium Thiosulfate	0.3 g
Phenol red	0.024 g
Agar	13.0 g
Distilled water	1,000 ml

Final pH (at 25 °C): 7.3 ± 0.2

A5.2.5.4 **Preparation of TSI medium**

1. Suspend ingredients in 1,000 ml distilled water.
2. Adjust pH to 7.3
3. Heat to boiling to completely dissolve the medium.
4. Mix well and distribute into test tubes.
5. Sterilize by autoclaving at 121 °C for 15 minutes or as per validated cycle.
6. Allow the medium to set in a sloped position to form a 2.5 cm butt and 3.8 cm slant.

A5.2.5.5 **Material for inoculation of TSI agar**

Medium: TSI agar slant/slope with a deep butt (2.5 cm)

Reagent: None

Storage conditions: Store at 4 °C, do not exceed the expiration date on the label.

Equipment and consumables: Sterile inoculating needle.

Specimen: Fresh 24-hour culture of organism to be identified.

A5.2.5.6 **Quality control of TSI agar**

QC testing is performed with each new lot or shipment of media.

Prior to use, the media is tested for sterility i.e., no growth in uninoculated medium following 48 hours of incubation at 35 °C.

Each new lot or shipment of media must also produce expected reactions with QC organisms (**Table A5.11**)

TABLE
A5.11

The following TSI test reactions must be observed with the QC organisms.

QC ORGANISM	SLANT	BUTT	GAS [^]	H ₂ S [*]
<i>Enterobacter aerogenes</i>	A	A	+	—
<i>Klebsiella pneumoniae</i> <i>Escherichia coli</i>	A	A	+	—
<i>Citrobacter freundii</i>	K	A	+	+
<i>Pseudomonas aeruginosa</i>	K	K	—	—
<i>Shigella</i> spp.; <i>Serratia</i> spp.	K	A	—	—
<i>Salmonella</i> spp. (NTS)	K	A	+	+
<i>Proteus</i> spp.	K	A	—	+
<i>Salmonella</i> Typhi	K	A	+	+
<i>Salmonella</i> Paratyphi A	K	A	—	—

Note: Phenotypically equivalent strains may be substituted based on local availability.

A, acid (yellow); K, alkaline (red)

*+, blackening; —, no blackening, trace, weak blackening

[^]+, bubbles on splitting of agar; —, agar intact, uninterrupted by bubbles

A5.2.5.7 Method for inoculation of TSI test

1. Harvest a small amount of growth with a sterile, straight inoculating needle.
2. Stab inoculate by making a single vertical stab straight into the centre of agar towards the butt and stop approximately 1 cm from the bottom of the tube.
3. Streak inoculate the surface of the agar slant.
4. Incubate tubes, with lids loosened, at 35–37 °C / in aerobic atmosphere (without CO₂ enrichment) / 18–24 hours.
5. Examine tubes for fermentation reactions, gas formation and H₂S production.
6. Record results.

Note: Do not use a loop to stab the agar as it will split the agar and give false positive indications of gas production.

A5.2.5.8 Interpretation of TSI test results and reporting

The interpretation of TSI results is explained in **Table A5.12** and recording in **Table A5.13** and **Figure A5.16**.

TABLE
A5.12

Interpreting the TSI test reactions.

TEST	POSITIVE	NEGATIVE
Gas production	gas bubbles in Agar or splitting of Agar	no bubbles or splitting of Agar
H₂S production	black colour along the streak or throughout the medium	no black colour
Carbohydrate fermentation	alkaline slant/alkaline butt – no sugars fermented	
	alkaline slant/acid butt – only glucose fermented	
	acid slant/acid butt – glucose fermented along with lactose and/or sucrose	

TABLE
A5.13

TSI test reactions are indicated and recorded using the following notations.

TEST	NOTATION	TEST REACTION
Carbohydrate	A (capital)	Acid production (yellow) in slant or butt
Fermentation	K (capital)	Alkaline reaction (red) in slant or butt
Gas production	g (lowercase, subscript)	Bubbles or splitting of agar
H₂S production	TR (superscript)	Trace amount of H ₂ S
	+ (superscript)	small to moderate amount of H ₂ S
	+++ (superscript)	copious amount of H ₂ S

The fermentation reactions on the slant and butt are recorded separated by a diagonal line.

FIGURE
A5.16

Visual examples of TSI test reaction outcomes.



Source: Photo courtesy of the Photo courtesy of the US-CDC.

A) Uninoculated TSI

B) K/A TR: alkaline slant / acid butt / no gas / trace H_2S (*S. Typhi*)

C) K/A, g +++: alkaline slant / acid butt / gas / H_2S positive (copious)
(e.g. *S. Newport*)

D) K/A: alkaline slant / acid butt / no gas / H_2S negative (e.g. *S. flexneri*)

E) A/A, g: acid slant / acid butt / copious gas / H_2S negative (e.g. *E. coli*)

F) Non-fermenter / no gas / H_2S negative (e.g. *P. aeruginosa*)

A5.2.5.9 Limitations of TSI agar

1. It is important to stab the butt of the medium. Failure to stab the butt invalidates this test. The integrity of the agar must be maintained when stabbing.
2. Caps must be loosened during this test or erroneous results will occur.
3. TSI agar must be read within the 18-24 hour stated incubation period. A false positive reaction may be observed if read too early. A false negative reaction may be observed if read later than 24 hours.
4. An organism that produces hydrogen sulfide may mask acid production in the butt of the medium. However, hydrogen sulfide production requires an acid environment, thus the butt portion should be considered acid.
5. TSI is not as sensitive in detecting hydrogen sulfide in comparison to other iron-containing media, such as sulfide indole motility medium.
6. Certain species or strains may give delayed reactions or completely fail to ferment the carbohydrate in the stated manner.

A5.2.6 Kligler's Iron agar (KIA) (17,18)

A5.2.6.1 Purpose

KIA is a solid culture medium used for differentiation of enteric Gram-negative bacilli on the basis of their ability to produce acid from carbohydrate (sugar) substrates, with or without gas production, and to produce H₂S from iron.

A5.2.6.2 Background

In KIA, casein and meat peptones supply nitrogenous compounds, amino acids and vitamins necessary for bacterial growth. Lactose and dextrose are carbohydrate sources. Phenol red is an indicator of carbohydrate fermentation. Fermentation reactions are read on the slant and in the butt, indicated by a colour change for red (alkaline) to yellow (acid). The concentration of lactose and dextrose present in the medium is at a ratio of 10:1, which serves to distinguish dextrose only fermenting organisms from those that also ferment lactose. The small amount of acid produced from dextrose fermentation oxidizes rapidly and causes the slant to revert to alkaline (red). The acid production in the butt remains yellow due to absence of oxygen. Lactose fermentations result in a yellow slant and butt because high level of acid production from both sugars maintains an acid pH under aerobic conditions. If the organism does not ferment dextrose, the slant and butt remain neutral (red). Ferric ammonium citrate is an indicator of H₂S production. If H₂S is produced from sodium thiosulfate, it reacts with ferric ammonium citrate to form a black precipitate (ferrous sulfate) in the medium. Gas production is indicated by bubbles, a splitting in the agar, or displacement of the medium.

A5.2.6.3 Composition of KIA medium

Ingredients	Weight or volume
Meat Peptone	10.0 g
Casein Peptone	10.0 g
Dextrose (or Glucose)	1.0 g
Lactose	10.0 g
Ferric Ammonium Sulfate	0.2 g
Sodium Chloride (NaCl)	5.0 g
Sodium Thiosulfate	0.3 g
Phenol Red	0.024 g
Agar	12.5 g
Distilled Water	1,000 ml

Final pH (at 25 °C) 7.4 ± 0.2

A5.2.6.4 **Preparation of KIA medium**

1. Suspend 49 g in 1,000 ml distilled water.
2. Heat to boiling with agitation to completely dissolve the medium.
3. Dispense into tubes.
4. Sterilize by autoclaving at 121 °C for 15 minutes or as per validated cycle.
5. Cool the tubes in sloped position to form a 2.5 cm butt and 3.8 cm slant.

A5.2.6.5 **Material for KIA test**

Medium: KIA slant/slope with a deep butt (25 mm)

Reagent: None

Storage conditions: Store at 4 °C, do not exceed the expiration date on the label.

Equipment and consumables: Sterile inoculating needle.

Specimen: Fresh 24-hour culture of organism to be identified.

A5.2.6.6 **Quality control of KIA test**

(See A5.2.5.4 Quality of control of TSI agar)

A5.2.6.7 **Method for inoculation of KIA test**

(See A5.2.5.7 Method for inoculation of TSI agar)

A5.2.6.8 **Interpretation of KIA test results and reporting**

(See A5.2.5.8 Interpretation of results and reporting for TSI agar)

A5.2.6.9 **Limitations of KIA test**

(See A5.2.5.9 Limitations of TSI agar)

A5.2.7 **Christensen's Urease test medium (Urea agar) (2,6,9,19,20)**

A5.2.7.1 **Purpose**

Urea agar is used to differentiate organisms based on urease activity.

This assay is used for phenotypic characterization of bacteria.

A5.2.7.2 **Background**

Urease activity is one of several phenotypic assays (biochemical tests) used in the identification and characterization of bacteria.

Identifications are based on interpreting multiple phenotypic tests.

Organisms which produce urease split urea into carbon dioxide and ammonia.

The ammonia combines with water to form ammonium carbonate which raises the pH of the medium. This pH shift is detected by the phenol red indicator.

A5.2.7.3 Composition of Urea medium

Ingredients	Weight or volume
Peptone	1 g
Dextrose	1 g
Sodium chloride	5 g
Potassium phosphate, monobasic	2 g
Urea	20 g
Phenol red	0.012 g
Agar	15 g to 20 g
Distilled Water	100 ml + 900 ml

A5.2.7.4 Preparation of Urea medium

1. Dissolve the first six ingredients in 100 ml distilled water and filter sterilize (0.45-mm pore size).
2. Suspend the agar in 900 ml of distilled water and heat to boiling to completely dissolve the medium.
3. Sterilize by autoclaving at 121 °C for 15 minutes or as per validated cycle.
4. Cool the agar to 50-55 °C
5. Aseptically add 100 ml filter sterilized urea base to the cooled agar solution and mix thoroughly.
6. Distribute 4–5 ml into sterile tubes (13 x 10 mm) and allow to solidify in a sloped position to create agar slants (long slant and short butt is optimal).

A5.2.7.5 Material for Urease test

Medium: Urea agar slant/slope (yellow-orange in colour)

Reagent: None

Storage conditions: Store at 4 °C. Once prepared, do not re-heat the medium as the urea will decompose. Do not use beyond expiry date.

Equipment and consumables: Sterile inoculating needle.

Specimen: Fresh 24-hour culture of organism to be identified.

A5.2.7.6 Quality control of Urea agar

QC testing is performed with each new lot or shipment of media.

Prior to use, the media is tested for sterility i.e., no growth on uninoculated medium following 48 hours of incubation at 35 °C.

Each new lot or shipment of media must also produce expected reactions with QC organisms (**Table A5.14**).

TABLE
A5.14

The following Urease test reactions must be observed with the QC organisms.

QC ORGANISM	RESULT	REACTION
<i>Providencia rettgeri</i>	Positive	intense pink colour on the slant
<i>Enterobacter aerogenes</i>	Negative	no colour change

Note: Phenotypically equivalent strains may be substituted based on local availability.

A5.2.7.7 Method for inoculation of Urease test

1. Harvest a small amount of pure culture with a sterile (1 µl) loop or needle.
2. Lightly inoculate the surface of the agar slant only (do not stab).
3. Incubate tubes with caps loosened at 35-37 °C / in aerobic atmosphere (without CO₂ enrichment).
4. Examine tubes and record results at (6 hours), 24 hours, 48 hours, and 5-7 days.

Note: Many *Proteus* spp. will exhibit a positive reaction within 6 hours.

A5.2.7.8 Interpretation of Urease test results and reporting

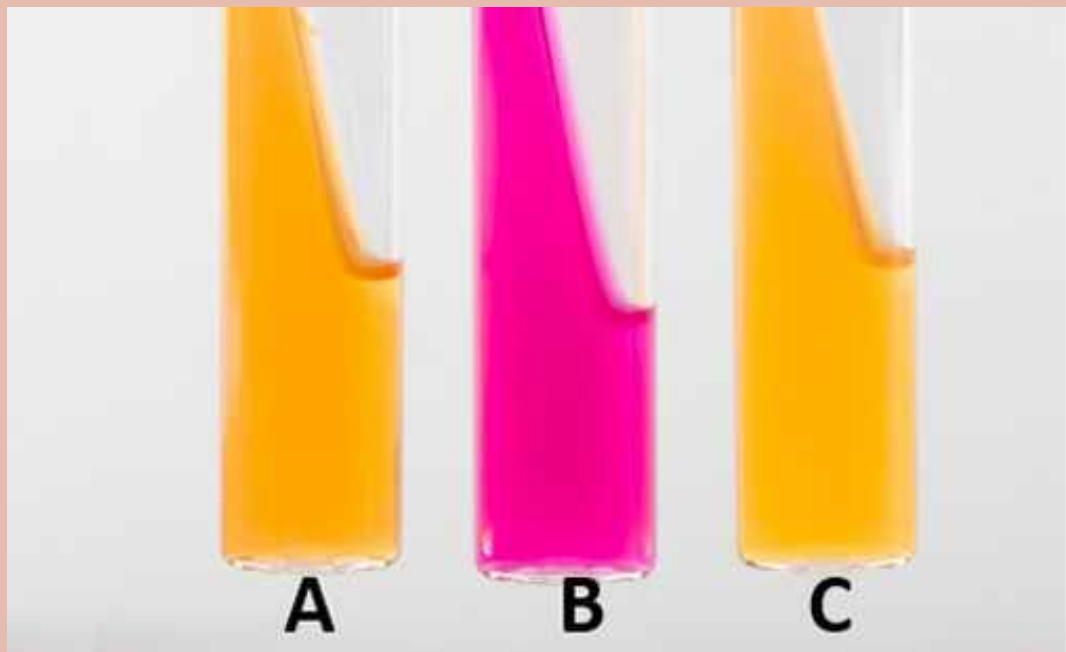
Urease production is indicated by a bright pink (fuchsia) colour on the slant that may extend into the butt. Note that any degree of pink is considered a positive reaction.

The culture medium will remain a yellowish colour if the organism is urease negative.

See A5.2.7.6 QC and QC Organisms in Table A5.14 and Figure A5.17

FIGURE
A5.17

Visual examples of Urease test reaction outcomes.



- A) Uninoculated Urea Agar
- B) Urease: positive; (*Proteus mirabilis*)
- C) Urease: negative (*E. coli*)

Source: Photo courtesy of the Photo courtesy of the US-CDC.

A5.2.7.9 Limitation of Urease test

1. Prolonged incubation may result in a false positive test due to hydrolysis of proteins in the medium.

A5.2.8 D-tartrate test (21)

A5.2.8.1 Purpose

D-tartrate broth is a liquid medium used in the differentiation of Enterobacterales based on tartrate utilization.

A5.2.8.2 Background

D-tartrate test medium contains peptone which supplies nitrogenous compounds and amino acids necessary for the growth of enteric Gram-negative bacilli. Sodium tartrate is a differential agent which is fermented by some enteric Gram-negative bacilli.

Although, *S. Paratyphi* B (SPB) and SPB biovar Java (*S. Java*) have the same antigenic formula, their usual clinical manifestations differ where SPB causes enteric fever and *S. Java* causes gastroenteritis. The two biovars can be differentiated based on their ability to ferment d-tartrate: *S. Java* is positive (3 days) and SPB is negative (6 days).

A5.2.8.3 Composition of d-tartrate medium

Ingredients	Weight or volume
Bacteriological Peptone (e.g. Oxoid LP0037)	10 g
Bromothymol blue solution, 1%	2.4 ml
Potassium sodium (+) tartrate	10 g
Distilled Water	1,000 ml

Final pH (at 25 °C) 7.4 ± 0.2 (adjusted with sodium hydroxide)

A5.2.8.4 Preparation of d-tartrate medium

1. Suspend the peptone in water and sterilize by autoclaving at 121 °C for 15 minutes.
2. Add 10 g potassium sodium tartrate (final concentration 10%).
3. Adjust pH with sodium hydroxide (NaOH) pH7.4.
4. Aseptically add 1% bromothymol blue solution, as an indicator (final concentration of 0.0024%).
5. Dispense 8 ml volumes in cotton wool stoppered test tubes or 3 ml volumes in screw-capped bijoux bottles.
6. Sterilize by autoclaving at 100 °C for 15 minutes, twice.

A5.2.8.5 Materials for d-tartrate test

Medium: d-tartrate broth

Reagent Preparation: Saturated Lead Acetate

Ingredients	Weight or volume
Lead Acetate (CH ₃ COO) ₂ Pb.3H ₂ O *MW: 379.3	25 g
Potassium hydroxide, 40%	10 ml
Distilled Water	40 ml

**Important to use lead acetate with the stated formula and molecular weight (MW)*

Always wear gloves when preparing and using lead acetate (toxic).

Storage conditions: Store at 4 °C. Shelf life ~6 to 8 weeks. Do not use beyond the expiration date.

Reagent: Store at room temperature. Shake well to resuspend lead acetate reagent before addition to d-tartrate medium.

Equipment and consumables: Sterile inoculating loop or needle.

Specimen: Fresh 24-hour culture of organism to be identified.

A5.2.8.6 Quality control of d-tartrate test medium

TABLE
A5.15

The following d-tartrate reactions must be observed with the QC organisms.

QC ORGANISM	RESULT	REACTION
S. Paratyphi B NCTC 3176	Negative	precipitate settled at base of tube/vial after 1 hour following lead acetate addition
S. Paratyphi B bv. Java NCTC 5706	Positive	fluffy, fine precipitate mainly dispersed throughout tube/vial after 1 hour following lead acetate addition
Uninoculated medium	Negative	

Note: Phenotypically equivalent strains may be substituted based on local availability.

A5.2.8.7 Method for the inoculation of the d-tartrate test

1. Collect 12 d-tartrate broths and label 4 pairs as shown:

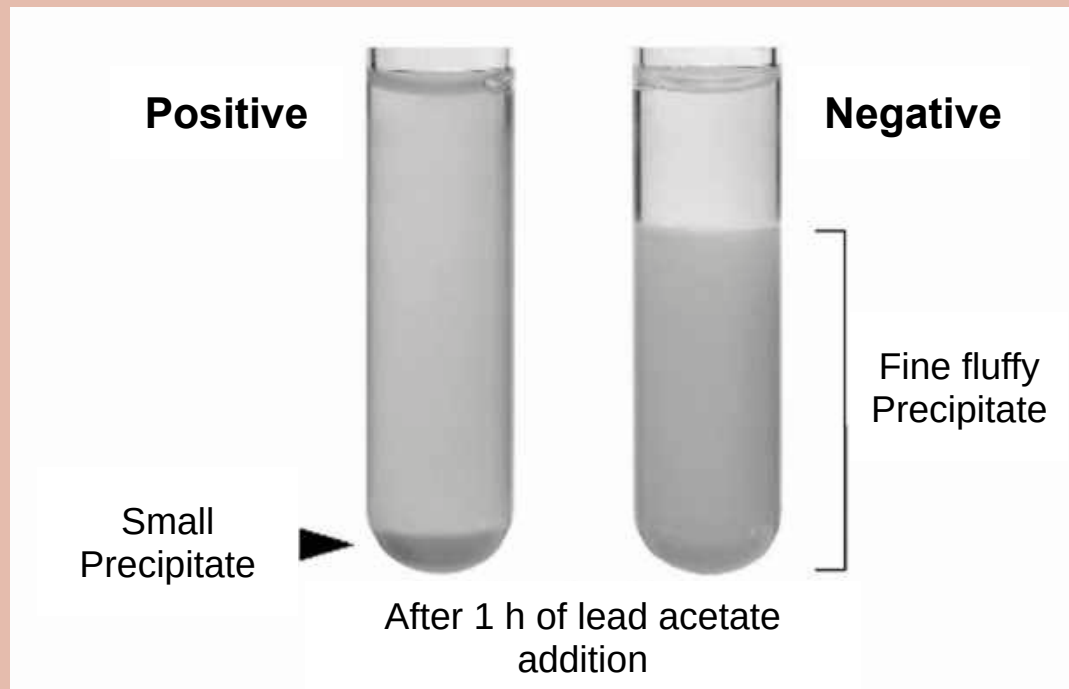
Control / Sample	Number of broths
Test Sample	2
S. Paratyphi B	2
S. Paratyphi B biovar Java	2
Uninoculated	2

2. Harvest a loopful of growth with a 1 µl sterile loop.
3. Inoculate broths with growth from a pure culture of the test sample and controls.
4. Incubate tubes, with caps loosened, at $35 \pm 2^\circ\text{C}$ / in an aerobic atmosphere, (optimally with 10% CO_2 enrichment for 48 hours).
5. Day 3, add 250 µl of lead acetate reagent (wear gloves) to one of each broth i.e. one test sample, two controls and one uninoculated broth.
6. Mix briefly (and carefully to avoid leakage).
7. Examine after 1 hour for precipitation and record results.
8. If the test is positive, discard all broths.
9. If the test is negative, re-incubate the sets of broths until Day 6 and follow steps 5 to 7. Discard all broths.

A5.2.8.8 Interpretation of d-tartrate test results and reporting

FIGURE
A5.18

Visual examples of d-tartrate test reaction outcomes.



Source: Adapted from (21).

A5.2.8.9 Limitations of d-tartrate test

1. Incubation times and conditions influence reliability of results and can give a high degree of false negative results.

A5.2.9 ONPG test (22, 23)

A5.2.9.1 Purpose

O-nitrophenyl-beta-D-galactopyranoside (ONPG) test is a broth medium used to determine the ability of an organism to produce β -galactosidase enzyme i.e. to identify late-lactose fermenters.

A5.2.9.2 Background

Lactose fermenting bacteria possess both lactose permease and β -galactosidase, two enzymes required for the production of acid in the lactose fermentation test. Permease allows lactose to enter the bacterium where it is then broken down into glucose and galactose by β -galactosidase. The glucose and galactose can then be metabolized by the bacteria. However, some organisms lack permease and appear as late-lactose-fermenters.

ONPG is an artificial substrate structurally similar to lactose with the exception that glucose is substituted with an o-nitrophenyl group. Unlike lactose, the substrate ONPG is capable of penetrating the bacterial cell without the presence of permease. The enzyme β -galactosidase cleaves ONPG into two residues: galactose and O-nitrophenol. ONPG is a colourless compound but O-nitrophenol is yellow, providing visual evidence of hydrolysis late-lactose fermentation.

Non-lactose fermenting bacteria lack both enzymes and are incapable of producing acid from lactose.

A5.2.9.3 Composition of ONPG test medium

Ingredients	Weight or volume
Bacteriological Peptone (e.g. Oxoid LP0037)	7.5 g
Sodium chloride NaCl	3.75 g
disodium hydrogen orthophosphate Na ₂ HPO ₄	0.355 g
O-Nitrophenyl- β -D-Galactopyranoside ONPG	1.5 g
Distilled Water	1,000 ml

Final pH (at 25 °C) 7.3 $\pm$ 0.2

A5.2.9.4 Preparation of ONPG broth medium

1. Suspend all components in 1,000 ml distilled water.
2. Mix well and distribute in tubes or flasks.
3. Dispense 3.0 ml into bijoux bottles.
4. Sterilize by autoclaving at 121 °C for 15 minutes.

A5.2.9.5 Materials for ONPG test

Medium: ONPG broth medium

Storage conditions: Store at 4 °C. Shelf life ~6 to 8 weeks. Do not use beyond the expiration date. Allow medium to warm to room temperature prior to use.

Equipment and consumables: Sterile inoculating loop.

Specimen: Fresh 24-hour culture of organism to be identified.

A5.2.9.6 Quality control of ONPG test

TABLE
A5.16

The following ONPG reactions must be observed with the QC organisms.

QC ORGANISM	RESULT	REACTION
<i>Escherichia coli</i> (ATCC 25922)	Positive	yellow colour development
<i>Proteus mirabilis</i> (ATCC 12453)	Negative	no colour development

Note: Phenotypically equivalent strains may be substituted based on local availability.

1. All organisms tested must be inoculated from a lactose-containing medium (e.g., TSI or MacConkey).
2. Cultures that naturally produce a yellow pigment cannot be tested with this media.
3. For complete identification biochemical, immunological, molecular, or mass spectrometry testing on colonies from pure culture is recommended.

A5.2.9.7 Method for the inoculation of the ONPG test

1. Allow the ONPG test medium to warm to room temperature prior to use.
2. Inoculate the test medium with heavy inoculum from a pure 18-24 h culture.
3. Incubate aerobically, with loosened caps, at $35 \pm 2^\circ\text{C}$ / aerobic atmosphere (without CO_2 enrichment).
4. Examine for a yellow colour development at 1 hour.
5. If no colour change seen after an hour of incubation, continue incubation for up to 24 hours.

A5.2.9.8 Interpretation of ONPG test results and reporting

In the broth method of testing, the organism is taken from a medium containing a high concentration of lactose and is inoculated into the ONPG broth. If the organism possesses beta-galactosidase, the enzyme will split the beta-galactosidase bond, releasing O-nitrophenol which is a yellow-coloured compound. This indicates a positive test.

In the disk method, the organism to be tested is taken from a medium containing a high concentration of lactose. A dense suspension (turbidity equivalent to a McFarland 3) is prepared. An ONPG disk is added to 0.5 ml of the suspension. If the organism possesses beta-galactosidase, the enzyme will split the beta-galactosidase bond, creating a yellow colour change in the suspension. Organisms with strong beta-galactosidase activity can produce a positive reaction a few minutes after inoculation of the ONPG medium; other organisms may take up to 24 hours.

FIGURE
A5.19

Visual example of ONPG test reaction outcomes



Source: Photo adapted from Microbe Online (<https://microbeonline.com/>).

A5.2.9.9 Limitations of ONPG agar

1. Further biochemical, molecular, immunological, or mass spectrometry testing on colonies from pure culture is advised for complete identification.
2. Cultures which naturally yield yellow colour cannot be tested in this media.
3. Inoculation of test organism should be done only from lactose-containing medium.

A5.2.10 Malonate test (24,25)

A5.2.10.1 Purpose

Malonate test medium is used to test the ability of an organism to utilize malonate as a sole source of carbon and energy for growth and to differentiate *Enterobacterales* organisms on the basis of malonate utilization.

A5.2.10.2 Background

An organism that simultaneously can utilize sodium malonate as its carbon source and ammonium sulfate as its nitrogen source produces an alkaline reaction due to the formation of sodium hydroxide. This changes the indicator (bromothymol blue) from its original green colour to light blue or Prussian blue. Organisms which cannot utilize malonate and ammonium sulfate and do not ferment dextrose produce no colour change. Organisms which are malonate negative but do ferment dextrose result in an acid reaction causing the pH indicator to turn from green to yellow.

A5.2.10.3 Composition of Malonate broth

Ingredients	Weight or volume
Yeast Extract (e.g. Oxoid LP0021)	1 g
Ammonium sulfate	2 g
Potassium dihydrogen orthophosphate K_2HPO_4	0.6 g
Potassium dihydrogen phosphate (KH_2PO_4)	0.4 g
Sodium chloride (NaCl)	2 g
Sodium malonate	3 g
Glucose	0.25 g
Bromothymol Blue (1%)	2.5 ml
Distilled Water	1,000 ml

Final pH (at 25 °C) 6.7 ± 0.2

A5.2.10.4 Preparation of Malonate medium

1. Suspend all components in 1,000 ml distilled water.
2. Heat, to boiling, to dissolve the medium completely.
3. Mix well and dispense 2 ml aliquots into bijoux bottles with screw-capped lids.
4. Sterilize by autoclaving at 121 °C for 15 minutes.
5. The uninoculated medium will be green due to the pH indicator (bromothymol blue) at neutral pH.

A5.2.10.5 Materials for Malonate test

Medium: Malonate broth medium

Storage conditions: Store at 4 °C. Shelf life ~6 to 8 weeks. Do not use beyond the expiration date.

Equipment and consumables: Sterile inoculating loop.

Specimen: Fresh 24-hour culture of organism to be identified.

A5.2.10.6 Quality control of Malonate test

TABLE
A5.17

The following Malonate test reactions must be observed with the QC organisms.

QC ORGANISM	RESULT	REACTION
<i>Klebsiella pneumoniae</i> ATCC 27736	Positive	abundant growth; colour change from green to blue (pH > 7.6)
<i>Serratia marscencens</i> ATCC 8100	Negative	no colour change i.e. green, (pH neutral) or yellow, if dextrose utilized (pH <6.0)
<i>E. coli</i> ATCC 25922		

Note: Phenotypically equivalent strains may be substituted based on local availability.

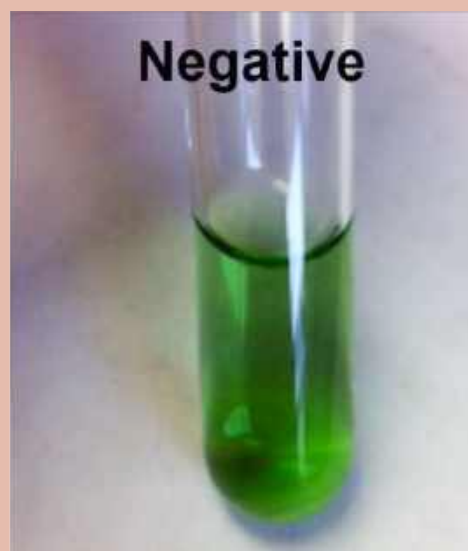
A5.2.10.7 Method for the inoculation of the Malonate medium

1. Harvest a heavy inoculum of pure culture with a sterile (1 µl) loop.
2. Inoculate broth aseptically into sterile vial of malonate broth.
3. Incubate tubes, with caps loosened, at 35 ± 2 °C / aerobic atmosphere (without CO₂ enrichment)
4. Examine tubes and record results at 24 hours and, if negative, at 48 hours.

A5.2.10.8 Interpretation of Malonate test results and reporting g

FIGURE
A5.20

Visual examples of Malonate test reaction outcomes.



Source: Photo sourced from <https://www.onlinebiologynotes.com/malonate-test-principle-requirements-procedure-and-result-interpretations/>

A5.2.10.9 Limitations of ONPG agar

1. False negative result may occur where inoculum is too light or if screw-capped vial is too tight (low oxygen atmosphere).
2. Some malonate positive organisms produce only slight alkalinity. Compare the results of the test isolate vial with an uninoculated malonate broth to verify weak positive reactions.
3. Any trace of blue denotes a positive results at the end of 48 hours incubation.
4. Some malonate negative organisms produce a yellow colour due to dextrose fermentation.

A5.2.11 Dulcitol broth medium (26)

A5.2.11.1 Purpose

Dulcitol broth is a liquid medium used to determine the ability of an organism to ferment dulcitol. This carbohydrate medium may be included in a panel of sugar fermentation tests to differentiate members of *Enterobacterales*.

A5.2.11.2 Composition of 1% Dulcitol Broth

Ingredients	Weight or volume
Bacteriological Peptone (e.g. Oxoid LP0037)	10 g
Sodium chloride NaCl	5 g
Dulcitol (or required carbohydrate)	10 g
Bromocresol Purple 0.2%	10 ml
Distilled Water	1,000 ml

Final pH (at 25 °C) 7.2 ± 0.2

A5.2.11.3 Preparation of Dulcitol medium

1. Dissolve 15 g in 1,000 ml distilled water.
2. Mix well and dispense 2 ml volumes in screw-capped bijoux bottles.
3. Sterilize by autoclaving at 121 °C for 15 minutes.
4. Note: When sterile solutions are to be added after autoclaving, reduce the volume of water for reconstitution by an equal amount.
5. The uninoculated medium will be a deep purple colour due to the bromocresol purple indicator.

A5.2.11.4 Materials for Dulcitol broth test

Medium: Dulcitol Broth Medium

Storage conditions: Store at room temperature. Shelf life ~6 to 8 weeks. Do not use beyond the expiration date.

Equipment and consumables: Sterile inoculating loop or needle.

Specimen: Fresh 24-hour culture of organism to be identified.

A5.2.11.5 Quality control of Dulcitol test

TABLE
A5.18

The following Dulcitol test reactions must be observed with the QC organisms.

QC ORGANISM	RESULT	REACTION
Non-typhoidal <i>Salmonella</i>	Positive	acid production, colour change to yellow from purple
<i>S. Typhi</i>	Negative	purple, no colour change

Note: Phenotypically equivalent strains may be substituted based on local availability.

A5.2.11.6 Method for the inoculation of Dulcitol medium

1. Harvest a small amount of growth with a sterile (1 µl) loop.
2. Inoculate broths with growth from a pure culture using the light inoculum (do not use a heavy inoculum).
3. Incubate vials, with caps loosened, at $35 \pm 2^\circ\text{C}$ / aerobic atmosphere (without CO_2 enrichment).
4. Examine vials and record results at 24 hours and 48 hours.

A5.2.11.7 Interpretation of Dulcitol test results and reporting

FIGURE
A5.21

Visual examples of Dulcitol test reaction can be found.



Source: Image taken from <https://www.sigmaaldrich.com/AU/en/technical-documents/technical-article/clinical-testing-and-diagnostics-manufacturing/cytology-and-microscopy/colour-identification-microorganisms>

A5.2.11.8 Limitations of Dulcitol agar

1. Sub-cultures may be necessary to ensure purity of the inoculant
2. Mixed or contaminated cultures will give false reactions

A5.2.12 Mucate test (27, 28)

A5.2.12.1 Purpose

Mucate broth is a liquid medium used to determine the ability of an organism to utilize mucate as a carbon source.

A5.2.12.2 Background

Mucate broth contains digest of animal tissue which provides carbon, nitrogen and other necessary nutrients to the organisms. Mucic acid is a saccharolactic acid which acts as a carbon source in the medium. Organisms which ferment mucic acid produce acid turning the medium yellow due to the pH indicator (bromothymol blue).

A5.2.12.3 Composition of Mucate medium

Ingredients	Weight or volume
Bacteriological Peptone (e.g. Oxoid LP0037)	10 g
Bromothymol Blue, 1%	2.4 ml
Mucic Acid	10 g
Sodium Hydroxide	100 ml
Distilled Water	900 ml

Final pH (at 25 °C) 7.2 ± 0.2

A5.2.12.4 Preparation of Mucate broth medium

1. Suspend all components in 900 ml distilled water; (add acid to water).
2. Mix well and dispense 2-3 ml aliquots into screw-capped bijoux bottles.
3. Sterilize by autoclaving at 121 °C for 15 minutes.
4. The uninoculated medium will be a deep forest green due to the neutral pH and the bromothymol blue indicator.

A5.2.12.5 Materials for Mucate test

Medium: Mucate broth medium

Storage conditions: Store at 4 °C. Shelf life ~6 to 8 weeks. Do not use beyond the expiration date.

Equipment and consumables: Sterile 3 mm inoculating loop.

Specimen: Fresh 18-24-hour culture, with well-isolated colonies, of organism to be identified.

A5.2.12.6 Quality control of Mucate test

TABLE
A5.19

The following Mucate test reactions must be observed with the QC organisms.

QC ORGANISM	RESULT	REACTION
<i>E. coli</i>	Positive	light green-yellow colour
<i>Morganella morganii</i>	Negative	no colour change i.e. blue
<i>Uninoculated broth</i>	Negative	no colour change i.e. blue

Note: Phenotypically equivalent strains may be substituted based on local availability.

A5.2.12.7 Method for the inoculation of Mucate medium

1. Inoculate a vial of TSB (see **Annex A5.5.5**) with a well-isolated colony
2. Incubate TSB at $35 \pm 2^\circ\text{C}$ / aerobic atmosphere (without CO_2 enrichment) / 18-24 hours.
3. Inoculate mucate medium with 3 mm loopful of overnight TSB culture, along with an uninoculated mucate broth (as a negative control).
4. Incubate vial at $35 \pm 2^\circ\text{C}$ / aerobic atmosphere (without CO_2 enrichment).
5. Examine and record results daily, up to 14 days.

A5.2.12.8 Interpretation of Mucate test results and reporting

FIGURE
A5.22

Visual examples of Mucate test reaction outcomes.



Source: Photo courtesy of the United States Centers for Disease Control and Prevention.

- A) Mucate: positive
- B) Mucate: negative
- C) Uninoculated mucate medium

A5.2.12.9 Limitations of Mucate broth medium

1. Occasionally an uninoculated control may turn green-yellow on standing, therefore, always include an uninoculated broth alongside the test sample as a negative control

A5.2.13 Lysine or Ornithine decarboxylase (29, 30)

A5.2.13.1 Purpose

Decarboxylase test media is used in the differentiation of Gram-negative enteric bacilli based on the production of lysine or ornithine decarboxylase.

A5.2.13.2 Background

Decarboxylase media contains peptone and beef extract to supply nitrogenous nutrients needed to support bacterial growth. Pyridoxal is an enzyme cofactor for the amino acid decarboxylase. Dextrose is a fermentable carbohydrate. Bromocresol purple and cresol red are pH indicators. The amino acids lysine and ornithine are added to the basal medium to detect production of the decarboxylase enzyme specific for these substrates.

When the medium is inoculated with a bacterium that is able to ferment dextrose, acids are produced that lowers the pH of the medium and change the colour of the indicator from purple to yellow. The acidic conditions also stimulates decarboxylase activity. If the organism produces the appropriate enzyme, decarboxylation results in alkaline end-products (amines), causing the medium to revert to its original purple colour. If the organism does not produce the appropriate enzyme, the medium remains acidic (yellow). Each isolate tested must also be inoculated into a vial of basal medium, which does not contain the amino acid. If the tube becomes alkaline, the test is invalid.

A5.2.13.3 Composition of Mucate medium

Ingredients	Weight or volume
Moellers Decarboxylase Broth (BBL 211430)	10.5 g
L-lysine hydrochloride/L-ornithine hydrochloride	10 g
Distilled Water	1,000 ml

Final pH (at 25 °C) 6.0 ± 0.2

A5.2.13.4 Preparation of lysine or ornithine or control decarboxylase medium

1. Suspend all components in 1,000 ml distilled water.
2. Mix well and dispense 2-3 ml volumes into screw-capped bijoux bottles.
3. Sterilize by autoclaving at 121 °C for 15 minutes.

Note: Slight colour variations may be observed within a batch of media. This colour variation should not affect performance.

A5.2.13.5 Materials for lysine or ornithine decarboxylase test

Medium: lysine or ornithine decarboxylase broth medium

Storage conditions: Store at 2–8 °C, in the dark. Avoid freezing or overheating. Do not open until ready to use. Minimize exposure to light. Shelf life ~6 to 8 weeks. Do not use beyond the expiration date. Allow medium to warm to room temperature prior to use. Do not use medium if there are signs of deterioration e.g. microbial contamination, discolouration, drying.

Equipment and consumables: Sterile inoculating 1 µl loop.

Reagents: Sterile mineral (paraffin) oil. Decarboxylation is an anaerobic reaction, and thus, the contents of each tube must be sealed with mineral or paraffin oil.

Specimen: Fresh 24-hour culture of organism to be identified.

A5.2.13.6 Quality control of lysine and ornithine decarboxylase test

Reaction	Appearance
Positive	alkaline, purple (or red)
Negative	acidic, yellow

TABLE
A5.20

The following Lysine/Ornithine decarboxylase reactions must be observed with the QC organisms.

QC ORGANISM	DECARBOXYLASE RESULT		
	Lysine	Ornithine	Control
<i>Klebsiella pneumoniae</i> ATCC 33495	Positive	Negative	Negative
<i>Enterobacter cloacae</i> ATCC 13047	Negative	Positive	Negative

Note: Phenotypically equivalent strains may be substituted based on local availability.

A5.2.13.7 Method for the inoculation of the lysine decarboxylase test

1. Inoculate a vial TSB (see Annex A5.5.5) with a well-isolated colony
2. Incubate TSB at $35 \pm 2^\circ\text{C}$ / aerobic atmosphere (without CO_2 enrichment) / 18-24 hours.
3. Allow the media to come to room temperature before inoculation,
4. Harvest a loopful of pure growth from the 18-24 hour TSB culture using a sterile 1 μl loop and inoculate the decarboxylase medium with the appropriate amino acid substrate (lysine or ornithine).
5. Harvest a second loopful of pure growth from the 18-24 hour TSB culture using a sterile 1 μl loop and inoculate a control decarboxylase medium (does not contain lysine). Overlay all vials (test samples and controls) with 1-2 ml sterile mineral oil.
6. Incubate vials, with lids tightened, at $35 \pm 2^\circ\text{C}$.
7. Examine and record results at 24, 48, 72 and 96 hours.

A5.2.13.8 Interpretation of lysine or ornithine decarboxylase test results and reporting

TABLE
A5.21

Visual examples of decarboxylase test reaction outcomes can be found in the following sections.

REACTION	LYSINE	ORNITHINE
Positive	Figure A5.2 C	Figure A5.3 E
Negative	Figure A5.3 C	Figure A5.2 E

A5.2.13.9 Limitations of lysine decarboxylase medium

1. For identification, the organism used must be in pure cultures.
2. Control decarboxylase test must be tested alongside test sample to validate the test.
3. Morphological, biochemical and/or serological tests must be performed for final identification.

A5.3 Serological tests and procedures

A5.3.1 Materials and Methods for Slide O and Vi Agglutinations and O group Determination (1,2,6,9,31)

A5.3.1.1 Reagents and equipment

1. Known O Antisera
 - a. polyvalent containing O:1-10 (O-groups A-E1)
 - b. specific: O:2, O:4, O:9, (O:7 for *S. Paratyphi C*), Vi
 - c. optional (specific): O:1, O:6,7, O:8, O:12
2. Pure bacterial culture on non-selective agar
3. Sterile normal saline (0.85%)
4. Glass microscope slides
5. Bacteriological straight wire (preferably nichrome) or disposable alternative
6. Disposable plastic loops: 1 µl and 10 µl
7. Micropipette and sterile tips
8. Disposable transfer pipettes
9. Black chinagraph (grease) pencil
10. Hand lens x10 magnification

A5.3.1.2 Safety considerations

1. Agglutination reactions are likely to generate aerosols and must be performed in a Class II microbiological/ biological safety cabinet (BSC-II). Where a BSC-II is not available, additional PPE should be worn i.e., disposable gloves, eye protection and face mask (to prevent ingestion of aerosols).
2. Some purchased antisera contain 0.5% phenol and 0.1% sodium azide.

Although the concentration is low, both are known to be toxic by ingestion and skin contact. Avoid ingestion of the reagents. If any contact with skin or eyes occurs, wash the area extensively by immediately rinsing with a copious volume water.

A5.3.1.3 Procedure for agglutination test

1. Use a greaseproof pencil to draw 4 or 5 squares/rectangles onto a glass microscope slide to divide into separate compartments. The grease pencil marking will create a barrier to prevent the individual agglutination reactions running into one another.
2. Dispense 10 µl saline using a micropipette (or use a larger plastic loop or fine-tipped, plastic transfer pipette), to place 2 drops onto 2 separate squares on the glass slide.
3. Use a loop to take one discrete colony from a pure, fresh culture of the organism and emulsify thoroughly into each drop of saline to obtain a uniform, turbid ('milky') suspension (~McFarland Standard 6).
4. Examine for auto-agglutination (clumping) to check if the suspension is 'rough' in saline. If auto-agglutination is observed, discard the slide because slide agglutination using antisera can only be performed on smooth suspensions. Proceed if the culture is smooth.
5. Prepare the required number of milky suspensions, plus one to act as an uninoculated/negative control, on the glass slide to test the nominated antisera.
6. Use a micropipette to add 20 µl polyvalent antiserum to one suspension only and mix by tilting the slide back and forth and rotating at least 5 times, then inspect for agglutination. Agglutinations will be seen more readily by using the light with a black background and the naked eye. Continue to rock the slide for up to 60 seconds to ensure the negative reactions remain negative. Any reaction occurring after 60 seconds must be considered negative (see **A5.3.1.5**). A 10x eyepiece can be used to view the reactions more closely.
7. Add the nominated O *Salmonella* antisera group for specific antigens separately to the antigen suspensions and examine for agglutinations (as above).
8. To interpret agglutination reactions, refer to **Table A5.22** and **Table A5.23** and **Figure A5.23**.
 - a. Positive result: distinct agglutination of the antigen suspension; no clumping in the saline control
 - b. Negative result: the suspension remains turbid (milky)
9. Record O agglutination results on the appropriate worksheet.
10. Discard slide/s appropriately into a discard jar, e.g., sharps container, when agglutinations are complete to avoid contamination of PPE or workbench with live bacterial culture.

TABLE
A5.22**Serogroup and antigenic formula for expected agglutination reactions for *S. Typhi*, *S. Paratyphi A*, *S. Paratyphi B* and *S. Paratyphi C* (31).**

Serotype	O-Serogroup	O Antigen	Antigenic formula (O : H1 : H2)
<i>S. Paratyphi A</i>	Group A	1,2,12	1,2,12 : a : [1,5] ^b
<i>S. Paratyphi B</i> ^c	Group B	1,4,[5],12	1,4,[5],12 : b : 1,2
<i>S. Paratyphi C</i> ^d	Group C1	6,7,[Vi]	6,7,[Vi] : c : 1,5
<i>S. Typhi</i>	Group D1	9,12,[Vi]	9,12,[Vi] : d : —

a, specific O factor **bolded**; [], antigens in parentheses are either weak or absent in some isolates; __, underlined O factors are determined by phage conversion. They are present only if the culture is lysogenized by the corresponding converting phage.

b, most strains of *Paratyphi A* possess a monophasic antigen phase 1, H1= a. In rare cases, diphasic strains with phase 2, H1,5=2, may be isolated. For this reason, [1,5] is mentioned in square brackets in the formula of this serovar.

c, L(+) tartrate /d-tartrate positive variant is often referred to as bv. (biovariant) Java. (see **Table A5.3**)

d, The same global antigenic formula with differing biochemical characteristics, pathogenicity and/or habitat have historically been given different serovar names. (see **Table A5.4**)

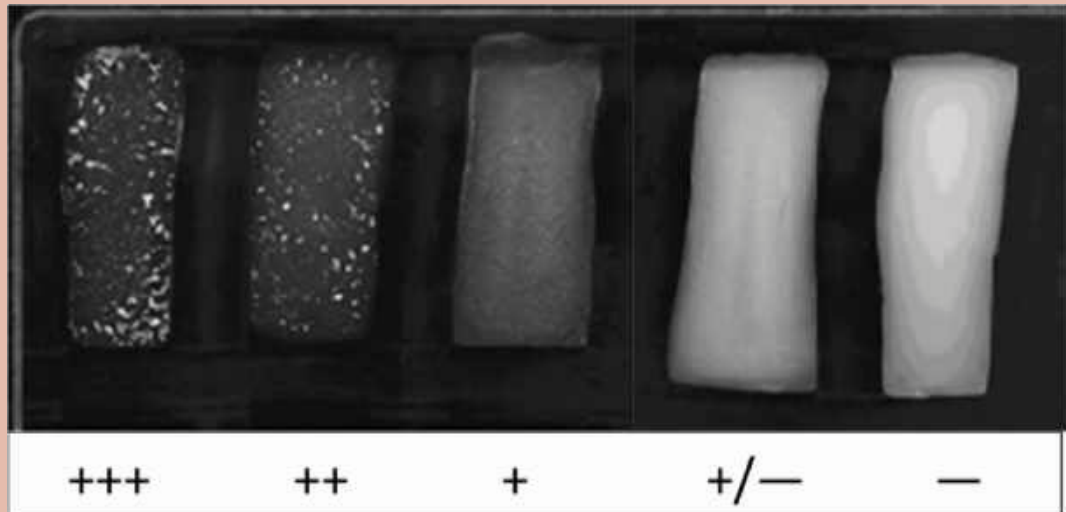
e, Rare strains can have, as phase 1, H:j instead of H:d (–261 nucleotide deletion in gene *fliC*). Independently, rare strains can have an additional phase H:z66 determined by a plasmid-borne gene.

TABLE
A5.23**Grading of O slide agglutination reactions (also see Figure A5.23).**

SEROTYPE	O-SEROGROUP	O ANTIGEN
—	no agglutination	suspension remains milky with no visible agglutination or clearing of the background
+/-	very weak	barely visible
+	sparse	weak, granular or fine
++	moderate	obvious with only slightly milky background
+++	heavy	strong with a clear background

FIGURE
A5.23

Visual representation for scoring grade of O agglutination reaction.



Source: Image adapted from <https://www.cdc.gov/meningitis/lab-manual/images/chapt-9figure07.gif>

A5.3.1.4 Interpretation

Compare results with the expected results shown in **Table A5.22** to determine the O group. At this stage, it is possible to report the *Salmonella* culture as belonging to an O group e.g.,

- (i) *Salmonella* that react with O antisera 2 belong to Serogroup/Group A
- (ii) *Salmonella* that react with O antisera 9 belong to Serogroup/Group D1

A5.3.1.5 Notes

1. The manufacturer's instructions indicate the volume of antisera to be used and if the antisera must be used neat in agglutination tests. If the antisera is to be used neat, omit the preparation of a turbid suspension in saline and emulsify the test organism directly into drops of antisera in separate compartments on the slide.
2. The length of time for agglutination to occur and become visible may vary with different antisera (refer to manufacturer's instructions). False positives may occur if the reaction is observed beyond the manufacturer's specified time frame.
3. The Vi antigen may mask the O antigen, therefore, Typhi may not agglutinate with O-9 antiserum. Vi antigen can be removed by preparing a milky suspension in sterile saline and boiling for 10 minutes and then repeating the agglutination tests to detect the heat stable O group D antigen/s.

A5.4 Storage of *Salmonella*

A5.4.1 Long-term storage of *Salmonella* isolates (9)

A5.4.1.1 Storage of isolates at room temperature

1. Isolates can be stored on non-carbohydrate rich media e.g., tryptic soy, nutrient or Brain Heart Infusion (BHI) agar slopes, or Dorset egg (DE) medium slopes (see A5.5.2 and A5.5.3, respectively).
 - a. Inoculate the slant (incubation is not required), or
 - b. Inoculate the DE slant and incubate 35-37° / aerobic atmosphere with 5% CO₂ / 18-24 hours.
2. Tighten the cap and cover with parafilm to make airtight.
3. Store away from light at room temperature (20-22 °C).
4. Isolates stored in this way generally remain viable for 2-3 years (10 years or more on DE if they remain hydrated) but may lose plasmid encoding antimicrobial resistance.

A5.4.1.2 Storage of isolates at -70 °C (or -80 °C)

1. Use a sterile swab stick or a rigid microbiological loop to harvest the growth on the entire agar plate to maximize the transfer of organisms, while avoiding to break the agar surface.
2. Place the heavy inoculum into a cryovial containing 1 ml of sterile:
 - a. TSB plus 15-20% glycerol, or
 - b. STGG (skim milk, tryptone, glucose, glycerol/glycerin broth)
3. Place vials in a designated, labelled box and freeze at -70 °C (or -80 °C)
4. Isolates stored in this way generally remain viable for many years and retain plasmid encoding antimicrobial resistance.

A5.4.1.3 Notes

1. If contamination is observed on the purity plate, discard the plate and subculture the organism again.
2. Freezers with automatic defrosters should never be used.

A5.5

Composition, preparation, inoculation and quality control of growth and storage media

Important note: Detailed instructions have been provided for preparation of media from first principals using individual components. However, follow the manufacturer's instructions where commercially available powders are employed.

A5.5.1 Nutrient agar (16,32)

A5.5.1.1 Purpose of using NA

NA is used as a general-purpose medium for the growth of a wide variety of non-fastidious microorganisms.

A5.5.1.2 Background

The solid medium consists of peptone, beef extract, and agar. This relatively simple formulation provides the nutrients necessary for the replication of a large number of non-fastidious microorganisms.

A5.5.1.3 Composition of NA medium

Ingredients	Weight or volume
Beef extract	3.0 g
Peptone	5.0 g
Sodium chloride	8.0 g
Agar	15.0 g
Distilled water	1,000 ml

Final pH 6.8±0.2 @ 25 °C

Note: Nutrient broth contains all these ingredients except agar.

A5.5.1.4 Preparation of NA medium

NA and broth are available from most suppliers of culture media in powdered (free-flowing, homogeneous) form.

1. Suspend 23.0 g in 1 L of distilled water.
2. Heat with frequent agitation and boil for one minute to completely dissolve powder.
3. Sterilize by autoclaving at 121°C (15 lbs) for 15 minutes.
4. Cool to 45 – 50 °C.
5. Dispense the medium into tubes i.e.
 - a. 3 ml in slanted position to make NA slopes
 - b. 5 ml in slanted position to make NA deeps, or
 - c. 20 ml in 9 cm diameter Petri dishes to make plates
6. Leave the agar medium to solidify.
7. Date the medium and give it a batch number.
8. Store in a cool dark place.

A5.5.1.5 Materials for inoculation of NA

Medium: NA in bottles or tubes or on a plate (or broth)

Reagents: None

Storage conditions: Dehydrated medium at 10-30 °C; use before expiry date on the label. Prepared medium in tubes or bottles stored 2-8 °C sealed in plastic bags to reduce the chances of contamination and in the dark. Up to 2 years providing there is no change in the appearance of the medium to suggest contamination or deterioration.

Equipment and consumables: Sterile inoculating loop.

Specimen: Fresh 24-hour culture of organism to be tested.

A5.5.1.6 Method for inoculation of NA

1. Lightly inoculate the agar using the test organism (or sample) of interest.
2. Incubate the tubes with caps loosened at 35-37 °C / in aerobic atmosphere for 24 hours (without CO₂ enrichment) / 18-24 hours.
3. Examine tubes for growth.

A5.5.1.7 Quality control of NA

Visible growth on the NA is an indication the medium is effective.

TABLE
A5.24

The following outcome on NA must be observed with the QC organisms.

QC ORGANISM	OBSERVATION
<i>Escherichia coli</i> ATCC 25922	Good to excellent growth
Uninoculated	No visible evidence of growth

Note: Phenotypically equivalent strains may be used based on local availability.

A5.5.1.8 Limitation of NA

1. Contamination, particularly with Gram-negative organisms other than *Salmonella* sp. cannot be discerned on NA

A5.5.2 Brain heart infusion agar (33-35)

A5.5.2.1 Purpose

BHI agar is an enriched non-selective solid medium for the isolation and cultivation of a wide variety of fastidious and non-fastidious microorganisms, including aerobic and anaerobic bacteria, yeast, and moulds from a variety of clinical and non-clinical specimens.

A5.5.2.2 Background

The medium contains proteose peptone and infusions from calf brain and beef heart.

The basic nutritive properties of this media are BHI from solids, meat peptone, and yeast extract which serve as sources of carbon, nitrogen, essential growth factors, amino acids and vitamins. Dextrose is used as a source of energy as it provides a carbohydrate source for the fermentative microorganisms. Disodium phosphate helps in maintaining the buffering action of the medium whereas sodium chloride maintains the osmotic equilibrium of the medium.

A5.5.2.3 Composition of BHI medium

The quantity of the ingredients may vary depending upon the manufacturer and the addition of some selective growth supplements; however, the basic ingredients remain the same.

Ingredients	Weight or volume
Brain Heart Infusion	17.5 g
Proteose peptone	10.0 g
Dextrose	2.0 g
Sodium chloride	5.0 g
Disodium phosphate	2.5
Yeast extract	5.0 g
Agar	15.0 g
OR Commercial BHI broth powder	37 g
Distilled water	1,000 ml

Final pH (@ 25 °C): 7.2 ± 0.2

A5.5.2.4 Preparation of BHI medium

The composition and method for preparation of media vary according to the manufacturer. Please refer to the manufacturer's instructions.

1. Suspend 37.0 g in 1 L of distilled water.
2. Heat if necessary to dissolve the medium completely.
3. Dispense into bottles or tubes as desired.
4. Sterilize by autoclaving at 121°C (15 lbs) for 15 minutes.
5. Label the side of each tube with the date of preparation and batch number.

A5.5.2.5 Materials for inoculation of BHI agar

Medium: BHI agar in bottles or plates

Reagents: None

Storage conditions: Dehydrated medium at 10-30 °C; use before expiry date on the label. Prepared medium in tubes or bottles stored 2-8 °C sealed in plastic bags to reduce the chances of contamination and in the dark.

Equipment and consumables: Sterile inoculating loop.

Specimen: Fresh 24-hour culture of organism to be tested.

A5.5.2.6 Method for inoculation of BHI agar

1. Allow medium to adjust to room temperature prior to inoculation.
2. Lightly inoculate the agar using the test organism (or sample) of interest.
3. Incubate the tubes with caps loosened at 35-37 °C / aerobic atmosphere for (without CO₂ enrichment) / 24-48 hours.
4. Examine BHI agar for growth.

A5.5.2.7 Quality control of BHI agar

Visible growth on the BHI agar is an indication medium supports is effective.

TABLE
A5.25

The following outcome on BHI agar must be observed with the QC organism/s.

QC ORGANISM	OBSERVATION
<i>Escherichia coli</i> ATCC 25922	Good to excellent growth
Uninoculated	No visible evidence of growth

Note: Phenotypically equivalent strains may be used based on local availability.

A5.5.3 Dorset egg medium (36-38)

A5.5.3.1 Purpose

DE medium is an enriched slope medium used for the cultivation, maintenance, transport and for long-term storage (at room temperature) of pure cultures of mycobacteria and other fastidious and non-fastidious organisms.

A5.5.3.2 Background

DE medium is a non-selective medium well suited for the growth and maintenance of pure cultures of mycobacteria. More recently, it has also been used for the maintenance and transport of other bacterial species, such as *Streptococcus pneumoniae*, *Neisseria meningitidis*, *Haemophilus influenzae*, and Enterotoxigenic *E. coli* at ambient temperature.

Beef extract and peptone provide nutrients such as amino acids and organic phosphates. Whole egg mass contains complex nutrients necessary for bacterial and mycobacterial growth and, additionally, neutralizes toxic compounds. The inspissation process during the preparation of the medium provides the necessary solidity of the medium and inactivates bactericidal compounds contained in eggs such as lysozyme.

A5.5.3.3 Composition of DE medium

Ingredient	Volume
Nutrient Broth	20 ml
Egg Mass (from whole fresh eggs)	80 ml

Final pH 7.4 ± 0.2 @ 25 °C

A5.5.3.4 Preparation of DE medium

1. Prepare and sterilize the nutrient broth.
2. Wash the eggs by scrubbing them carefully with soap and water followed by rinsing in clean running water.
3. Immerse the egg in 70% v/v ethanol (alcohol) for 10 minutes.
4. As aseptically possible, break the eggs into a sterile flask (pre-marked to hold 80 ml), which contains a few sterile beads. Mix well until egg yolks and whites are completely homogenized.
5. Strain the egg mixture through sterile gauze or muslin into a sterile bottle.
6. When the nutrient broth has cooled to 45-50 °C, add it aseptically to the sterile egg mixture and mix well.
7. Aseptically dispense the nutrient broth-egg medium in 3 ml amounts in sterile bijou bottles or screw cap tubes.
8. Heat (inspissate) at 75-80 °C with the bottles in a sloped position until the medium has solidified.

A5.5.3.5 Materials for inoculation of DE medium

Medium: DE medium in 3 ml bijou bottles or tubes

Reagents: None

Storage conditions: Store the slopes with caps screwed tightly in a cool (2-8 °C), dark place. Do not use if medium shows evidence of microbial contamination, discolouration, drying, cracking or other signs of deterioration.

Equipment and consumables: Inoculating loop.

Specimen: Fresh 24-hour culture of organism to be transported or stored.

A5.5.3.6 Method for inoculation of DE medium

1. Allow medium to adjust to room temperature prior to inoculation.
2. Lightly inoculate the agar using the organism (or sample) of interest.
3. Incubate the tubes with caps loosened at 35-37 °C / aerobic atmosphere for (without CO₂ enrichment) / 24-48 hours.
4. Examine DE agar for growth.
5. Tighten lids securely and store, in consecutive order, in the dark at room temperature.

A5.5.3.7 Quality control and interpretation

Inoculate samples of this medium with the strain mentioned below or similar local strains as available.

TABLE
A5.26

The following outcome on DE medium must be observed with the QC organisms.

QC ORGANISM	OBSERVATION
<i>Escherichia coli</i> (ATCC 25922)	Good to excellent growth
Uninoculated	Cream to white, opaque

Note: Phenotypically equivalent strains may be used based on local availability.

A5.5.3.8 Limitations of DE medium

1. DE medium is not intended to be used directly with clinical specimens.
2. This medium is not selective and, therefore, does not inhibit the growth of contaminants.
3. The transport of mixed cultures on DE medium is not recommended since the species present in mixed cultures might interfere with each other resulting either supplementation or inhibition. However, it is an excellent medium for transport of pure cultures to reference laboratories and for the maintenance of strains.
4. When used as a transport medium, extreme temperatures during transport must be avoided. It has been shown that room temperature (18 to 23 °C) is optimal for survival.

A5.5.4 Trypticase Soy broth with 20% glycerol (TSB20G) (39,40)

A5.5.4.1 Purpose

TSB, a nutritious medium supplemented with glycerol, may be used as a maintenance medium for the preservation of bacterial cultures. This medium is used in the long-term frozen storage of bacterial strains.

A5.5.4.2 Background

Enzymatic digests of protein substrates act as protective colloids. Glycerol is a cryoprotective agent that provides intracellular and extracellular protection against freezing and thawing.

A5.5.4.3 Composition of TSB20G medium

Ingredient	Volume
Casein peptone	17 g
Sodium chloride	5 g
Soy peptone	3 g
Dipotassium phosphate	2.5 g
Dextrose	2.5 g
Glycerol	200 ml
Distilled water	800 ml

Final pH 7.3 ± 0.2 @ 25 °C

Note: Recipe also available online for TSB with 15% glycerol (TSB15G) (40).

A5.5.4.4 Preparation of TSB20G medium

1. Suspend the first five ingredients in 800 ml of distilled water and add 200 ml glycerol.
2. Dispense into 1.5 ml aliquots into 2.0 ml screw-capped cryotubes.
3. Sterilize by autoclaving at 121°C for 15 minutes or as per validated cycle.

A5.5.4.5 Materials for inoculation of TSB20G

Medium: TSB with 20% Glycerol

Reagents: None

Storage conditions: Store uninoculated TSB20G with caps screwed tightly in a cool (2-8 °C), dark place. Do not use if medium shows evidence of turbidity i.e., microbial contamination.

Equipment and consumables: Sterile inoculating loop or sterile swab. Vortex.

Specimen: Fresh 24-hour culture on a non-selective agar plate (or slope) of organism to be stored.

A5.5.4.6 Method for inoculation of TSB20G

1. Using a sterile swab or inoculating loop, harvest all the growth from a fresh subculture on a non-selective agar plate (or sloped) medium.
2. Suspend in TSB20G to achieve the desired concentration of viable cells, usually extremely heavy/thick suspension.
3. Screw lids on tightly and vortex, in a BSC-II if possible and vortex to homogenize the suspension.
4. Freeze suspension immediately at ideally at -70 °C or below.

A5.5.4.7 **Method for resuscitation from TSB20G**

1. Remove the selected stored organism from the -70 °C freezer.
2. Allow to thaw slightly. (Minimizing freeze thaw cycles maximizes viability of stored bacterial cells.)
3. Using a sterile swab, rub the surface of the frozen sample then use the 'loaded' swab to inoculate the non-selective medium e.g., BA or broth medium.
4. Agar plates: incubate at 35-37 °C / aerobic atmosphere (without CO₂ enrichment) / 24-48 hours. Broth: incubate at 35-37 °C / aerobic atmosphere (without CO₂ enrichment) / 24-48 hours.
5. Next day, subculture on to agar plates and incubate accordingly.
6. Return the TSB20G as soon as possible to -70 °C freezer.

A5.5.4.8 **Limitation of TSB20G**

1. Growth on subculture of isolates preserved in TSB20G may depend on inoculum size, duration of storage and fastidious nature of the organism.

A5.5.5 **TSA or trypticase soy broth (TSB) (41,42,43)**

A5.5.5.1 **Purpose**

TSA or TSB is used as a general-purpose medium for the growth of a wide variety of non-fastidious microorganisms (both Gram-positive and Gram-negative bacteria). TSA is not used for the isolation of pathogens from clinical specimens but may be used for maintaining or sub-culturing bacterial strains. TSA is mainly used as an initial growth medium for the purposes of observing colony morphology, achieving a pure culture and preparing sufficient growth for further biochemical testing and culture storage. TSA slants are used to store and ship bacterial cultures.

A5.5.5.2 **Background**

Tryptic Soy medium contains digests of casein and soybean meal. The combination of casein and soy peptones provides a nutritious medium supplying organic nitrogen, particularly amino acids and longer-chain peptides. Sodium chloride is added to maintain the osmotic equilibrium.

A5.5.5.3 **Composition of TSA medium**

Ingredient	Weight or volume
Pancreatic digest of casein	17.0 g
Papaic digest of soyabean meal	3.0 g
Sodium chloride	5.0 g
Dextrose (Glucose)	2.5 g
Dipotassium hydrogen phosphate	2.5 g
Agar	15.0 g
Distilled water	1,000 ml

Final pH 7.3±0.2 @ 25 °C.

Note: The medium may be supplemented with blood to facilitate growth of more fastidious bacteria or antimicrobial agents to permit the selection of various microbial groups from mixed microbiota.

TSB contains all the ingredients above except agar.

A5.5.5.4 **Preparation of TSA medium**

TSA and TSB are available from most suppliers of culture media in powdered (free-flowing, homogeneous) form.

1. Suspend 45.0 g in 1 L of distilled water.
2. Heat to boiling to completely dissolve powder.
3. Dispense into tubes
 - a. 3 ml to make NA slopes
 - b. 5 ml to make NA deeps, or
4. Sterilize by autoclaving at 121°C for 15 minutes.
5. Cool to 45-50 °C.
6. For agar plates, mix well and dispense into 20 ml into 9 cm sterile Petri dishes.
7. For slopes or deeps, allow the agar medium to solidify in a slanted position.
8. Date the medium and give it a batch number.
9. Store in a cool dark place.

A5.5.5.5 **Materials for inoculation of TSB20G**

Medium: TSA in bottles, tubes or plates (or broth in bottles or tubes)

Plates should be warmed to room temperature and the agar surface should be dry prior to inoculating.

Reagents: None

Storage conditions: Store the dehydrated medium at 10-30 °C and use before the expiry date on the label. Store the prepared broth away from light at room temperature.

Equipment and consumables: Sterile inoculating loop.

Specimen: Fresh 24-hour culture of organism to be tested.

A5.5.5.6 **Method for inoculation of TSA (or TSB)**

1. Lightly inoculate the agar (or broth) using the test organism (or sample) of interest.
2. For agar plates, streak out to achieve single colonies.
3. Incubate the bottles/tubes with caps loosened or plates at 35 °C ± 2 °C / in aerobic atmosphere for 24 hours (without CO₂ enrichment) / 18-24 hours.
4. Examine agar or broth for growth.

A5.5.5.7 Quality control of TSA (or TSB)

Visible growth on the agar (or turbidity in the broth) is an indication the medium is effective.

TABLE
A5.27

The following outcome on TSA (or TSB) must be observed with the QC organisms.

QC ORGANISM	OBSERVATION
<i>Escherichia coli</i> ATCC 25922	Good to excellent growth
Uninoculated	No visible evidence of growth

Note: Phenotypically equivalent strains may be used based on local availability.

A5.5.5.8 Limitation of TSA (or TSB)

1. Contamination, particularly with Gram-negative organisms other than *Salmonella* sp. cannot be discerned on TSA or TSB.
2. It cannot be used to isolate pathogens from clinical specimens.

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Annex 6

Antimicrobial susceptibility testing

Note: CLSI and EUCAST guidelines prescribe the use of commercially-formulated, dehydrated Mueller-Hinton (MH) base that are prepared according to the manufacturer's instruction (or to purchase commercially prepared MH plates). The AST procedures, including the MH plates, must be followed precisely to eliminate inconsistency and to obtain valid results.

A6.1

Disk diffusion and minimum inhibitory concentration (MIC)

A6.1.1 Materials required for disk and gradient MIC strips (E-test) AST

1. Normal sterile saline medium (1.5 ml quantities)
2. Mueller-Hinton agar (see Mueller-Hinton Agar – Composition, Principle, Uses and Preparation)
3. Antimicrobial disks and E-test strips
4. Sterile cotton swabs
5. 0.5 McFarland (BaSO₄) turbidity standard
6. Sterile forceps / needle / disk dispenser
7. Sliding callipers / ruler
8. QC reference strains

(see https://www.eucast.org/ast_of_bacteria/quality_control/)

A6.1.2 **Methods: Preparation of inoculum for disk susceptibility and MIC**

Each culture to be tested should be streaked onto a non-inhibitory agar medium (e.g., BA [Annex A4.1], BHI agar [Annex A5.5.2], or TSA [Annex A5.5.4]); see also Annex 4) to obtain isolated colonies.

After incubation at $35^{\circ}\text{C} \pm 2^{\circ}\text{C}$ overnight, select four or five well-isolated colonies with a sterile inoculating needle or loop.

Transfer the growth to a tube of sterile saline or non-selective broth (e.g., Mueller-Hinton broth [MHB], heart infusion broth, or TSB see A5.5.2 and A5.5.4, respectively) and vortex thoroughly.

The bacterial suspension should then be compared to the 0.5 McFarland turbidity standard. This comparison can be made more easily if the tubes are viewed against a Wickerham reading card or a sheet of white paper on which sharp black lines are drawn (Figure A6.1).

FIGURE
A6.1

Comparing test suspension (right) with 0.5 McFarland standard (left) against a Wickerham reading card.



Source: Taken from (1).

Note: The turbidity standard should be agitated on a vortex mixer immediately prior to use. If the bacterial suspension does not appear to be the same density as the 0.5 McFarland turbidity standard, the turbidity can be reduced by adding sterile saline or broth or increased by adding more bacterial growth.

Alternatively, the growth method may be used to prepare the inoculum.

1. Pick four or five colonies from overnight growth on agar and inoculate them into broth (MHB, heart infusion broth, or TSB).
2. Incubate the broth at $35^{\circ}\text{C} \pm 2^{\circ}\text{C}$ until turbid (usually 16–24 hours).
3. Adjust the turbidity to the proper density.

A6.1.3 Inoculation procedure

Within 15 minutes after adjusting the turbidity of the inoculum suspension, follow these steps:

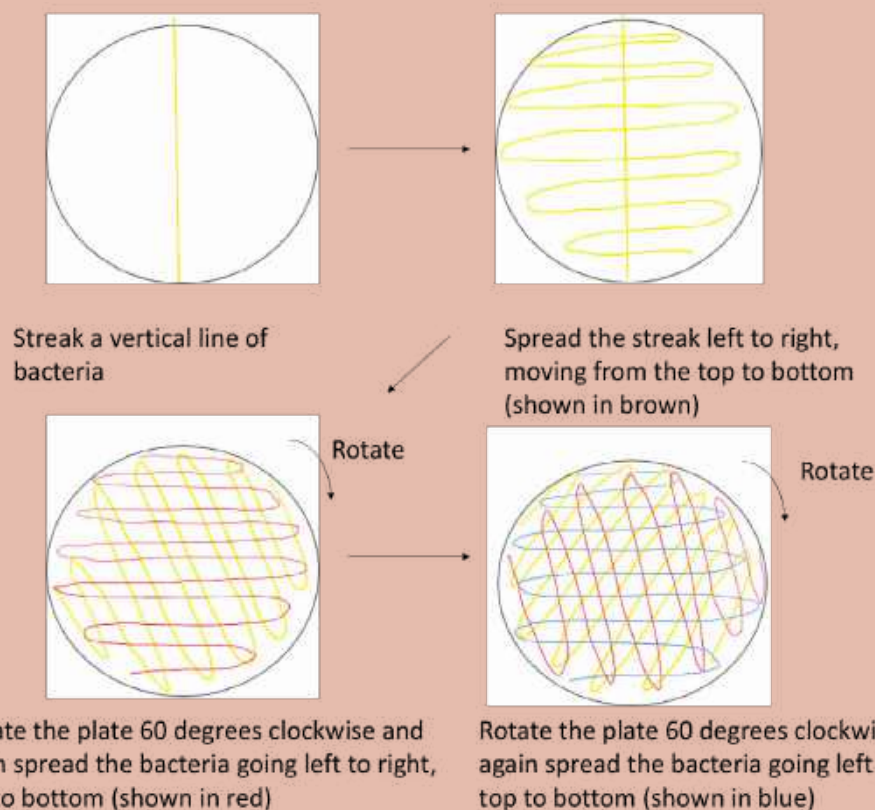
1. Dip a sterile cotton swab into the suspension. Pressing firmly against the inside wall of the tube just above the fluid level, rotate the swab to remove excess liquid.
2. Streak the swab over the entire surface of the medium three times, rotating the plate approximately 60 degrees after each application to ensure an even distribution of the inoculum (Figure A6.2).
3. Finally, swab around the entire edge of the agar surface.

Note: Inoculation of the agar plates needs to take place within 15 minutes of preparation to avoid large changes in the density of the suspension.

For MHA plate, dry open agar plate 30 minutes in $35\text{--}37^{\circ}\text{C}$ incubator before use.

FIGURE
A6.2

Spreading inoculum in three directions (60 degrees apart) using sterile swab.



Note: If the bacterial colonies used to prepare the suspension are picked off a plate containing mixed growth (i.e., if isolated colonies are picked from a plate that does not contain pure culture, as may occur when working with cultures from stool specimens), laboratorians should prepare a purity plate (see Annex A6.1.3.1) to ensure the suspension used for AST is pure.

A6.1.3.1 Preparation of a purity plate for the inoculum:

Within 15 minutes after adjusting the turbidity of the inoculum suspension, follow these steps:

1. After inoculating the MHA plate for confluent growth, label (a portion of) a separate TSA plate (or other non-selective medium).
2. Use the same swab of suspension with which the MHA was inoculated to streak for isolation; do not place the swab back into the suspension.
3. Several inocula can be streaked on different sections of a properly labelled purity plate, but the streaks must not overlap.
4. If mixed growth (indicates multiple bacterial species/contamination) appears after incubating overnight at 37 °C, discard the MHA plates and perform AST again.

A6.1.4 Disk diffusion procedure

The working supply of antimicrobial disks should be stored in a refrigerator (4 °C). Upon removal of the disks from the refrigerator, the package containing the cartridges should be left unopened at room temperature for approximately 1 hour to allow the temperature to equilibrate; this reduces the amount of condensation on the disks. If a disk-dispensing apparatus is used, it should have a tight-fitting cover, be stored in the refrigerator, and be allowed to warm to room temperature before use. Apply the antimicrobial disks to the plates as soon as possible, but no longer than 15 minutes, after inoculation of the suspension.

In general, no more than 12 disks should be placed on a 150 mm plate and no more than four disks should be placed on a 100 mm plate to prevent overlapping of the zones of inhibition and possible resultant error in measurement. Also, the placement of antimicrobial agents on the agar surface should be arranged depending on their type, as several larger zones fused together may result in erroneous interpretation. Diffusion of the drug in the disk begins immediately; therefore, once a disk contacts the agar surface, the disk should not be moved or removed.

A6.1.5 Setting up a plate for disk diffusion

1. Ensure the plate surface is dry, with no liquid remaining.
2. Place the disks individually with sterile forceps or with mechanical dispensing apparatus, and then gently press down onto the agar.
3. After the disks are placed on the agar, replace the lid and invert the plate.
4. Incubate at 35 °C ± 2 °C / aerobic atmosphere (without CO₂ enrichment) / 16 to 18 hours; if a purity plate was prepared, incubate under the same conditions.

A6.1.6 Reading and interpreting results

1. After 16 to 18 hours of incubation, measure the diameter of the zones of complete inhibition (including the diameter of the disk) (Figure A6.3) and record in millimetres.
2. Interpret the results using the recommended guidelines.

The measurements can be made with callipers or a ruler on the under surface of the plate without opening the lid. With sulfonamides and trimethoprim-sulfamethoxazole, a slight amount of growth may occur within the inhibition zone; in this instance, slight growth (approximately 80% inhibition) should be ignored and the zone diameter should be measured to the margin of heavy growth. *S. Typhi* and *Paratyphi*, zones of growth inhibition should be compared with the zone size interpretative table for Enterobacterales (see the guidelines provided by CLSI or EUCAST), and recorded as susceptible, intermediate, or resistant to each drug tested.

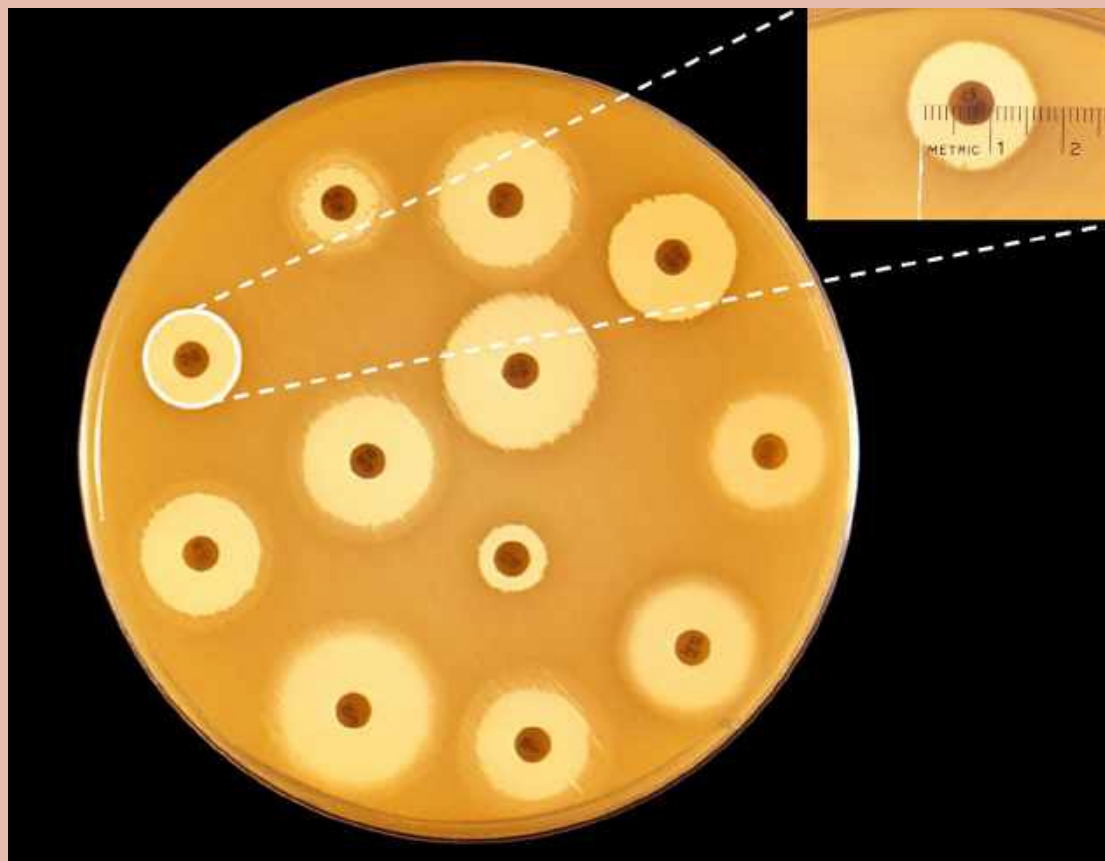
Guidelines:

CLSI: <https://clsi.org/standards/products/microbiology/documents/m100/>

EUCAST (Enterobacterales) http://www.eucast.org/clinical_breakpoints/

FIGURE
A6.3

Zones of inhibition after incubation



Source: Photo credit: US Centers for Disease Control and Prevention. Public Health Image Library. Public Domain.

The MIC is the lowest possible concentration of a molecule (typically an antimicrobial agent) needed to prevent the growth of any given microorganism. Below are the protocols for MIC testing using both E-test and broth microdilution.

A6.2.1 MIC using E-test

Inoculum preparation, selection of agar medium and inoculation of bacterial cells for E-test is similar to the disk diffusion method (Kirby-Bauer) described in Chapter 6.2.2 and Annex A6.1. Once antimicrobial strips are placed accordingly, the agar plate should be incubated in an inverted position at $35 \pm 2^\circ\text{C}$ / aerobic atmosphere (without CO_2 enrichment) / 16–18 hours.

Note: Following inoculation, the test strip should be placed in the middle of the plate if just one antimicrobial agent is being tested. For plates that are bigger in size (15 cm), up to six test strips can be employed at once by arranging them in the form of a star. If testing multiple antimicrobial agents, strips containing the maximum concentration must be placed towards the edge of the agar plate.

A6.2.2 Setting up a plate for MIC testing using E-test strips

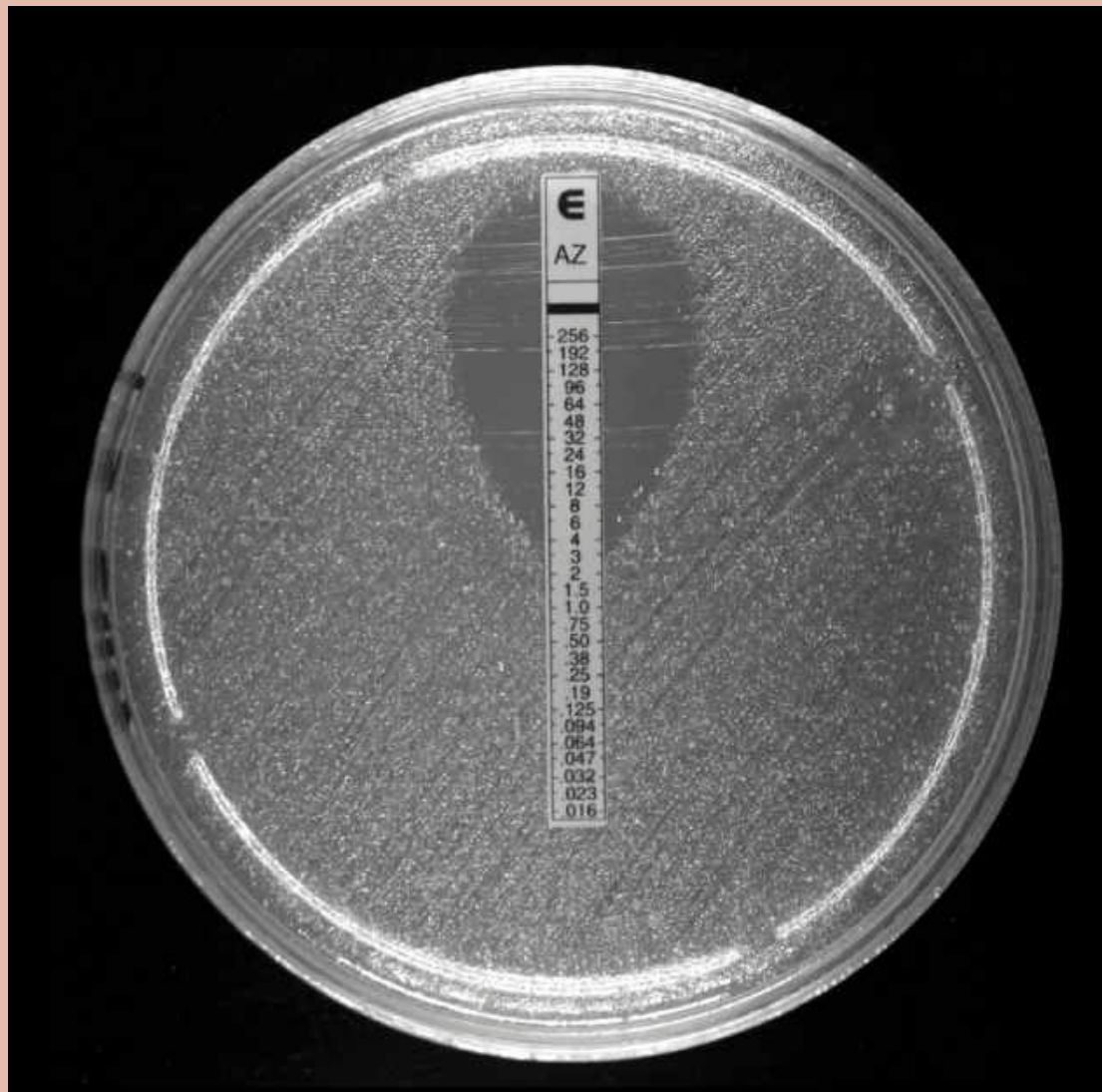
1. Ensure the plate surface is dry, with no liquid remaining.
2. Place the strips individually with sterile forceps and then gently press down, from the lower to the higher concentration, onto the agar. Diffusion of the drug in the strip begins immediately; therefore, once placed on the agar surface, the strip should not be moved or removed.
3. After the strips are placed on the agar, replace the lid and invert the plate. Incubate at $35 \pm 2^\circ\text{C}$ / aerobic atmosphere (without CO_2 enrichment) / 16–18 hours.

A6.2.3 Reading and interpreting results

After the required period of incubation, MIC value can be read where the lower portion of the ellipse intersects the surface of the test strip. If there is any difference between the point of intersection (right versus left side of the strip), the higher value should always be read and recorded (**Figure A6.4**).

FIGURE
A6.4

MIC of azithromycin (AZ) for a blood culture positive isolate of *S. Typhi*. MIC interpretation 3 µg/ml.



Source: Image Courtesy of Child Health Research foundation (CHRF).

Any presence of mixed, light or heavy growth on the plate should be considered as erroneous and the test must be repeated once again. As previously stated, CLSI or EUCAST guidelines should be followed for MIC breakpoints and interpretation criteria.

A6.3 MIC using broth microdilution

Inoculum preparation for this test is prepared as previously described (see Annex A6.1.3) but antimicrobial agent preparation and inoculation procedure is distinct

A6.3.1 Materials required for MIC by broth microdilution

1. Normal sterile saline medium
2. Cation-adjusted Mueller-Hinton broth (MHCab)
3. Antibiotics (suspension or in powdered form)
4. Sterile 96 well plates
5. 0.5 McFarland (BaSO₄) turbidity standard
6. Normal and multichannel pipettes
7. QC reference strains

A6.3.2 Setting up a plate for broth microdilution

The following protocol to perform broth microdilution requires a starting concentration as dictated by the routines applied in the laboratory for the use of the antimicrobial agent in question. The calculations can be done as suggested below:

1. First, the concentration at a volume of 90 µl should be calculated using the following formula:

$$C_{1(100)}V_{1(100)}=C_{2(90)}V_{2(90)}$$

2. To calculate the volume of antimicrobial agents that should be added, the same formula can be used:

- a. let's assume we have an antimicrobial agent suspension of a 10 mg/ml concentration

$$C_{1(ab)}V_{1(ab)}=C_{2(90)}V_{2(90)}$$

- b. Based on the calculations, 139 µl of MHCab must be added to the first column, 90 µl must be added from columns 2 to 11, and 100 µl should be added to column 12 that serves as a negative control.
3. To column 1, 41 µl of the 10 mg/ml antimicrobial agent should be added to have a total volume of 180 µl, then serial dilution takes place by mixing 90 µl from column 1 and transferring it into column 2, 2 to 3, further on until column 10 where after mixing, 90 µl are discarded.
 4. After that, columns 1 to 10 contains 90 µl of antimicrobial agents and broth, column 11 contains 90 µl of broth only, and column 12 contains 100 µl of broth only.
 5. To prepare a bacterial suspension of 5 x10⁶ cfu/ml:
 - a. A bacterial suspension of 0.5 McFarland (equivalent to 108 CFU) should be prepared by adding 1-2 colonies to 2 ml MHCab and this is measured using a densitometer.
 - b.

$$C_1V_1=C_2V_2$$

$$10^8 \times V_1 = 5 \times 10^6 \times 15$$

$$V_1 = 750 \mu\text{l}$$

→ 750 µl of the 0.5 McFarland bacterial suspension should be added to 14,250 µl of MHCab to reach a bacterial suspension of 5x10⁶ CFU/ml (**Note:** the final volume, which is 15 ml here, isn't fixed and can be changed based on your needs).

6. From the diluted bacterial suspension, add 10 µl to columns 1 to 11 to reach a final volume of 100 µl in all the wells.

After inoculum and antimicrobial agent preparation is done:

1. Inoculate the panels containing 0.1 ml MHB and desired concentration of antimicrobial agents with approximately 5×10^5 CFU/ml *S. Typhi*/Paratyphi cells.
2. Following inoculation, cover the plates with a sterile lid/film immediately to prevent evaporation and contamination.
3. Incubate in an upright position at $35^\circ\text{C} \pm 2^\circ\text{C}$ / aerobic atmosphere (without CO₂ enrichment) / 16–18 hours.

Note: Do not invert the plates as it will result in leakage of broth, leading to subsequent failure or even accidental exposure to the pathogen.

A6.3.3 Reading and interpreting results

Once the incubation at desired conditions is over, microtitre plate should be taken out of the incubator and examined carefully for bacterial growth. If the broth inside any well turns cloudy and/or a layer of cells can be observed at the bottom, then it indicates bacterial growth. In that case, MIC for that specific panel should be the lowest concentration of antimicrobial agents where no growth is visible/detectable. For interpretation and antimicrobial breakpoints, please see CLSI or EUCAST guidelines.

A6.3.4 Critical considerations

Various factors, including physical parameters, medium components, bacterial inoculum and growth conditions can affect the accuracy of the test. Therefore, it is of utmost importance to strictly adhere to the instructions provided by the manufacturer or appropriate guidelines.

Bacterial inoculum: Inoculum density must be adequately measured and maintained. Any faulty measurement can lead to false interpretation. Turbidity standard can be adjusted by comparing with an in-house or commercially produced McFarland 0.5 standard. Similarly, a spectrophotometer can also be used to adjust the optical density of the suspension at 600 nm wavelength.

Bacteriological medium: Growth media used for AST should have the appropriate pH, formulation, thickness/consistency and thymidine content. Prepared bacteriological agar media are usually susceptible to dehydration and chemical degradation. During storage of agar plates, dehydration can be reduced by properly wrapping them with plastic bags. However, it is also important to get rid of moisture from the plates during the test to reduce the chances of any contamination. Aseptic preparation is another necessary element to minimize contamination. Usually, prepared bacteriological broth (MHB) can be stored at $2-8^\circ\text{C}$ for up to a year under sterile conditions. Agar plates (MHA) should also be stored at $2-8^\circ\text{C}$ but, the shelf life does not typically exceed more than a month.

Growth conditions: Conditions of growth depends on the type of organism. As a Gram-negative facultative anaerobe belonging to the family *Enterobacteriales*, *S. Typhi* and Paratyphi are receptive to high temperature and typically die at around 70°C . Therefore, it is important to maintain a constant temperature all around the incubation period.



Antimicrobial agents: Moisture content, storage temperature, solubility are just some of the important conditions which must be considered for optimal performance of the antimicrobial agents. Working stock of susceptibility disks should be stored at 2–8 °C with desiccants for up to the period recommended by the manufacturer. For long-term storage, they should be kept at –20 °C. Desiccants must be used during storage to minimize moisture build-up. Not every antimicrobial agent (for broth microdilution) is water soluble and in some cases additional compound, such as acid may be needed to dissolve them properly. For this reason, appropriate solvent or chemicals recommended by the manufacturer must be used. Certain antimicrobial molecules can be damaged by the presence of light. Stock solution of light sensitive antimicrobial agents should be wrapped with aluminium paper to avoid long-term light exposure.

QA: The objectives of quality measurement are to confirm the precision of the test being used. In order to achieve that, overall performance of the method, reagents and personnel who perform these tests should be taken into consideration. The guidelines for routine performance assessment and standard QC parameters have been elaborately described in **Chapter 9** of this manual.

Please refer to EUCAST (www.eucast.org) or CLSI (www.clsi.org) for recommended bacterial strains, expected disk and MIC values, and QC organisms versus antimicrobials.

QC testing is essential to assess the accuracy of both disk diffusion and MIC results. If QC results are out of range, patient results are invalid and must not be reported

References for Annex 6

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Annex 7

Molecular testing

A7.1

Guidance on the Code of Practice for Molecular Suites

The information provided here is guidance on the best practice that is reasonably practicable with the infrastructure available in your laboratory. A lot of the information provided here are from SOPs from UKHSA but can be used as a guide for users to adapt as appropriate.

The key message is to have a clean area to set-up your reagents that does not come into contact with PCR products, that clean gloves and PPE must be worn and areas are wiped down with appropriate decontamination products (surface decontaminant that eliminates unwanted DNA and DNase) to prevent any cross contamination with each PCR batch.

A7.1.1 Room set-up and requirements

Ideally there would be four rooms available:

1. one Specimen Extraction room: for the specimen handling and DNA extraction
2. one Pre-PCR Clean room: for making up the PCR reagents
3. one PCR set-up room: for adding the DNA template
4. one Post-PCR room: for performing the PCR

When only three rooms are available, the Specimen Extraction room and PCR set-up room (ideally with a PCR cabinet) can be merged together. When only two rooms are available, merging the post-PCR room with the Specimen Extraction room is not ideal as the post-PCR products may contaminate new extractions. Only having one room is not recommended as contamination of PCR products with new reagents would be highly likely.

This Code of Practice is for the use of a Molecular Facility that includes a Pre-PCR Clean room, Specimen Extraction room, PCR set-up room and the post-PCR room. Employees movement in the molecular laboratory room should follow unidirectional walk flow.

Note: Laboratory coats must only ever be worn in their designated laboratory. This system is to prevent cross contamination of molecular work between rooms. On leaving a laboratory area, laboratory coats must be removed and hung up in the laboratory (one coat per peg). Hands must be washed before leaving each laboratory.

1. Pre-PCR clean room:

No DNA amplicons, live organisms or contaminated material permitted. Reagents for molecular test (including PCR) are stored and prepared in this specific area.

2. Specimen extraction room

No DNA amplicons permitted

All work with pure cultures or samples known or potentially containing infectious material **MUST** be performed within a microbiological safety cabinet by trained and competent staff.

DNA extracts are stored within dedicated fridges and freezers within this laboratory and must be clearly labelled.

3. PCR set-up room

No DNA amplicons permitted.

DNA extracts are stored within dedicated fridges and freezers within this laboratory and must be clearly labelled with department name and date of discard. DNA template prepared from pathogens is added to PCR master mix using laminar flow PCR hoods. Areas should be wiped down before and after with a DNase based product to remove any potential DNA contaminants. In addition, PCR hoods should have the UV light activated after use to decontaminate the area.

4. Post-PCR room

DNA amplicons permitted

PCR products are generated using Real-Time PCR platforms including TaqMan 7500 Fast, Vii7 and Rotor-Gene Q, Biorad CFX 96. Alternatively, block-based PCR platforms can be used and PCR products can be separated using stained agarose gels.

DNA amplicons are stored in dedicated fridges and freezers within this laboratory and must be clearly labelled with department name and date of discard.

A7.1.2 Risk assessments

A risk assessment is the written findings of the process whereby hazards in the workplace are identified and control measures put in place to reduce the risk from those hazards as far as reasonably practicable.

Risk assessments must only be undertaken by suitably trained and competent nominated assessors.

All laboratory SOPs are underpinned by risk assessments (see **Chapter 2**)

Note: The specific control measures identified by the risk assessment should be highlighted in your local SOPs.

It is important that staff carrying out these procedures have a full understanding of the hazard control measures in place for that procedure. This should be explained during initial training and competency assessment.

Any new method introduced into routine diagnostic use must be risk assessed first before any work commences. If any method is changed in any way, then the risk assessment must be reviewed and any changes to control measures must be highlighted in the amended SOP.

In summary, risk assessments must be carried out before implementing or introducing into the workplace for all of the following:

1. new procedures
2. new equipment
3. new chemicals or reagent
4. any change to an existing procedure, including relocation

A7.2

Environmental controls

In every batch of PCR assay, controls are used to monitor the PCR procedure:

1. Non-template control (NTC) monitors contamination during extraction
2. PCR Mastermix negative control (NEG) monitors contamination during PCR and PCR set-up
3. Positive control (POS) monitors the primers (and probes) specificity and the good performance of the PCR assay. If this fails, then the test is invalid and needs to be repeated
4. When the POS control shows amplification AND either or both the NTC and NEG controls show amplification, then there is evidence of nucleic acid/ DNA contamination. This indicated that the test is invalid and needs to be repeated after decontaminating the areas

A7.2.1 How contamination occurs

1. Micro-aerosols between samples or reactions in wells
2. The material used or the work surface are contaminated
3. The reagents or controls used are contaminated
4. There is contamination of PCR products in the 'clean' rooms

A7.2.2 Decontamination procedures and aseptic techniques

1. Always wear gloves
2. Wipe the work surfaces with a surface decontaminant that eliminates unwanted DNA and DNase across the laboratory and especially in your working area.
3. Wipe the pipettes, marker pens, racks, and tips boxes (outside surfaces) with surface decontaminant after each experiment.
4. Discard all the PCR water in use according to your local risk assessment
5. Discard primers and probe in use to your local risk assessment
6. In the PCR UV cabinet, repeat the top 4 bullet points above
7. Put the equipment, dismantle if necessary, in the UV cabinet and decontaminate with the UV light according to the manufacturer instructions. A minimum of 10 minutes following a contamination event is recommended.
8. Also decontaminate racks, pens, markers and pipettes, if necessary, by repeating the UV decontamination, rotating the items to expose all surfaces to UV if necessary
9. Repeat PCR assay with new water and newly prepared reagents and assess NTC and NEG controls.
10. Follow recommendations of using 'pre-PCR clean' and 'post-PCR' rooms and do not move laboratory coats, PPE or pipettes between rooms.

A7.3

Storage/reconstitution of probes, primers and preparation of Mastermix for PCR

Note, although volumes, specific reagents and reagents are specified, these are just a guide. Different brands, different platforms, different Mastermix reagents with different chemistry parameters can be used. Laboratories can use and adapt current PCR platforms and methods with the primers and probes stated and it is always recommended that laboratories always validate and optimize any new PCR implemented into the laboratory.

A7.3.1 Storage of probes, primers and Takyon

TABLE
A7.1

Storage and duration of probes, primers and Takyon

Oligonucleotides	Format	Storage (temperature)	Stability duration
Probes	Dried (powder form)	Room temp. (5 °C to 15 °C)	18 months
	Added PCR water (liquid form)	Freezer (-20 °C)	24 months
Primers	Dried (powder form)	Room temp. (5 °C to 15 °C)	18 months
	Added PCR water (liquid form)	Freezer (-20 °C)	24 months
Takyon (Low Rox)	Liquid form	Fridge (4 °C)	6 months
		Freezer (-15 °C to -25 °C)	24 months

Reference: Eurogentec – Research – Reconstitution and Storage.

A7.3.2 Receiving probes and primers delivery

1. Double check the sequence of probes and primers received match your order
2. Keep all the freeze-dried probes and primers in their corresponding labelled boxes at room temperature away from the light
3. Record all the received probes and primers in the laboratory inventory

A7.3.3 Reconstitution of probes and primers

1. Spin the tube briefly to collect the DNA in the bottom of the tube. Add an appropriate volume (according to manufacturer instructions) of nuclease free water. Allow the tube to stand for a few minutes at room temperature, ensure it is gently and thoroughly mixed.
2. Prepare the stock solution at 100 μ M concentration, which can be achieved by adding a volume of sterile nuclease free water (follow the Data Technical Sheet for the volume that needs to be added to make 100 μ M) and distribute in aliquots (to avoid multiple freeze thaw cycles) using the screwcap microtube 1.5 ml. All reconstituted reagents should be frozen (-20oC) when not in use as they will degrade at fridge temperature.
3. Record the batch number and preparation of primers, probes and other reagents for PCR for audit trail purposes.
4. Label the aliquot lids using a system to easily distinguish, such as colour label stickers according to the specific probe and primer.
5. Label the tube by writing the following information:
 - a. Primer/probe name
 - b. Batch number
 - c. Preparation date and prepared by
 - d. Concentration (μ l)
 - e. First use (1st batch) or last use (last batch)

Note* Fluorescently labelled probes are light sensitive. Therefore, it is better to store them in an opaque tube or wrap a small piece of silver foil paper around the tube.

A7.3.4 Preparation of Mastermix PCR

Use the 1.5 ml microtube and label the lid with the specific PCR Mastermix using white lid label sticker and the tube with the following information:

1. Name of Mastermix PCR
2. Date prepared
3. Prepared by
4. First use (1st batch) or last use (last batch)

A7.3.5 *Salmonella* typhoidal Real-Time PCR Multiplex Composition

TABLE
A7.2

Primer and probe dilution for RT-PCR multiplex.

Working solution:

Primer Dilution 20 μ M

Primer	10 μ l (from 100 μ M)	80 μ l (from 100 μ M)	180 μ l (from 100 μ M)
PCR Water	40 μ l	320 μ l	720 μ l
TOTAL	50 μL	400 μL	900 μL

Probe Dilution 5 μ M

Probe	2.5 μ l (from 100 μ M)	20 μ l (from 100 μ M)	50 μ l (from 100 μ M)
PCR Water	47.5 μ l	380 μ l	950 μ l
TOTAL	50 μL	400 μL	1000 μL

Specific Mastermix: 22.5 μ l + sample DNA: 2.5 μ l = 25 μ l/well

- Record batches of Mastermix made on a local document (for quality control auditing purposes) including the Mastermix name, the batches of reagents for audit trial purposes.
- 1 ng/ μ l of DNA should be sufficient but optimize your experiments to assess what works best in your laboratory.
- Ensure the probes and primers are validated prior to use with routine samples and record for audit trial purposes.

TABLE
A7.3**Master mix 1, 2 and 3 working concentrations and sample preparation.****MASTERMIX 1**

WORKING CONCENTRATION		NO. OF SAMPLES						
		1X	10X	20X	30X	40X	50X	60X
Nuclease Free Sterile PCR Water		2	20	40	60	80	100	120
sseJ2 forward	20 μ M	2.66	27	53	80	106	133	160
sseJ2 reverse	20 μ M	2.66	27	53	80	106	133	160
sseJ2 Probe	5 μ M	2.66	27	53	80	106	133	160
Takyon Low Rox or equivalent reagent for your PCR machine		12.5	125	250	375	500	625	750

MASTERMIX 2

WORKING CONCENTRATION		NO. OF SAMPLES						
		1X	10X	20X	30X	40X	50X	60X
Nuclease Free Sterile PCR Water		4.96	50	99	149	198	248	298
tviB forward	20 µM	0.5	5	10	15	20	25	30
tviB reverse	20 µM	0.5	5	10	15	20	25	30
tviB Probe	5 µM	1	10	20	30	40	50	60
SPC0869 forward	20 µM	0.53	6	11	16	21	26.5	32
SPC0869 reverse	20 µM	0.53	6	11	16	21	26.5	32
SPC0869 Probe	5 µM	1.06	11	21	32	42	53	64
ttr3 forward	20 µM	0.22	2	4	7	9	11	13
ttr3 reverse	20 µM	0.22	2	4	7	9	11	13
ttr3 Probe	5 µM	0.46	5	9	14	18	23	28
Takyon Low Rox or equivalent reagent for your PCR machine		12.5	125	250	375	500	625	750

MASTERMIX 3

WORKING CONCENTRATION		NO. OF SAMPLES						
		1X	10X	20X	30X	40X	50X	60X
Nuclease Free Sterile PCR Water		5.36	54	107	161	214	268	322
StaG Forward	20 μM	0.22	2	4	7	9	11	13
StaG Reverse	20 μM	0.22	2	4	7	9	11	13
StaG Probe	5 μM	0.46	46	9	14	18	23	28
SrfJ Forward	20 μM	0.53	53	11	16	21	26.5	32
SrfJ Reverse	20 μM	0.53	53	11	16	21	26.5	32
SrfJ Probe	5 μM	1.06	11	21	32	42	53	64
SPA2308 Forward	20 μM	0.4	4	8	12	16	20	24
SPA2308 Reverse	20 μM	0.4	4	8	12	16	20	24
SPA2308 Probe	5 μM	1.06	11	21	32	42	53	64
Takyon Low Rox or equivalent reagent for your PCR machine		12.5	125	250	375	500	625	750

TABLE
A7.4Suggested 96-well plate set-up for *Salmonella* typhoidal PCR.

		1	2	3	4	5	6	7	8	9	10	11	12
Mastermix 1	A	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	Sample 6	Sample 7	Sample 8	Sample 9	Sample 10	Sample 11	Sample 12
Mastermix 2	B	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	Sample 6	Sample 7	Sample 8	Sample 9	Sample 10	Sample 11	Sample 12
Mastermix 3	C	Sample 1	Sample 2	Sample 3	Sample 4	Sample 5	Sample 6	Sample 7	Sample 8	Sample 9	Sample 10	Sample 11	Sample 12
	D												
	E												
Mastermix 1	F	Sample 13	Sample 14	Sample 15	Sample 16	Sample 17	Sample 18	Sample 19	Sample 20			Paratyphi C (NCTC96)	Negative Control
Mastermix 2	G	Sample 13	Sample 14	Sample 15	Sample 16	Sample 17	Sample 18	Sample 19	Sample 20			Paratyphi C (NCTC96)	Negative Control
Mastermix 3	H	Sample 13	Sample 14	Sample 15	Sample 16	Sample 17	Sample 18	Sample 19	Sample 20	Typhi (NCTC 85)	Paratyphi A (NCTC 1180)	Paratyphi C (NCTC96)	Negative Control

Note: Red wells specify positive NCTC controls known to contain genes of interest, but any local positive control can be used as long it contains the genes you are testing in your Mastermix.

A7.4

Positive controls

Positive controls for each target should be included with each batch of PCR assay set-up. Positive controls can be obtained from culture collections or previously positive samples fully characterized in your laboratory.

Below are examples of controls that can be used available from The National Collection of Type Cultures (NCTC).

TABLE
A7.5

Examples of controls that can be used with each batch of PCR assay set-up.

ORGANISM	REFERENCE NUMBER	EXPECTED TEST RESULTS
<i>S. Typhi</i>	NCTC 8385	<i>ttr</i> +, <i>tviB</i> +, <i>staG</i> +
<i>S. Paratyphi A</i>	NCTC 11803	<i>ttr</i> +, <i>SPA2308</i> +
<i>S. Paratyphi B</i>	NCTC 8299	<i>ttr</i> +, <i>srfJ</i> +,
<i>S. Paratyphi C</i>	NCTC 96	<i>ttr</i> +, <i>sseJ</i> +, <i>SPC0869</i> +, <i>tviB</i> +

A7.5

Real-Time PCR

A7.5.1 Performance characteristics

To ensure validity of results, it is important that all controls have worked successfully. This includes positive control strains representing targets of interest. A negative control containing nuclease free water and, showing no amplifications, should also be used.

A7.5.2 Type of sample

Salmonella cultures may be tested directly from DNA extracted from incoming agar slopes or agar plates after overnight growth.

A7.5.3 Equipment and reagents

Please note that these are a guide and other systems do exist and can be used.

1. Applied Biosystems 7500, ViiA-7 Fast Sequence Detector or QuantStudio 5 or equivalent Real-Time PCR machine
2. Heating block
3. Plate centrifuge
4. Micropipettes (P1000, P200, P20 and/or P10) and aerosol resistant tips (200 µl, 20 µl) and/or shaftguard filter tips
5. Fast Taqman 96-well reaction plate
6. MicroAmp self-adhesive optical covers
7. 1.5 ml microfuge tube
8. 2.0 ml microfuge tube
9. Protective gloves
10. Primers and probes
11. Takyon Rox Mix or equivalent as appropriate to your local PCR platforms
12. NCTC control strains
13. Water (sterile-filtered, BioReagent, suitable for cell culture)

Note: If you don't have the exact equipment or reagents above, equivalent consumables can be used.

A7.5.4 Calibration procedures and maintenance

PCR efficiency should be tested on all PCR machines for each of the targets on an annual basis. Ideally the PCR machine should be serviced annually.

A7.5.5 Quality control procedures

There are multiple QC procedures used throughout this assay. All results and data should be recorded on document-controlled worksheets or dedicated software packages and stored in an easily accessible way.

A5.5.5.1 Internal quality controls

Positive controls strains used as internal QC for *Salmonella* PCR should be prepared once a month from cryobeads or equivalent storage media, which are stored in a -80 °C freezer in CL2 or CL3 (depending on the control strain) and plated onto an appropriate agar plate medium, which is then incubated at 37 °C overnight. A sterile 1 µl loop is used to inoculate a pure colony of the control strain into 490 µl of nuclease free water in a 1.5 ml microfuge tube. Thereafter, the control sample is heat inactivated at 100 °C for 20 minutes. The PCR control strain is then validated against appropriate *Salmonella* PCR test and CT values are recorded. Once the PCR control has passed QC, it is recorded in and stored in the fridge for up to 2 months. Severn Biotech Ltd. Nuclease free water is used as the negative control for *Salmonella* PCR procedures.

A7.5.6 Procedural steps

A7.5.6.1 DNA crude extraction

1. 1.5 ml screwcap tube with 490 µl nuclease free water are pre-aliquoted ready for use and are kept in the fridge.
2. Using a sterile disposable loop take a sweep of the surface of the original culture, and carefully resuspend the organism in the 490 µl pre-aliquoted 1.5 ml screwcap tube.
3. Place the tubes into a pre-heated boiling water bath for 20 minutes to generate 'lysates'.

Note: Samples are always inactivated by boiling before transferring lysates out of the laboratory as a safety precaution.

4. Lysates are stored in the fridge until the PCR is undertaken.
5. After the PCR has been set-up, lysates are transferred to the fridge for up to a week.
6. If you want to keep the lysate for repeat testing, lysate are transferred to the freezer for up to a month.

A7.5.6.2 Preparing working stock solution for primers and probes

See **Annex A7.6 'Probe and primer sequences'** list for information on genes and sequences.

TABLE
A7.6

Working concentrations for the primers and probes which should be stored in the freezer at -20°C.

PRIMER/PROBE	WORKING CONC.	PRIMER/PROBE	WORKING CONC.
ttr_F	20µM	sseJ2_F	20 µM
ttr_R	20 µM	sseJ2_R	20 µM
ttr_P	5 µM	sseJ2_P	5 µM
SPA2308_F	20 µM	srfJ_F	20 µM
SPA2308_R	20 µM	srfJ_R	20 µM
SPA2308_P	5 µM	srfJ_P	5 µM
staG_F	20 µM	tviB_F	20 µM
staG_R	20 µM	tviB_R	20 µM
staG_P	5 µM	tviB_P	5 µM
SseJ2_F	20 µM	SPC0869_F	20 µM
sseJ2_R	20 µM	SPC0869_R	20 µM
sseJ2_P	5 µM	SPC0869_P	5 µM

1. For 20 μM working concentration: add 20 μl of stock solution (100 μM) to 80 μl distilled nuclease free water and mix well.
2. For 10 μM working concentration: add 10 μl of stock solution to 90 μl Sterile distilled water and mix well.
3. For 2.5 μM working concentration: prepare a 10 μM solution as above and then add 25 μl of this solution to 75 μl SDW and mix well.

A7.5.6.3 Preparing Mastermix for PCR

1. In a clean pre-PCR room, follow all the appropriate Guidance on the Code of Practice for Molecular Suites in **Annex A7.1** to avoid contamination and to follow safety and risk protocols.
2. A master mix of nuclease free water, primers, probes and Takyon Low Rox reagent (or enzyme/dNTP mix as appropriate) is made up depending on the number of samples and controls. Please refer to **Annex A7.3.4** for guidance on preparation of the PCR Mastermix.
3. Defrost the required PCR reagents.
4. Calculate the Mastermix per reaction as in **Annex A7.3.4**.
5. Always include appropriate positive controls (**Annex A7.4**) with each target of interest and negative controls. In each batch of triplexes/duplexes, prepare enough Mastermix for the number of samples to test, the 2 positive controls, the extraction NTC, a negative control for the PCR reaction (water instead of DNA) and prepare enough reagent for at least 2 more samples to make up for loss in pipetting.
6. Dispense the Mastermix into a 2.0 ml microtube.
7. Ensure that the Mastermix is thoroughly mixed by gently pipetting.
8. Place a qPCR 96-well reaction plate in a suitable holder, ensuring that the plate is oriented the correct way, i.e., numbers across the top of the plate, letters down the left-hand side of the plate. If your real-time machine is not plate based, then follow local manufacturer's instructions.
9. Add 22.5 μl of the Mastermix into each required well of the PCR plate.
10. Seal the plate with an optical seal to prevent contamination when moving to the next room.
11. Wipe down the PCR hood with surface decontaminant that eliminates unwanted DNA and DNase and switch on the UV light when finished.

A7.5.6.4 Adding the DNA template

In DNA template loading room – Gloves should be worn during the PCR set-up to prevent contamination with DNA.

1. Add 2.5 μl (or 1 ng/reaction) of DNA template to the reaction mix in the qPCR plate.
2. Add 2.5 μl (or 1 ng/reaction) of DNA template of the positive controls (vortex before adding) to the reaction mix in the qPCR plate and the same of Severn Biotech Ltd Nuclease Free water and cover the plate with an optical seal.
3. Wipe down the PCR hood with surface decontaminant and switch on the UV light when finished

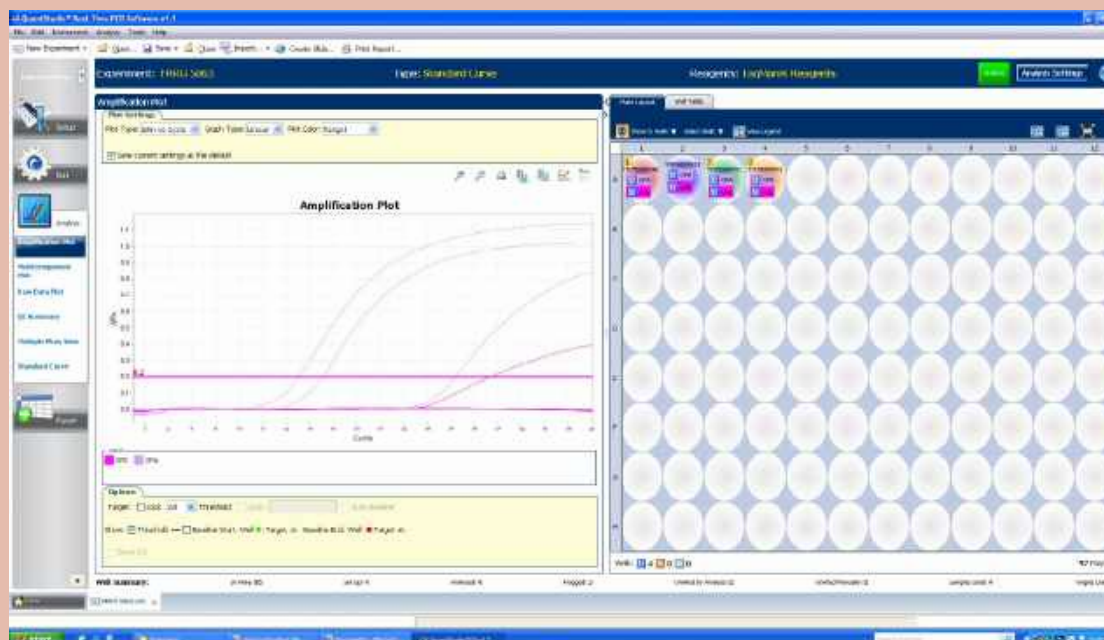
A7.5.6.5 Setting up the PCR

In the post-PCR room:

1. Put the sealed reaction plate in a suitable holder and place in the centrifuge. Put a balance plate (also in a holder) in the opposite bucket and pulse spin for 10 seconds.
2. Turn on the Real-Time PCR machine and its associated laptop/computer. Open the reaction compartment of the machine and place the reaction plate inside. On the computer, click OK at the password prompt and launch the software. Log in and follow local manufacturer's instructions for your machine.
3. The conditions for the PCR are an initial activation of 95 °C for 3 minutes, followed by 40 cycles of denaturation at 95 °C for 30 seconds, annealing at 60 °C for 30 seconds, and extension at 72 °C for 10 seconds. Do check the manufacturer instructions for recommended conditions according to the reagents used and do optimize your conditions.
4. You will need to 'Define Targets' for the experiments as per your local set-up, for example, this could be *ttr* FAM NFQ-MGB, *sseJ2* FAM NFQ-MGB, *srffJ* NFQ-MGB, *tvxB* JOE NFQ-MGB, SPC0869 CY5 NFQ-MGB, SPA2308 CY5 NFQ-MGB and *staG* FAM NFQ-MGB.
5. Start the run as per the manufacturer's instructions, the run time is approximately 45 minutes.
6. Once the run is complete, analyse the data as per the manufacturer's instructions, you will need to set threshold values as per your local validation experiments. The following are a suggested starting point: *ttr* -0.25, *sseJ2* -0.25, *srffJ* -0.25, *tvxB* -0.25, SPC0869 -0.25, SPA2308 and *staG* -0.25.
7. It is important to check the amplification curves and thresholds for each sample and not just rely on CT values before your interpretation. For example, you may have a poor amplification curve or a strong one under the threshold and will get a false negative if they are not checked (**Figure A7.1**).

FIGURE
A7.1

Screen capture example of gene positive amplification curves.



The first two curves are typical strong early reactions (and would show a low CT value) and ideally how you would like curves to be. The third curve shows a strong late reaction and the fourth curve shows a late and weak reaction but still crosses the threshold and is positive (this would need to be repeated).

8. You will need to validate positive CT values with your machine and procedures but in general a CT value of between 11 and 30 indicates a strong positive for the different genes (*though you want to adjust your ranges depending on your validation data as appropriate*).
9. Internal QC data of positive controls should be recorded with each PCR assay to ensure they are within range before results are released. Ideally, trend analysis of QC performance of the control should be undertaken monthly to ensure the PCR is performing optimally.
10. Internal QC data can be recorded in MedLabQC – A complete QC programme that is freely available (<https://www.smartlabtools.com/medlabqc.html>). The internal QC data is taken from the result output of the Real-Time PCR machine and inputted into the MedLabQC application against each target gene.
11. It is important to use aseptic techniques and decontamination procedures to avoid contamination (**Annex 7.1**).

A7.5.6.6 Reportable interval of examination results

Results are interpreted as follows:

TABLE
A7.7

Gene profiles for the identification of *S. Typhi* and *S. Paratyphi* from other serovars.

PROFILE	SALMONELLA IDENTIFICATION	TTR	SSEJ	TVIB#	SRFJ	SPC0869	SPA2308	STAG
1	<i>S. Typhi</i>	+	-	+/-	-	-	-	+
2	<i>S. Paratyphi</i> A	+	-	-	-	-	+	-
3	<i>S. Paratyphi</i> B	+	-	-	+	-	-	-
4	<i>S. Paratyphi</i> C	+	+	+	-	+	-	-
5	<i>Salmonella</i> spp. (NTS serovar*)	+	+/-	+/-	+/-	+/-	+/-	+/-
6	<i>Salmonella</i> not isolated (Non- <i>Salmonella</i> spp.)	-	-	-	-	-	-	-

#*tviB* + means the strain is genotypically *Vi* positive, this is rare (<1%).

*Footnote: A proportion of NTS serovars will be positive for the *ttr* gene and a combination of targets that do not match any of the typhoidal *Salmonella* profile

A7.6

Probe and primer sequences

TABLE
A7.8Probe and primer sequences for *Salmonella*.

NAME	SEQUENCE 5'- 3'	NUMBER OF BASES	5' MOD	3' MOD	GENE TARGETED
ttr_F	CTCACCAGGAGATTACAACATGG	23			ttr
ttr_R	AGCTCAGACCAAAAGTGACCATC	23			
ttr_P	CACCGACGGCGAGACCGACTTT	22	FAM	BHQ1	
*sseJ_2_F	CCCTCCTATGGTCAATACT	19			sseJ_2
sseJ_2_R	CCTAAGAAGTGTGGAGATGAT	21			
sseJ_2_P	CGGAAGGTTTACTAATGGATTACC	25	FAM	BHQ1	
srfJ_F	CTGTCTGTATAGCGTGAAGAG	22			srfJ
srfJ_R	GTCCACCAGGCCATCTTTAT	20			
srfJ_P	CGGCAGGGTATGGATGAGATGGAG	25	JOE	BHQ1	
tviB_F	TGTGGTAAAGGAACTCGGTAAA	23			tviB
tviB_R	GACTCCGATACCGGATAATG	23			
tviB_P	TGGATGCCGAAGAGGTAAGACGAGA	26	JOE	BHQ1	
SPC0869_F	CTGGCTGACACATGAACAAATC	23			SPC0869
SPC0869_R	CCTGAGAACGAGTCAGGTTTAC	23			
SPC0869_P	TGTACGACTGCAAACGCCAAAGTC	25	CY5	BHQ1	
SPA2308_F	ACGATGATGACTGATTTATCGAAC	24			SPA2308
SPA2308_R	TGAAAAGATATCTCTCAGAGCTGG	24			
SPA2308_P	CCCATACAATTTCACTTATTGAGAATGCGC	32	CY5	BHQ2	
staG_F	CGCGAAGTCAGAGTCGACATAG	22			staG
staG_R	AAGACCTCAACGCCGATCAC	20			
staG_P	CATTGTTCTGGAGCAGGCTGACGG	25	FAM	BHQ1	

**Note: The sseJ gene in Salmonella is more variable than the other gene targets, so the binding strength varies between strains (1). It has also been shown that some rare strains of Salmonella (e.g., S. Mishmarhaemek) do not bind to the sseJ primer set despite sseJ being present due to a mutation in the binding region, this can result in a false S. Paratyphi B PCR profile. The originally validated SSEJ primer set 1 has been modified resulting in the SSEJ2 primer set and the CY5 probe replaced with FAM probe resulting in a higher efficacy of the test. It is currently recommended that the SSEJ2 primer/probe set is used. Regular re-validation of any PCR/probes sets and comparing them against an additional gold standard test are always recommended to ensure they still detect or account for any emerging variants.*

References for Annex 7

1. Nair S, Patel V, Hickey T, Maguire C, Greig DR, Lee W, Godbole G, Grant K, Chattaway MA. Real-Time PCR Assay for Differentiation of Typhoidal and Non-typhoidal *Salmonella*. J Clin Microbiol. 2019 Jul 26;57(8):e00167-19.

Annex 8

Advanced molecular testing – Analysis of whole-genome sequencing data

A8.1

Enterobase

Enterobase is a user-friendly platform (<http://enterobase.warwick.ac.uk/>) where you have a presumptive identification of a *Salmonella* strain. This platform can be used for the identification, typing, searching publicly available data, setting up your own 'Workspace' and comparing your strain to publicly available sequence data. It assembles and quality checks your reads for further analysis. If your sequence data is of poor quality or the data is not a *Salmonella* strain, it will not be available for further analysis (details can be checked on your 'jobs' page).

There are multiple services available in Enterobase, this chapter will focus on:

1. Identification of the *Salmonella* serotype by multi-locus sequence typing (MLST) and SeqSero
2. Genomic analysis to look at strain relatedness for outbreak detection via core genome multi-locus sequence typing (cgMLST)
3. Genomic analysis to look at strain relatedness for outbreak detection via single nucleotide polymorphism (SNP) typing.

A8.1.1 Getting started

Enterobase is freely available to the public and runs entirely online, all you require is an updated web browser, however, it is not compatible with all browsers (such as Internet Explorer), an example of a compatible browser is Chrome or Microsoft Edge. Enterobase is available at <http://enterobase.warwick.ac.uk/>. Below is a summary of the basic requirements taken from the website, but for up to date information, it is worth going directly to the website <https://enterobase.readthedocs.io/en/latest/>

The main page presents available databases (*Salmonella*, *Escherichia*, *Moraxella* and *Yersinia*) with an overview of the number of records. From here you can:

1. Access a database by clicking the Database Home link (green box).
2. Access User account options, such as logging in (blue box), registering a new account (red box) and changing email/passwords are found in the top right.

Available Databases



Anonymous users do not have access to some important features, such as downloading files (assembly and annotations files and save metadata to local files), creating GrapeTree and SNP projects and saving and sharing workspaces. It is thus recommended to register and log in before starting to use Enterobase.

A8.1.2 Registering and logging in

To Register on Enterobase, visit the main webpage (<http://enterobase.warwick.ac.uk>) and click Register in the top right, where it will direct you to the registration form where you should fill in a username, password, email and details about yourself. Once the form is filled, click **Register**.

This will send a verification email to your specified email address. Click the link in this email to confirm your registration and you can then log into the Enterobase website.

Having created a login, the best way to get familiar with using the Enterobase website is to look at some of the tutorials

(<https://enterobase.readthedocs.io/en/latest/enterobase-tutorials/tutorials.html>).

These tutorials go into further details to the guidance presented in this chapter. Or if you prefer, you can just start by trying to use the website, particularly if you have used similar websites in the past.

A8.2 Uploading data

In order to identify a *Salmonella* strain, you will need to upload firstly, your raw reads with metadata for analysis, once uploaded, Enterobase will undertake quality checks of the data, assemble the raw reads which will be processed via multiple pipelines using a variety of bioinformatic tools. The output will enable you to identify and type your dataset1. Secondly, identification can then be undertaken by accessing the processed data and the predicted serovar via MLST

A8.2.1 Adding reads

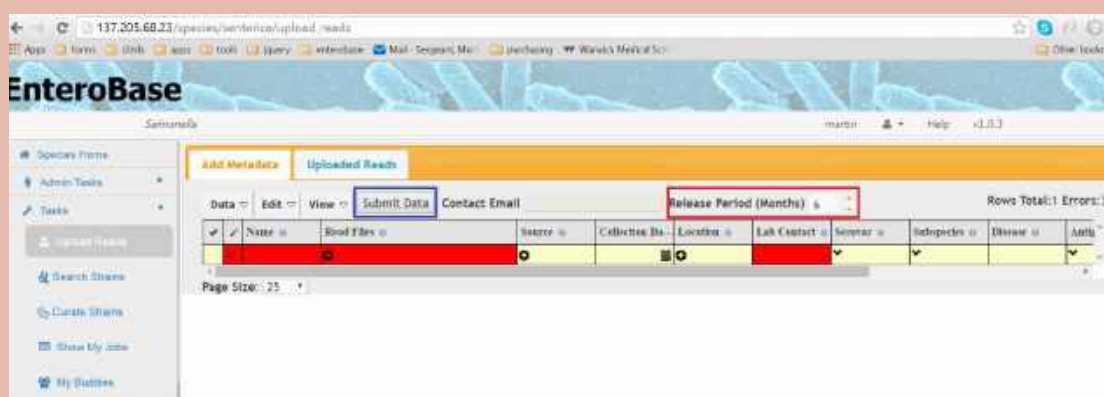
Before any reads can be uploaded, metadata concerning the reads need to be entered into Enterobase. This is done at the Upload Reads page (**Tasks -> Upload Reads** in the left-hand menu).

A8.2.2 Adding metadata manually

When the page first loads, a single blank row is displayed (**Figure A8.1**) where you can enter metadata concerning the reads you are going to upload. Any compulsory fields are shown in red. Clicking on individual cells allows you to enter data either by typing directly or selecting from a drop-down box, depending on the field. Extra rows can be added by **Edit -> Add Blank Row** or by right clicking on the table and selecting Insert Row.

FIGURE
A8.1

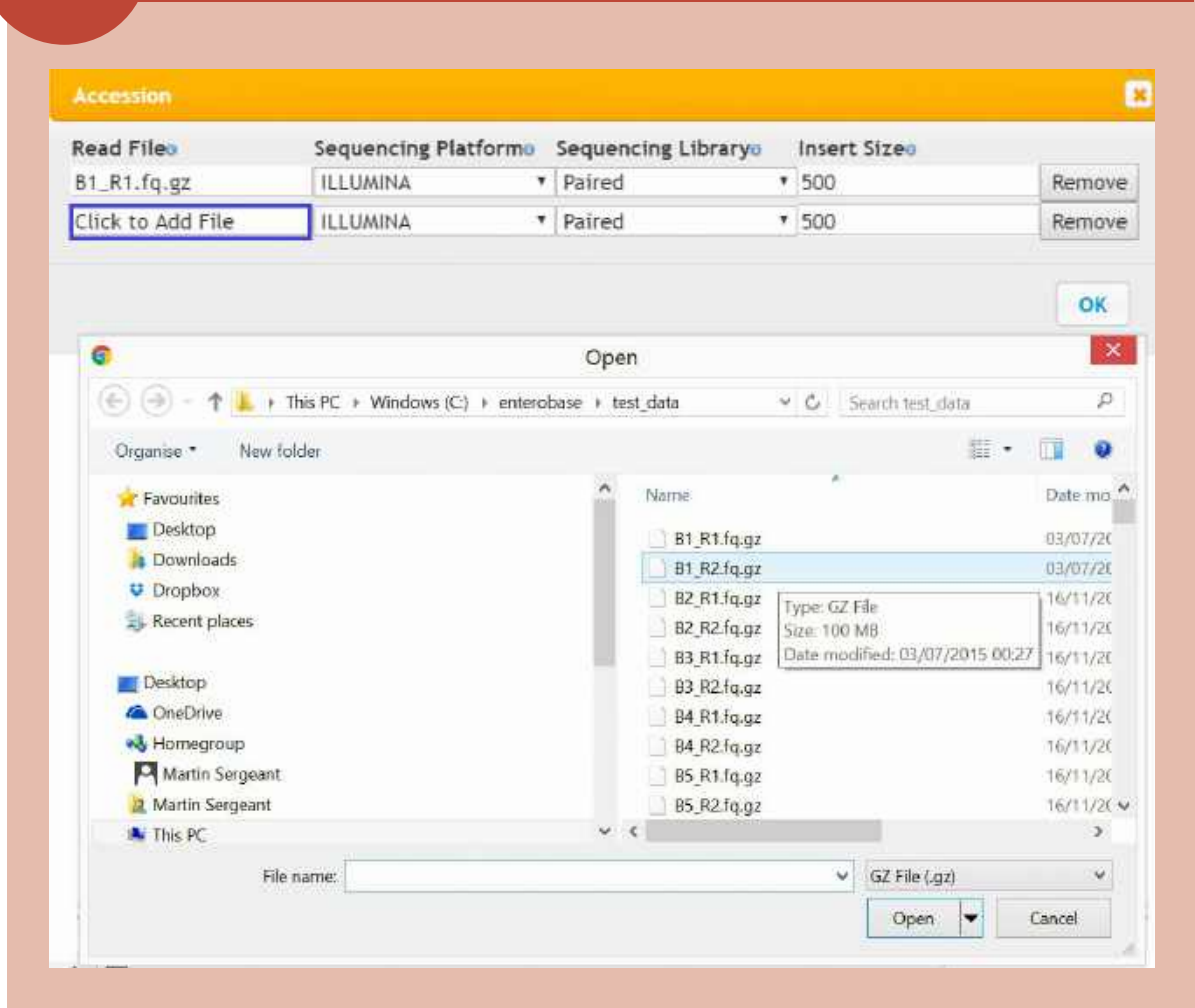
Screen capture of the EnteroBase platform, illustrating how to upload reads.



To enter information about the reads themselves, click on the 'Read Files' cell and a dialogue will appear (**Figure A8.2**).

FIGURE
A8.2

Screen capture of the read files.



By default, the reads are Illumina, paired with an insert size of 500. Click to Add File (blue box – **Figure A8.2**) and a file dialogue will appear, select a read file from a local directory and press OK. Preliminary checks on the read file, such as size and fastq format will be carried out and you will be informed if the read is not acceptable. You can also right click on the ‘Read File’ box and type the file name in directly, in this case the file will not be checked until it is uploaded.

In addition to Illumina reads, high quality complete genomes can be added by trusted users. To do this, in the reads dialogue remove one of the read entries and change Sequencing Platform to ‘Complete Genome’ and Sequencing Library to ‘Single’ and then add the appropriate file (.fasta) as above.

When all the data has been filled in correctly, the Submit Data button will become active and can be pressed (blue box – **Figure A8.1**). You will then be taken to the Uploaded Reads tab.

A8.2.3 Adding metadata from a file

If you have many reads to upload, it may be easier to add the metadata from a file generated in excel or using a script. The file needs to be text delimited and a template is available at https://enterobase.warwick.ac.uk/static/example/upload_template.txt. Both read files, separated by a comma need to go in the 'Read Files' column. The file format is flexible, not all columns need to be present, the column order does not matter, and any columns not present in the database will be ignored. For example, for the bare minimum (which depends on the criteria of the individual database), the following file would be acceptable:

Name	Read Files	Lab Contact
bug1	bug1 S7 L001 R1 001.fastq.gz,bug1 S7 L001 R2 001.fastq.gz	Sanger Centre

By default, reads are Illumina and paired. To specify a different read type, the Sequencing Platform and Library type must be specified in the 'Read Files' column, separated by a semi-colon. For example, to specify a complete genome, which is the only other type available to date (please ask for permission), the following file would be used:

Name	Read Files	Lab Contact
bug1	bug1.fasta;Complete Genome;Single	Sanger Centre

The date should be in three columns, Collection Year (4 digits), Collection Month (1-12) and Collection Day (1-31). The reason for this is that excel will format date columns in unpredictable ways. Also, if only one column represents the date, and only the year is entered, default values for month and day are added. If they are separate columns, it is much easier to add partial dates.

To load the file, go to data Load Local File and open the file. The data should appear in the table. If there are errors, put the mouse over the red cells to see what the problem is. When all errors have been corrected, the Submit Data (blue box – **Figure A8.1**) should become active.

A8.2.4 Release period

The Release Period input (red box – **Figure A8.1**) specifies the amount of time in months that your reads will remain private. This means that assemblies and annotation files cannot be downloaded by others unless you give them permission.

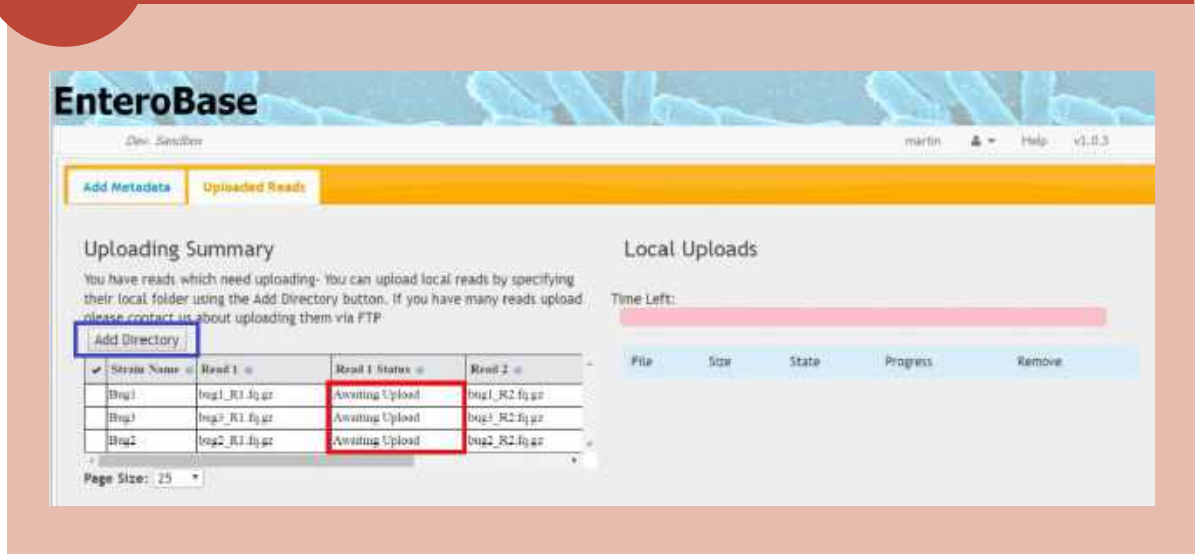
If you want to make your genome/s private, you need to alert the "Release Period" input and set the number of months which you want to keep the genome/s private (up to 12 months).

A8.2.5 Uploading reads

Once the metadata has been submitted, you will be taken to the Uploaded Reads tab and information about read status will be shown the left-hand table (red box – **Figure A8.3**).

FIGURE
A8.3

Screen capture of the EnteroBase when uploading reads.



Read status can be one of four values:

1. Awaiting Upload – metadata has been added but the read has not been uploaded.
2. Local Upload – The read has been found on the local file system and is ready to be uploaded.
3. Uploaded – The read has been uploaded but has not been processed, usually this is because the second read in the pair is absent.
4. Processing – The read has been uploaded and is being assembled.

If you clicked on a file in a file dialogue, the status will be Local Upload, however, if you loaded the name in a file or typed the name in the file dialogue, the status will be Awaiting Upload. There are two ways to find the files on your local system:

1. Click the Add Directory button (blue box – Figure A8.3). A file chooser should then appear. In some browsers, you can just select the folder where the files reside, in others you have to select all the individual files that you want.
2. Drag and Drop Files onto the right-hand area of the page.

Once the files have been located, they should appear on the right-hand side of the screen with the status 'pending' and the Start Upload button will initiate the transfer.

Bars which show the progress of each upload are displayed, as well as a bar which displays overall progress. You need to remain on the page until all the reads are uploaded. However, if there is a loss in connectivity or another interruption, you can simply refresh the page and upload any remaining reads.

Note: Due to security restrictions, if you refresh or return the page you will have to re-associate your reads by either the Add Directory button or by drag and drop.

To delete metadata that you have uploaded, select the rows you want to delete (shift + click to select multiple reads) and right click Delete Selected Row. You cannot delete rows associated with reads that are already being processed.

A8.2.6 Uploading reads via file transfer protocol (FTP)

If you have many reads to upload, please contact enterobase@warwick.ac.uk after you have added the metadata to arrange an upload to an FTP site.

A8.2.7 Associate new reads with an existing strain

If your assembly failed QC, or you have better quality reads for a strain that you previously uploaded into EnteroBase, then you have two options:

1. You can enter the metadata again, specifying the same strain name but different read files. The strain record will also be updated by any new/changed metadata that you add.
2. On the 'Main Search' page, in the right-hand grid, right click on the assembly that you own and select Associate New Reads. Enter the names of the new read files in the dialogue and press submit. These reads can then be uploaded by going to the Upload Page (see above) in the normal way.

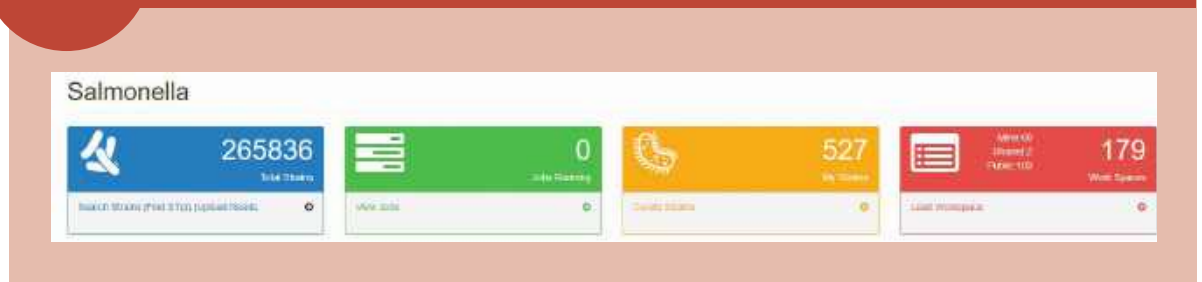
A8.2.8 Trouble shooting

If your reads have successfully uploaded, you can check they are being assembled by going to the 'View Jobs' page. If there are no jobs displayed here, go back to the Upload Reads tab on the Upload Page and check whether the status is Awaiting Upload. If so, the reads have not been uploaded and you need to retry uploading them. If after you have associated local files (by clicking Add Directory or drag and drop), and they do not appear on the right-hand side of the page, then check that the names of the files in the left-hand table exactly match the ones you are trying to associate

A8.2.9 Analysis of data

Once the reads and metadata has been uploaded you can check the status of the assembly and analysis by clicking on the 'View Jobs' section (in green) on the front page of the database (**Figure A8.4**).

FIGURE A8.4 Screen capture of the status of the assembly and analysis.



The jobs page will give you details of the programmes being run as well as whether the job is complete and if it passed or failed. Once complete, you can identify your isolates (**Figure A8.5**).

FIGURE
A8.5

Screen capture of analysis.

	ID	Status	Pipeline	Date	Input	Output
	0304049	COMPLETE	MSTree	2020-07-13 0...	Reads:5204	
	0304048	COMPLETE	MSTree	2020-07-13 0...	Reads:5202	
	0304047	COMPLETE	MSTree	2020-07-13 0...	Reads:5200	
	0304046	COMPLETE	MSTree	2020-07-13 0...	Reads:5199	
	0304045	COMPLETE	MSTree	2020-07-13 0...	Reads:5196	
	0401110	COMPLETE	SeqRead	2020-07-01 0...	SRASAL_IC1848AA_AS	
	0401109	FAILED	assembly...	2020-07-01 0...	SRASAL_IC1848AA_AS	
	0401108	COMPLETE	assembly...	2020-07-01 0...	SRASAL_IC1848AA_AS	
	0401107	COMPLETE	SeqRead_2	2020-07-01 0...	SRASAL_IC1848AA_AS	
	0401106	COMPLETE	assembly...	2020-07-01 0...	SRASAL_IC1848AA_AS	
	0401105	COMPLETE	assembly...	2020-07-01 0...	SRASAL_IC1848AA_AS	
	0401104	COMPLETE	assembly...	2020-07-01 0...	SRASAL_IC1848AA_AS	
	0401103	COMPLETE	SeqRead_2	2020-07-01 0...	SRASAL_IC1848AA_AS	
	0401102	COMPLETE	prokka_m...	2020-07-01 0...	SRASAL_IC1848AA_AS	
	0401101	COMPLETE	assembly...	2020-07-01 0...	SRASAL_IC1848AA_AS	
	0401100	COMPLETE	assembly...	2020-07-01 0...	SRASAL_IC1848AA_AS	
	0401099	COMPLETE	assembly...	2020-07-01 0...	SRASAL_IC1848AA_AS	
	0401108	COMPLETE	prokka_m...	2020-07-01 0...	SRASAL_IC1848AA_AS	
	0401012	COMPLETE	QAssembly	2020-07-01 0...	Reads:937403_salmonella typing sequencing proc...	Assembly: 937403.Assembly
	0401009	COMPLETE	QAssembly	2020-07-01 0...	Reads:937403_salmonella typing sequencing proc...	
	0401012	COMPLETE	QAssembly	2020-07-01 0...	Reads:937403_salmonella typing sequencing proc...	
	0401005	COMPLETE	QAssembly	2020-06-30 1...	Reads:937403_salmonella typing sequencing proc...	

A

B

C

A- ID of your isolate
B – Status of the programme and whether it passed or failed
C – The name pipeline being run as multiple pipelines are used.

A8.3

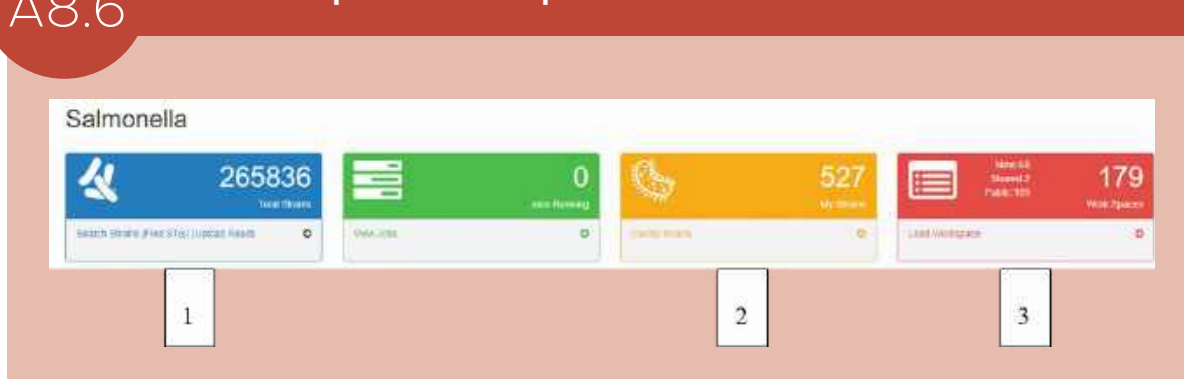
Identification and typing of *Salmonella*

There are a couple of ways you can access the data for analysis, go to the species homepage:

1. Search specific strains or isolates in public database (note your strains will not be there if you have not made them public and will only become public after an agreed amount of time – standard is six months).
2. You can go to the Curate Strains page (yellow in **Figure A8.6**. Note, this will only contain strains you have personally uploaded yourself).
3. You can go to the 'load Workspace' page (note, you will need to add strains of interest first to a Workspace – described in **Annex A8.3.6**).

FIGURE
A8.6

Screen capture of the options to assess data.



A8.3.1 Analysis by searching strains

The example here will be by specific isolates of interest, but you can search for strains in different ways, including serovar or sequence type (Figure A8.7). Go to the search strain page, under 'Field' select what you are searching for, in this case specific isolate names, under operator select how you want to search, in this case, it has been changed from 'contains' to 'in' which allows you to paste multiple entries at once, then submit.

FIGURE
A8.7

Screen capture of ways to search for strains.

Your searches will look like the example in Figure A8.8, note under the Experimental Data drop-down list on the top right it has defaulted to the assembly status data type. The metadata uploaded with the isolates is shown on the left-hand panel and will not change. The Experimental Data information will be shown on the right-hand panel and will change depending on the data you are assessing.

FIGURE
A8.8

Screen capture of the results from searching the strain data.

Strain ID	Name	Source	Location	Isolate	Subspecies	Antigen	Sub-Contact	Assembly	Coverage	NTS	Length	Quality
ISAL_KA071AA	174389	Human	United Kingdom	Adjuva	1	13.23+1.8	Public Health England (M. Chatterjee)	Assembly	75	14000	4018728	Submerged
ISAL_KA071AA	167130	Human	United Kingdom	Adjuva	1	13.23+1.8	Public Health England (M. Chatterjee)	Assembly	68	140115	4017709	Submerged
ISAL_KA071AA	164130	Human	United Kingdom	Adjuva	1	13.23+1.8	Public Health England (M. Chatterjee)	Assembly	74	141476	4019247	Submerged
ISAL_KA071AA	167774	Human	United Kingdom	Adjuva	1	13.23+1.8	Public Health England (M. Chatterjee)	Assembly	24	248117	4025883	Submerged
ISAL_KA071AA	167040	Human	United Kingdom	Adjuva	1	13.23+1.8	Public Health England (M. Chatterjee)	Assembly	58	143121	4026481	Submerged
ISAL_KA071AA	167117	Human	United Kingdom	Adjuva	1	13.23+1.8	Public Health England (M. Chatterjee)	Assembly	18	140044	4024981	Submerged
ISAL_KA071AA	167215	Human	United Kingdom	Adjuva	1	13.23+1.8	Public Health England (M. Chatterjee)	Assembly	13	138110	4012805	Submerged
ISAL_KA071AA	146601	Human	United Kingdom	Adjuva	1	13.23+1.8	Public Health England (M. Chatterjee)	Assembly	17	141186	4019558	Submerged
ISAL_C0057AA	174389	Human	Ireland	Adjuva	1	13.23+1.8	Canadian Food Inspection Agency	Assembly	1	4387111	4070111	Submerged
ISAL_C0054AA	167130	Human	United Kingdom	Adjuva	1	13.23+1.8	Canadian Food Inspection Agency	Assembly	1	4478112	4078112	Submerged
ISAL_C0010AA	167130	Human	Ireland	Adjuva	1	13.23+1.8	Canadian Food Inspection Agency	Assembly	1	4478108	4074108	Submerged

To identify the isolates via MLST, change the Experimental Data type to Achtman 7 Gene MLST and you will get the sequence type (ST), the eBURST group (eBG), the predicted ST serotype and the allelic profile of the ST as shown in **Figure A8.9**.

**FIGURE
A8.9**

Screen capture of identifying isolates via MLST.

ST	eBG	Serotype	Lineage	Subspecies	aroC	dnaN	hemK	hisD	purF	sucA	thrA
3929	421	Adjame			127	287	289	94	731	199	126
3929	421	Adjame			127	287	289	94	731	199	126
3929	421	Adjame			127	287	289	94	731	199	126
4023	421	Adjame			127	651	289	94	731	199	126
4023	421	Adjame			127	651	289	94	731	199	126

To identify an isolate via the antigen genes, change the Experimental Data to Serotype (**Figure A8.10**).

**FIGURE
A8.10**

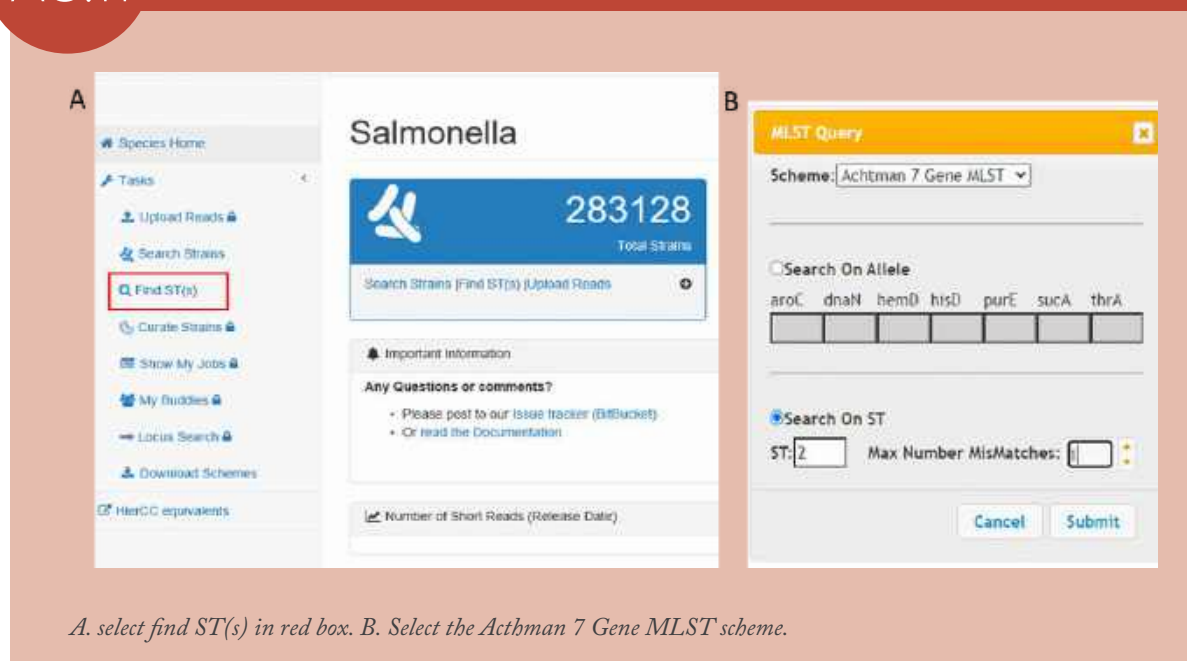
Screen capture of identifying isolates via serotype.

SISTR1 Serovar	SISTR1 Ser...	SISTR1 H1	SISTR1 H2	SeqSero2 Serovar	SeqSero2 S...	SeqSero2 ...	SeqSero2 ...
Adjame	G	r	1,6	Adjame	13	r	1,6
Adjame	G	r	1,6	Adjame	13	r	1,6
Adjame	G	r	1,6	Adjame	13	r	1,6
Adjame	G	r	1,6	Adjame	13	r	1,6
Adjame	G	r	1,6	Adjame	13	r	1,6
Adjame	G	r	1,6	Adjame	13	r	1,6
Adjame	G	r	1,6	Adjame	13	r	1,6
Adjame	G	r	1,6	Adjame	13	r	1,6
Adjame	G	r	1,6	Adjame	13	r	1,6
Adjame	G	r	1,6	Adjame	13	r	1,6

A8.3.2 Analysis by searching sequence types

To identify an isolate by ST or look up a group of ST to put your data into the global context, select find STs on your homepage (Figure A8.11).

FIGURE A8.11 Searching sequencing types.



A. select find ST(s) in red box. B. Select the Achtman 7 Gene MLST scheme.

Type in the ST of interest and ensure that the Achtman 7 Gene MLST scheme is selected. If you are interested in looking at the eBURST group (eBG) then allow for 1 allele mismatch.

In this example (Figure A8.12), you can see that ST2 with 1 allele mismatch falls into eBG13 and that ST1 is also part of this group. In the database, you can see what the institute originally identified the organism as based on their methods (under serovar), at the top you have the rows total, which is the number of strains. In this example, ST2 is *Salmonella* Typhi.

FIGURE A8.12 Screen capture of a database search based on sequence type.

Name	Location	Serovar	Subspecies	Antigenic Form	Lab Contact	ST	eBG	Serotype	Isolate	Indicates
Sal_ZAR	11085	Salmonella	Salmonella	Salmonella	Salmonella	2	13	Typhi		1
Sal_ZAR	11090	Salmonella	Salmonella	Salmonella	Salmonella	2	13	Typhi		1
Sal_ZAR	11091	Salmonella	Salmonella	Salmonella	Salmonella	2	13	Typhi		1
Sal_ZAR	812189	Salmonella	Salmonella	Salmonella	Salmonella	2	13	Typhi		1
Sal_ABO	814866	Salmonella	Salmonella	Salmonella	Salmonella	2	13	Typhi		1
Sal_ABO	811182	Salmonella	Salmonella	Salmonella	Salmonella	2	13	Typhi		1
Sal_ABO	811180	Salmonella	Salmonella	Salmonella	Salmonella	2	13	Typhi		1
Sal_ABO	811153	Salmonella	Salmonella	Salmonella	Salmonella	2	13	Typhi		1
Sal_ABO	822475	Salmonella	Salmonella	Salmonella	Salmonella	2	13	Typhi		1
Sal_ABO	822928	Salmonella	Salmonella	Salmonella	Salmonella	2	13	Typhi		1
Sal_ABO	823122	Salmonella	Salmonella	Salmonella	Salmonella	2	13	Typhi		1

A8.3.3 Genetic analysis of isolates

To identify an isolate via the core genome to see if strains are related, change the Experimental Data to cgMLST.

FIGURE
A8.13

Screen capture of results to identify an isolate via cgMLST.

Rows Total: 11 Filtered: 11

Edit Mode: ☐ Experimental Data: cgMLST V2 + HierCC V1

ST	HC0 (Indist...	HC2	HC5	HC10	HC20	HC50	HC100	HC200	HC400
81284	81284	81284	81284	81284	81284	81284	81284	71258	71258
71258	71258	71258	71258	71258	71258	71258	71258	71258	71258
81264	81264	81264	81264	81264	81264	72103	71258	71258	71258
81270	81270	81269	81269	81269	75355	72103	71258	71258	71258
81270	81270	81269	81269	81269	75355	72103	71258	71258	71258
81270	81270	81269	81269	81269	75355	72103	71258	71258	71258
81264	81264	81264	81264	81264	81264	72103	71258	71258	71258

The cgST can be used to see if strains are genetically related. If they are the same cgST number then they are identical and will also have the same Hierarchical Clustering (HC) 0 number. To see strains that match within two alleles (of the 3002 genes) look at the HC2 column, for 5 alleles, the HC10 column and so on. If you right click on the HC number of interest, you can search for other related strains at this level in the global database. For outbreaks, the allelic differences can vary but in general, 0-5 alleles are very closely related phylogenetically, up to 10 alleles can also be part of the same outbreak and more than 20 alleles is less likely to have a related source. However, validation needs to be undertaken in each laboratory to define allelic differences, and epidemiology must be undertaken in conjunction with outbreak investigations. cgMLST is a useful technique to look at genetic relatedness to inform epidemiological investigations. For a more in-depth example, go the EnteroBase document page:

<https://enterobase.readthedocs.io/en/latest/enterobase-tutorials/search-agama.html?highlight=searching%20strains>

A8.3.4 Visualizing datasets

Once you have your data analysed by cgMLST + HierCC, you can visualize the data by clicking on the MLST GrapeTree icon or the SNP project icon.

For MLST GrapeTree, you will have several algorithms to choose from including Minimal Spanning Tree methods or neighbour joining methods. Select the method appropriate for your analysis and name the tree, this will be saved to your Workspace. Example of how these methods can be used are currently published by Zhou et al. (1,2).

FIGURE
A8.14

Screen capture to create a GrapeTree file to visualize datasets.



The screenshot shows a 'Create GrapeTree' dialog box with the following fields and values:

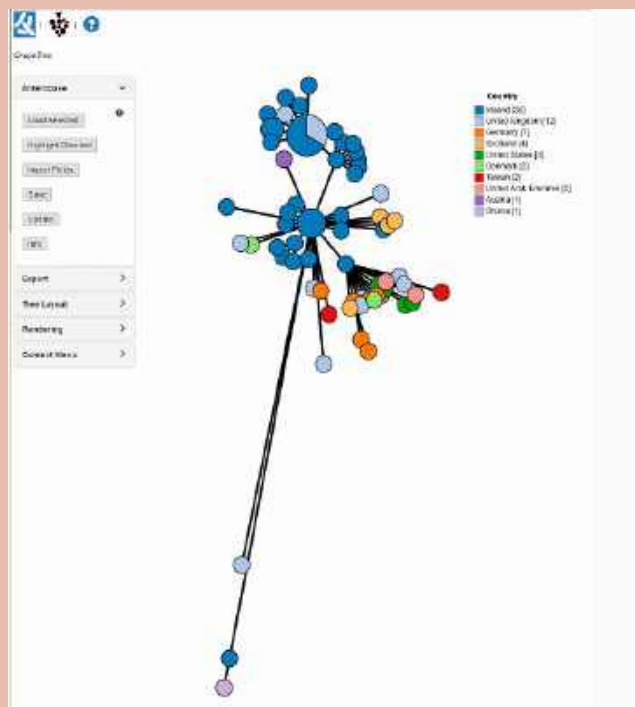
Field	Value
Name	cgMLST V2 + HierCC V1_Tree_2020-
Algorithm	MSTree V2
Selected Only	☐
Scheme	cgMLST V2 + HierCC V1
Workspace	None
No. Strains (With STs)	32
Number of Nodes (STs)	19

A 'Submit' button is located at the bottom right of the dialog box.

You can load the GrapeTree from your Workspace and colour code isolates according to your preference. For more details on formatting the GrapeTree, refer to <https://enterobase.readthedocs.io/en/latest/grapetree/grapetree-tutorial-1.html>

FIGURE
A8.15

Example of GrapeTree analysis.



A8.3.5 SNP typing

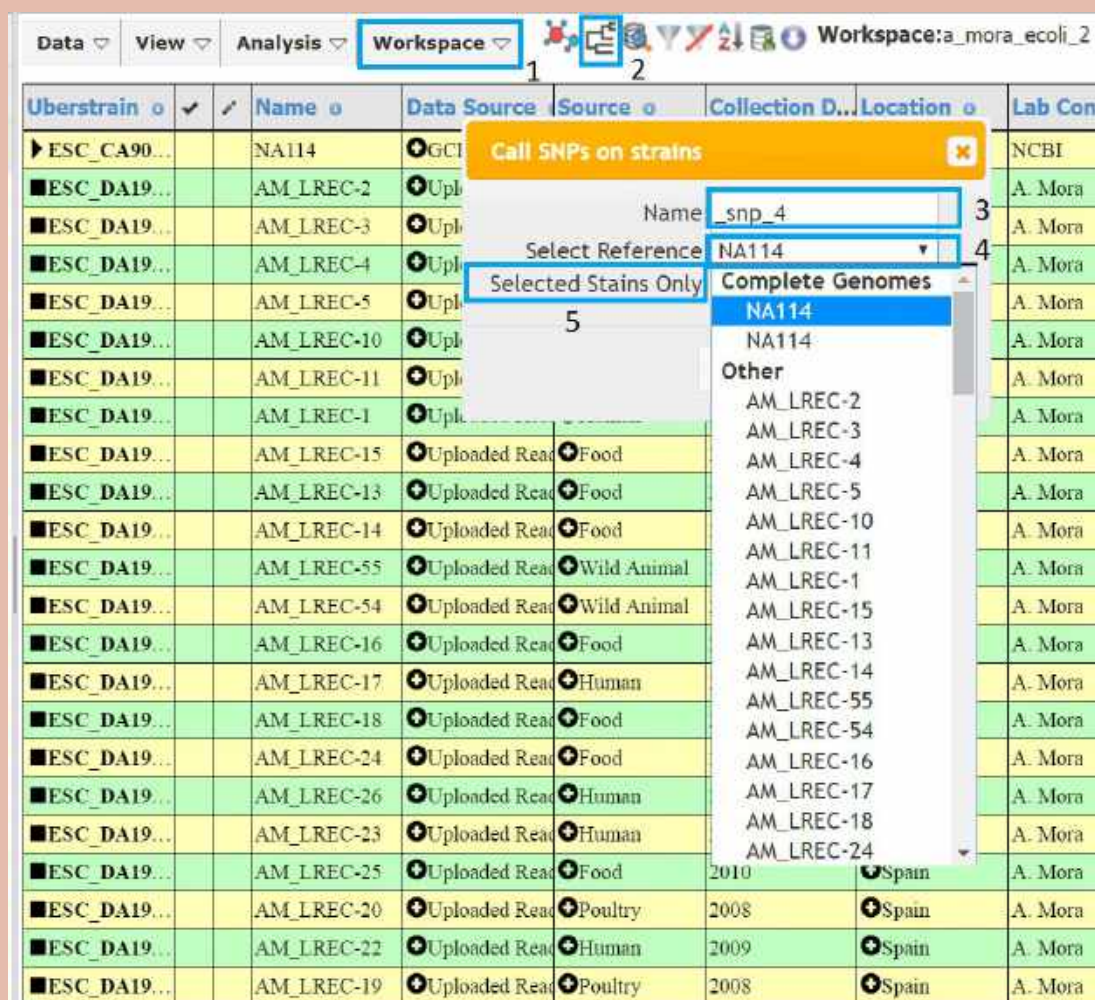
For SNP projects, SNPs are called using assembled genomes.

A8.3.5.1 Creating an SNP project

Like MS Trees, SNP Projects can either be created as stand-alone or part of a Workspace from the main search page, but you need to be logged in.

FIGURE
A8.16

Creating an SNP project. (1-5 indicate the steps to take).



A8.3.5.2 Setting up stand-alone projects

To create a stand-alone project ensure you have all the strains you want in the table (Figure A8.16). You can copy /paste strains between searches using the **Workspace -> Copy Selected** and **Workspace -> Paste Strains** (Figure

A8.16, 1). Click the SNP icon (Figure A8.16, 2) and a dialogue will appear with a drop-down allowing you to pick a reference genome (Figure A8.16, 4) and add a suitable name (Figure A8.16, 3). It is best to pick a complete genome as these are usually a single contig and well annotated. You can also just create an SNP project from a subset of strains in the table by selecting the strains you want and checking **Selected Strains Only** in the dialogue (Figure A8.16, 5). Once you are ready, you can press submit.

A8.3.5.3 Project linked to a Workspace

First, create a Workspace. Once created an 'Analysis' toolbar should appear in the main menu. Choose the 'Create SNP Project' under the sub-menu items and a dialogue box should appear as above. The process is now exactly the same as a stand-alone project (follow the instructions as above). The only difference is that the project will now be connected to the Workspace.

A8.3.5.4 Loading a project

SNP projects usually take a while to run, you can follow the progress of the SNP calling with Show My Jobs. The four jobs that need to be completed for an SNP project are:

1. **refMasker**: finds repeat regions in the reference which are ignored for SNP calls.
2. **refMapper**: maps contigs of all the query strains against the reference and work out SNPs.
3. **refMapperMatrix**: combines all the SNPs and filters out those that are in repeat regions or are in regions that are missing in 10% or more of the all the query strains.
4. **matrix_phylogeny**: The SNP profiles are then used to create a RAxML tree.

Once all four jobs are complete your project can be loaded. You will be informed if you try and load a project and it is not yet complete.

A8.3.5.5 Accessing stand-alone projects

1. Clicking the load Workspace panel on the top right of the database homepage.
2. On the main search page menu click **Workspace -> load**
3. Going directly to the tree link e.g., <https://enterobase.warwick.ac.uk/>. If the project is not public and you are not logged in, you will be asked to do so.

A8.3.5.6 Projects connected to a Workspace

1. First load the Workspace and then click **Analysis -> load SNP Project** or you can click on the Project in the summary dialogue.
2. Go directly to the link e.g., <https://enterobase.warwick.ac.uk/>

A8.3.5.7 Visualizing SNPs

When you load an SNP project, two windows should open (be sure that your browser allows popups from this site). One of the windows (Enterobase%20SNPs%20JBrowse) is an instance of [JBrowse], which allows you to visualize the SNPs with respect to the reference. The reference will be annotated by the [prokka_annotation] pipeline plus any MLST schemes that have been called and if it is a complete genome, the [GenBank annotation] as well. Each of the strains in the project will have a track (left-hand panel) showing the SNPs. It is hoped in the future to display where the contigs of each strain mapped to the reference and which regions were masked for repeats.

For further details on working with trees, go to:

<https://enterobase.readthedocs.io/en/latest/features/snp-projects.html>

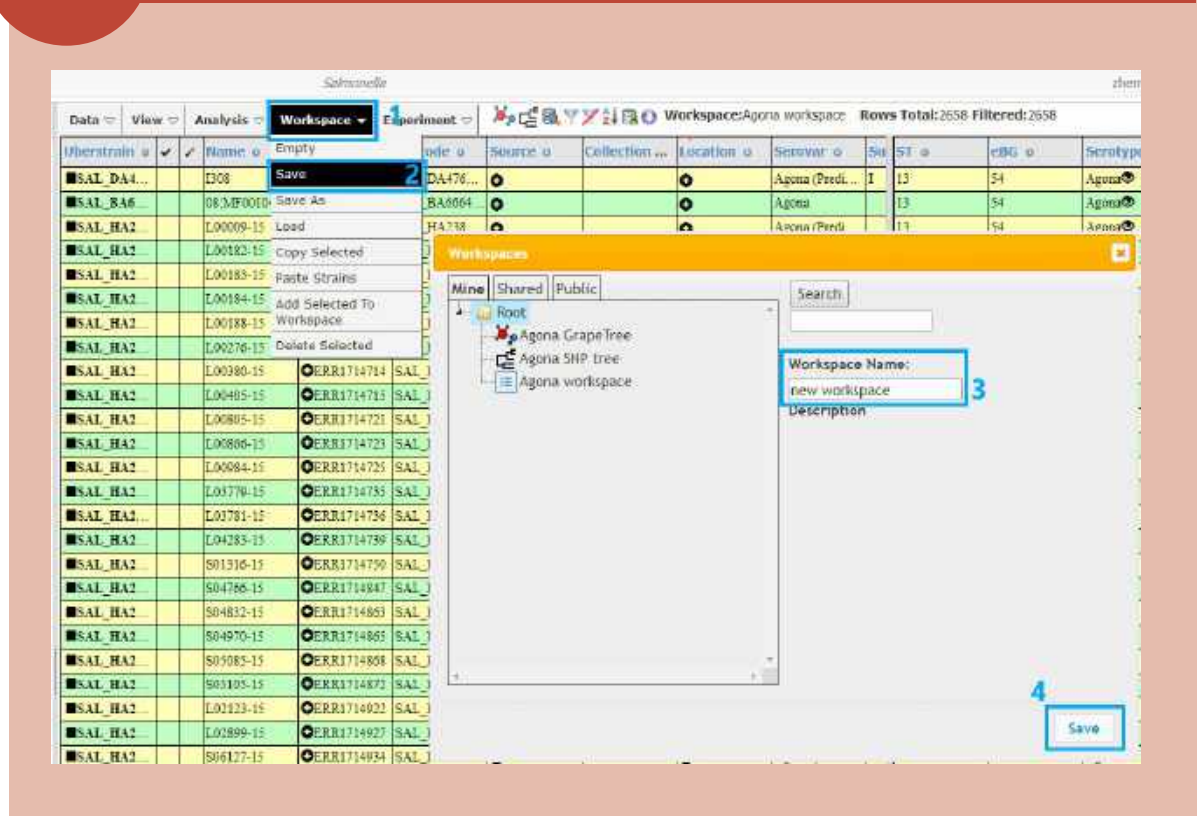
A8.3.6 Workspaces

Workspaces allow you to define a set of strains in a defined group, which you can revisit at any time. Such groups should be considered as a list of strains used in a given study and can be shared with other users (this functionality is covered in Managing your buddies). Workspaces are also the main method to define input data sets for certain analysis pipelines in EnteroBase.

Take an existing set of search results and save a Workspace via (Figure A8.17):

1. From the menu above the strain records, go to the Workspace menu.
2. Select **Workspace -> Save**.
3. Enter a memorable name for this Workspace.
4. Click save button.

FIGURE A8.17 Step by step instructions on how to create a Workspace.

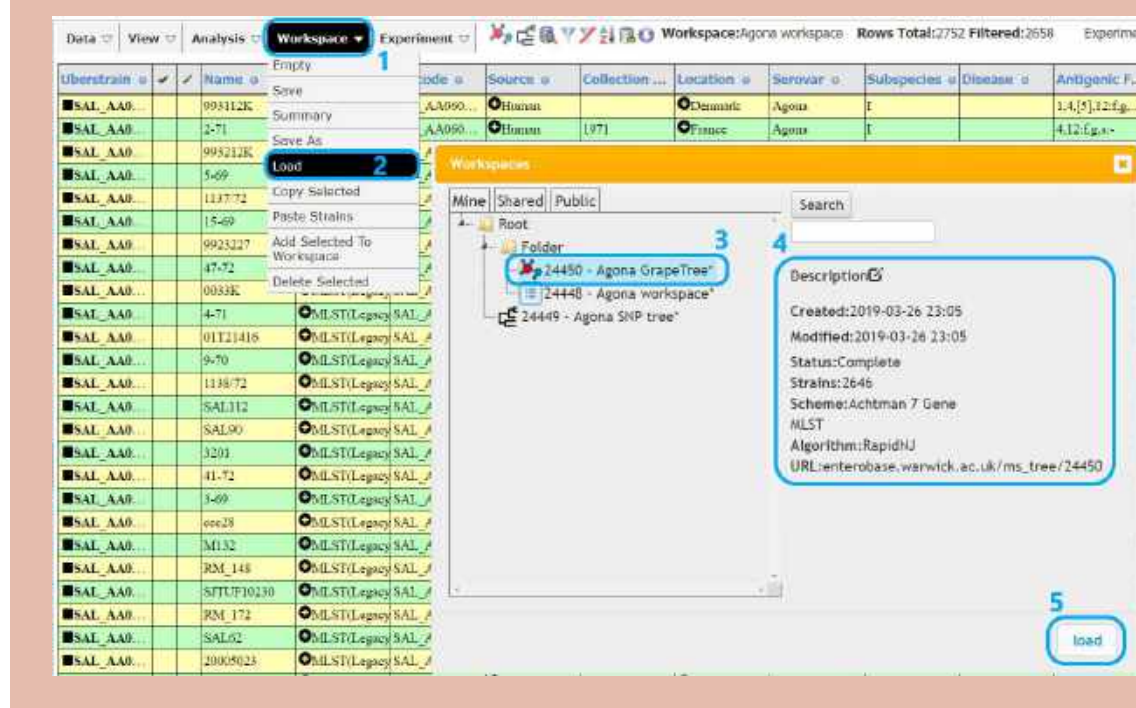


To retrieve a Workspace (Figure A8.18):

1. Go to the Workspace menu.
2. Select **Workspace -> Load**
3. Choose from the names of Workspaces.
4. Check the brief description of the Workspace.
5. Click load button.

FIGURE
A8.18

Step by step instructions on retrieving a Workspace.



Figures A8.17 and A8.18 are brief descriptions of tools in EnteroBase, for more in-depth guidance, please refer to <https://enterobase.readthedocs.io/en/latest/about.html>

A8.4 Center of Genomic Epidemiology

The aim of the Center for Genomic Epidemiology (CGE) is to provide access to bioinformatics resources, which is also available for those with limited experience, and thereby allowing all countries, institutions and individuals to take advantage of the novel sequencing technologies (<http://www.genomicepidemiology.org/>). By doing so, it aims to facilitate more open data sharing around the world and provide more equal opportunities for all. CGE is entirely non-commercial and operates a number of free online bioinformatics services.

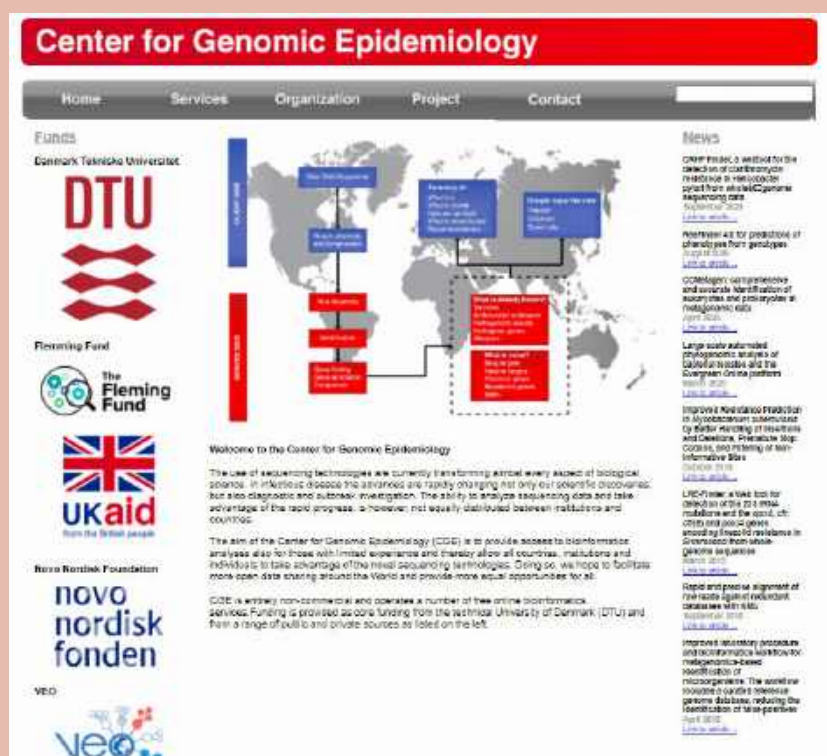
There are multiple services available, this section will focus on:

1. Identification of the strain to a species level using the KmerFinder Platform (can be any bacterial genus)
2. MLST of the strain (there are multiple schemes of different pathogens you can select from)
3. *Salmonella* pathogenicity island detection
4. Identification of acquired antimicrobial resistance genes using ResFinder

A8.4.1 Getting started

CGE runs entirely online and can be accessed via this link <https://www.genomicepidemiology.org/>. The main page contains an overview of the platform, if you select the service tab, it will bring you to the full list of services.

FIGURE A8.19 Screen capture of the Center for Genomic Epidemiology platform.



A8.4.2 Registering and logging in

In the services tab you can select 'New' and set-up a new user profile or log in if you are already registered. If you are only checking one dataset, you do not need to log in but if you want to work with batches of data then you should register an account and log in to access the data.

FIGURE
A8.20

Overview of services and log in to the Center for Genomic Epidemiology platform.

The screenshot displays the 'Center for Genomic Epidemiology' website. At the top, there is a red header with the site name and a navigation bar with 'Home', 'Services' (selected), and 'Datasets' tabs. Below the navigation bar, the page is titled 'Overview of Services'. The content is organized into four main categories: Phenotyping, Typing, Phylogeny, and Metagenomics, each with a list of tools and their functions.

Center for Genomic Epidemiology

Home Services Datasets

Overview of Services

Phenotyping

- RasFinder**
Identification of acquired antibiotic resistance genes.
- SesFinderEG**
Identification of functional metagenomic antibiotic resistance determinants.
- LRE-finder**
Identification of genes and mutations leading to tetracycline resistance.
- KmerResistance**
Identification of acquired antibiotic resistance genes using Kmers.
- PathogenFinder**
Prediction of a bacterium's pathogenicity towards human hosts.
- VirulenceFinder**
Identification of acquired virulence genes.
- RestrictionModificationFinder**
Determination of Restriction-Modification sites (based on [REBASE](#)).
- SPFinder**
SPFinder identifies Salmonella Pathogenicity Islands.
- ToxFinder**
ToxFinder identifies genes involved in mycotoxin synthesis.

Typing

- MLST**
Multi Locus Sequence Typing (MLST) from an assembled genome or from a set of reads.
- PlasmidFinder**
PlasmidFinder identifies plasmids in total or partial sequenced isolates of bacteria.

Phylogeny

- MINI_VIEST**
Identification of SNPs with automatic filtering, masking and site validation together with inferred phylogeny based on both long and short sequencing data.
- CSI-Phylogeny**
CSI-Phylogeny calls SNPs, filters the SNPs, does site validation and infers a phylogeny based on the concatenated alignment of the high quality SNPs.
- NCItree**
NCItree constructs phylogenetic trees from Single-End or Pair-End FASTQ files.
- Evergreen**
Evergreen generates a forest of constantly updated phylogenetic trees with publicly available whole-genome sequencing data from foodborne bacterial isolates that were deposited in the short sequencing read archive (NCBI SRA/ENA).
- TreeViewer**
Phylogeny Tree Viewer.

Metagenomics

- CCMetagen**
CCMetagen: Comprehensive and accurate identification of eukaryotes and prokaryotes in metagenomic data.

PCR-tools

- RUCS**
RUCS: Rapid identification of PCR Primers Pairs for Unique Core Sequences.

A8.5

Identification and typing of *Salmonella*

Unlike EnteroBase, you do not need to know the presumptive identification of your strain to use the Kmer ID, so a useful tool if you are not sure if the isolate you have is *Salmonella*. In order to identify your strain, you will need to 1) select the programme of interest, and then 2) upload your reads, they can be raw or assembled.

A8.6

Identification of species and subspecies

A8.6.1 **KmerFinder – Prediction of bacterial species using a fast Kmer algorithm.**

1. Select KmerFinder under the 'Typing' section on the services page.
2. Select the bacterial database.
3. Browse for your files by clicking on the Isolate File Icon.
4. Press upload once you have added your files.

FIGURE
A8.21

Screen capture of the processes to set-up KmerFinder.

KmerFinder 3.2
 Software version: 3.0.2 (2020-10-30)
 Database: [Available here](#)
 View the [version history](#) of this server.

Select database
 Bacteria organisms

Upload file(s)
 To input the sequences, upload a single FASTA file, or one/two FASTQ file(s), or one interleaved FASTQ file on your local disk by using the applet below. Both assembled genome (in FASTA format) and raw reads single end or paired end (in FASTQ format) are supported. Skipped FASTA/FASTQ files are also supported.
 If you get an "Access forbidden: Error 403", make sure the start of the web address is https and not just http. Fix it by clicking [here](#).

1 isolate file

Name	Size	Progress	Status
1051363_Sample1_salmonella-typing_ngsservice_processed.R1.fastq.gz	153.52 MB		
1051363_Sample1_salmonella-typing_ngsservice_processed.R2.fastq.gz	155.05 MB		

Upload Remove

5. Enter your email address, the link to your results will be sent once processed. Remember to check your junk mail.

FIGURE
A8.22

Screen capture of process to set-up KmerFinder 3.2.

Your job is being processed

Wait here to watch the progress of your job, or fill in the form below to get an email message upon completion.

To get notified by email:

This page will update itself automatically.

6. Once the results are ready, click on the link sent to your email.
7. View your data and look at the quality metrics to assign identification, the first hit will be the best match with the data submitted and will be your identification. In this example, the result is *Salmonella* Typhi.

FIGURE
A8.23

Screen capture of the results from data submitted in KmerFinder.

KmerFinder-3.2 Server - Results

KmerFinder 3.2 results:

Template	Num	Score	Expected	Template length	Query Coverage	Template Coverage	Depth	tot. query Coverage	tot. template Coverage
NZ_LR590081.1 Salmonella enterica subsp. enterica serovar Typhi str. Ty2 strain 4318STDY8599669 genome assembly, chromosome 1	11623	14324699	71	156483	96.03	96.51	91.54	96.03	96.51
NZ_CP015088.1 Escherichia coli O25b:H4 extrachromosomal sequence	14079	902	31	997	0.01	1.50	0.90	0.49	43.93

A8.6.2 Multi-Locus Sequence Typing from an assembled genome or from a set of reads.

1. Select MLST under the 'Typing' section on the services page.
2. Select the *Salmonella enterica* database.
3. Select the type of data input.
4. Browse for your files by clicking on the Isolate File Icon.
5. Press upload once you have added your files.
6. Enter your email address, the link to your results will be sent once processed. Remember to check your junk mail.

FIGURE
A8.24

Screen capture of the process to do Multi-Locus Sequence Typing.

MLST 2.0 (Multi-Locus Sequence Typing)

Software version: [2.0.1 \(2019-05-08\)](#)
Database version: [2.0.0 \(2020-12-28\)](#)

Select MLST configuration

MLST allele sequence and profile data are obtained from [PubMLST.org](#).

Please note that for four organisms, two or three different MLST schemes are available:

- *Acinetobacter baumannii* (*Acinetobacter baumannii* #1 [1], *Acinetobacter baumannii* #2 [link])
- *Escherichia coli* (*Escherichia coli* #1 [4], *Escherichia coli* #2 [5])
- *Pasteurella multocida* (*Pasteurella multocida* #1 (RIFDC), *Pasteurella multocida* #2 (multihost))
- *Leptospira* (*Leptospira* #1, *Leptospira* #2, *Leptospira* #3).

Note: *Campylobacter coli* and *Campylobacter jejuni* are considered together.

Select type of data input
Only data from one single isolate should be uploaded. If raw sequencing reads are uploaded KMA will be used for mapping. KMA supports the following sequencing platforms: Illumina, Ion Torrent, Roche 454, SOLID, Oxford Nanopore, and PacBio.

Please note that "Assembled Genomes/Contigs" should be selected, if you have already assembled your short sequencing reads into one continuous genome or into several contigs. It is indifferent which type of short sequence reads were used to produce the genome/contigs.

Name	Size	Progress	Status
1051363_Sample1_salmonella-typing.ngsservice.processed.R1.fastq.gz	153.62 MB		
1051363_Sample1_salmonella-typing.ngsservice.processed.R2.fastq.gz	155.86 MB		

1. Once the results are ready, click on the link sent to your email
2. View your data and look at quality metrics to assign identification, the MLST will give you the allele number and also state the ST if the allelic profile is recognized.
3. However, to assign a serotype based on the ST, you will need to use another programme such as EnteroBase, please see the section A8.3.2 on how to search by ST for identification. In this example, the result is *Salmonella* Typhi.

FIGURE
A8.25

Screen capture of MLST data using sequence type.

MLST-2.0 Server - Results

mlst Profile: *senterica*

Organism: *Salmonella enterica*

Sequence Type: 2

Locus	Identity	Coverage	Alignment Length	Allele Length	Gaps	Allele
aroC	100	100	501	501	0	aroC_1
dnaN	100	100	501	501	0	dnaN_1
hemD	100	100	432	432	0	hemD_2
hisD	100	100	501	501	0	hisD_1
purE	100	100	399	399	0	purE_1
sucA	100	100	501	501	0	sucA_1
thrA	100	100	501	501	0	thrA_5

[extended output](#)

Input Files: 1051363_Sample1_salmonella-typing.ngsservice.processed.R1.fastq 1051363_Sample1_salmonella-typing.ngsservice.processed.R2.fastq

A8.7

Identification of *Salmonella* serotypes

A8.7.1 SeqSero

SeqSero predicts the *Salmonella* serotype of either the pre-assembled or raw read sequence data provided to the service.

1. Select SeqSero under the 'Typing' section on the services page.
2. Select the type of data input.
3. Browse for your files by clicking on the Isolate File Icon.
4. Press upload once you have added your files.
5. Enter your email address, the link to your results will be sent once processed. Remember to check your junk mail.

FIGURE
A8.26

Screen capture of an example of SeqSero analysis.

SeqSero 1.2

SeqSero predicts the *Salmonella* serotype of either the pre-assembled or raw read sequence data provided to the service.

Important: SeqSero only accepts the following file extensions: `.fasta`, `.fna`, `.fsa`, `.fq`, `.fastq`, `.fq.gz` and the additional `.gz` extension if the file has been compressed by gzip (e.g. `.fq.gz`).

Note: This service is hosted by CGE but all credit and scientific questions should be given to the original authors from [Deng Lab \(SeqSero\)](#).

Data type
Data can either be pre-assembled or raw read data.
If you are uploading read data please specify if it is paired-end or single-end.
Paired-end reads (single file) ▼

⌵ Isolate File

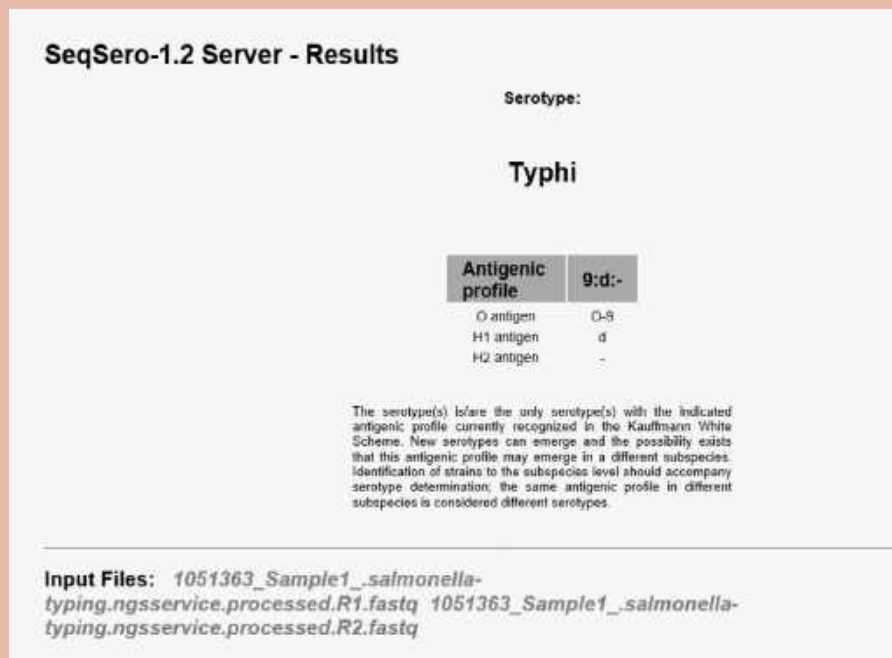
Name	Size	Progress	Status
1051363_sample1_salmonella-typing.ngsservice.processed.R1.fastq.gz	153.62 MB		
1051363_sample1_salmonella-typing.ngsservice.processed.R2.fastq.gz	155.86 MB		

⬆ Upload ⬆ Remove

1. Once the results are ready, click on the link sent to your email.
2. View your data and look at the somatic and flagella antigens, there will also be an interpretation of the serotype. In this example, the result is *Salmonella* Typhi.

FIGURE
A8.27

Example of the results ran using SeqSero-1.2.



A8.8 Identification of *Salmonella* pathogenicity islands

A8.8.1 SPIFinder

SPIFinder identifies *Salmonella* pathogenicity islands.

1. Select SPIFinder under the Phenotyping section on the services page.
2. Select the *Salmonella* pathogenic island database.
3. Select the minimum identity and coverage, default is as 95% and 60%, respectively.
4. Select the type of your reads.
5. Browse for your files by clicking on the Isolate File Icon.
6. Press upload once you have added your files.
7. Enter your email address, the link to your results will be sent once processed. Remember to check your junk mail.

FIGURE
A8.28 Screen capture example of using the SPIFinder for *Salmonella* pathogenicity islands.

SPIFinder 2.0

SPIFinder identifies *Salmonella* Pathogenicity Islands in total or partial sequenced isolates of bacteria.
Fasta file with text sequence: [Test sequence](#)

Select database
Select multiple items, with Ctrl-Click (or Cmd-Click on Mac)
Salmonella Pathogenic Island

Select threshold for minimum % identity
95 %

Select minimum % coverage
60 %

Select type of your reads
Only data from one single isolate should be uploaded. If raw sequencing reads are uploaded KMA will be used for mapping. KMA supports the following sequencing platforms: Illumina, Ion Torrent, Roche454, SOLiD, Oxford Nanopore, and PacBio.
Raw Sequencing Reads (fastq)

*Please note that "Assembled Genomes/Contigs" should be selected if you have already assembled your short sequencing reads into one continuous genome or into several contigs. "Assembled Genomes/Contigs" is defined as one or several contigs in one FASTA file (one entry per contig). It is indifferent which type of short sequence reads were used to produce the genome/contigs.

Isolate File

Name	Size	Progress	Status
1051363_Sample1_salmonella-typing.ngsservice.processed.R1.fastq.gz	163.62 MB		
1051363_Sample1_salmonella-typing.ngsservice.processed.R2.fastq.gz	155.86 MB		

Upload **Remove**

8. Once the results are ready, click on the link sent to your email.
9. View your data and look at quality metrics to assign identification. You will get a line list of the different SPI found in that strain and you can access the accession number for comparison analysis.

FIGURE
A8.29

Screen capture of results from the SPIFinder *Salmonella* pathogenic island.

Detection: *Salmonella* Pathogenic Island

Salmonella Pathogenic Island									
SPI	Identity	Query / Template length	Contig	Position in contig	Organism	Insertion Site	Category Function	Note	Accession number
SPI-1	96.02	249 / 251	NA	NA_NA	Salmonella enterica-Typhimurium-OEHS13		27		JX252841
SPI-1	99.99	41891 / 41891	NA	NA_NA	Salmonella enterica-Typhi-CT10	thiA-mutS	5	gi16758993.2858736-2900506 - go to http://www.ncbi.nlm.nih.gov/ncbi/about_paidb.php for link to partial flatfile	NC_003198
SPI-10	99.98	32941 / 32934	NA	NA_NA	Salmonella enterica-Typhi-CT10	IRNA-leu	6	gi16758993.4693005-4716538 - go to http://www.ncbi.nlm.nih.gov/ncbi/about_paidb.php for link to partial flatfile	NC_003198
SPI-12	98.94	11078 / 11075	NA	NA_NA	Salmonella enterica-Choleraesuis-SC_B67	IRNA-pro	8	gi62178570.2354604-2365678 - go to http://www.ncbi.nlm.nih.gov/ncbi/about_paidb.php for link to partial flatfile	NC_006806
SPI-2	99.97	41811 / 41810	NA	NA_NA	Salmonella enterica-Typhi-Ty2	IRNA-val	14	gi29140543.1314607-1356216 - go to http://www.ncbi.nlm.nih.gov/ncbi/about_paidb.php for link to partial flatfile	NC_008531
SPI-3	99.05	738 / 738	NA	NA_NA	Salmonella enterica-Typhimurium-14028s	IRNA_selC	15	N.B. This record is part of a PAI which includes several records: AF106566.Y13894	AJ000509
SPI-3	99.98	16939 / 16941	NA	NA_NA	Salmonella enterica-Typhi-CT10	IRNA-pro	15	gi16758993.3883613-3900593 - go to http://www.ncbi.nlm.nih.gov/ncbi/about_paidb.php for link to partial flatfile	NC_003198
SPI-4	99.99	23390 / 23391	NA	NA_NA	Salmonella enterica-Typhi-CT10	ssb	17	gi16758993.4322993-4346383 - go to http://www.ncbi.nlm.nih.gov/ncbi/about_paidb.php for link to partial flatfile	NC_003198
SPI-5	99.95	7496 / 7496	NA	NA_NA	Salmonella enterica-Typhi-CT10	IRNA-ser	18	gi16758993.1085068-1092563 - go to http://www.ncbi.nlm.nih.gov/ncbi/about_paidb.php for link to partial flatfile	NC_003198
SPI-6	99.99	58667 / 58666	NA	NA_NA	Salmonella enterica-Typhi-CT10	IRNA-asp	19	gi16758993.302092.360767 - go to http://www.ncbi.nlm.nih.gov/ncbi/about_paidb.php for link to partial flatfile	NC_003198
SPI-7	99.37	133638 / 133638	NA	NA_NA	Salmonella enterica-Typhi-CT10	IRNA-phe	20	gi16758993.4409511-4543148 - go to http://www.ncbi.nlm.nih.gov/ncbi/about_paidb.php for link to partial flatfile	NC_003198
SPI-8	99.99	6885 / 6885	NA	NA_NA	Salmonella enterica-Typhi-CT10	IRNA-phe	21	gi16758993.3132530-3139414 - go to http://www.ncbi.nlm.nih.gov/ncbi/about_paidb.php for link to partial flatfile	NC_003198
SPI-9	99.97	15696 / 15696	NA	NA_NA	Salmonella Typhi-CT10	Not_published	22	gi16758993.2743495-2759190 - go to http://www.ncbi.nlm.nih.gov/ncbi/about_paidb.php for link to partial flatfile	NC_003198

A8.9

Identification of antimicrobial resistance determinants

A8.9.1 ResFinder

Identification of acquired antimicrobial resistance genes using ResFinder

1. Select ResFinder under the Phenotyping section on the services page.
2. Select for mutations of interest, such as chromosomal mutations or acquired resistance.
3. Select the antimicrobial panel of interest for acquired resistance, default is all.
4. Select the threshold for %ID and minimum length, default is as 90% and 60%, respectively.
5. Select the *Salmonella* species.
6. Select the type of your reads.
7. Browse for your files by clicking on the Isolate File Icon.
8. Press upload once you have added your files.
9. Enter your email address, the link to your results will be sent once processed. Remember to check your junk mail.

FIGURE
A8.30

Screen capture of the ResFinder database.

ResFinder 4.1

ResFinder identifies acquired genes and/or finds chromosomal mutations mediating antimicrobial resistance in total or partial DNA sequence of bacteria.

The database is curated by: Frank Møller Aarestrup (click to contact)

Updates

ResFinder and PointFinder software: (2020-10-21)
 ResFinder database: (2020-12-01)
 PointFinder database: (2019-07-02)

Chromosomal point mutations ☒

Select threshold for %ID
 90 %

Select minimum length
 60 %

☐ Show unknown mutations, not found in the database

Acquired antimicrobial resistance genes ☒

Select Antimicrobial configuration
Select multiple items with Ctrl-Click (or Cmd-Click on Mac) – or default all databases is selected.

Aminoglycoside
 Beta-lactam
 Colistin
 Fluoroquinolone
 Fusidomycin
 Fusidic Acid

Select threshold for %ID
 90 %

Select minimum length
 60 %

Acquired disinfectant resistance genes ☐

Select species
Salmonella spp. 
Chromosomal point mutation database exists

Select type of your reads
Illumina - paired end reads 

If you get an "Access forbidden: Error 403". Make sure the start of the web address is https and not just http. Fix it by clicking [here](#).

 Isolate File

Name	Size	Progress	Status
1051363_Sample1_salmonella-typing.ngsservice.processed.R1.fastq.gz	153.62 MB		
1051363_Sample1_salmonella-typing.ngsservice.processed.R2.fastq.gz	155.86 MB		

 Upload  Remove

Confidentiality:
The sequences are kept confidential and will be deleted after 48 hours.

10. Once the results are ready, click on the link sent to your email.
11. View your data to analyse resistance, you will get a line list of the resistance markers found, what antimicrobial agents they predict resistance to and where the mutations occur. Unknown mutations should be validated phenotypically.

FIGURE
A8.31

Screen capture of the results from the ResFinder platform.

Antimicrobial		Class	WG3-predicted phenotype	Genetic background	
amikacin		aminoglycoside	Resistant	aac(6)-Iaa (aac(6)-Iaa_NC_003197)	
tigecycline		tetracycline	No resistance		
tobramycin		aminoglycoside	Resistant	aac(6)-Iaa (aac(6)-Iaa_NC_003197)	
cefepime		beta-lactam	No resistance		
chloramphenicol		phenicol	No resistance		
cefoxitin		beta-lactam	No resistance		
ampicillin		beta-lactam	No resistance		
ampicillin+clavulanic acid		beta-lactam	No resistance		

Aminoglycoside									
Resistance gene	Identity	Alignment Length/Gene Length	Position in reference	Contig or Depth	Position in contig	Phenotype	PMID	Accession no.	Notes
aac(6)-Iaa	97.26	438/438	1,439	76.2	None, None	amikacin,tobramycin	11577609	NC_003197	Chromosomal, Cryptic gene in Salmonella sp.

No class defined					
Mutation	Nucleotide change	Amino acid change	Phenotype	PMID	Notes
parC, p.A620T	gcg -> acg	a -> t	Unknown phenotype	-	Phenotype not found in database
pmrB, p.A111T	gcg -> acg	a -> t	Unknown phenotype	-	Phenotype not found in database

A8.10 Analysis of batch data

The examples given above do not require registration, after 48 hours, the data will no longer be accessible. In order to keep the data for future analysis, you will need to upload the batch uploader, which you can access by clicking on your profile name once you are logged in. The Batch Upload section will direct your strains through multiple pipelines including ContigAnalyzer, KmerFinder, MLST, ResFinder, VirulenceFinder, PlasmidFinder and pMLST.

FIGURE
A8.32

Screen capture of the Batch Upload analysis.

Bacterial Analysis Pipeline - Batch Upload

The CGE Bacterial Analysis Pipeline executes a workflow of services with predefined parameters and stores the submitted data and result in the database at the user's disposal.

Disclaimer

The pipeline is not maintained and updated with the newest CGE services and databases. A new pipeline is under development. Meanwhile if you need to run the newest services you will have to use our single services.

Services included in the Bacterial Analysis Pipeline:

- ContigAnalyzer-1.0
- KmerFinder-2.1
- MLST-1.6
- ResFinder-2.1
- VirulenceFinder-1.2
- PlasmidFinder-1.2
- pMLST-1.4

1. Download 'Metadata Template'.
2. Fill out template, look at the second tab for guidance, please note that the metadata file must be filled out and is sensitive to capitals. It is recommended to copy the exact wording from the guidance tab and paste as required.
3. Upload metadata file, if there are any issues with the format, this will come up as an error and guidance will be given to update the file.
4. Select the files, they must match the file names on the metadata template including the file type, e.g., fasta.
5. Submit the data.

FIGURE
A8.33

Screen capture of the step by step process for the batch analysis.

STEP 1: [Download Metadata Template](#) Template

STEP 2: Fill out template

STEP 3: [Remove Metadata File](#)

STEP 4: [Select Files](#)

STEP 5: [Submit](#)

Supported browsers

Browser	Windows 7	Windows 8	Windows 10	Ubuntu
≥ V 8.0				✓
V 10.0		✓	✓	
edge		✓	✓	
≥ V 36.0		✓	✓	✓
≥ V 41.0	✓	✓	✓	✓

Progress Overview

Name	Size	Progress	Status
1051363_Sample1_contigs.fasta	4.60 MB		

- You can view your results by clicking on your name after signing in on the top right corner.
- Select Sample manager from the drop-down list. Then select your sample of interest and analyse. Below is an example of *Salmonella* Typhi. This is also a useful tool if you want to analyse multiple strains and come back to them for analysis in the future.

FIGURE
A8.34

Screen capture of the bacterial analysis report.

Home Services Instructions Output Article abstract

Bacterial Analysis Summary Report

Pipeline Version: 1.1
Submission Date: 2020-12-29
Sample Name: 1051363_Sample1_contigs.fasta

Contigs Analysis

Assembly File	No. of contigs	No. of bases	N50
1051363_Sample1_contigs.fasta.gz	140	4744088	144747

Taxonomy

Predicted Lineage:
cellular organisms; Bacteria; Proteobacteria; Gammaproteobacteria; Enterobacteriales; Enterobacteriaceae; Salmonella; Salmonella enterica; Salmonella enterica subsp. enterica; Salmonella enterica subsp. enterica serovar Typhi; Salmonella enterica subsp. enterica serovar Typhi str. CT18

Predicted Species: *Salmonella enterica*
Closest Template: *Salmonella enterica* subsp. *enterica* serovar Typhi str. CT18
Template Coverage: 0.99

MLST Scheme[ST]: enterica [ST-2]

Plasmid[pMLST]

Resistance Genes

A8.11 Assembling raw data files

A8.11.1 Assembling data

Sometime the batch uploader or certain pipelines require you to upload contigs (joined assemblies) rather than raw data files. This can be done using the SPAdes programme.

A8.11.2 SPAdes

SPAdes assembles raw fasta files to produce assembled contigs for further analysis.

1. You need to go to the link – <https://home.cc.umanitoba.ca/~psgends/doc/spades/manual.html#sec1>
2. Select the type of your reads to assemble.
3. Browse for your files by clicking on the Isolate File Icon.
4. Press upload once you have added your files.
5. Enter your email address, the link to your results will be sent once processed. Remember to check your junk mail.

FIGURE
A8.35 Screen capture of the SPAdes 3.9 platform.

The screenshot displays the SPAdes 3.9 web interface. At the top, it says 'SPAdes 3.9' and provides a link to the version history. Below this, there is a 'Sequencing Platform' dropdown menu set to 'Illumina - paired end reads'. A 'Choose careful mode' checkbox is checked, with a note explaining its purpose. A file upload section shows two files being processed: '1061063_Sample1_salmonella-typing.ngsservice.processed.R1.fastq.gz' (153.62 MB) and '1061063_Sample1_salmonella-typing.ngsservice.processed.R2.fastq.gz' (155.86 MB). Each file has a progress bar and a status column. At the bottom, there are 'Upload' and 'Remove' buttons, and a note about the distribution of the SPAdes-3.9 assembler.

Name	Size	Progress	Status
1061063_Sample1_salmonella-typing.ngsservice.processed.R1.fastq.gz	153.62 MB		
1061063_Sample1_salmonella-typing.ngsservice.processed.R2.fastq.gz	155.86 MB		

6. Once the results are ready, click on the link sent to your email.
7. Click on the contigs button and save contig for further analysis.

FIGURE A8.36

Example of the results from the SPAdes analysis.

SPAdes-3.9 Server - Results

Input Files: 1051363_Sample1_salmonella-typing.ngsservice.processed.R1.fastq 1051363_Sample1_salmonella-typing.ngsservice.processed.R2.fastq

Configs

References for Annex 8

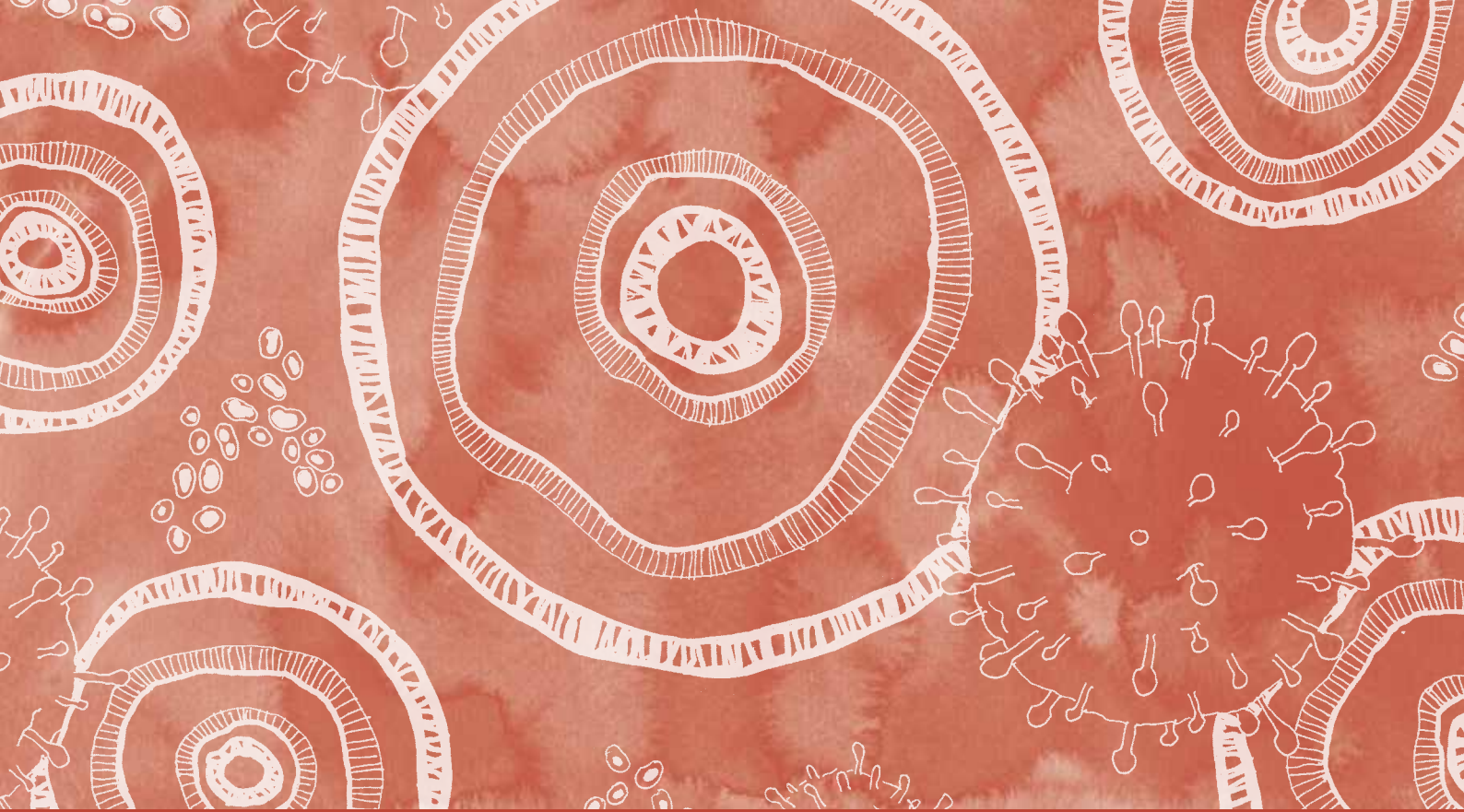
1. Zhou Z, Alikhan NF, Mohamed K, Fan Y; Agama Study Group, Achtman M. The Enterobase user's guide, with case studies on *Salmonella* transmissions, *Yersinia pestis* phylogeny, and *Escherichia* core genomic diversity. *Genome Res.* 2020 Jan;30(1):138-152. doi: 10.1101/gr.251678.119. Epub 2019 Dec 6. PMID: 31809257; PMCID: PMC6961584.
2. Zhou Z, Alikhan NF, Sergeant MJ, Luhmann N, Vaz C, Francisco AP, Carriço JA, Achtman M. GrapeTree: visualization of core genomic relationships among 100,000 bacterial pathogens. *Genome Res.* 2018 Sep;28(9):1395-1404. doi: 10.1101/gr.232397.117. Epub 2018 Jul 26. PMID: 30049790; PMCID: PMC6120633.

Additional reading

There are multiple references for using these tools with a summary below. However, for correct citation please go onto the website and look up the appropriate reference depending on the tools you are using.

1. Alikhan NF, Zhou Z, Sergeant MJ, Achtman M. A genomic overview of the population structure of *Salmonella*. *PLoS Genet.* 2018 Apr 5;14(4):e1007261. doi: 10.1371/journal.pgen.1007261.
2. Hasman H, Saputra D, Sicheritz-Ponten T, Lund O, Svendsen CA, Frimodt-Møller N, Aarestrup FM. Rapid whole-genome sequencing for detection and characterization of microorganisms directly from clinical samples. *J Clin Microbiol.* 2014 Jan;52(1):139-46. doi: 10.1128/JCM.02452-13.
3. Larsen MV, Cosentino S, Lukjancenko O, Saputra D, Rasmussen S, Hasman H, et al. Benchmarking of methods for genomic taxonomy. *J Clin Microbiol.* 2014 May;52(5):1529-39. doi: 10.1128/JCM.02981-13.
4. Larsen MV, Cosentino S, Rasmussen S, Friis C, Hasman H, Marvig RL, et al. Multilocus sequence typing of total-genome-sequenced bacteria. *J Clin Microbiol.* 2012 Apr;50(4):1355-61. doi: 10.1128/JCM.06094-11.
5. Zhang S, Yin Y, Jones MB, Zhang Z, Deatherage Kaiser BL, Dinsmore BA, et al. *Salmonella* serotype determination utilizing high-throughput genome sequencing data. *J Clin Microbiol.* 2015 May;53(5):1685-92. doi: 10.1128/JCM.00323-15.
6. Bartual SG, Seifert H, Hippler C, Luzon MA, Wisplinghoff H, Rodríguez-Valera F. Development of a multilocus sequence typing scheme for characterization of clinical isolates of *Acinetobacter baumannii*. *J Clin Microbiol.* 2005 Sep;43(9):4382-90. doi: 10.1128/JCM.43.9.4382-4390.2005.

7. Griffiths D, Fawley W, Kachrimanidou M, Bowden R, Crook DW, Fung R, et al. Multilocus sequence typing of *Clostridium difficile*. *J Clin Microbiol*. 2010 Mar;48(3):770-8. doi: [10.1128/JCM.01796-09](https://doi.org/10.1128/JCM.01796-09).
8. Lemee L, Dhalluin A, Pestel-Caron M, Lemeland JF, Pons JL. Multilocus sequence typing analysis of human and animal *Clostridium difficile* isolates of various toxigenic types. *J Clin Microbiol*. 2004 Jun;42(6):2609-17. doi: [10.1128/JCM.42.6.2609-2617.2004](https://doi.org/10.1128/JCM.42.6.2609-2617.2004).
9. Wirth T, Falush D, Lan R, Colles F, Mensa P, Wieler LH, et al. Sex and virulence in *Escherichia coli*: an evolutionary perspective. *Mol Microbiol*. 2006 Jun;60(5):1136-51. doi: [10.1111/j.1365-2958.2006.05172.x](https://doi.org/10.1111/j.1365-2958.2006.05172.x).
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